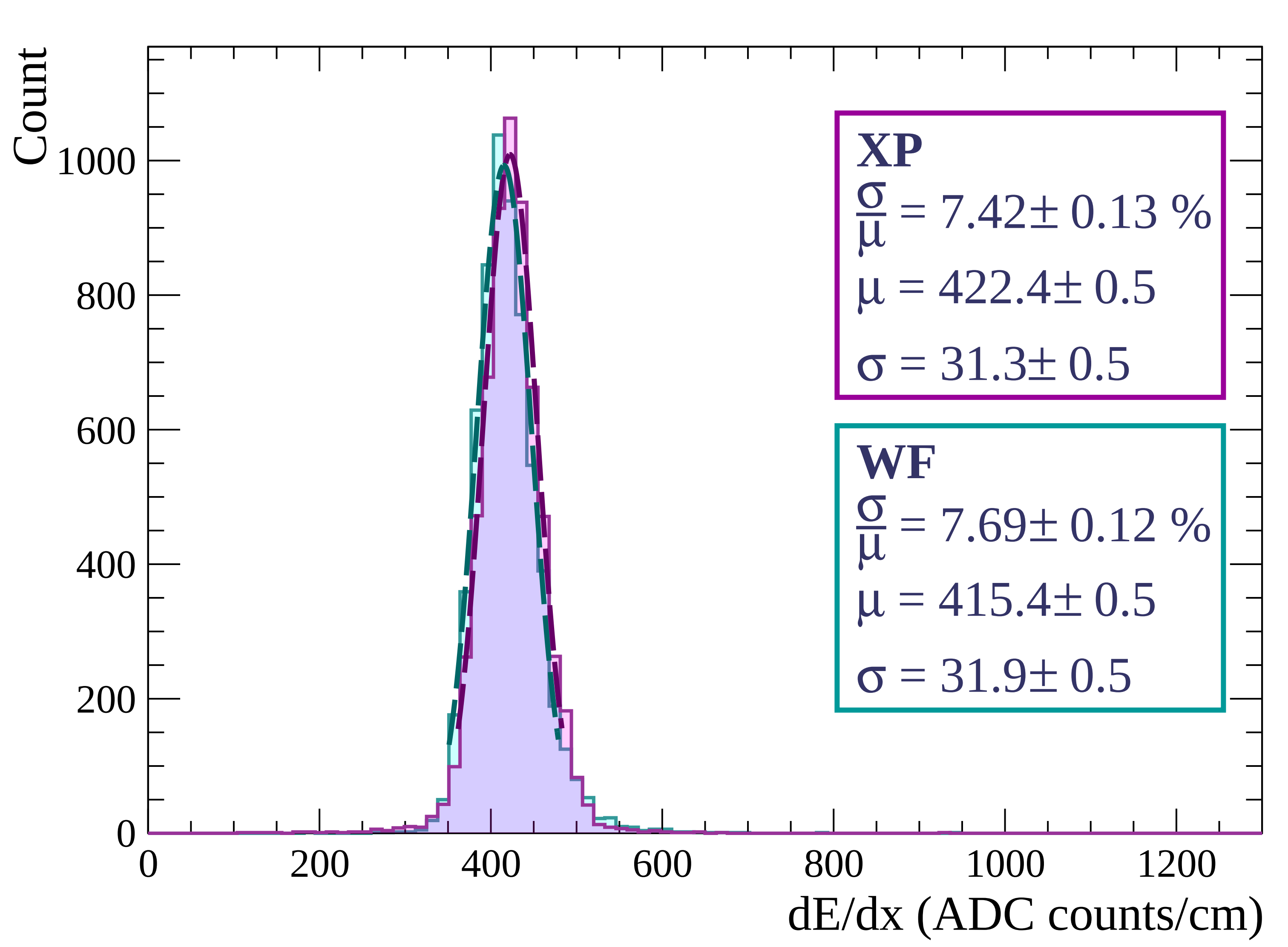
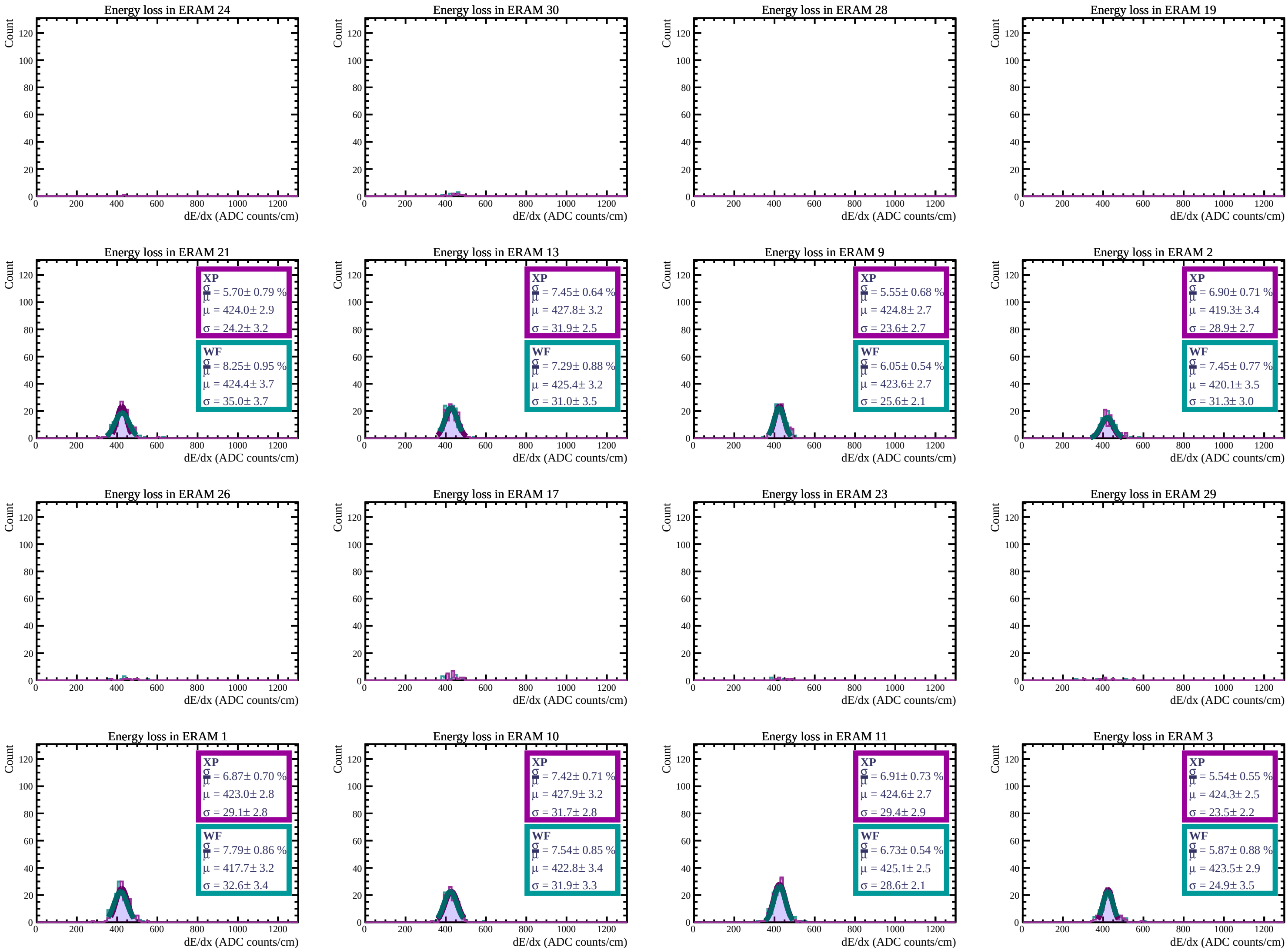
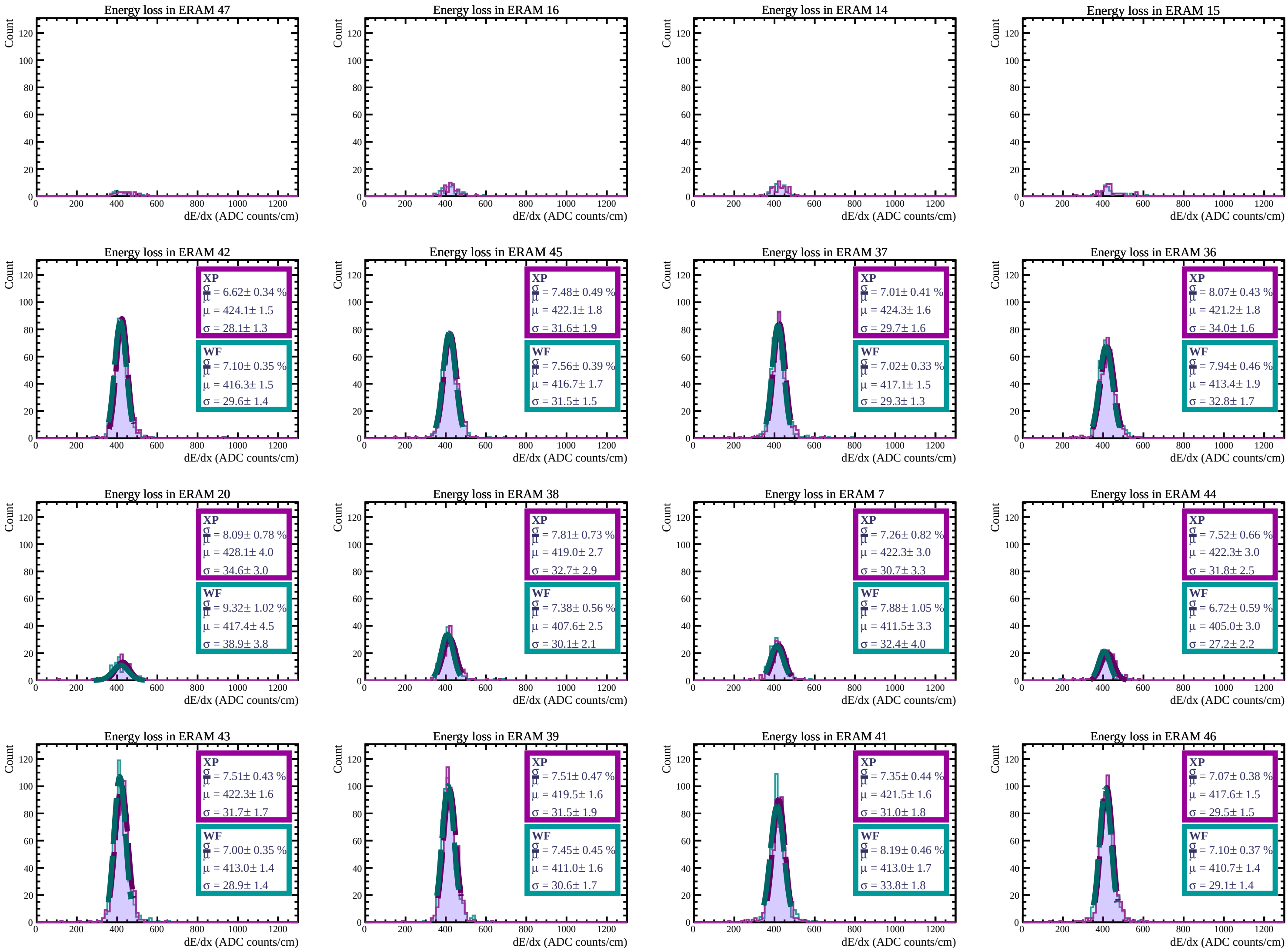
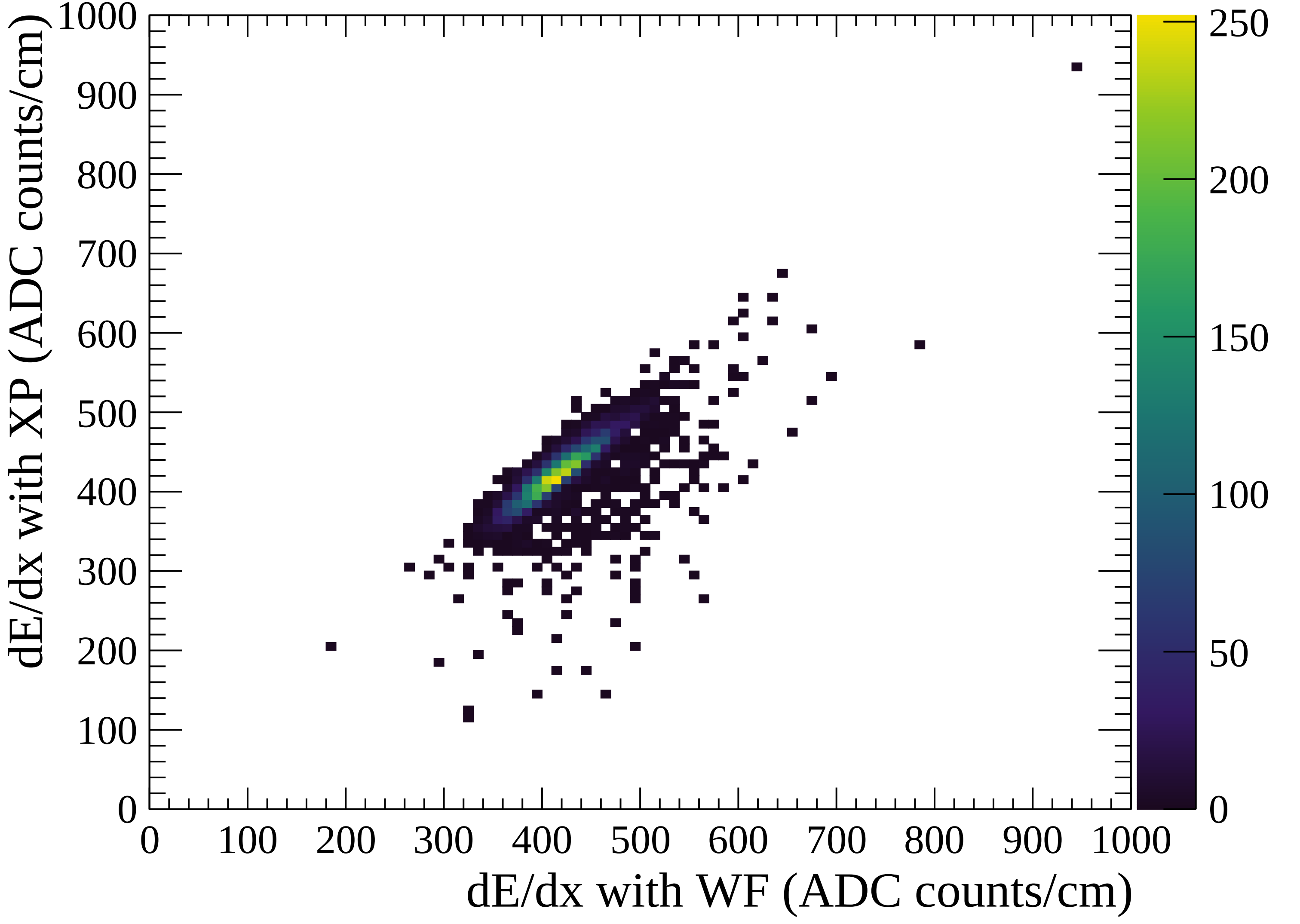


Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 7.42± 0.13 %
μ = 422.4± 0.5
σ = 31.3± 0.5
WF
$\frac{\sigma}{\mu}$ = 7.69± 0.12 %
μ = 415.4± 0.5
σ = 31.9± 0.5

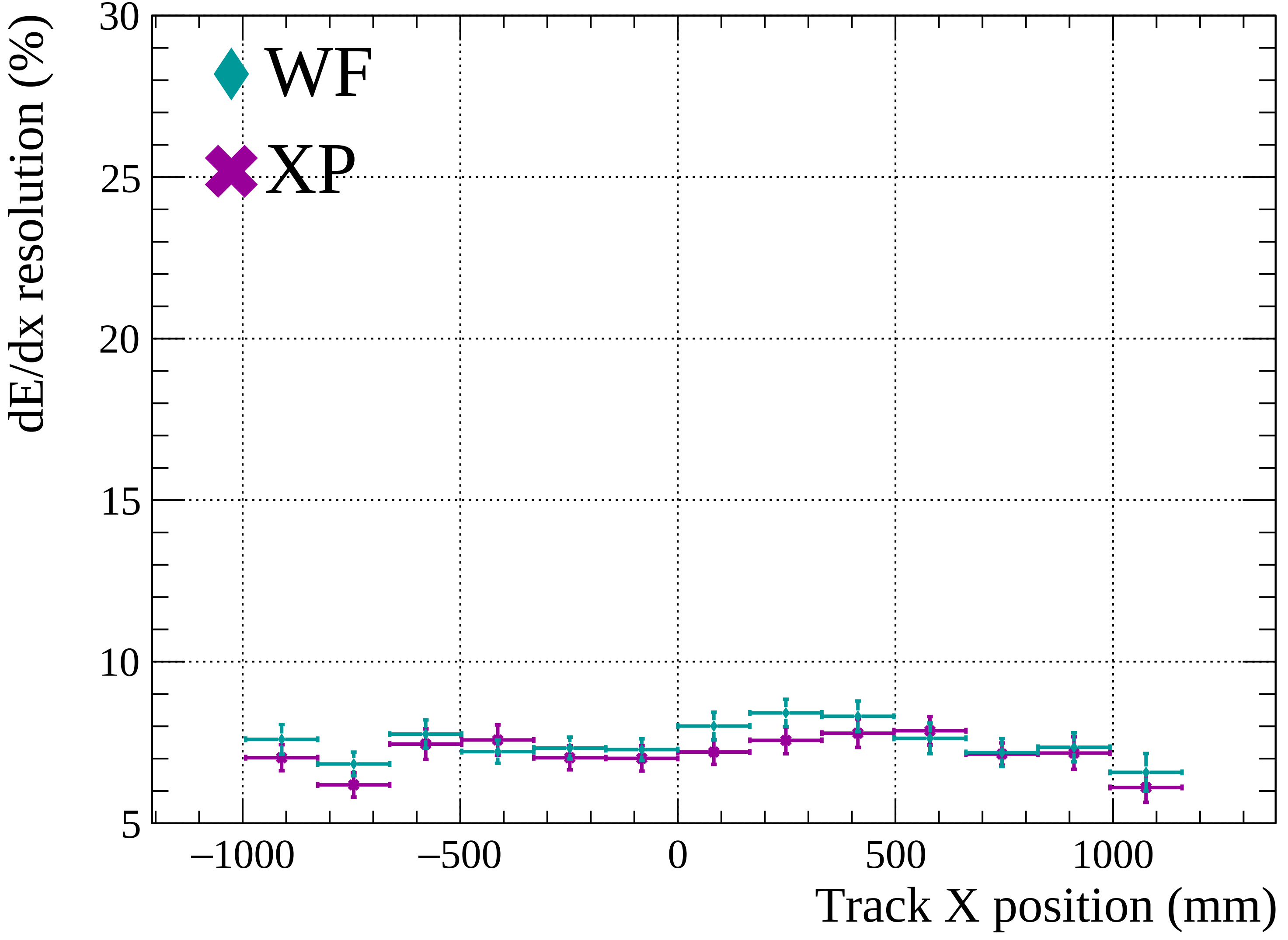


Energy loss in ERAM 24
Energy loss in ERAM 30
Energy loss in ERAM 28
Energy loss in ERAM 19
Energy loss in ERAM 21
XP
σ/μ = 5.70± 0.79 %
μ = 424.0± 2.9
σ = 24.2± 3.2
WF
σ/μ = 8.25± 0.95 %
μ = 424.4± 3.7
σ = 35.0± 3.7
Energy loss in ERAM 13
XP
σ/μ = 7.45± 0.64 %
μ = 427.8± 3.2
σ = 31.9± 2.5
WF
σ/μ = 7.29± 0.88 %
μ = 425.4± 3.2
σ = 31.0± 3.5
Energy loss in ERAM 9
XP
σ/μ = 5.55± 0.68 %
μ = 424.8± 2.7
σ = 23.6± 2.7
WF
σ/μ = 6.05± 0.54 %
μ = 423.6± 2.7
σ = 25.6± 2.1
Energy loss in ERAM 2
XP
σ/μ = 6.90± 0.71 %
μ = 419.3± 3.4
σ = 28.9± 2.7
WF
σ/μ = 7.45± 0.77 %
μ = 420.1± 3.5
σ = 31.3± 3.0
Energy loss in ERAM 26
Energy loss in ERAM 17
Energy loss in ERAM 23
Energy loss in ERAM 29
Energy loss in ERAM 1
XP
σ/μ = 6.87± 0.70 %
μ = 423.0± 2.8
σ = 29.1± 2.8
WF
σ/μ = 7.79± 0.86 %
μ = 417.7± 3.2
σ = 32.6± 3.4
Energy loss in ERAM 10
XP
σ/μ = 7.42± 0.71 %
μ = 427.9± 3.2
σ = 31.7± 2.8
WF
σ/μ = 7.54± 0.85 %
μ = 422.8± 3.4
σ = 31.9± 3.3
Energy loss in ERAM 11
XP
σ/μ = 6.91± 0.73 %
μ = 424.6± 2.7
σ = 29.4± 2.9
WF
σ/μ = 6.73± 0.54 %
μ = 425.1± 2.5
σ = 28.6± 2.1
Energy loss in ERAM 3
XP
σ/μ = 5.54± 0.55 %
μ = 424.3± 2.5
σ = 23.5± 2.2
WF
σ/μ = 5.87± 0.88 %
μ = 423.5± 2.9
σ = 24.9± 3.5
Count
dE/dx (ADC counts/cm)

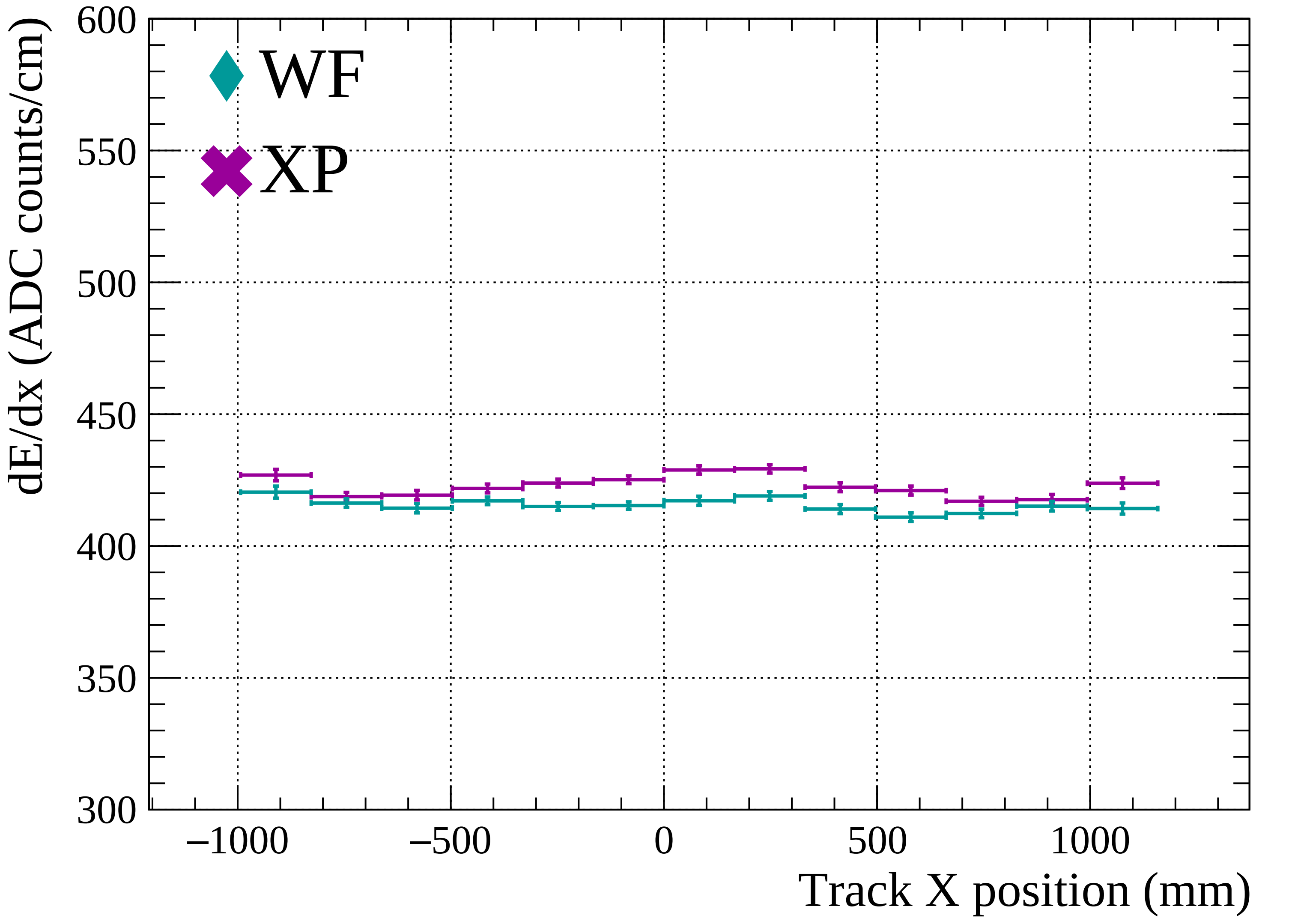

Energy loss in ERAM 47
Energy loss in ERAM 16
Energy loss in ERAM 14
Energy loss in ERAM 15
Energy loss in ERAM 42
XP
σ/μ = 6.62± 0.34 %
μ = 424.1± 1.5
σ = 28.1± 1.3
WF
σ/μ = 7.10± 0.35 %
μ = 416.3± 1.5
σ = 29.6± 1.4
Energy loss in ERAM 45
XP
σ/μ = 7.48± 0.49 %
μ = 422.1± 1.8
σ = 31.6± 1.9
WF
σ/μ = 7.56± 0.39 %
μ = 416.7± 1.7
σ = 31.5± 1.5
Energy loss in ERAM 37
XP
σ/μ = 7.01± 0.41 %
μ = 424.3± 1.6
σ = 29.7± 1.6
WF
σ/μ = 7.02± 0.33 %
μ = 417.1± 1.5
σ = 29.3± 1.3
Energy loss in ERAM 36
XP
σ/μ = 8.07± 0.43 %
μ = 421.2± 1.8
σ = 34.0± 1.6
WF
σ/μ = 7.94± 0.46 %
μ = 413.4± 1.9
σ = 32.8± 1.7
Energy loss in ERAM 20
XP
σ/μ = 8.09± 0.78 %
μ = 428.1± 4.0
σ = 34.6± 3.0
WF
σ/μ = 9.32± 1.02 %
μ = 417.4± 4.5
σ = 38.9± 3.8
Energy loss in ERAM 38
XP
σ/μ = 7.81± 0.73 %
μ = 419.0± 2.7
σ = 32.7± 2.9
WF
σ/μ = 7.38± 0.56 %
μ = 407.6± 2.5
σ = 30.1± 2.1
Energy loss in ERAM 7
XP
σ/μ = 7.26± 0.82 %
μ = 422.3± 3.0
σ = 30.7± 3.3
WF
σ/μ = 7.88± 1.05 %
μ = 411.5± 3.3
σ = 32.4± 4.0
Energy loss in ERAM 44
XP
σ/μ = 7.52± 0.66 %
μ = 422.3± 3.0
σ = 31.8± 2.5
WF
σ/μ = 6.72± 0.59 %
μ = 405.0± 3.0
σ = 27.2± 2.2
Energy loss in ERAM 43
XP
σ/μ = 7.51± 0.43 %
μ = 422.3± 1.6
σ = 31.7± 1.7
WF
σ/μ = 7.00± 0.35 %
μ = 413.0± 1.4
σ = 28.9± 1.4
Energy loss in ERAM 39
XP
σ/μ = 7.51± 0.47 %
μ = 419.5± 1.6
σ = 31.5± 1.9
WF
σ/μ = 7.45± 0.45 %
μ = 411.0± 1.6
σ = 30.6± 1.7
Energy loss in ERAM 41
XP
σ/μ = 7.35± 0.44 %
μ = 421.5± 1.6
σ = 31.0± 1.8
WF
σ/μ = 8.19± 0.46 %
μ = 413.0± 1.7
σ = 33.8± 1.8
Energy loss in ERAM 46
XP
σ/μ = 7.07± 0.38 %
μ = 417.6± 1.5
σ = 29.5± 1.5
WF
σ/μ = 7.10± 0.37 %
μ = 410.7± 1.4
σ = 29.1± 1.4
Count
dE/dx (ADC counts/cm)



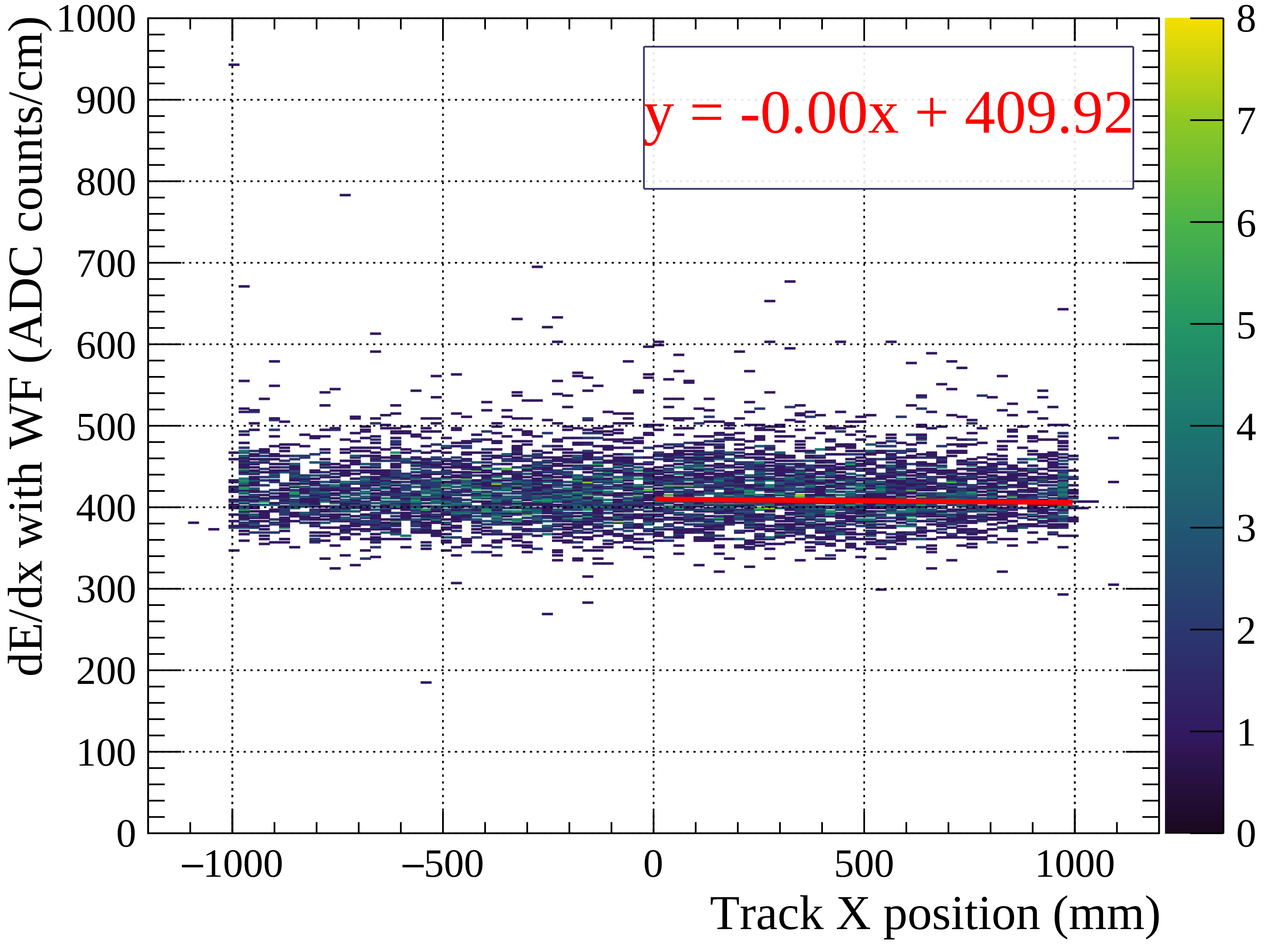
dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
1000
0
50
100
150
200
250



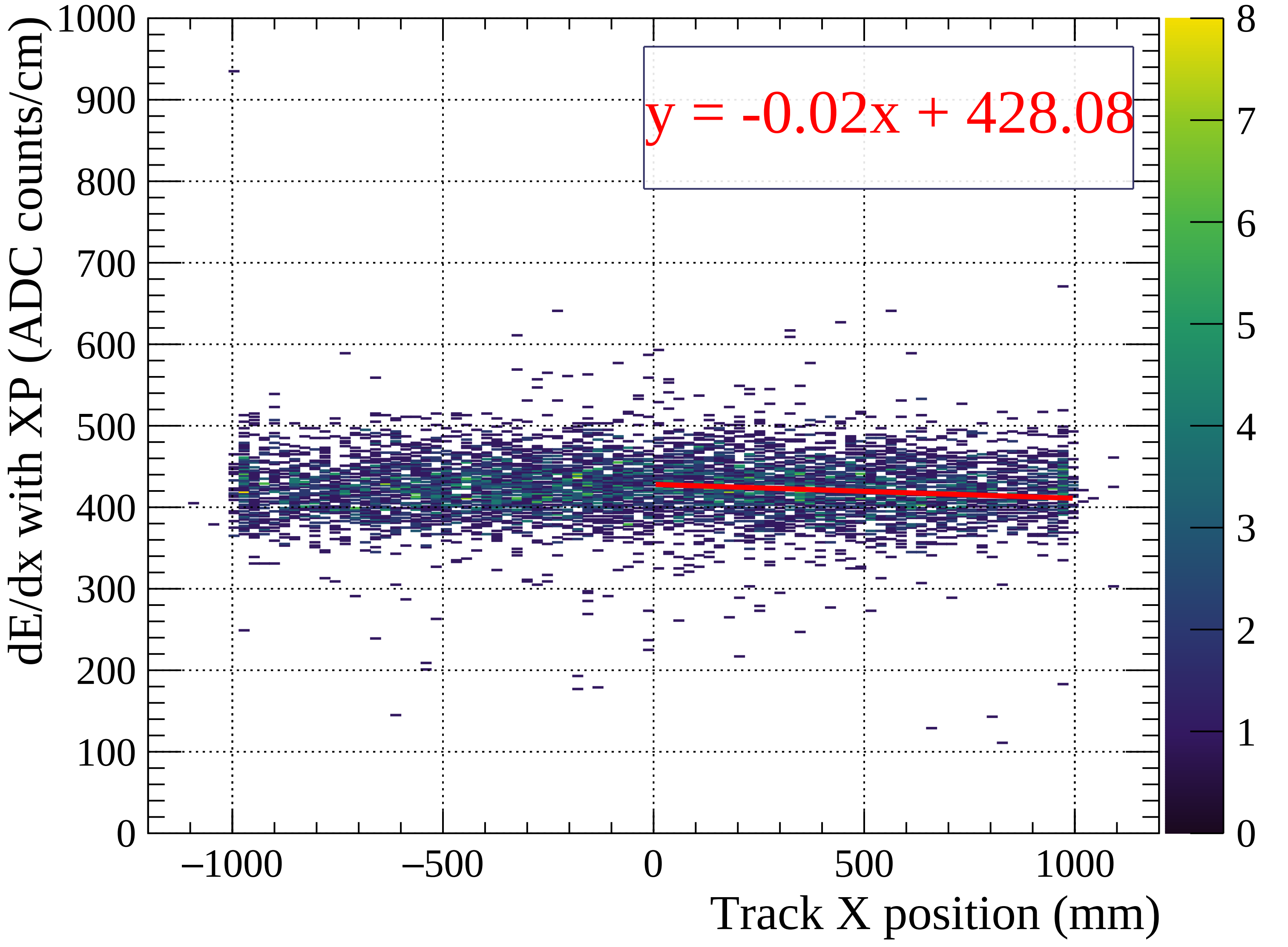
WF
XP
dE/dx resolution (%)
30
25
20
15
10
5
−1000
−500
0
500
1000
Track X position (mm)



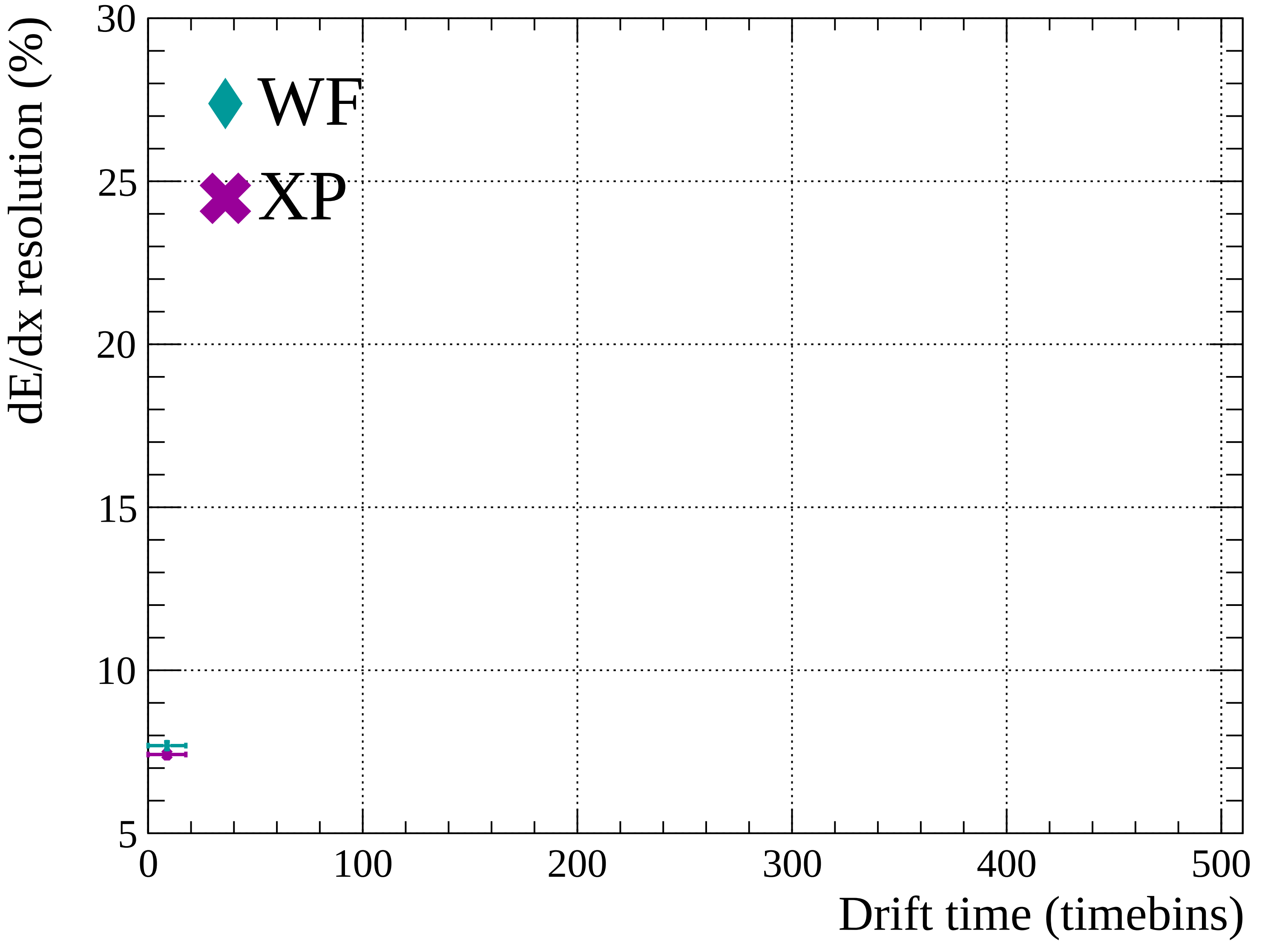
WF
XP
dE/dx (ADC counts/cm)
600
550
500
450
400
350
300
−1000
−500
0
500
1000
Track X position (mm)



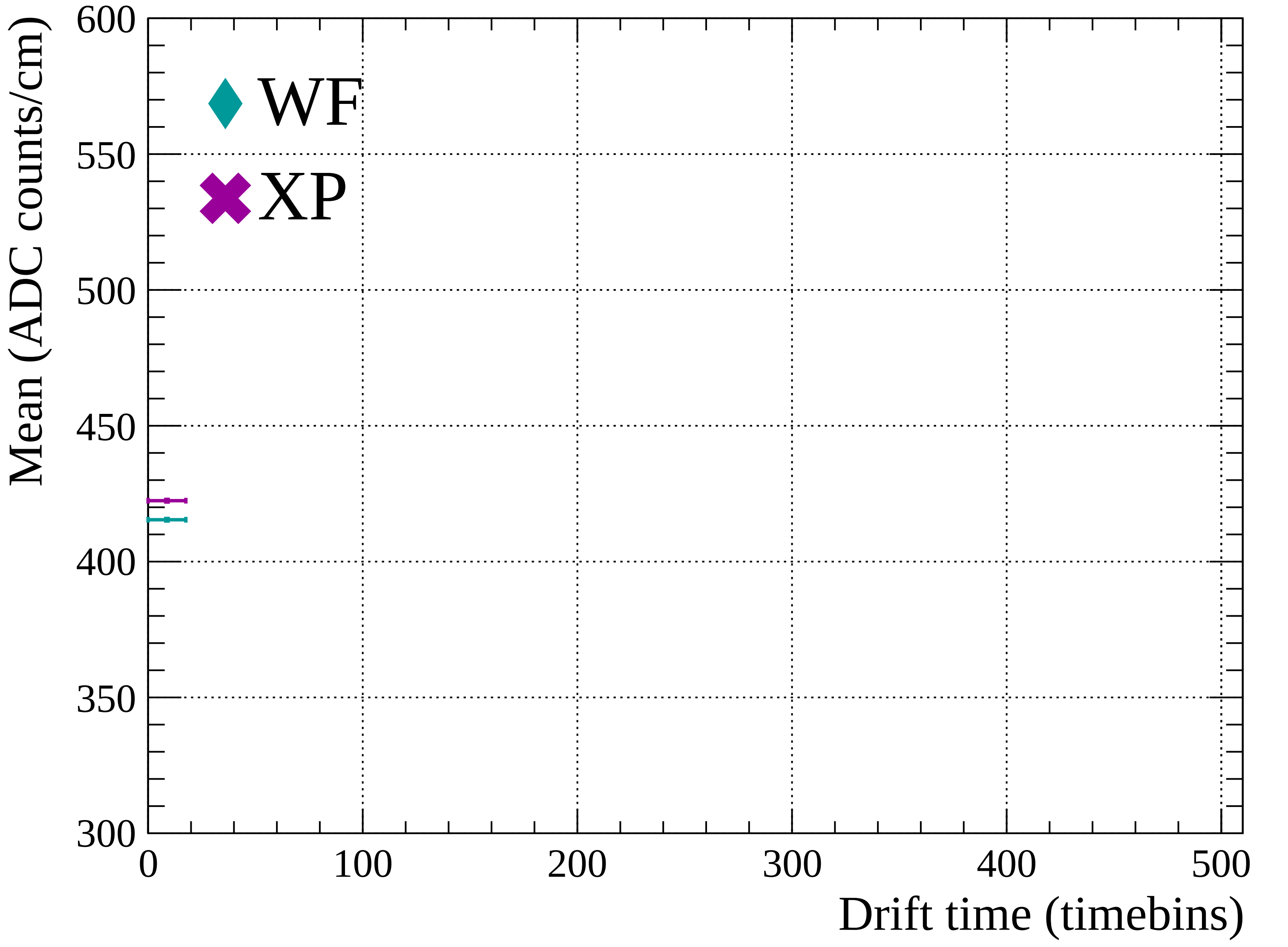
y = -0.00x + 409.92
dE/dx with WF (ADC counts/cm)
Track X position (mm)
1000
900
800
700
600
500
400
300
200
100
0
−1000
−500
0
500
1000
8
7
6
5
4
3
2
1
0



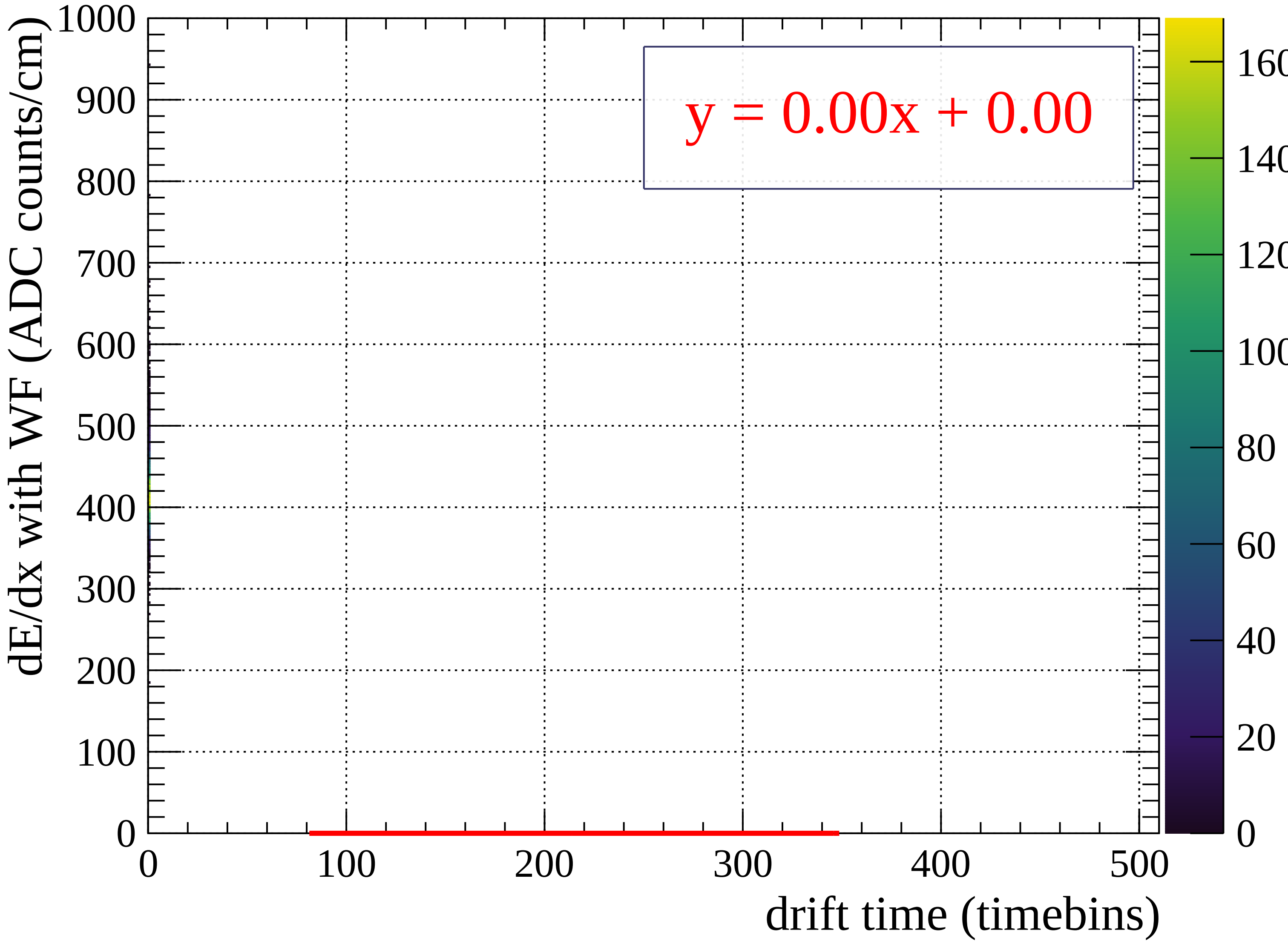
y = -0.02x + 428.08
dE/dx with XP (ADC counts/cm)
Track X position (mm)
0
100
200
300
400
500
600
700
800
900
1000
−1000
−500
0
500
1000
0
1
2
3
4
5
6
7
8



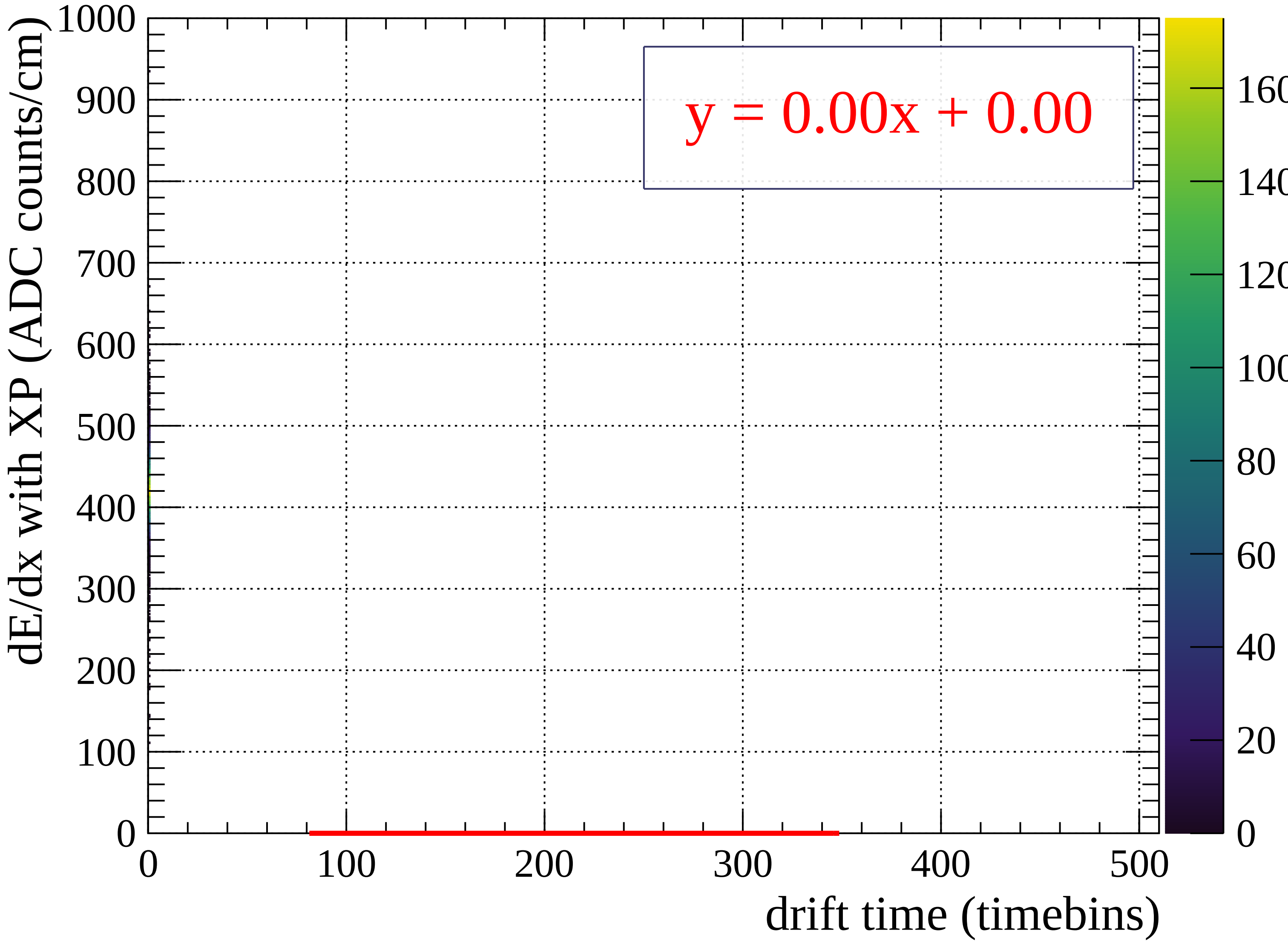
dE/dx resolution (%)
30
25
20
15
10
5
WF
XP
0
100
200
300
400
500
Drift time (timebins)

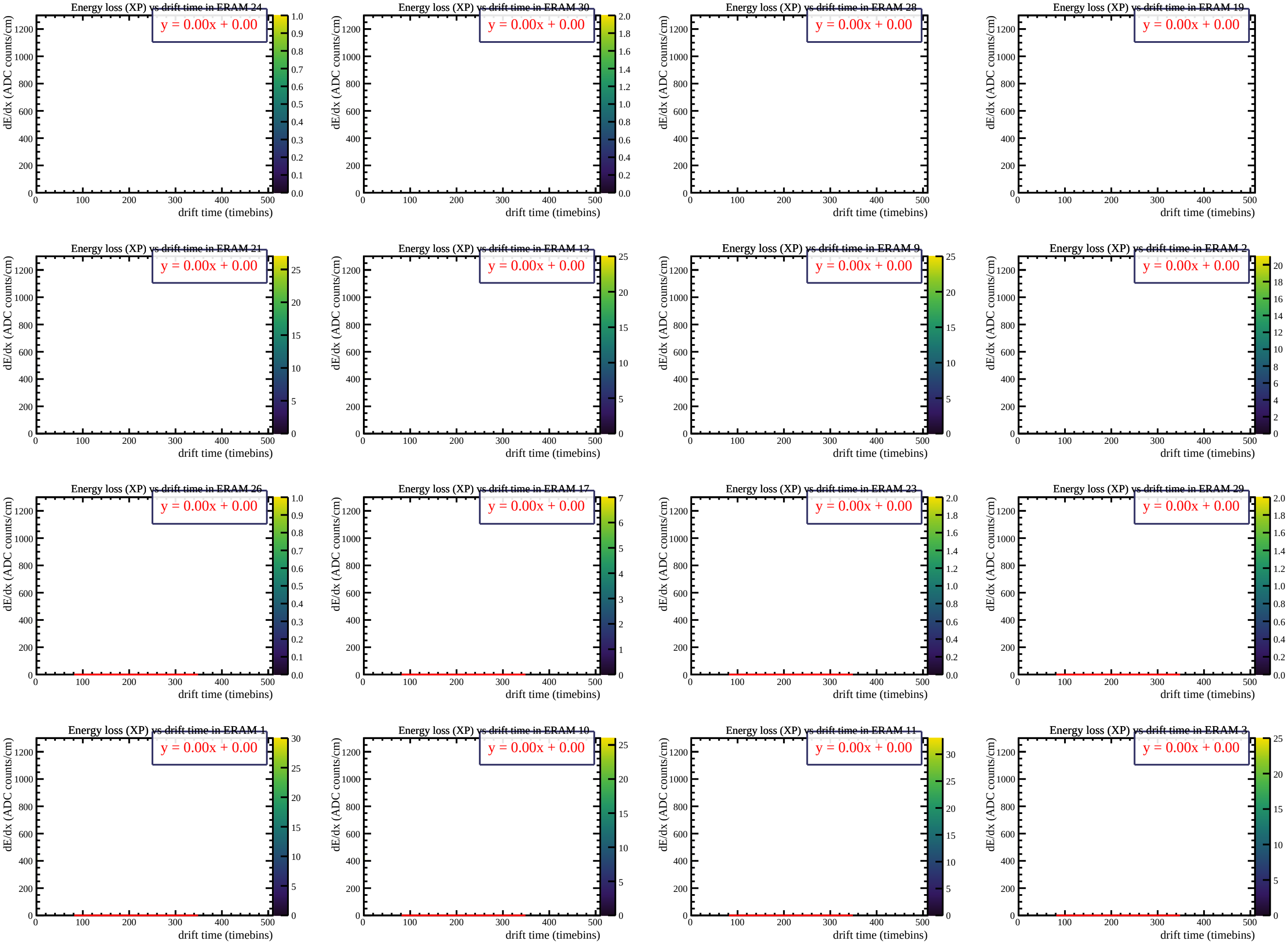


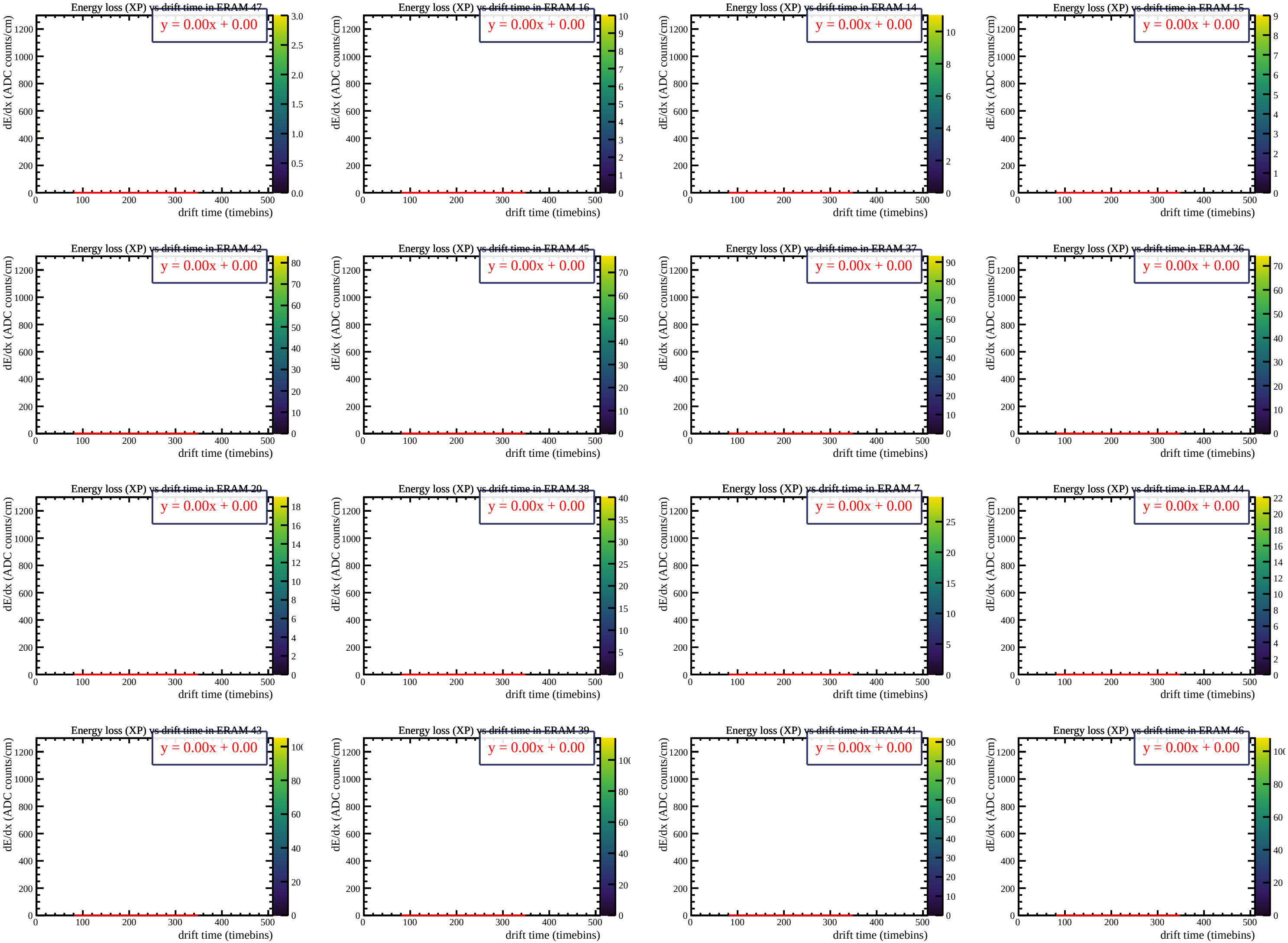
WF
XP
Mean (ADC counts/cm)
Drift time (timebins)
600
550
500
450
400
350
300
0
100
200
300
400
500



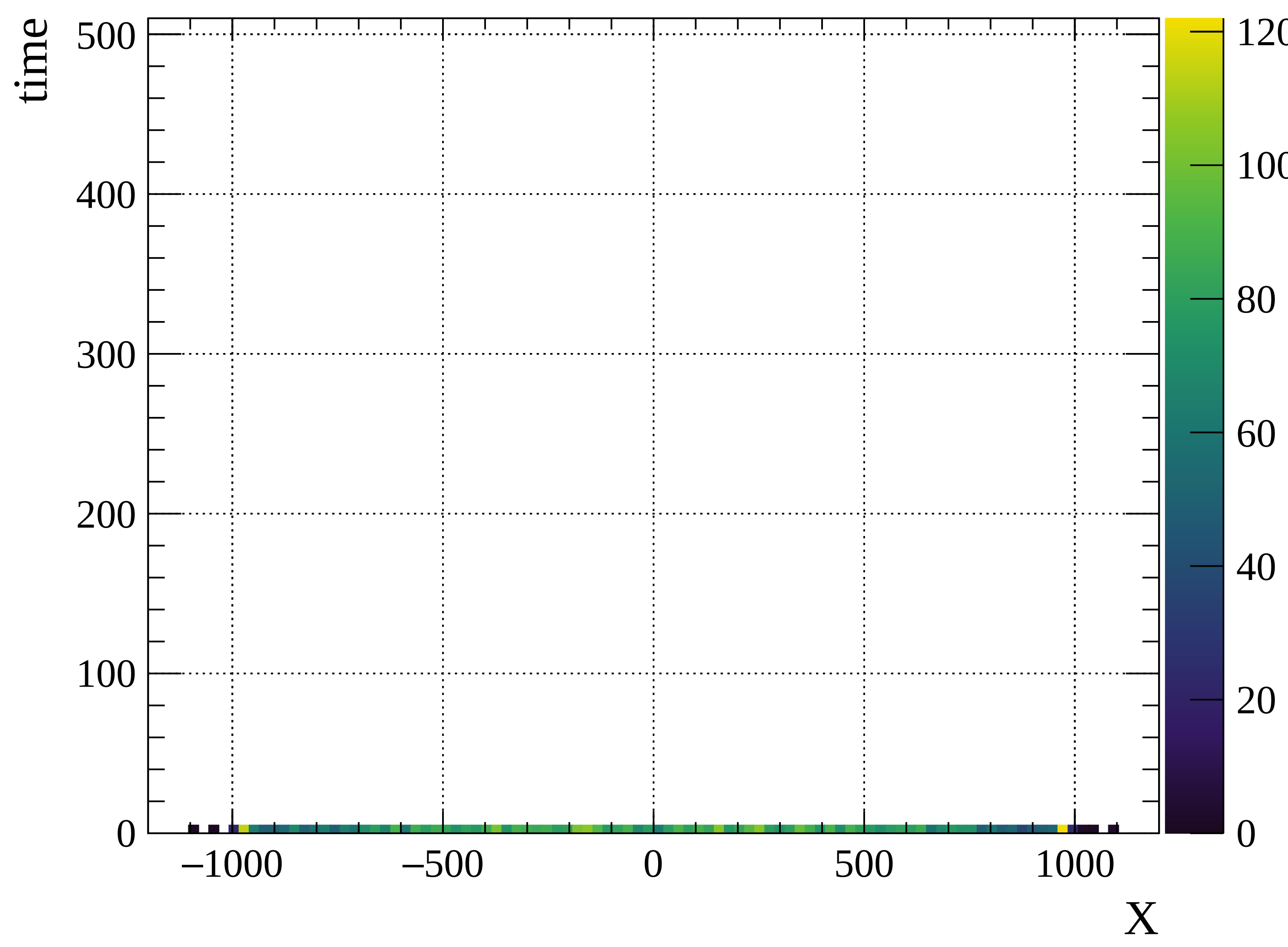
y = 0.00x + 0.00
dE/dx with WF (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
20
40
60
80
100
120
140
160

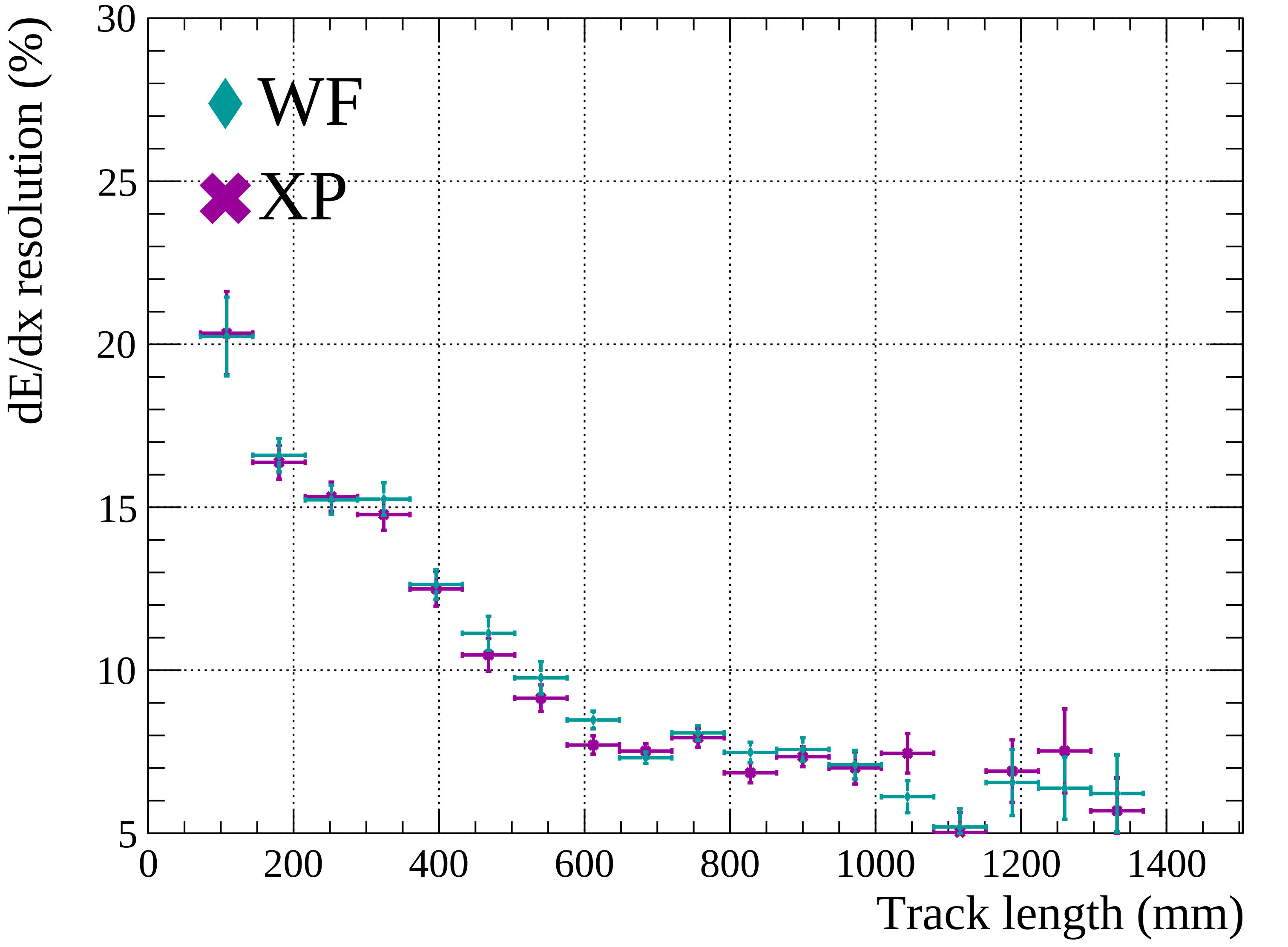

y = 0.00x + 0.00
dE/dx with XP (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
20
40
60
80
100
120
140
160


Energy loss (XP) vs drift time in ERAM 24
Energy loss (XP) vs drift time in ERAM 30
Energy loss (XP) vs drift time in ERAM 28
Energy loss (XP) vs drift time in ERAM 19
Energy loss (XP) vs drift time in ERAM 21
Energy loss (XP) vs drift time in ERAM 13
Energy loss (XP) vs drift time in ERAM 9
Energy loss (XP) vs drift time in ERAM 2
Energy loss (XP) vs drift time in ERAM 26
Energy loss (XP) vs drift time in ERAM 17
Energy loss (XP) vs drift time in ERAM 23
Energy loss (XP) vs drift time in ERAM 29
Energy loss (XP) vs drift time in ERAM 1
Energy loss (XP) vs drift time in ERAM 10
Energy loss (XP) vs drift time in ERAM 11
Energy loss (XP) vs drift time in ERAM 3
y = 0.00x + 0.00
dE/dx (ADC counts/cm)
drift time (timebins)

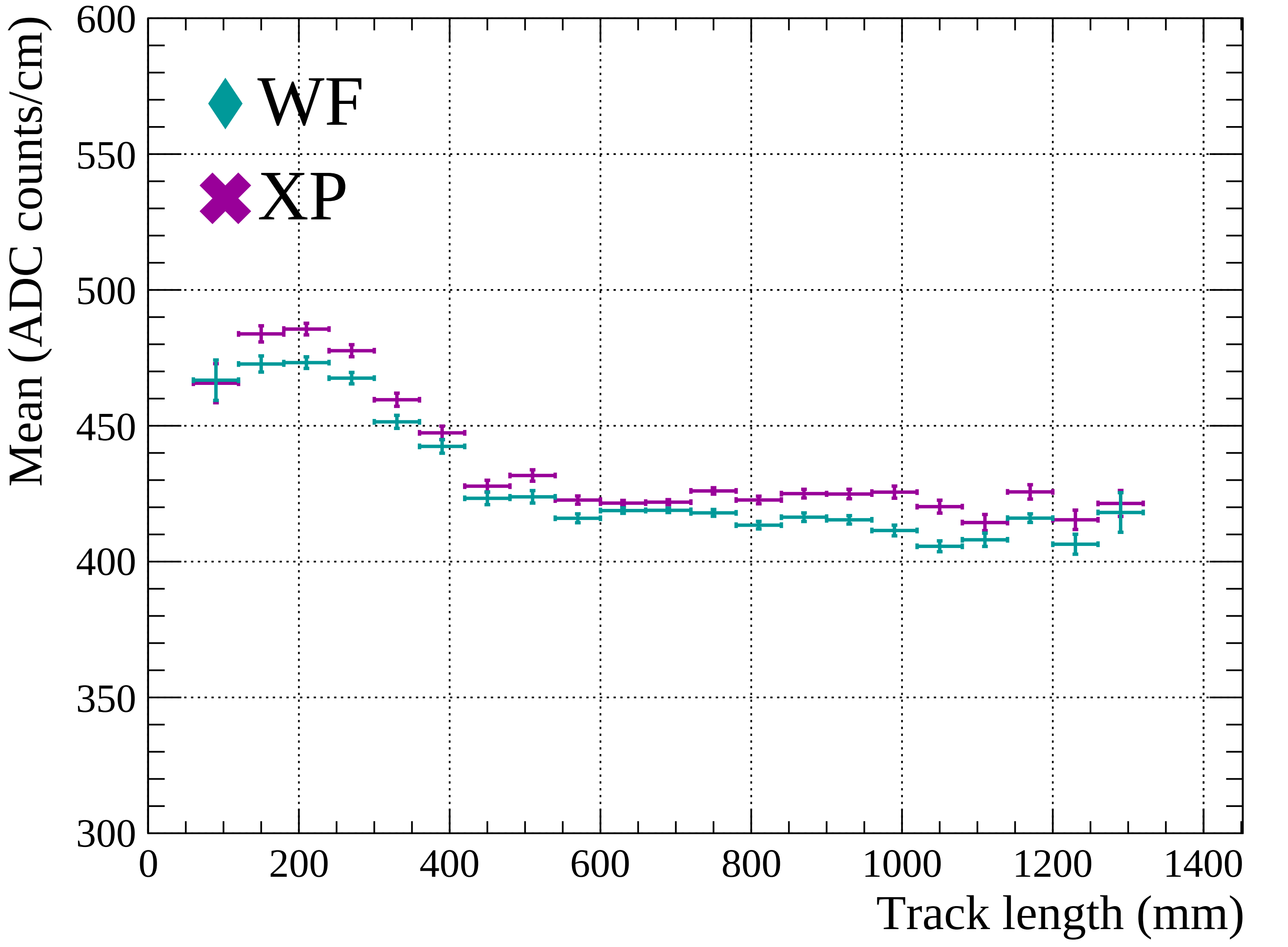


Energy loss (XP) vs drift time in ERAM 47
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 16
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 14
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 15
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 42
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 45
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 37
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 36
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 20
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 38
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 7
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 44
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 43
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 39
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 41
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 46
y = 0.00x + 0.00
dE/dx (ADC counts/cm)
drift time (timebins)

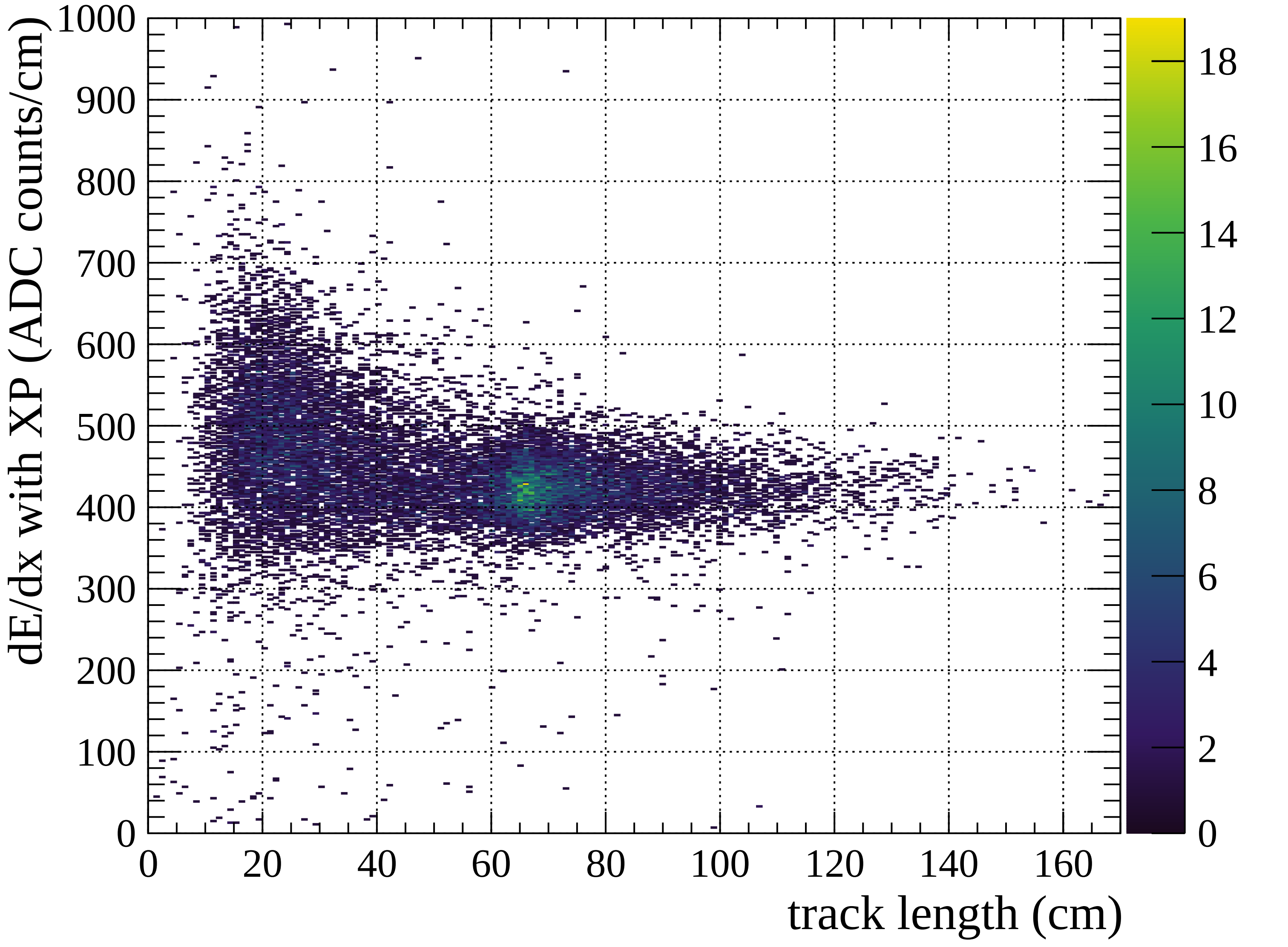

time
500
400
300
200
100
0
−1000
−500
0
500
1000
X
120
100
80
60
40
20
0

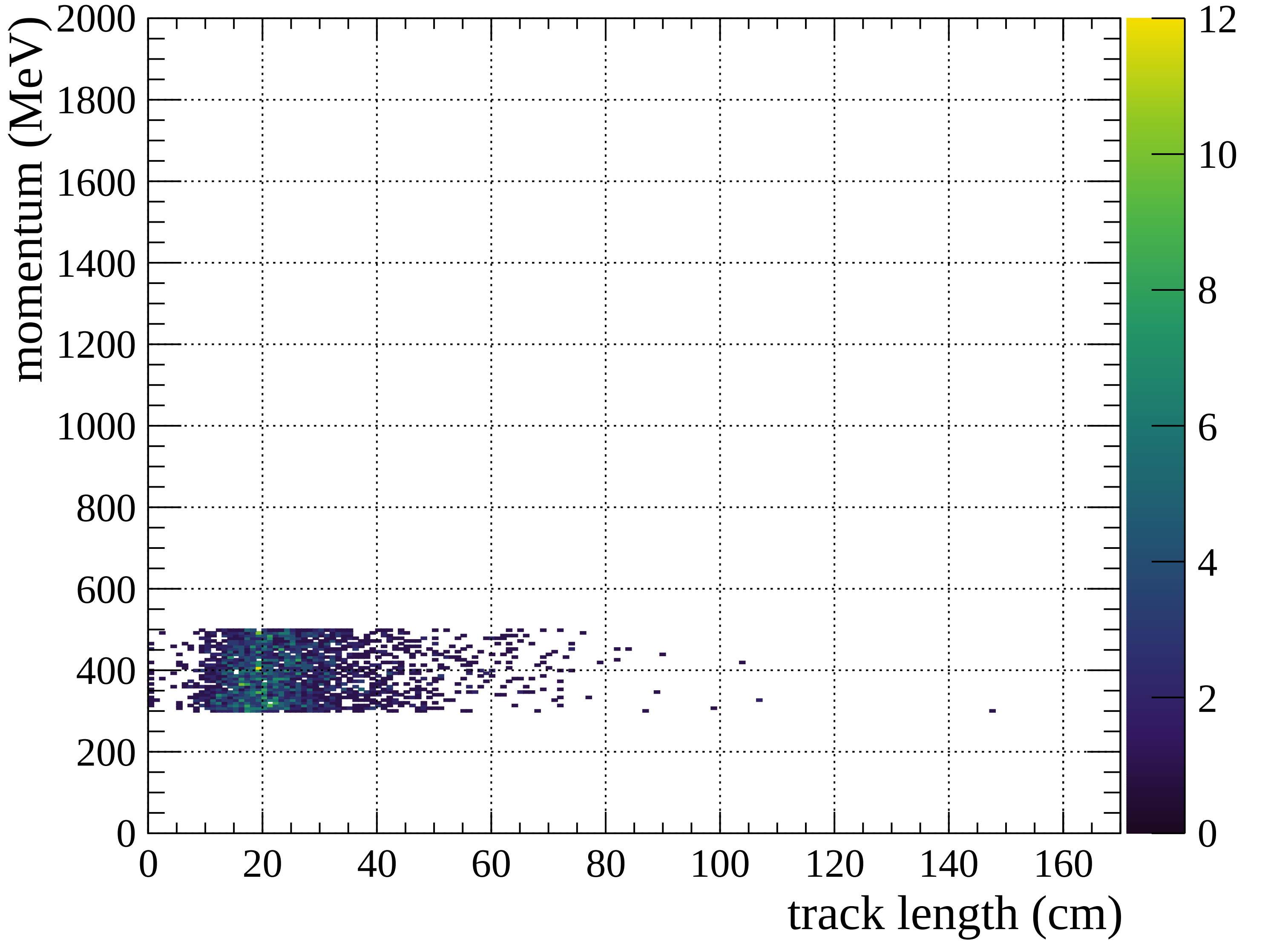


WF
XP
dE/dx resolution (%)
Track length (mm)
30
25
20
15
10
5
0
200
400
600
800
1000
1200
1400

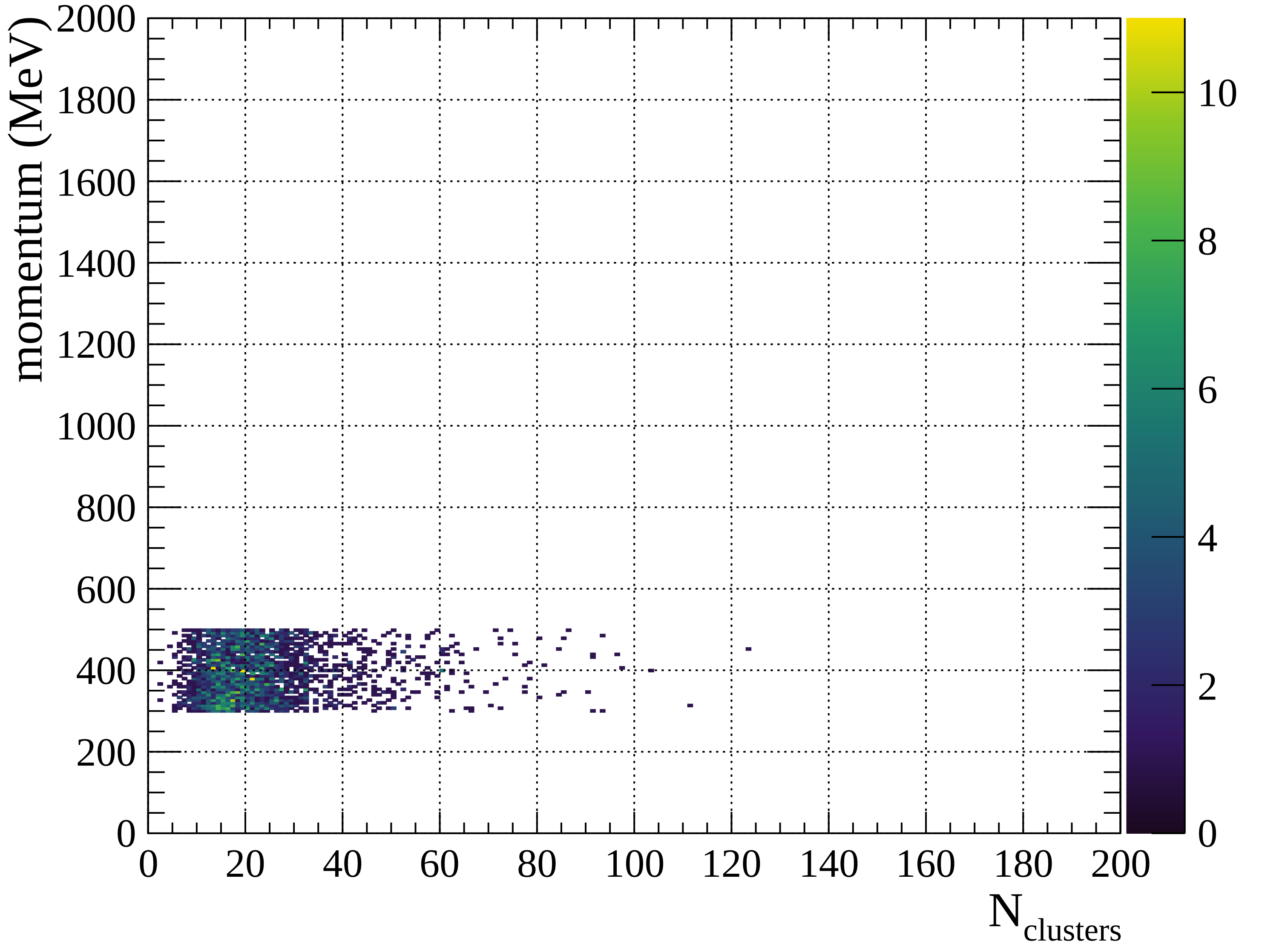


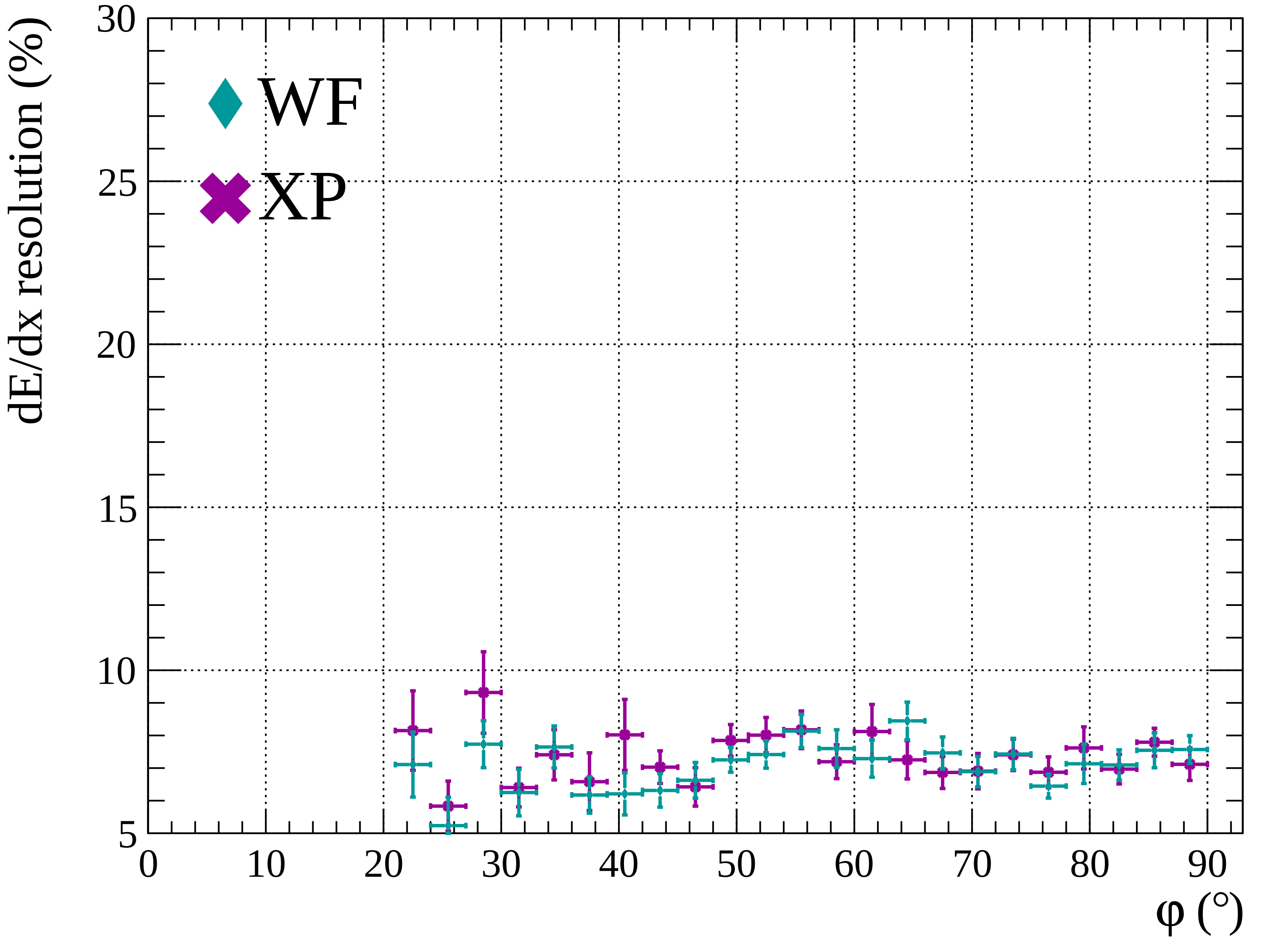
WF
XP
Mean (ADC counts/cm)
Track length (mm)
300
350
400
450
500
550
600
0
200
400
600
800
1000
1200
1400

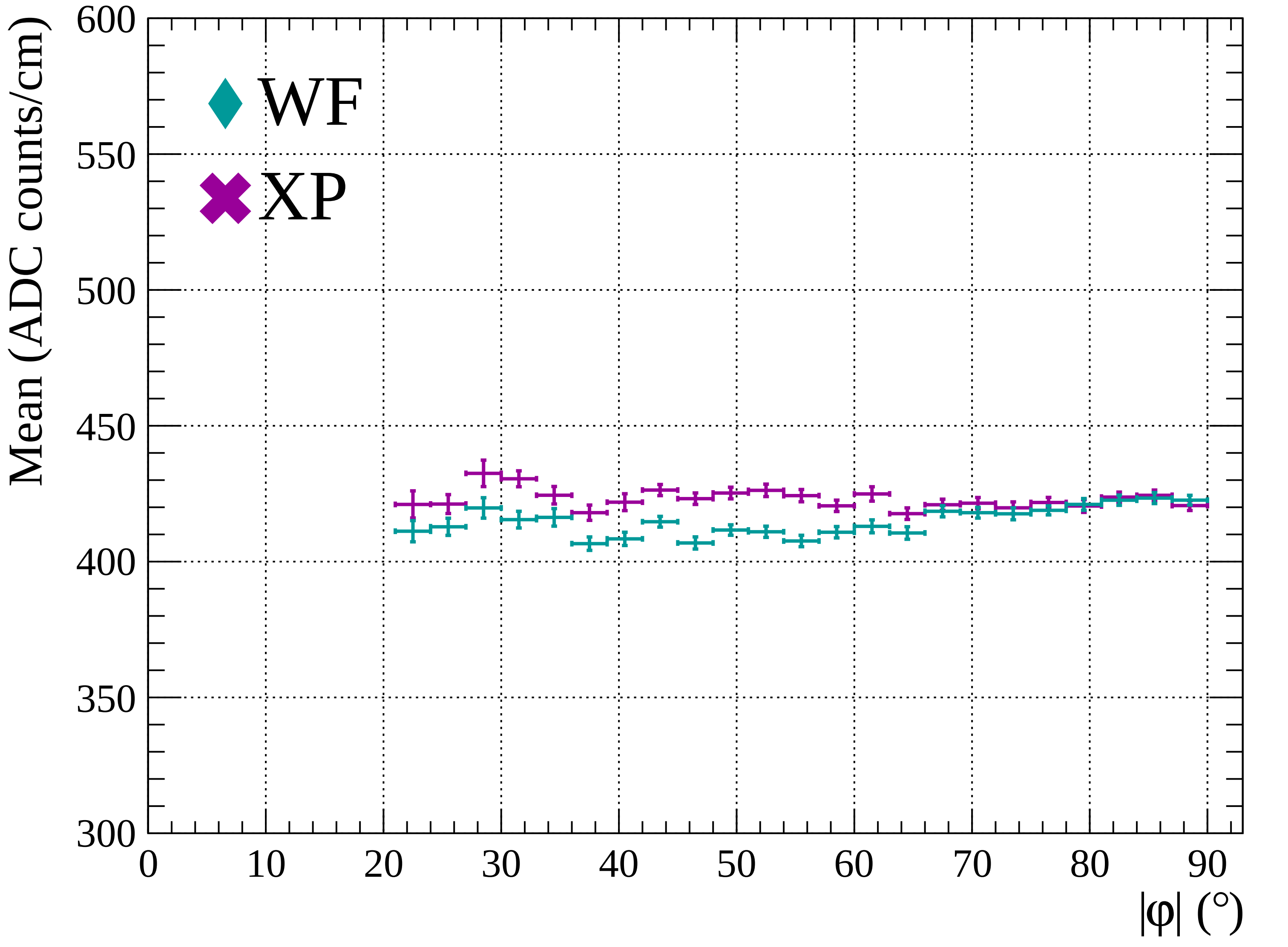

dE/dx with XP (ADC counts/cm)
track length (cm)
0
100
200
300
400
500
600
700
800
900
1000
0
20
40
60
80
100
120
140
160
0
2
4
6
8
10
12
14
16
18



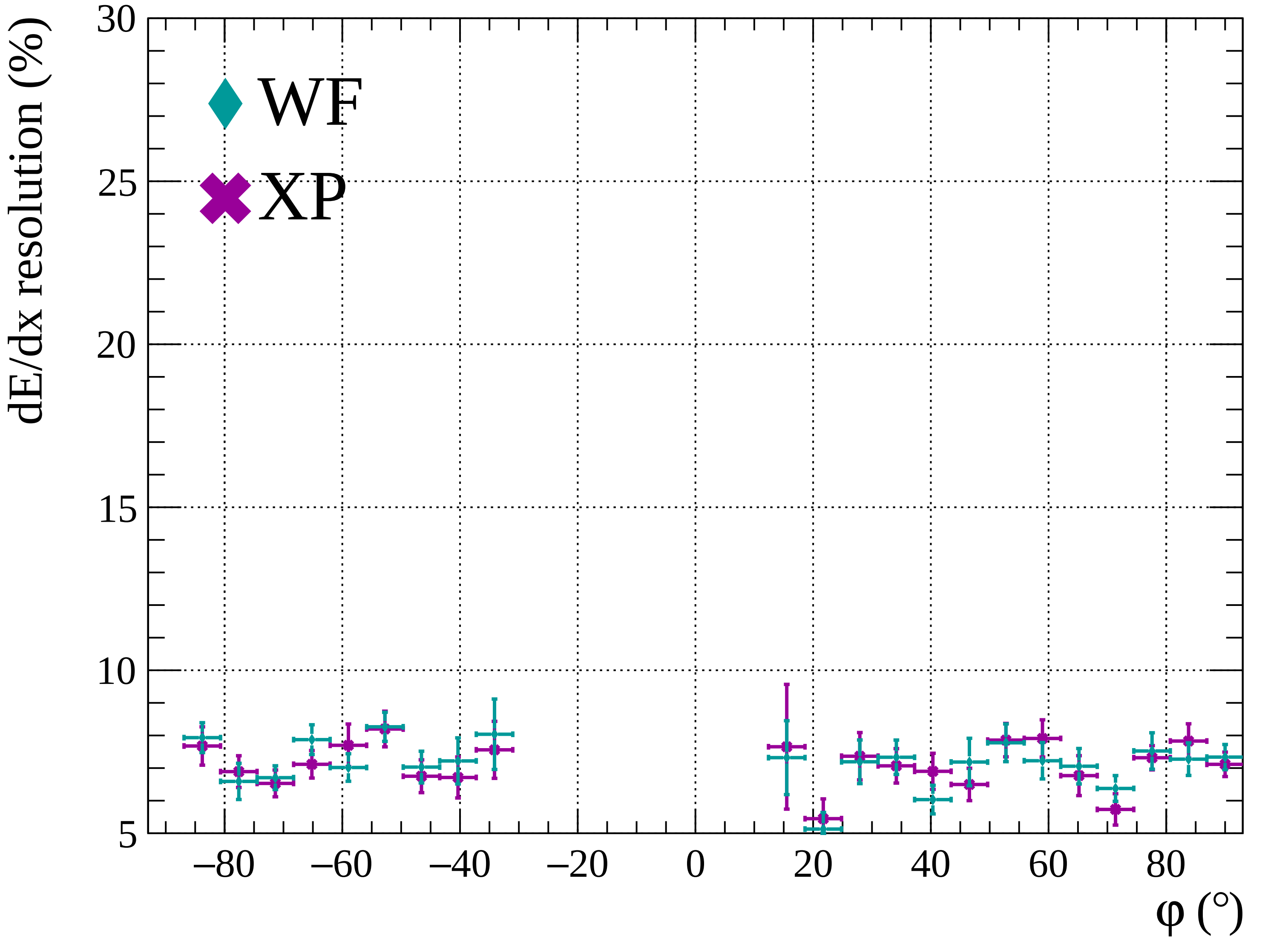
momentum (MeV)
2000
1800
1600
1400
1200
1000
800
600
400
200
0
0
20
40
60
80
100
120
140
160
track length (cm)
12
10
8
6
4
2
0


momentum (MeV)
2000
1800
1600
1400
1200
1000
800
600
400
200
0
0
20
40
60
80
100
120
140
160
180
200
N_clusters
10
8
6
4
2
0

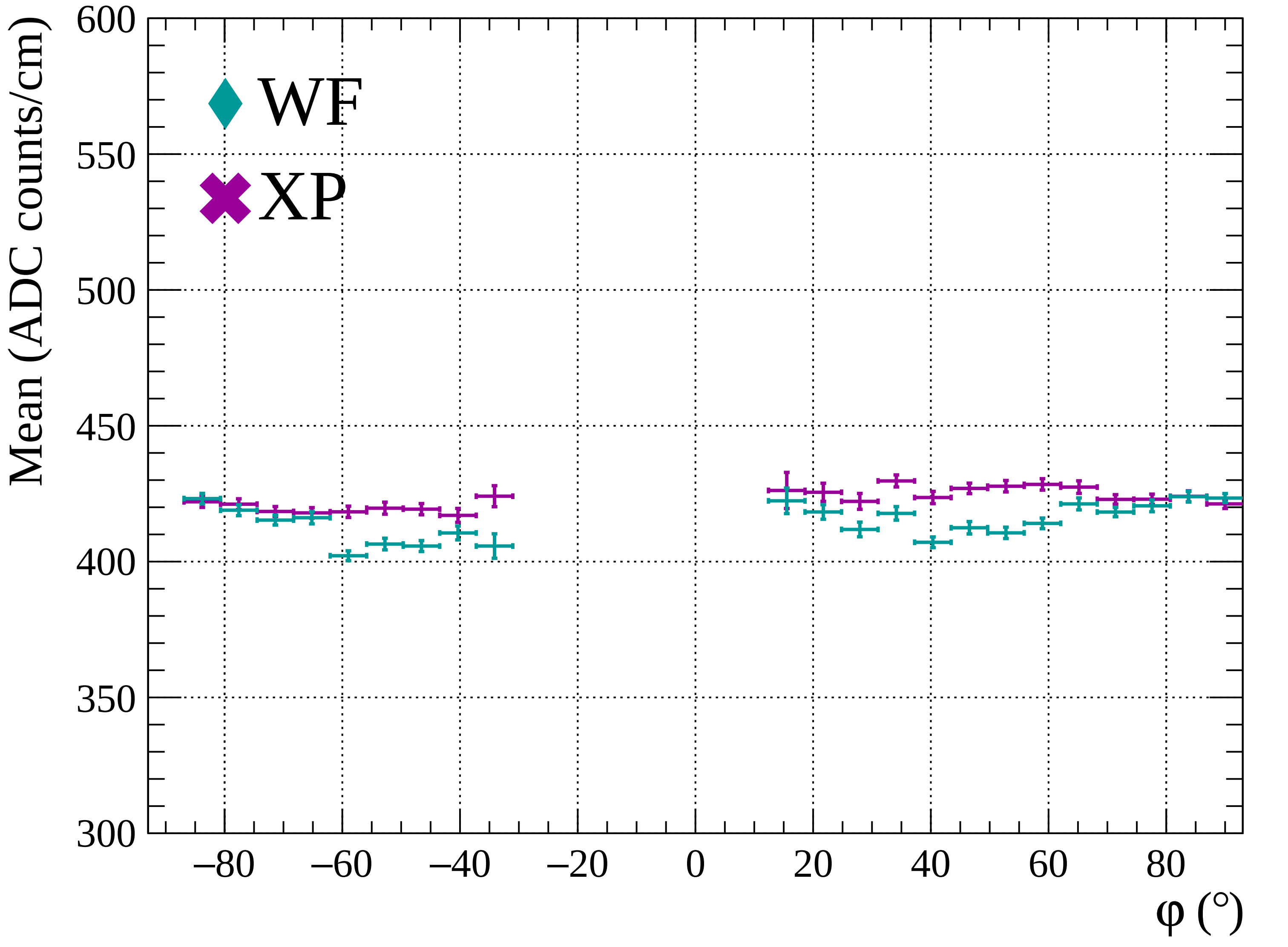

WF
XP
dE/dx resolution (%)
φ (°)
5
10
15
20
25
30
0
10
20
30
40
50
60
70
80
90



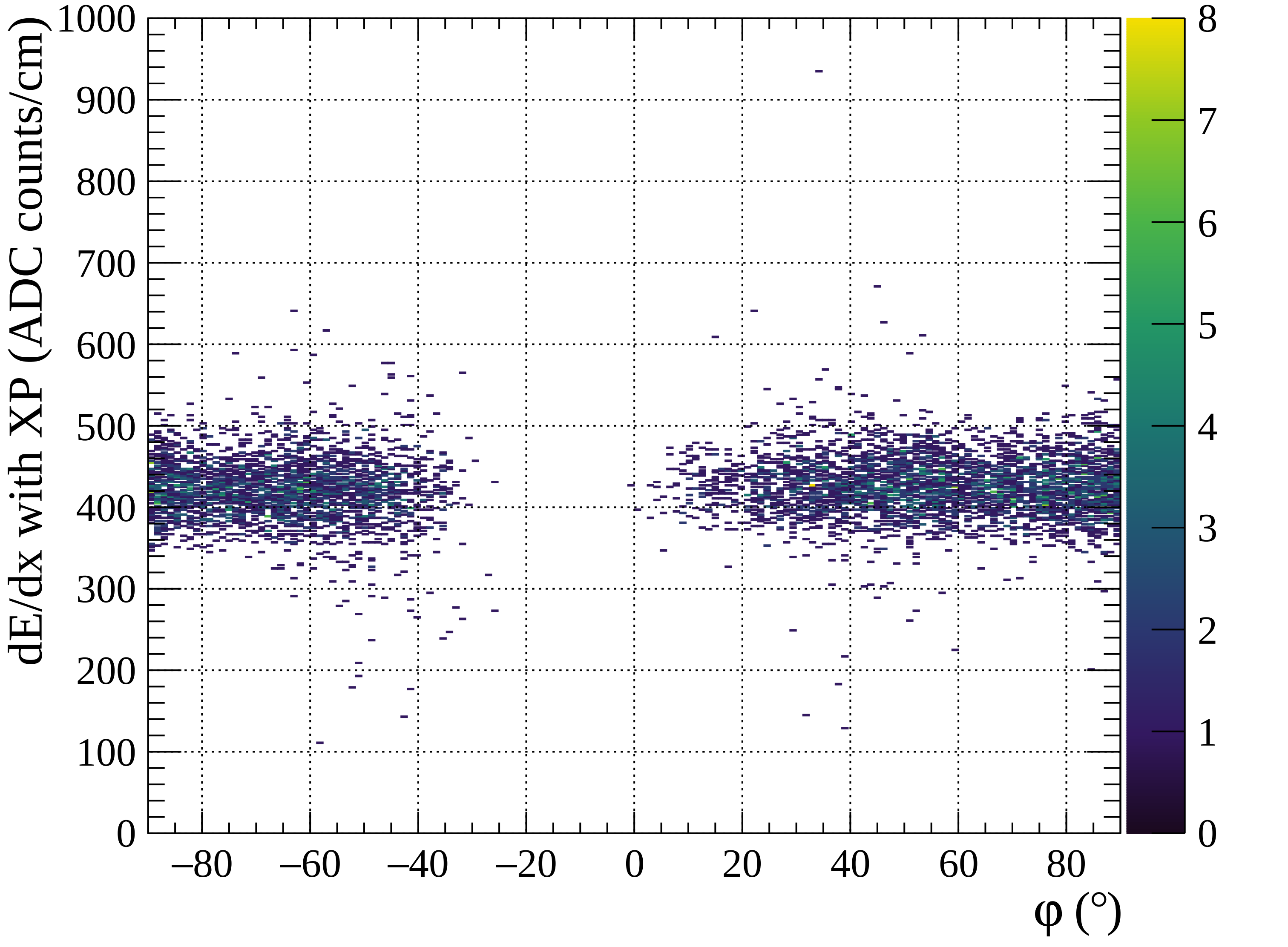
Mean (ADC counts/cm)
600
550
500
450
400
350
300
WF
XP
0
10
20
30
40
50
60
70
80
90
|φ| (°)

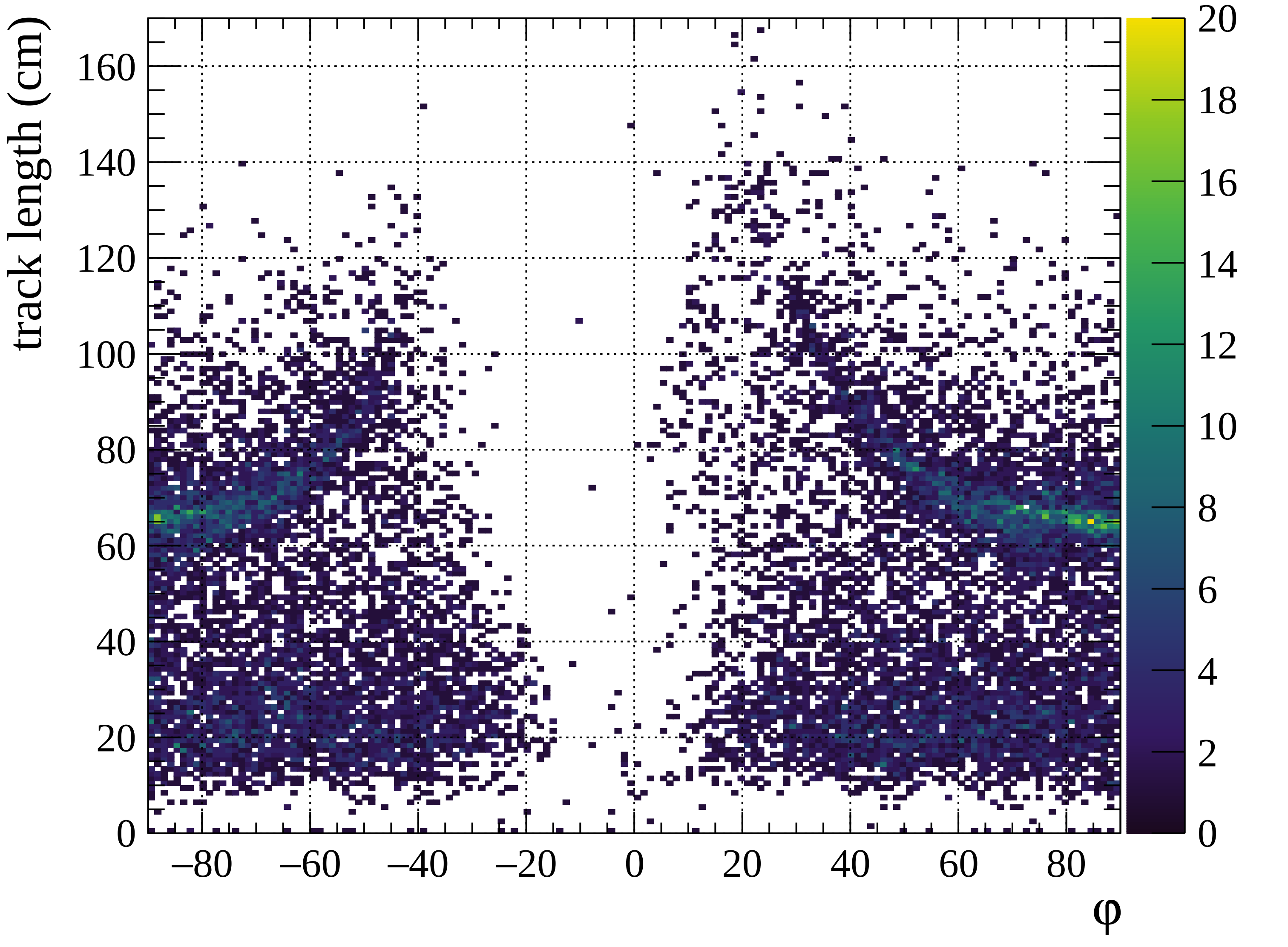


dE/dx resolution (%)
30
25
20
15
10
5
WF
XP
−80
−60
−40
−20
0
20
40
60
80
φ (°)

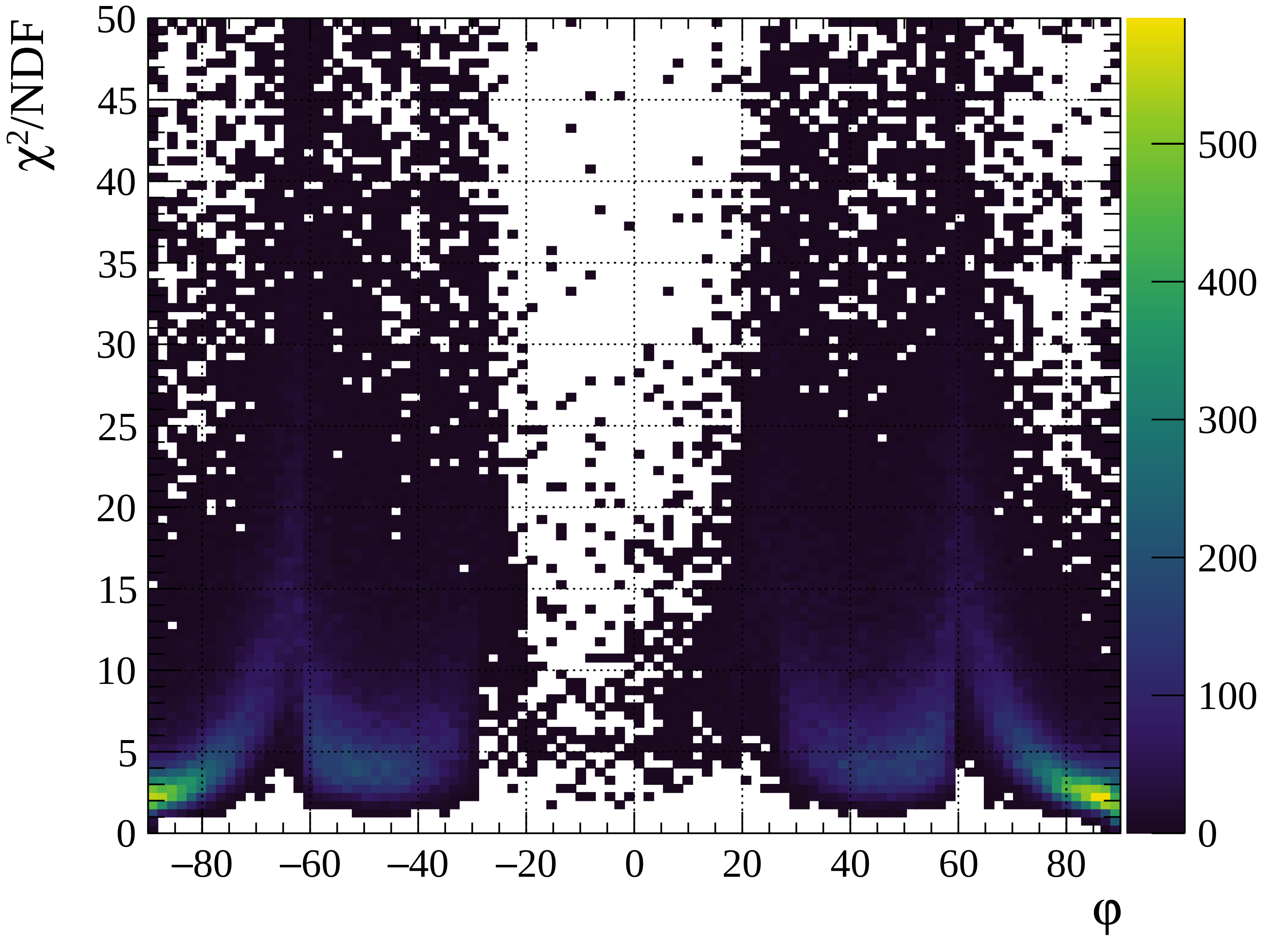


WF
XP
Mean (ADC counts/cm)
600
550
500
450
400
350
300
−80
−60
−40
−20
0
20
40
60
80
φ (°)

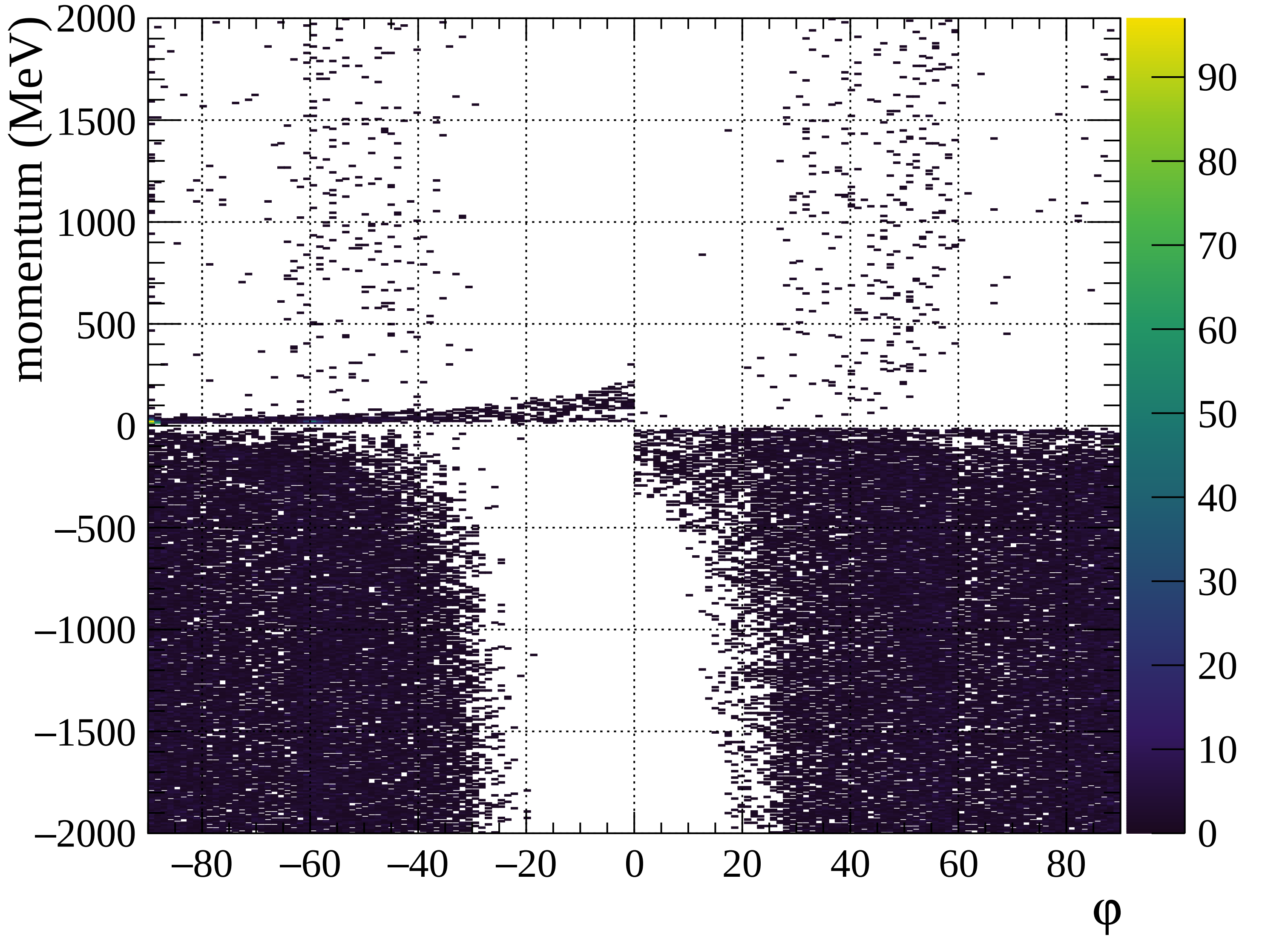

dE/dx with XP (ADC counts/cm)
φ (°)



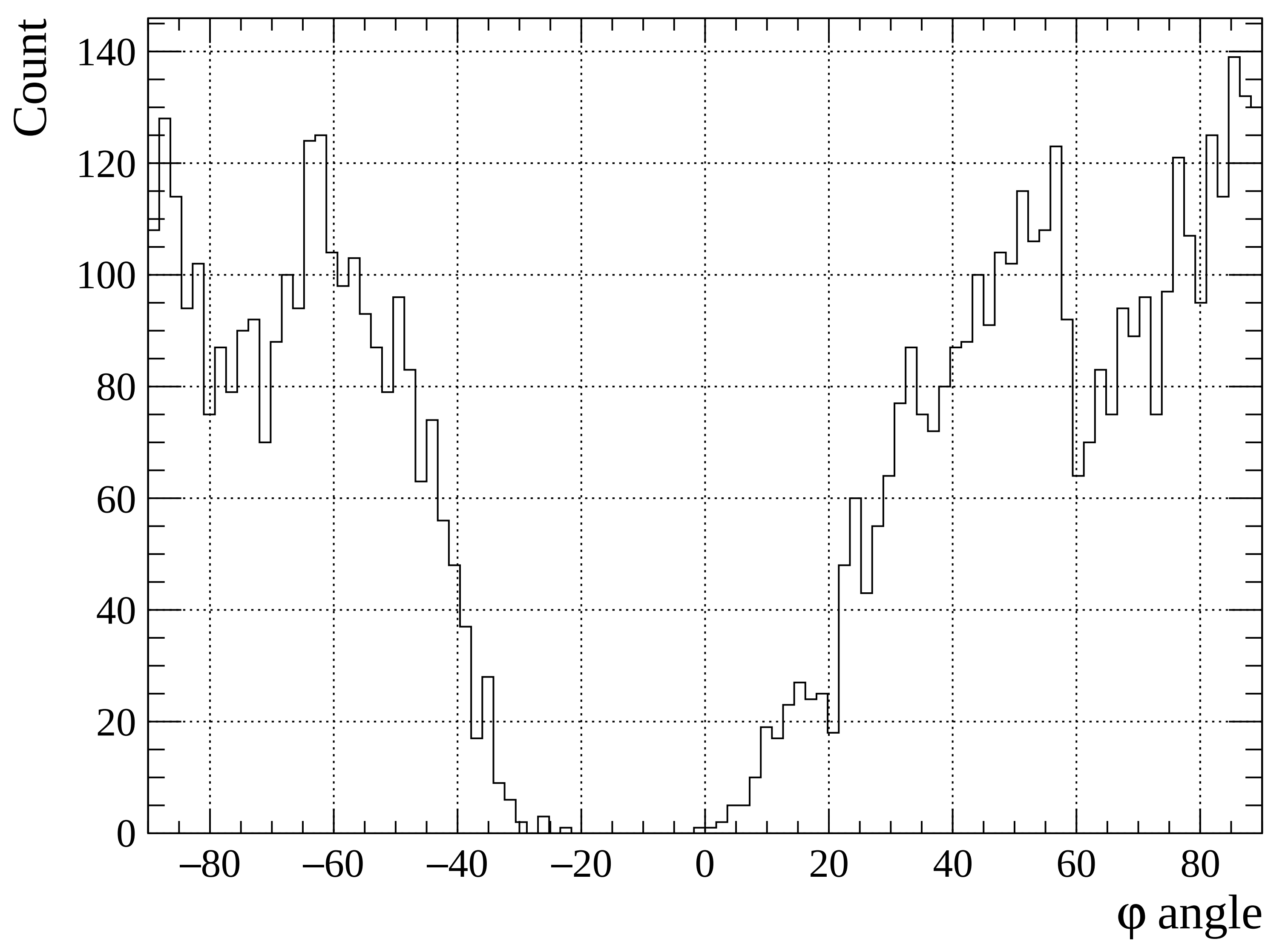
track length (cm)
0
20
40
60
80
100
120
140
160
−80
−60
−40
−20
0
20
40
60
80
φ
0
2
4
6
8
10
12
14
16
18
20



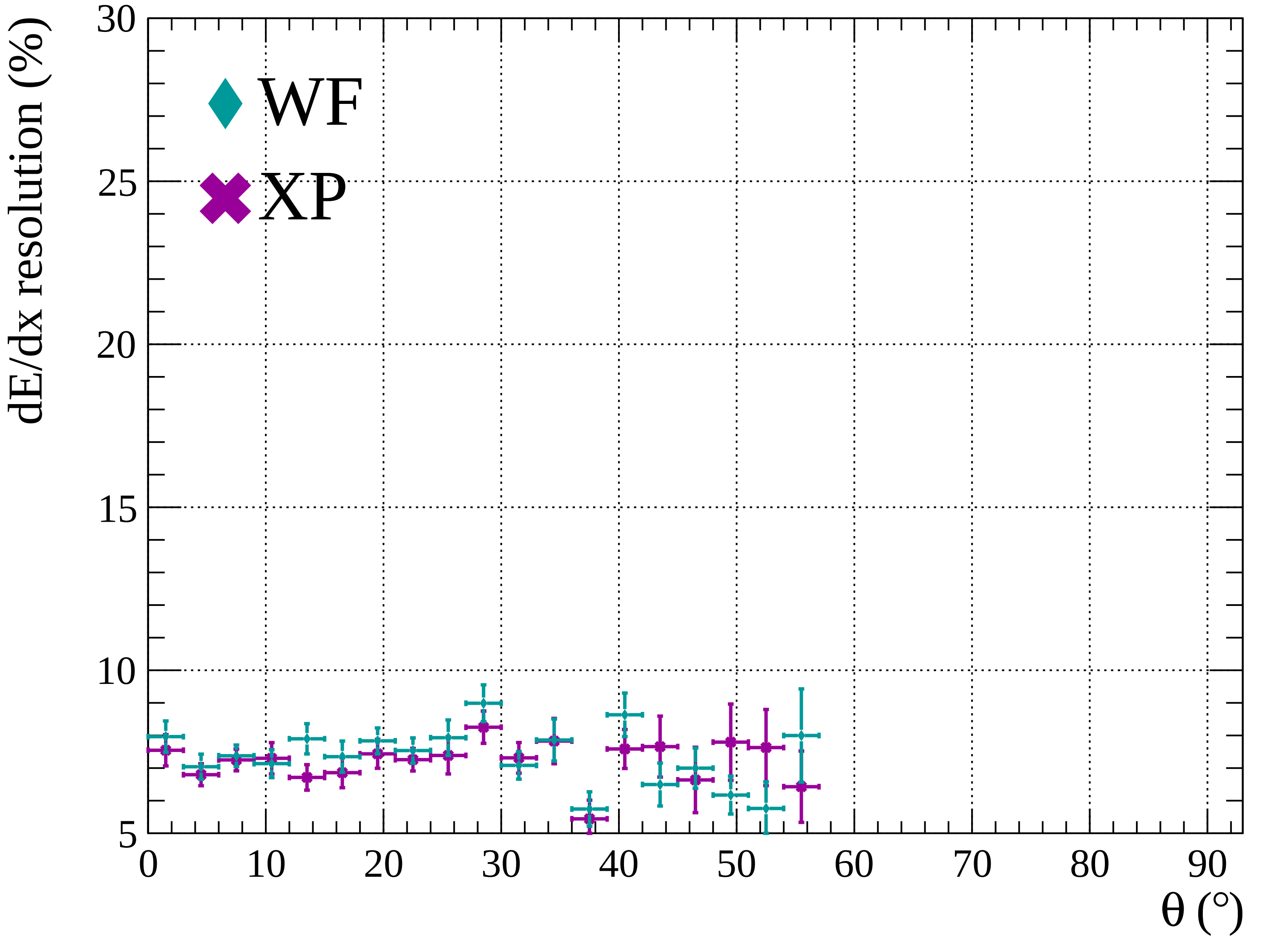
χ^2/NDF
φ
50
45
40
35
30
25
20
15
10
5
0
−80
−60
−40
−20
0
20
40
60
80
500
400
300
200
100
0

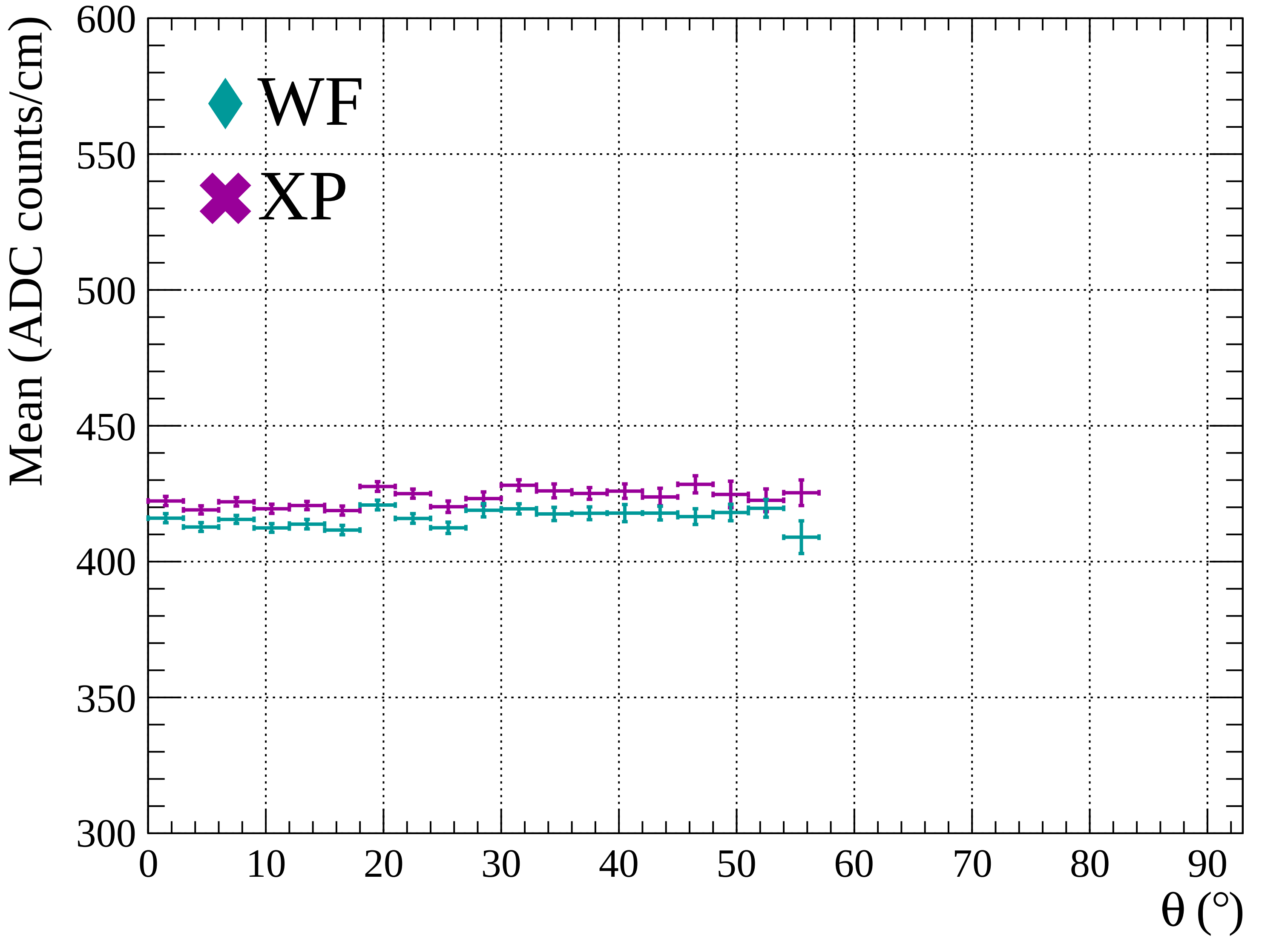


momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
φ
90
80
70
60
50
40
30
20
10
0

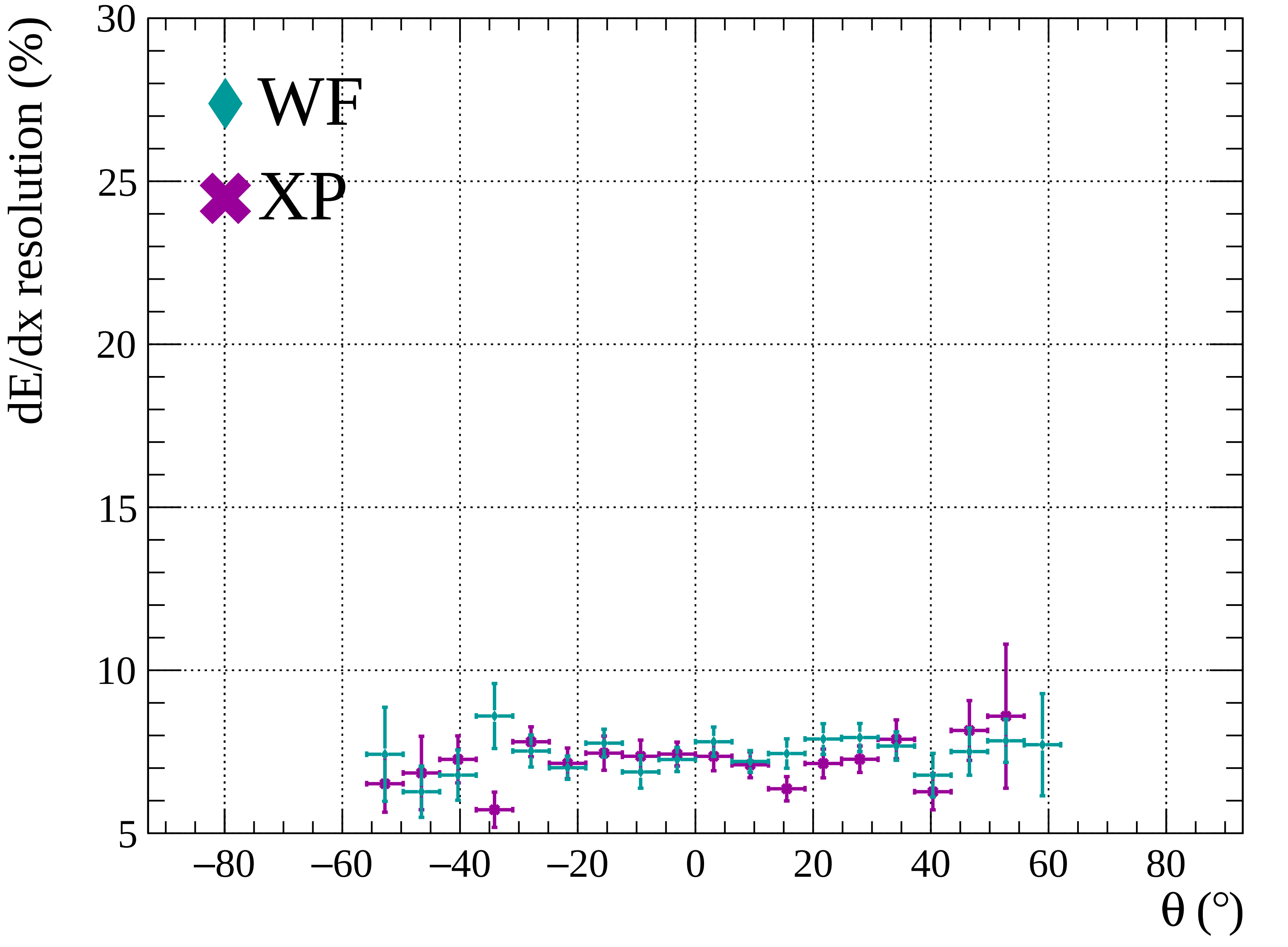


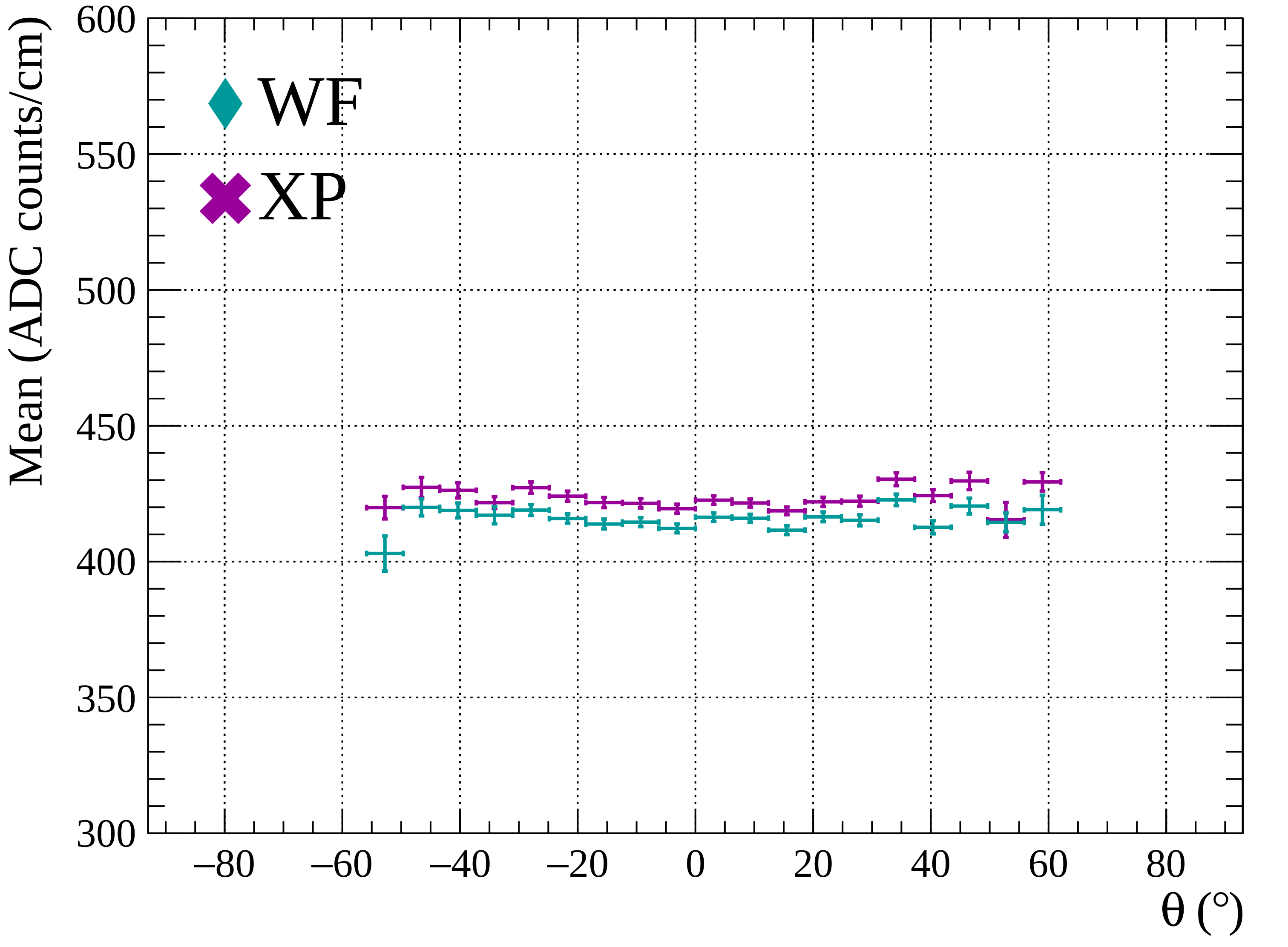
Count
140
120
100
80
60
40
20
0
−80
−60
−40
−20
0
20
40
60
80
φ angle

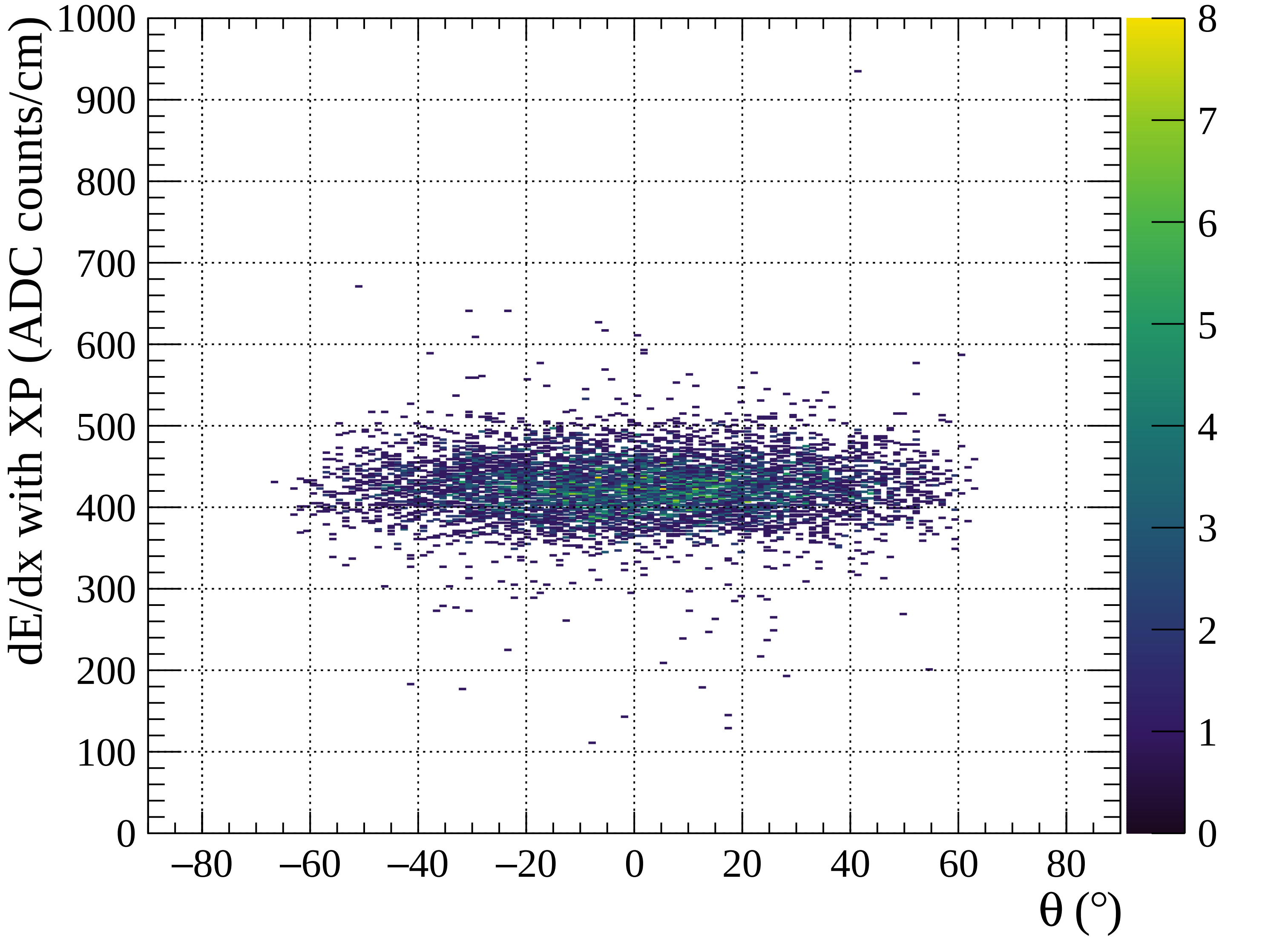

WF
XP
dE/dx resolution (%)
θ (°)
5
10
15
20
25
30
0
10
20
30
40
50
60
70
80
90



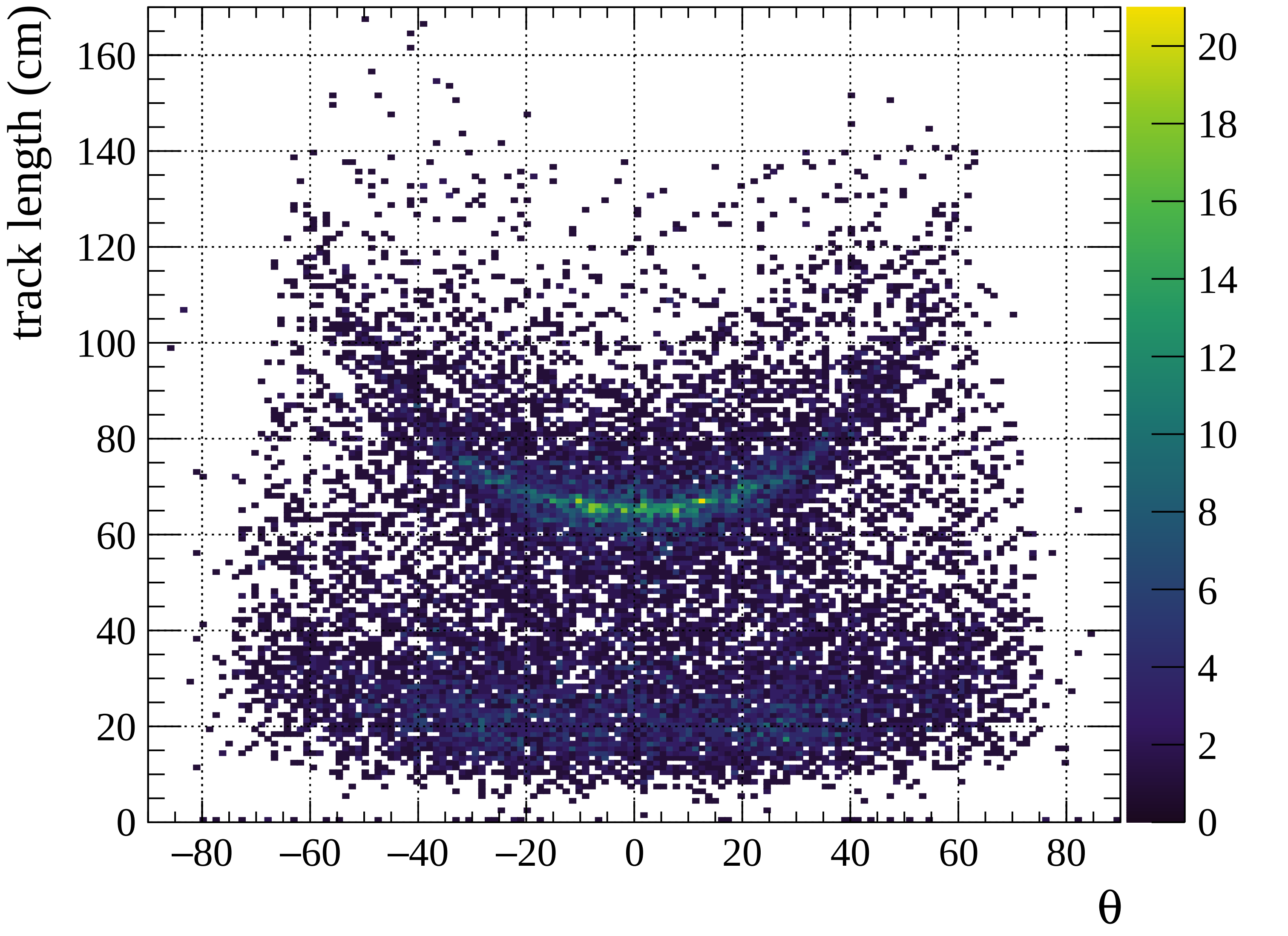
Mean (ADC counts/cm)
θ (°)
WF
XP
600
550
500
450
400
350
300
0
10
20
30
40
50
60
70
80
90


WF
XP
dE/dx resolution (%)
θ (°)
30
25
20
15
10
5
−80
−60
−40
−20
0
20
40
60
80

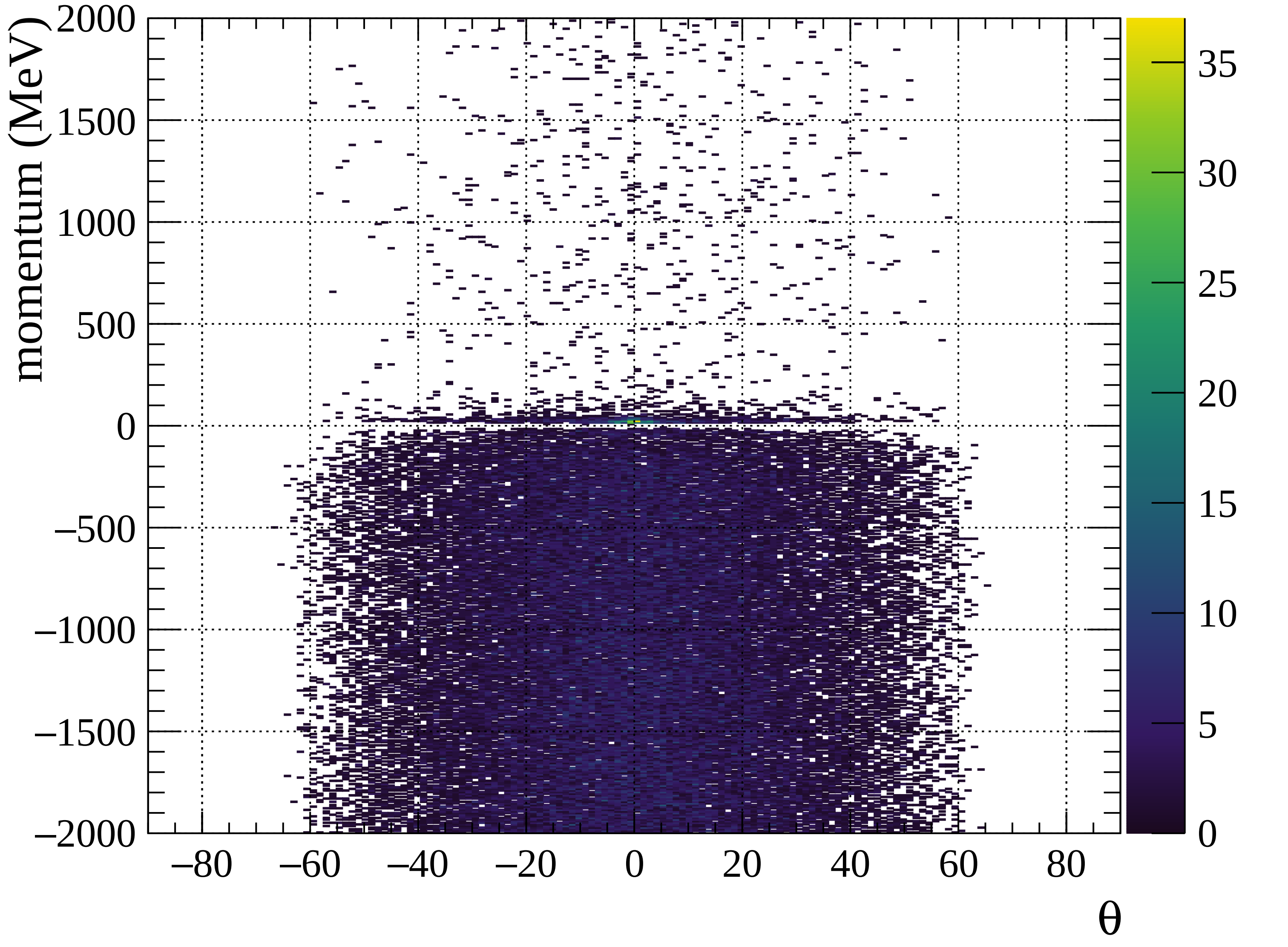

WF
XP
Mean (ADC counts/cm)
θ (°)
600
550
500
450
400
350
300
−80
−60
−40
−20
0
20
40
60
80



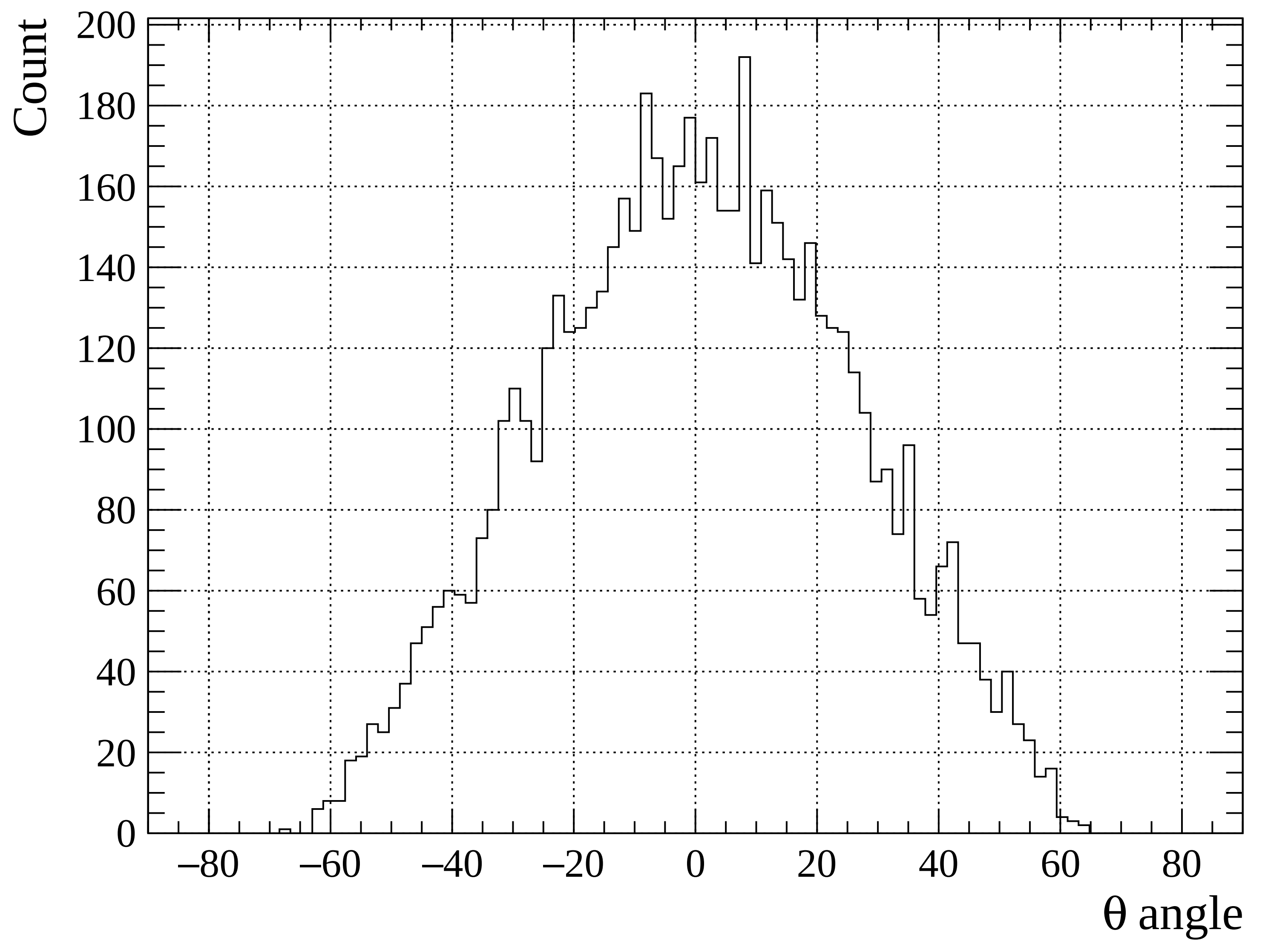
dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
θ (°)
8
7
6
5
4
3
2
1
0



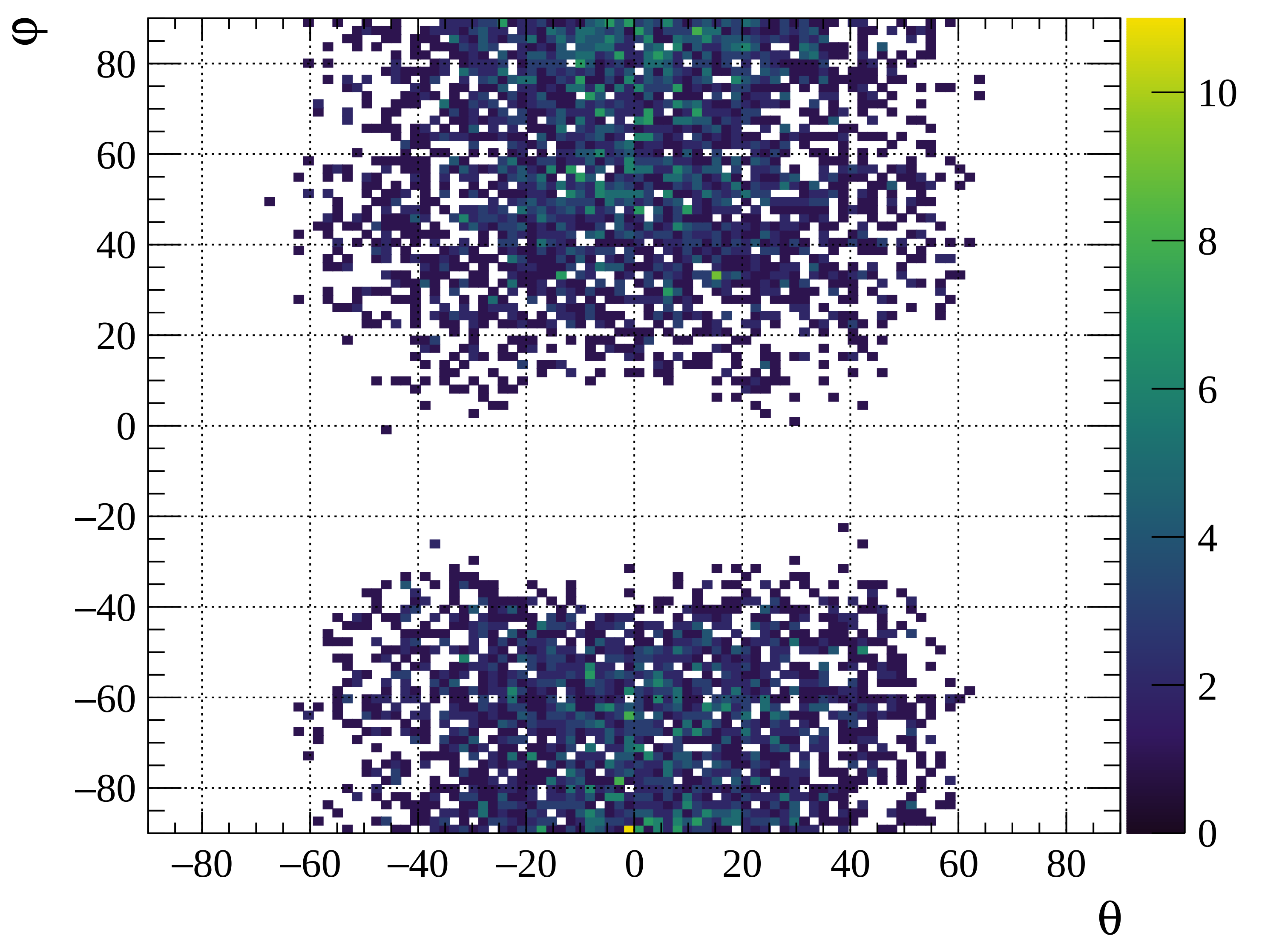
track length (cm)
160
140
120
100
80
60
40
20
0
−80
−60
−40
−20
0
20
40
60
80
θ
20
18
16
14
12
10
8
6
4
2
0



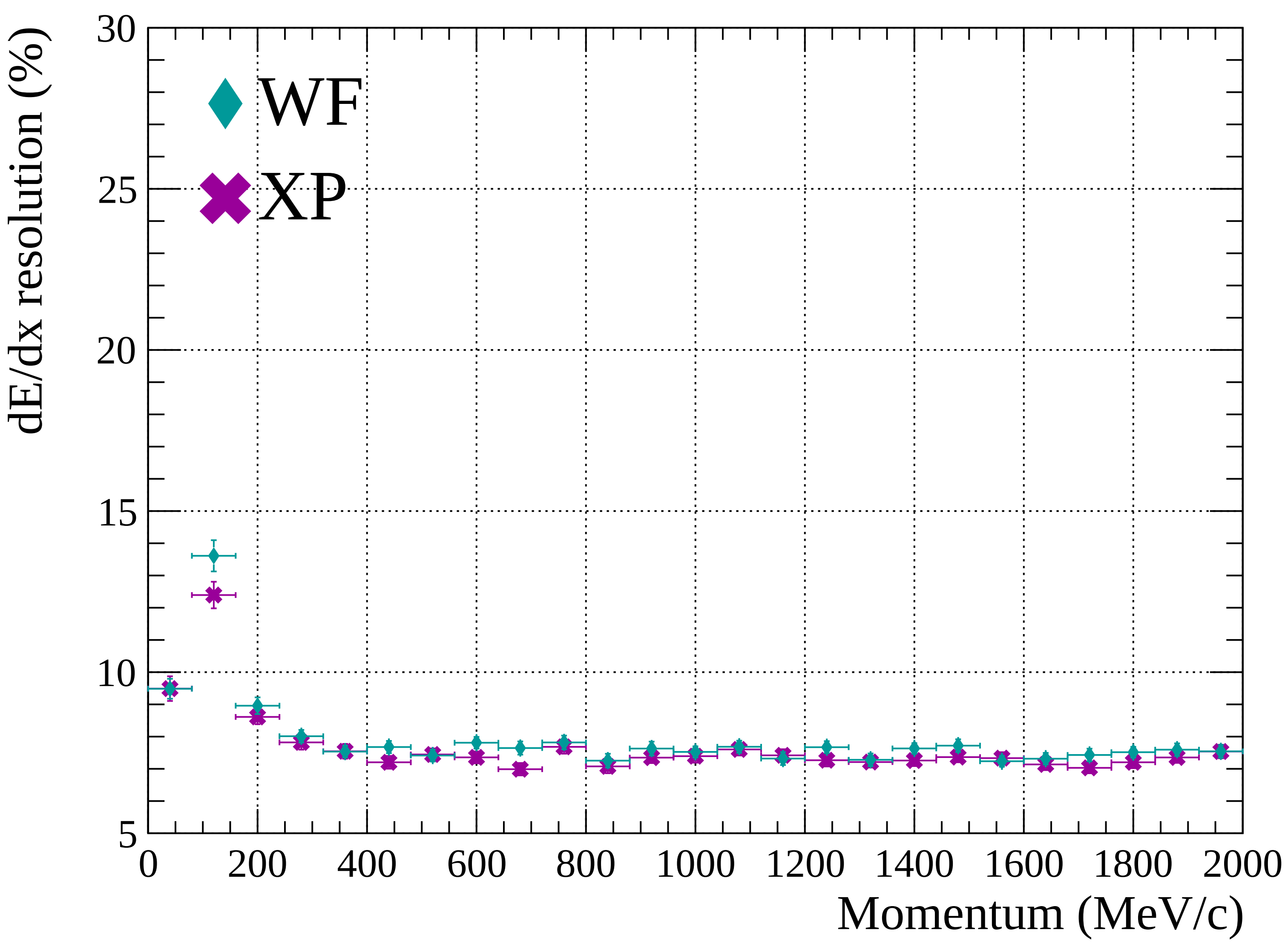
momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
θ
35
30
25
20
15
10
5
0



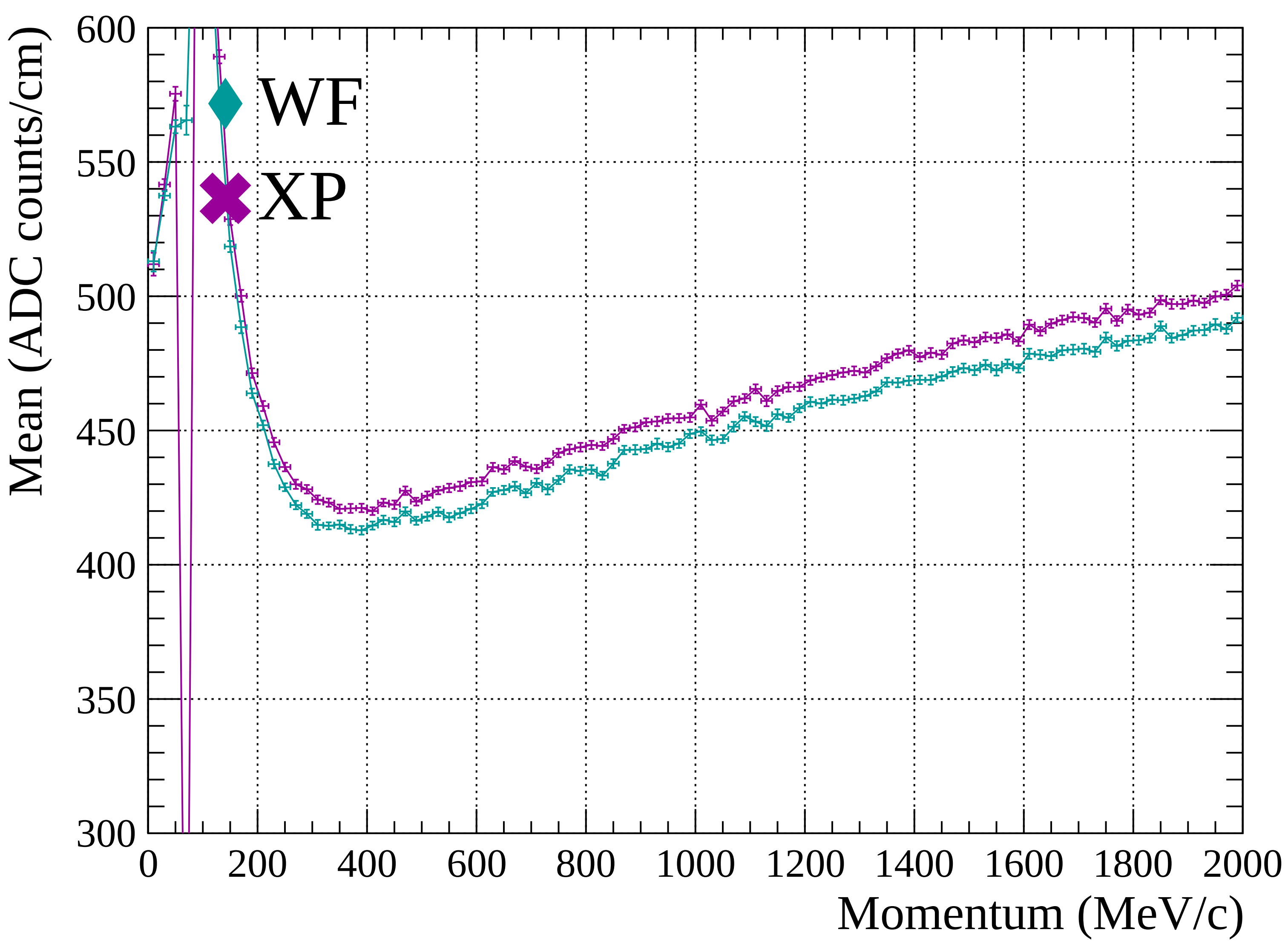
Count
200
180
160
140
120
100
80
60
40
20
0
−80
−60
−40
−20
0
20
40
60
80
θ angle



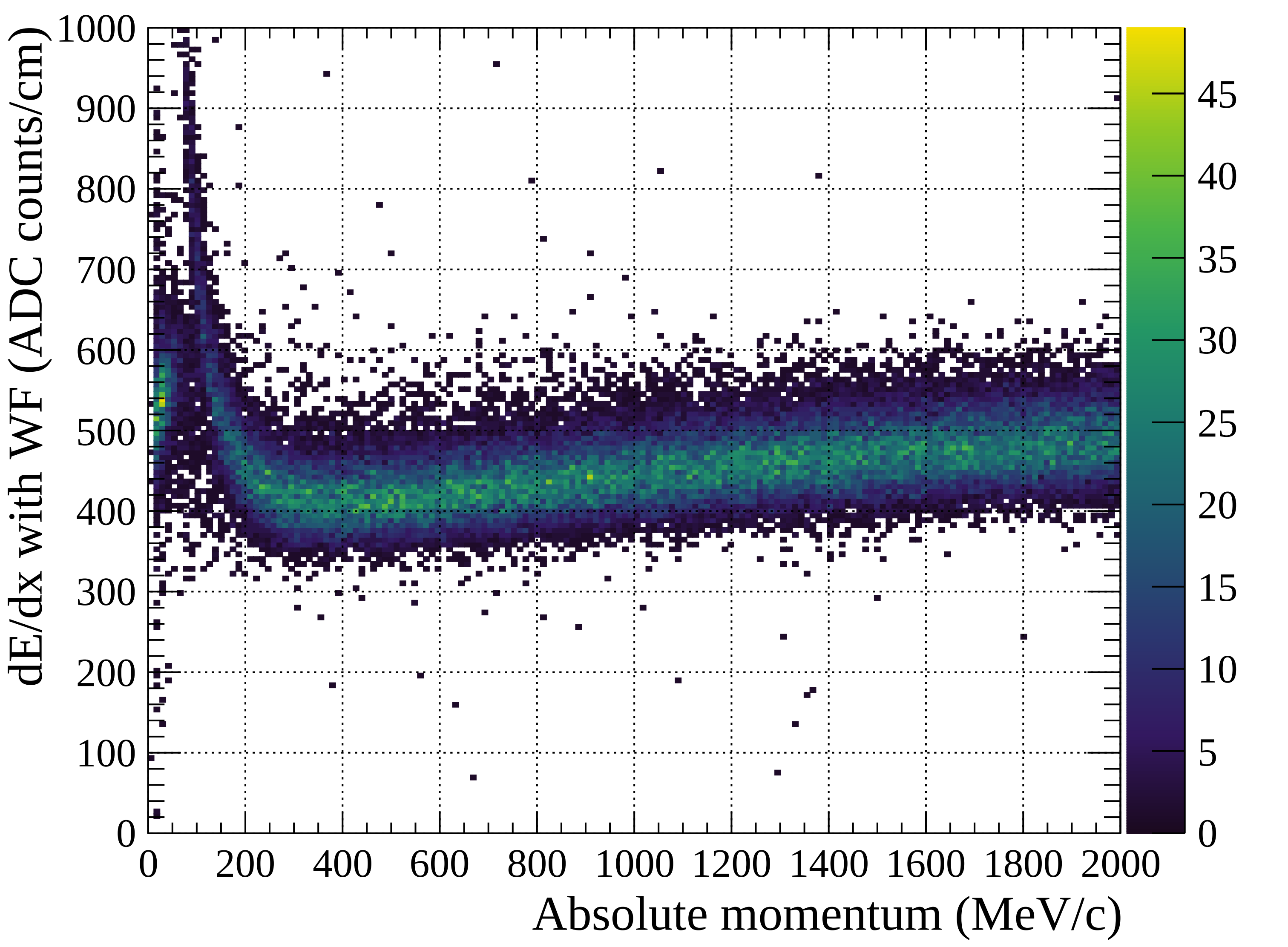
φ
80
60
40
20
0
−20
−40
−60
−80
−80
−60
−40
−20
0
20
40
60
80
θ
10
8
6
4
2
0



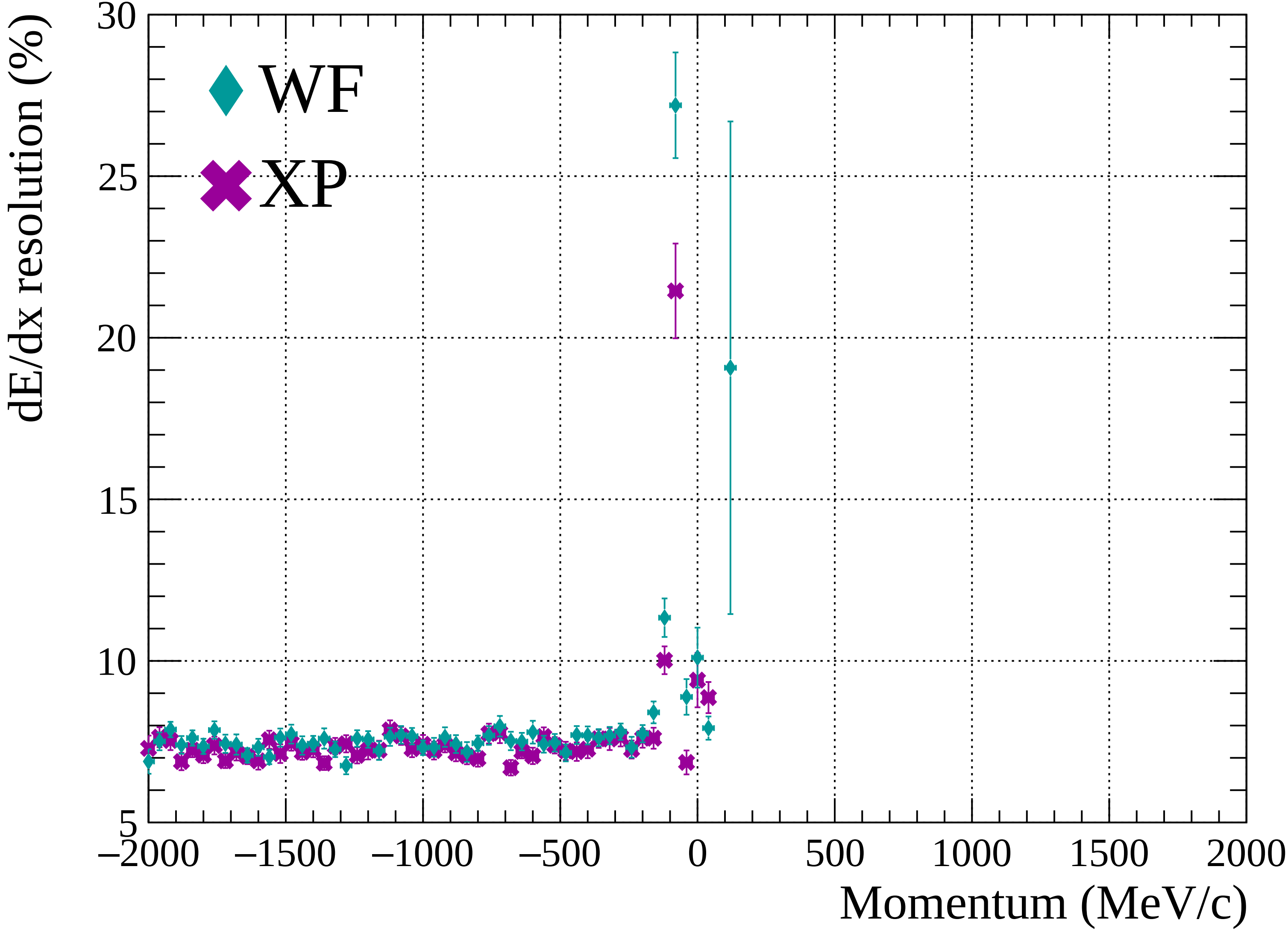
dE/dx resolution (%)
30
25
20
15
10
5
0
200
400
600
800
1000
1200
1400
1600
1800
2000
Momentum (MeV/c)
WF
XP



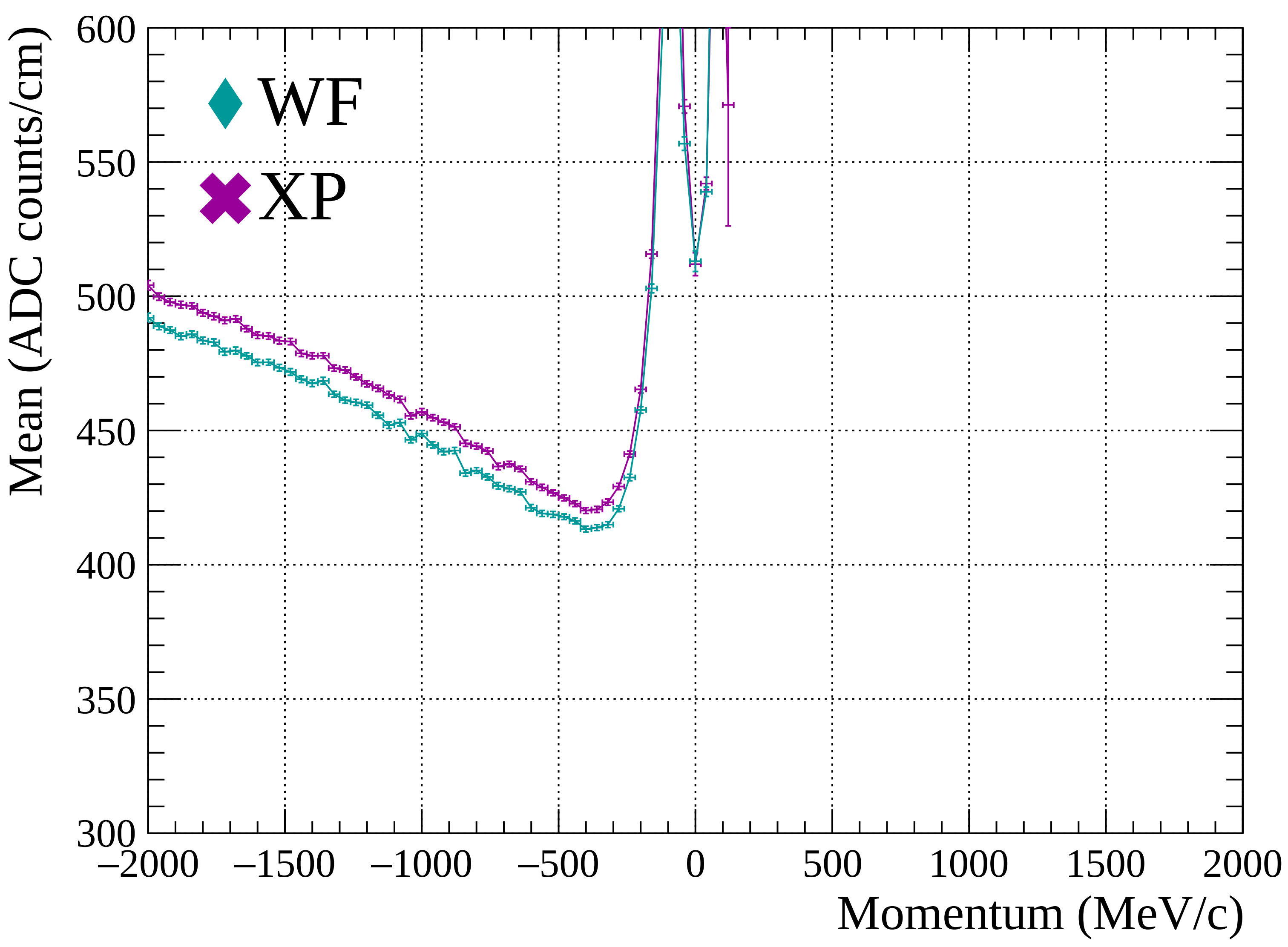
Mean (ADC counts/cm)
Momentum (MeV/c)
WF
XP
600
550
500
450
400
350
300
0
200
400
600
800
1000
1200
1400
1600
1800
2000



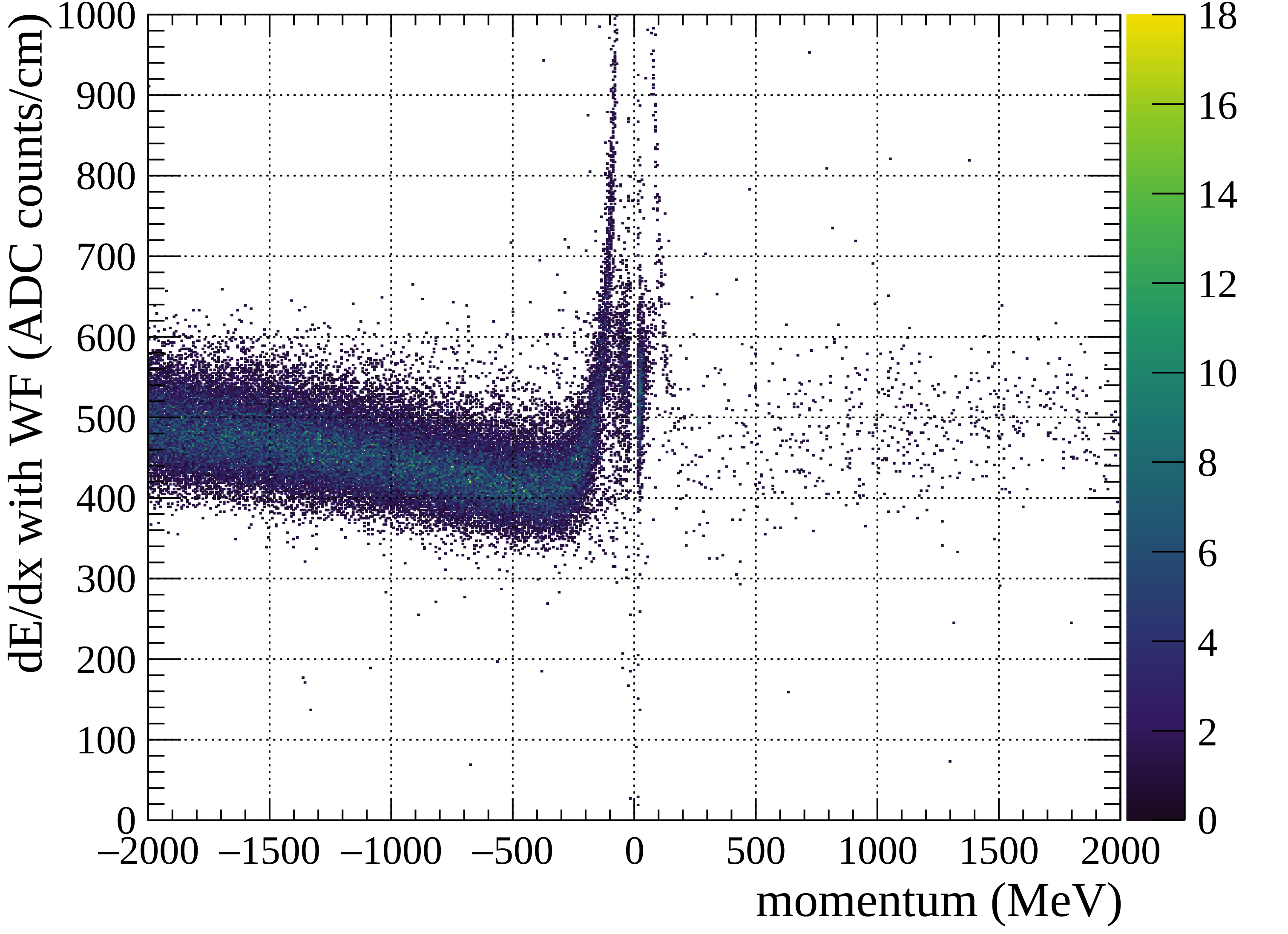
dE/dx with WF (ADC counts/cm)
Absolute momentum (MeV/c)
0
100
200
300
400
500
600
700
800
900
1000
200
400
600
800
1000
1200
1400
1600
1800
2000
5
10
15
20
25
30
35
40
45

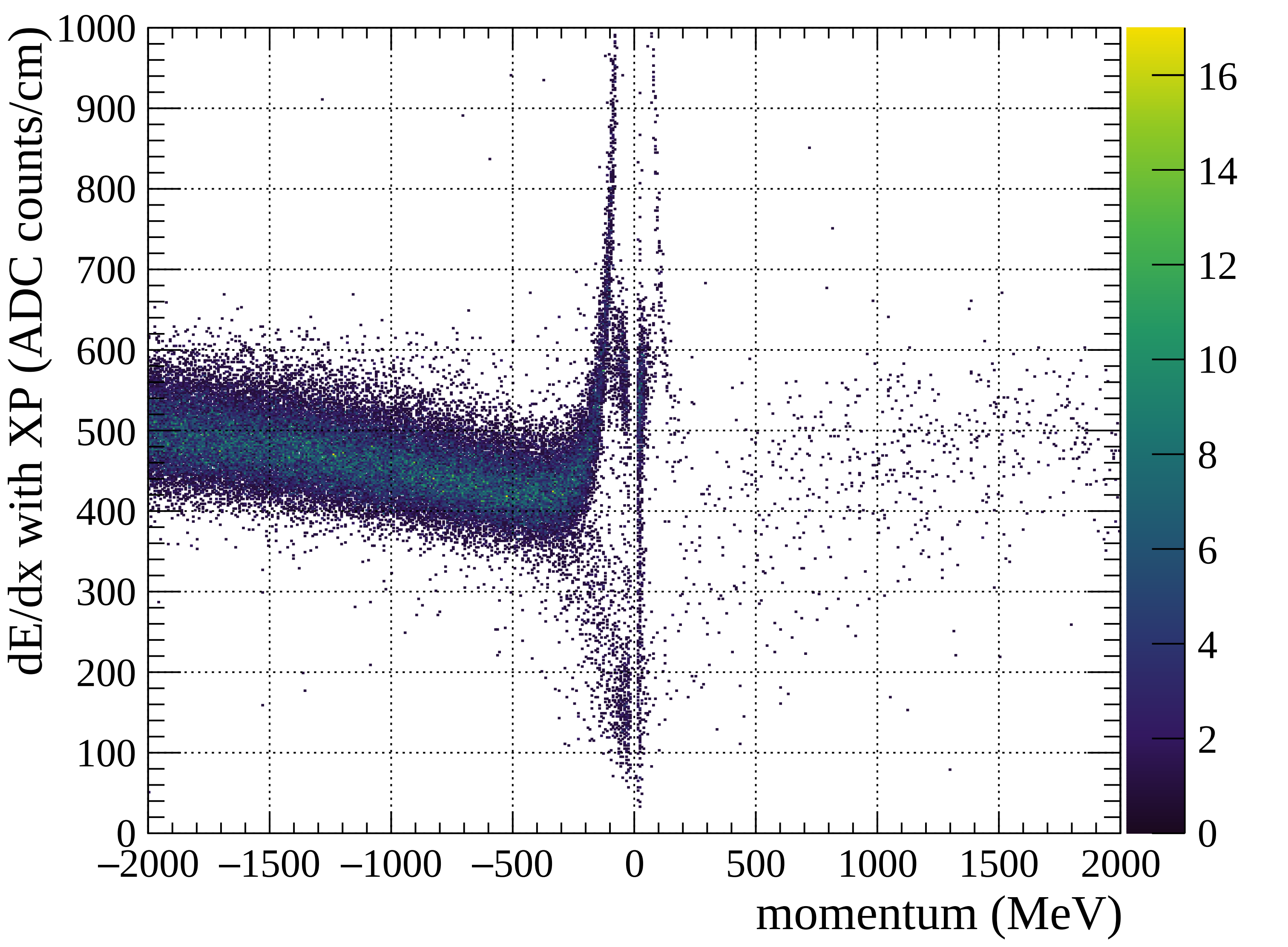


WF
XP
dE/dx resolution (%)
Momentum (MeV/c)
30
25
20
15
10
5
−2000
−1500
−1000
−500
0
500
1000
1500
2000

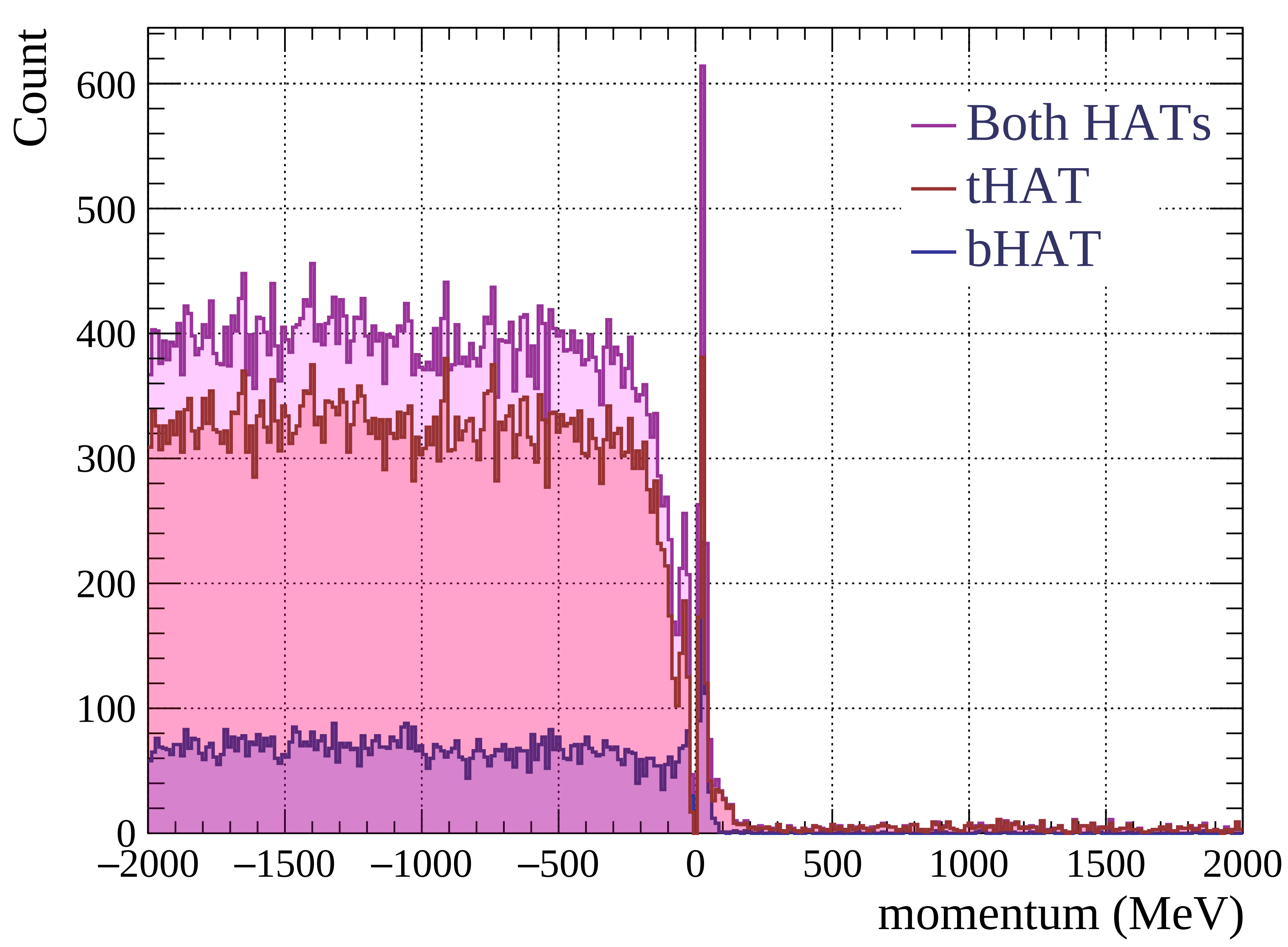


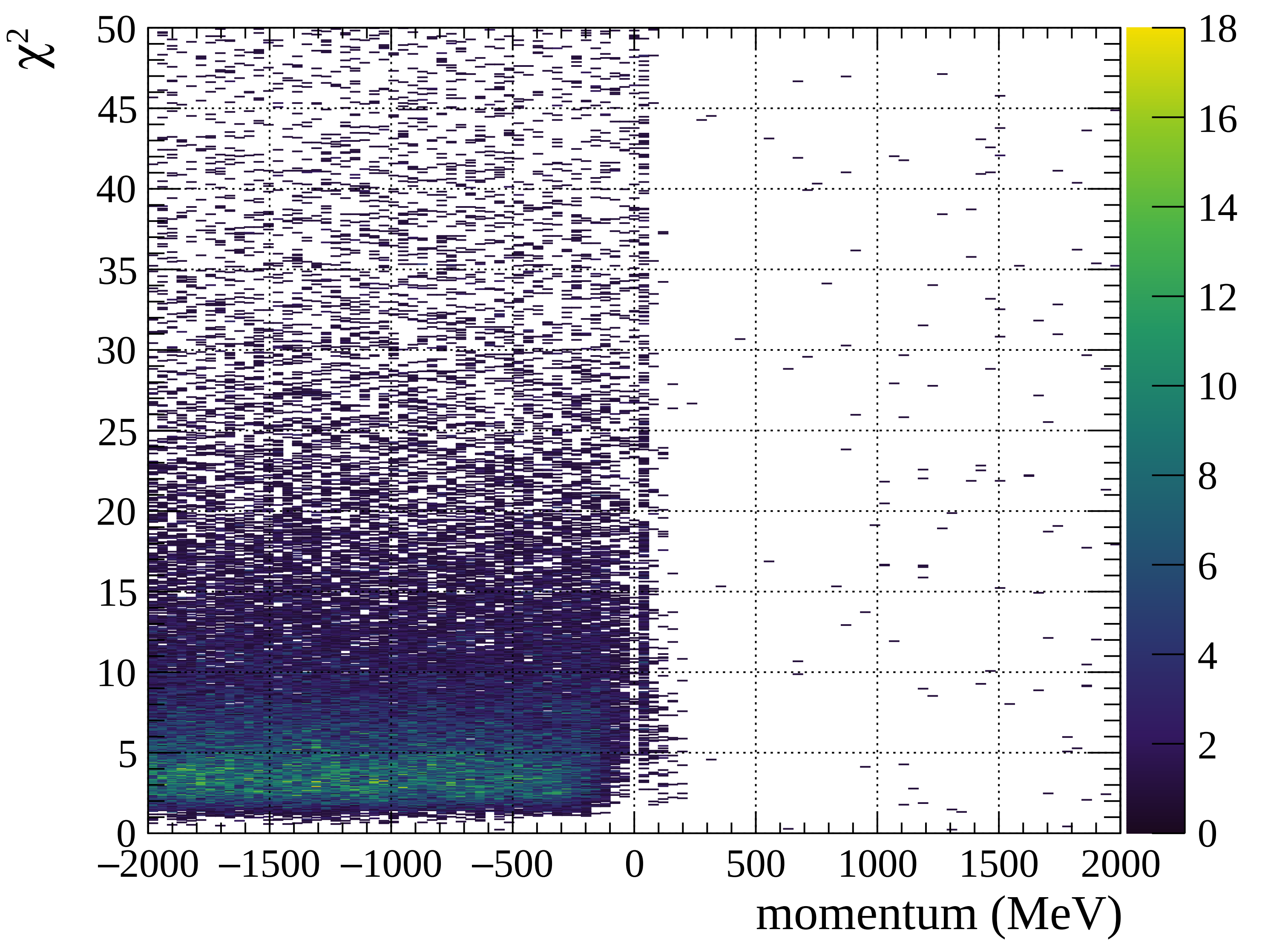
WF
XP
Mean (ADC counts/cm)
Momentum (MeV/c)
600
550
500
450
400
350
300
−2000
−1500
−1000
−500
0
500
1000
1500
2000


dE/dx with WF (ADC counts/cm)
momentum (MeV)
1000
900
800
700
600
500
400
300
200
100
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
18
16
14
12
10
8
6
4
2
0

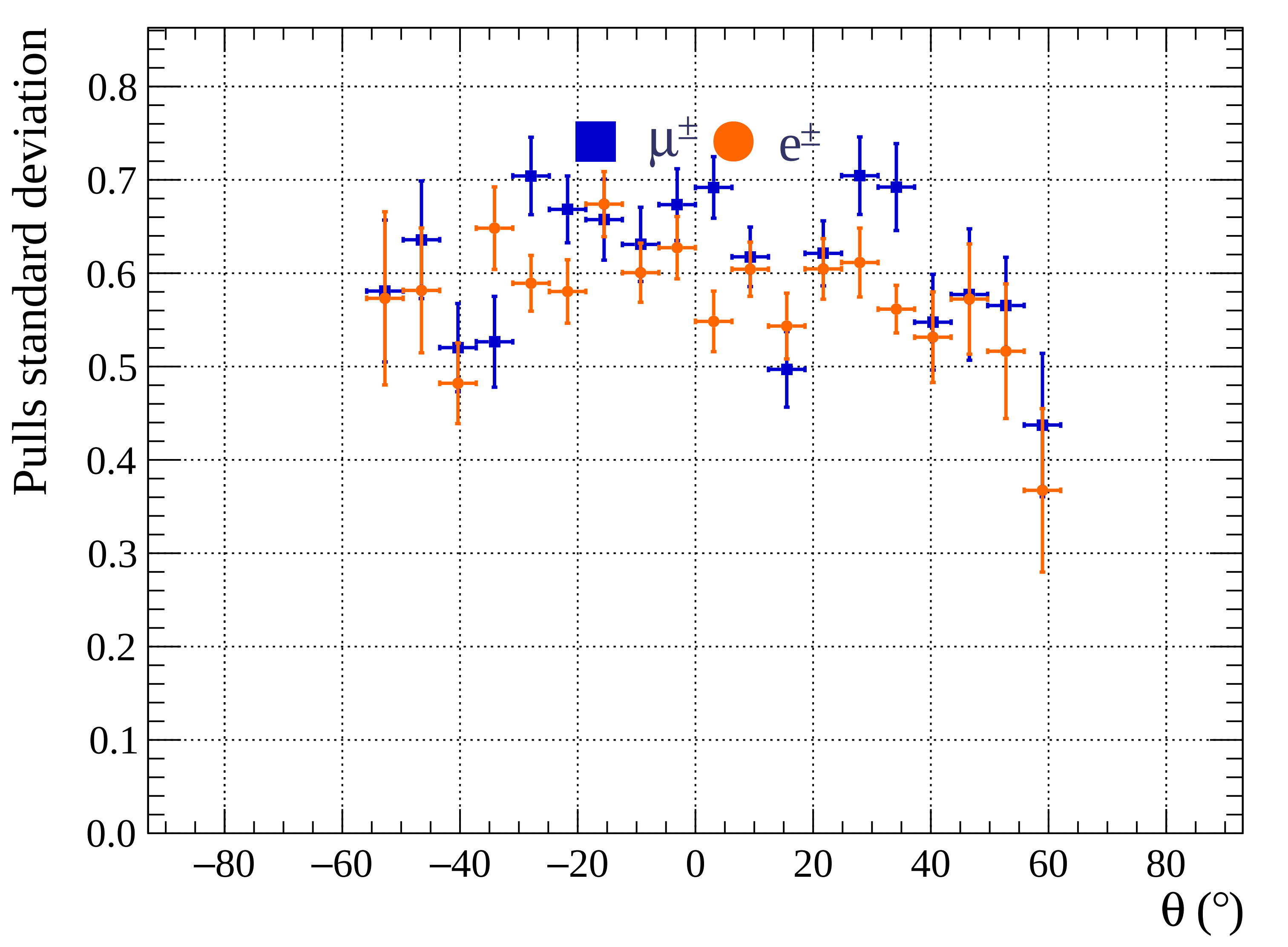


dE/dx with XP (ADC counts/cm)
momentum (MeV)
0
100
200
300
400
500
600
700
800
900
1000
−2000
−1500
−1000
−500
0
500
1000
1500
2000
0
2
4
6
8
10
12
14
16

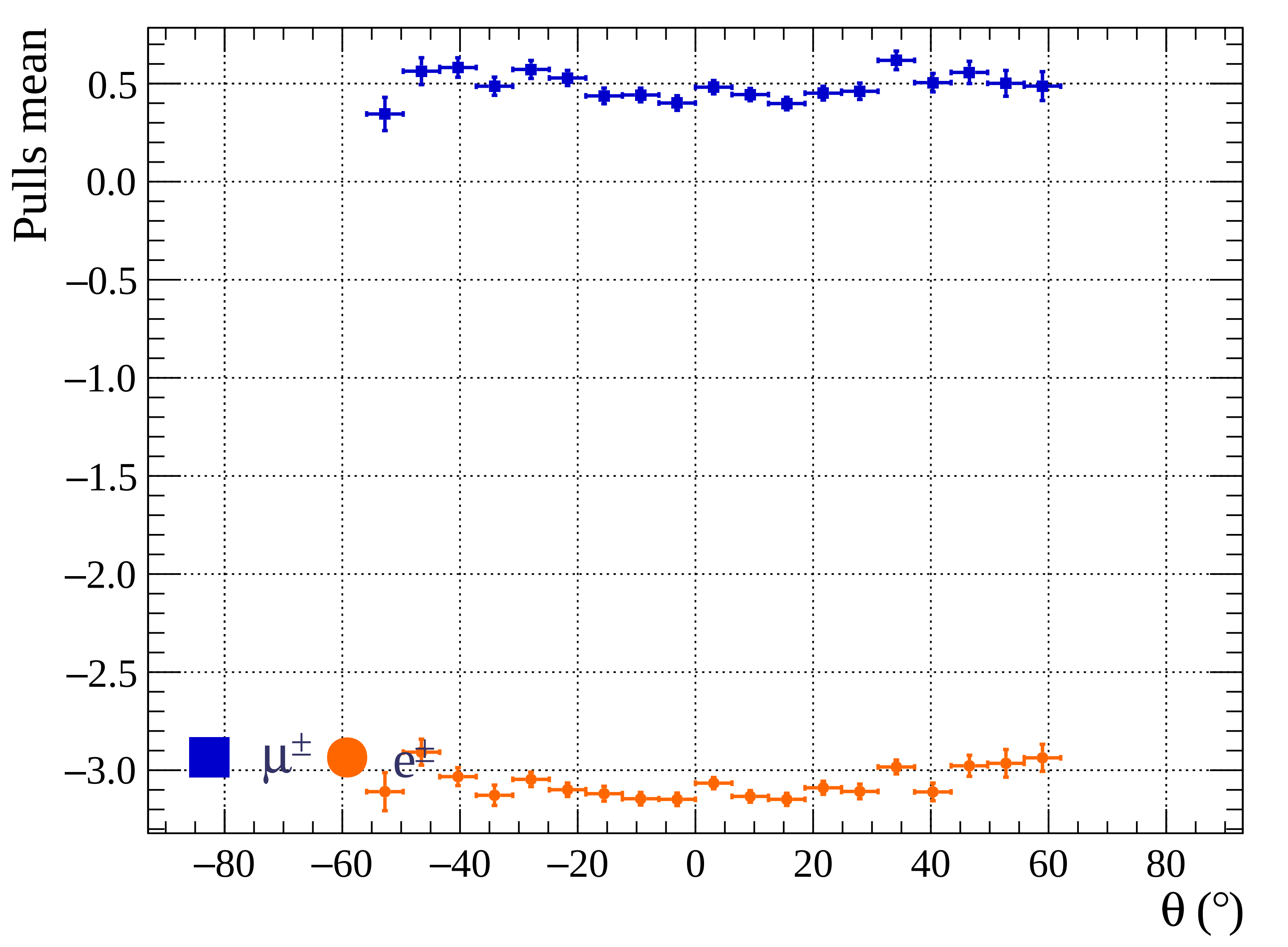

Count
600
500
400
300
200
100
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
Both HATs
tHAT
bHAT



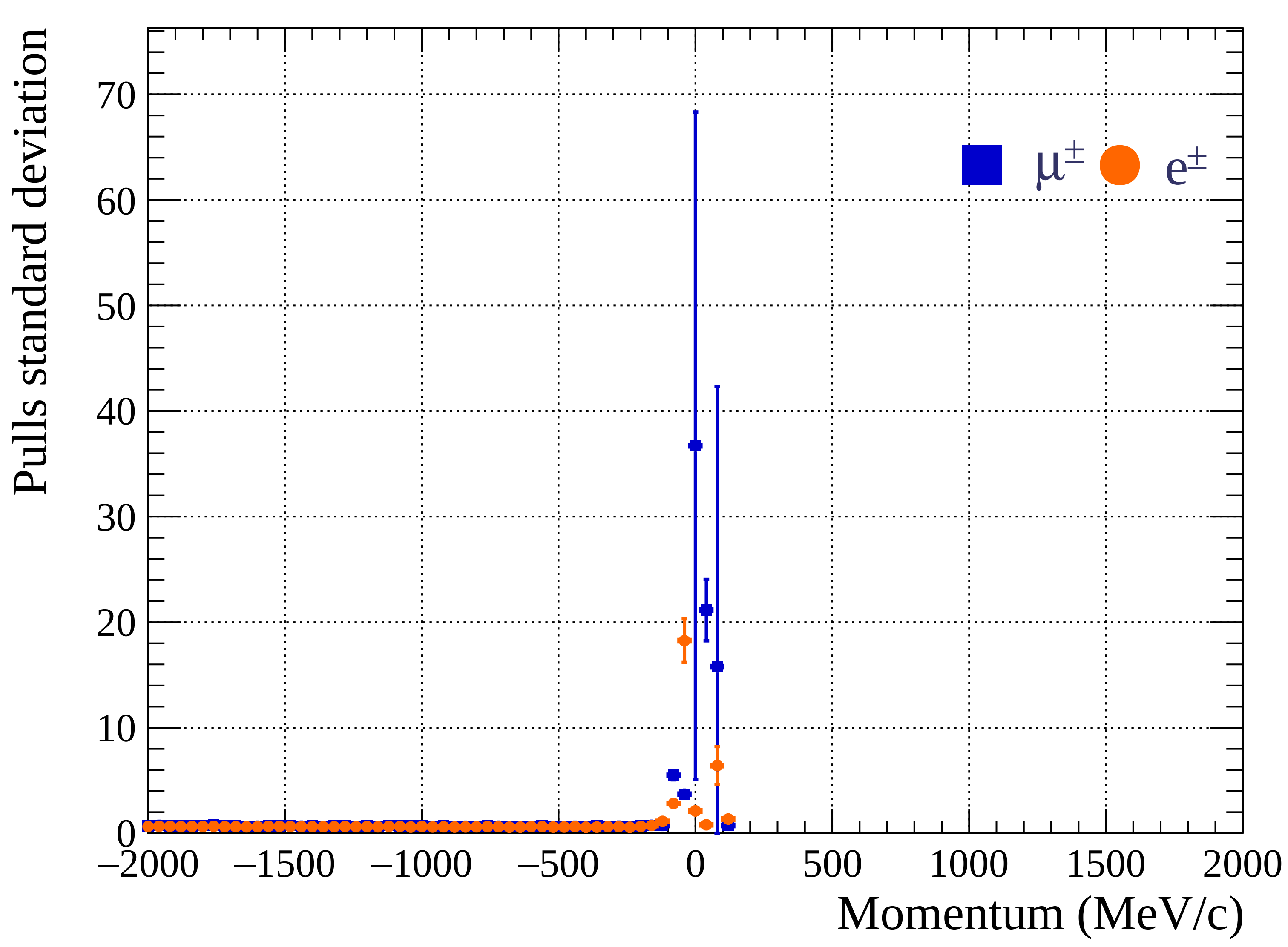
χ^2
momentum (MeV)



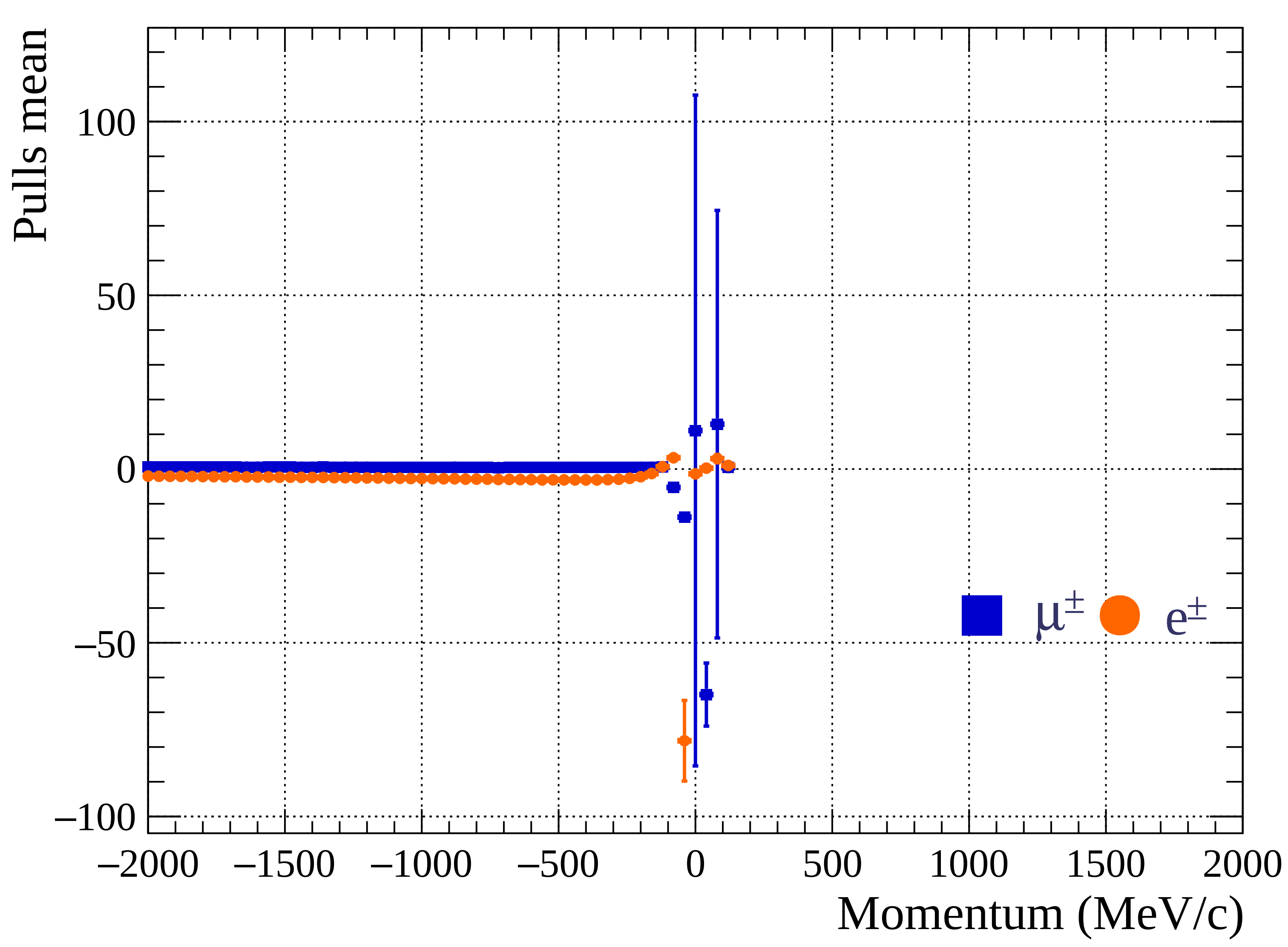
Pulls standard deviation
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
−80
−60
−40
−20
0
20
40
60
80
θ (°)
μ±
e±

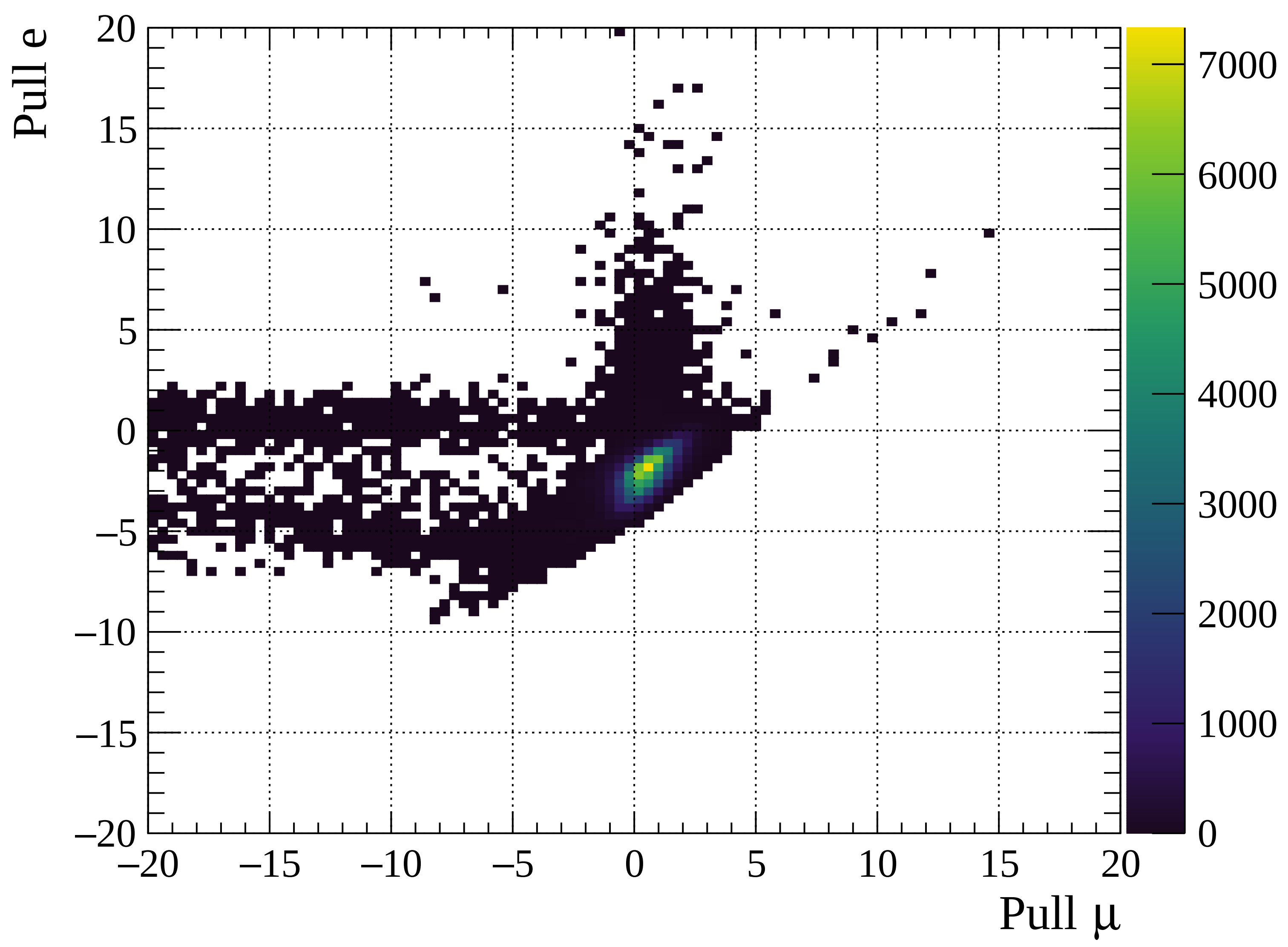


Pulls mean
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
−3.0
−80
−60
−40
−20
0
20
40
60
80
θ (°)
μ±
e±

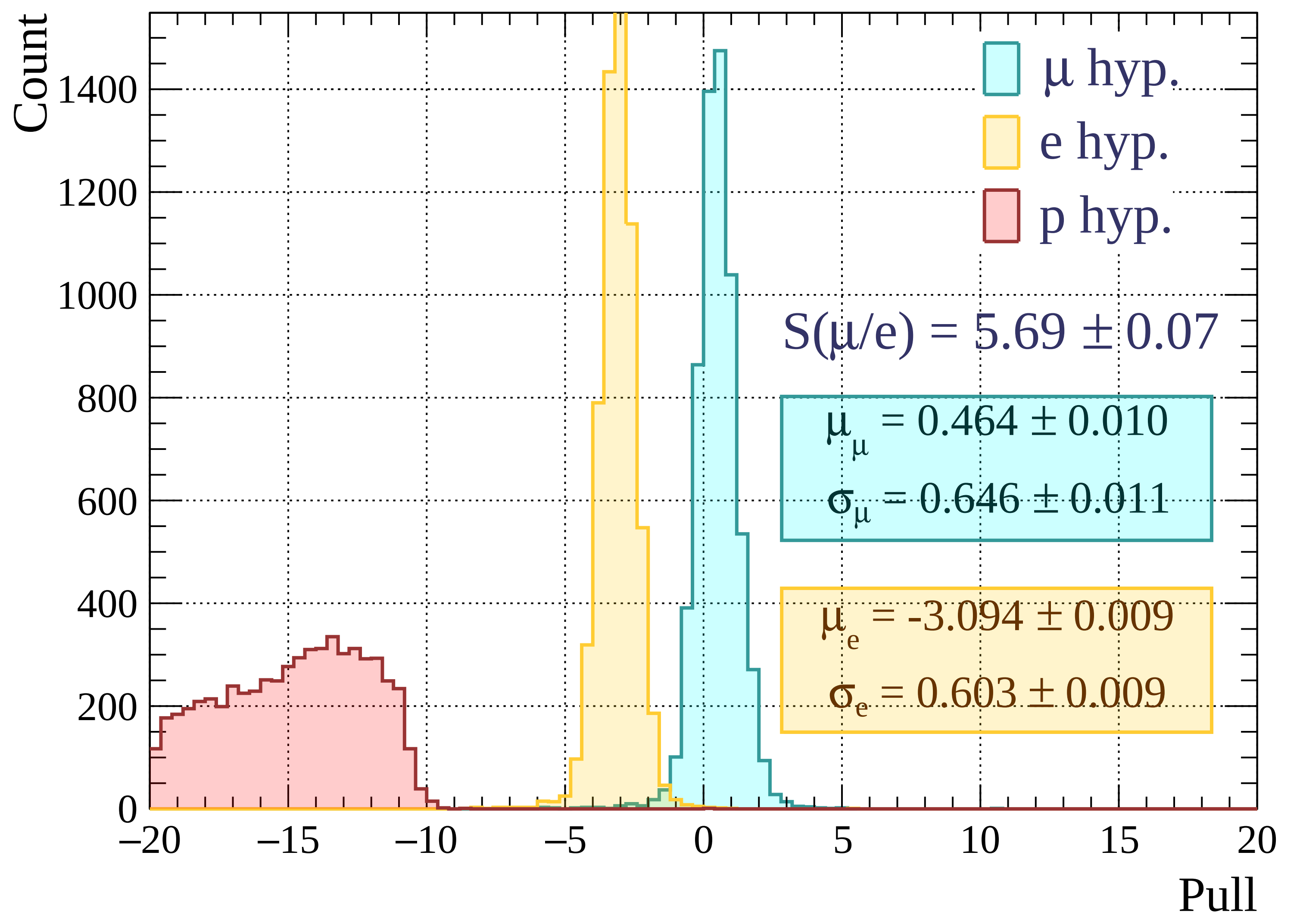


Pulls standard deviation
70
60
50
40
30
20
10
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)
$\mu^{\pm}$
$e^{\pm}$

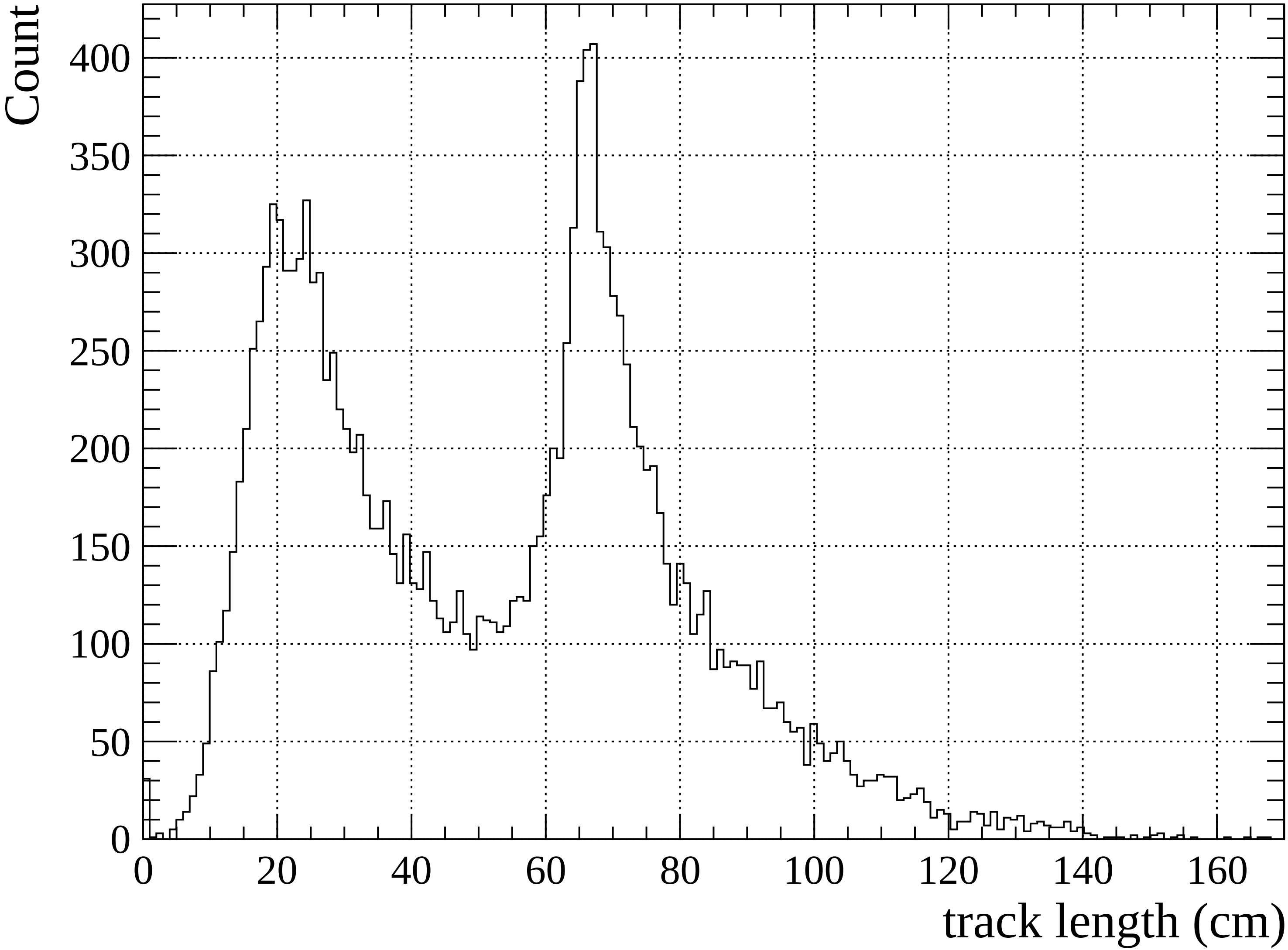

Pulls mean
100
50
0
−50
−100
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)
$\mu^{\pm}$
$e^{\pm}$



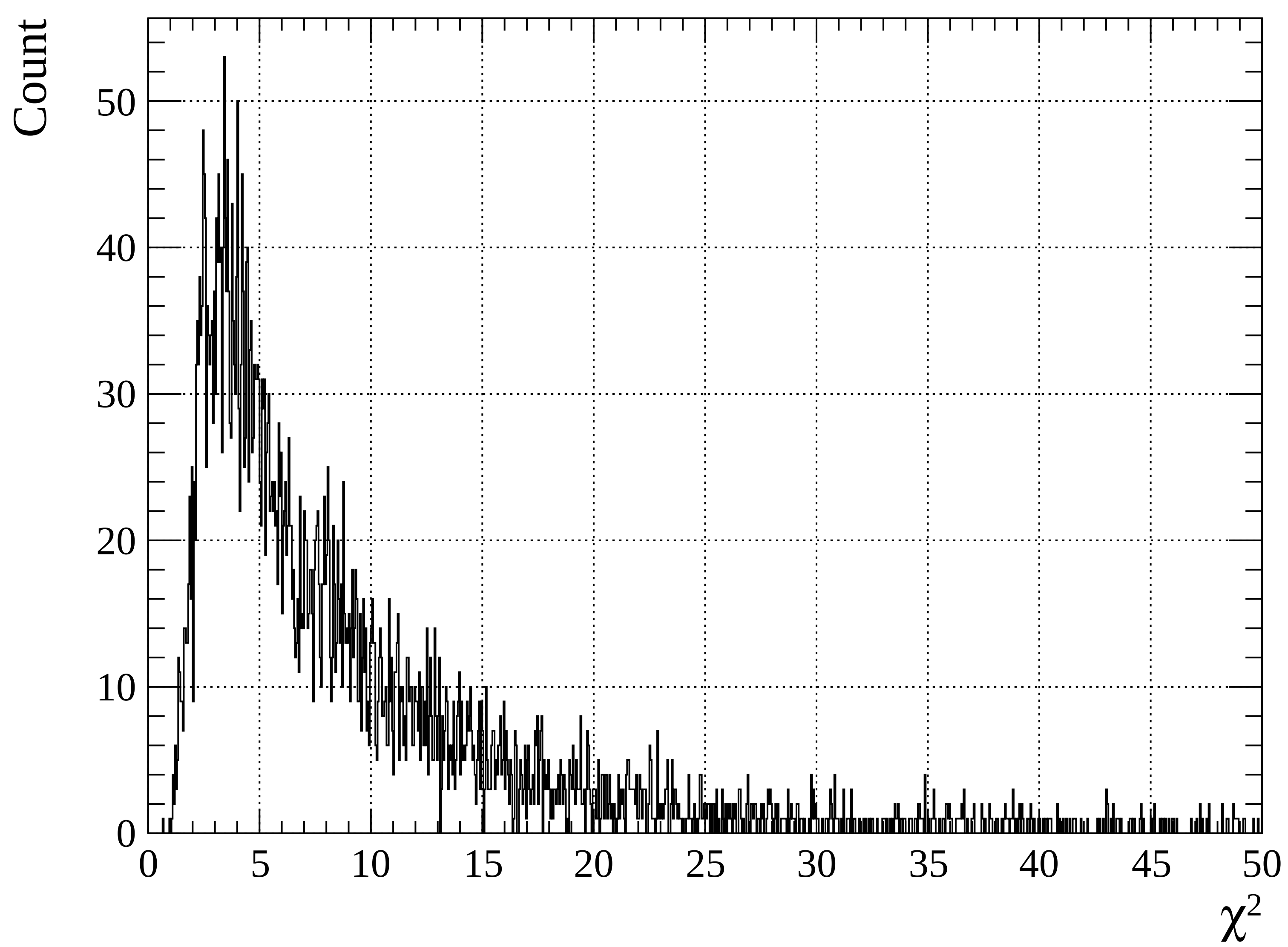
Pull e
20
15
10
5
0
−5
−10
−15
−20
−20
−15
−10
−5
0
5
10
15
20
Pull μ
7000
6000
5000
4000
3000
2000
1000
0



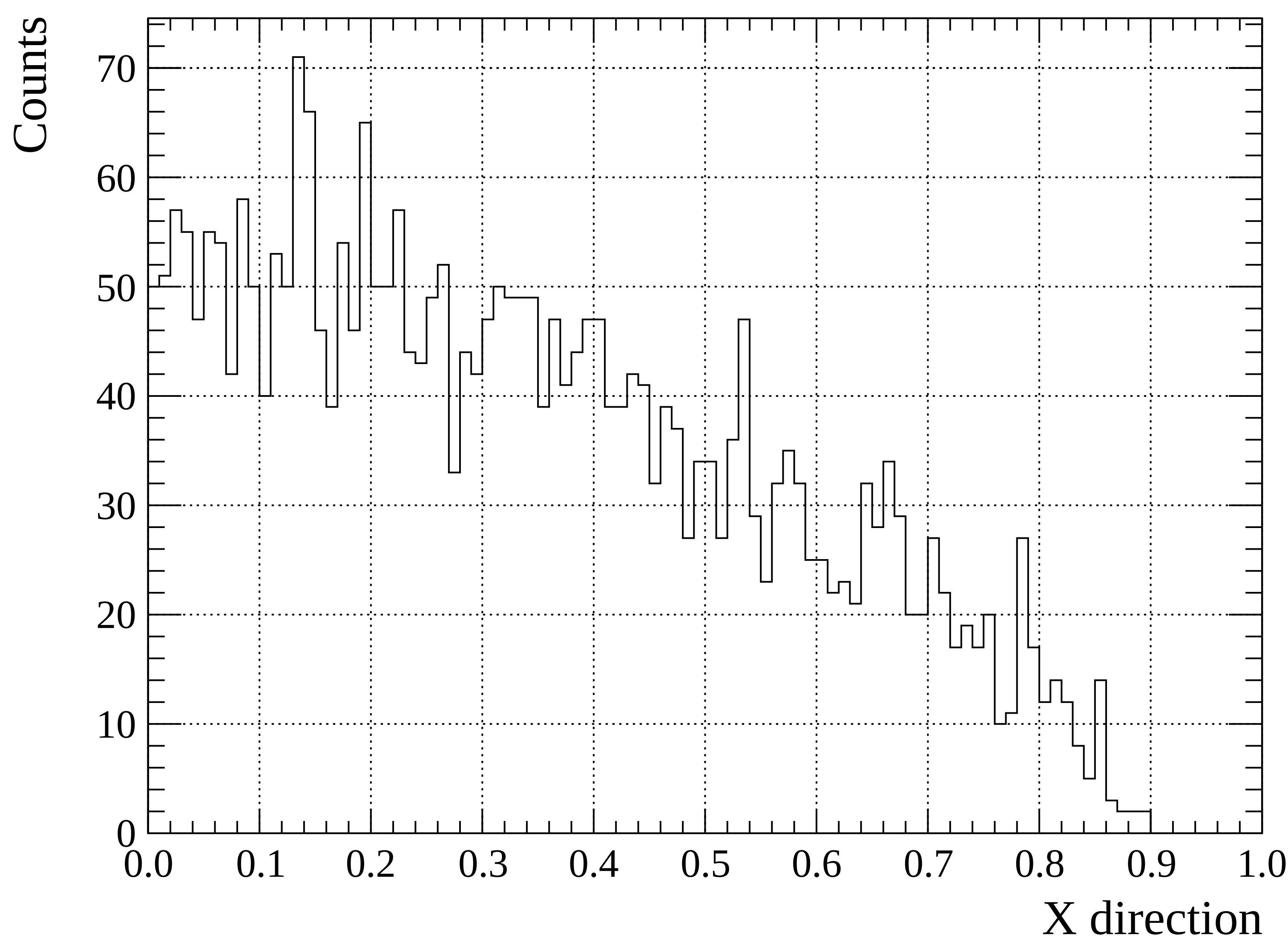
Count
Pull
μ hyp.
e hyp.
p hyp.
S(μ/e) = 5.69 ± 0.07
μμ = 0.464 ± 0.010
σμ = 0.646 ± 0.011
μe = -3.094 ± 0.009
σe = 0.603 ± 0.009



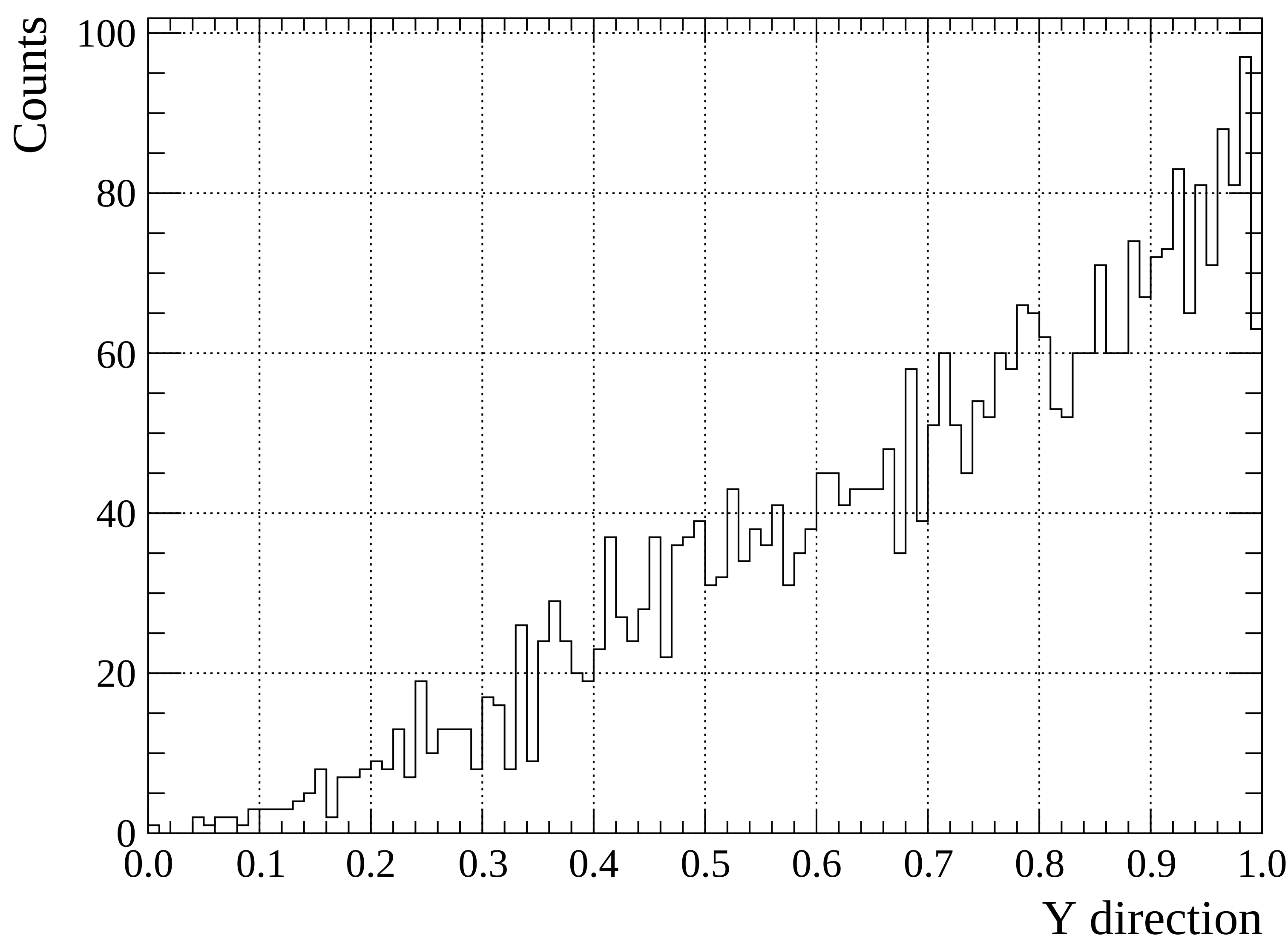
Count
400
350
300
250
200
150
100
50
0
0
20
40
60
80
100
120
140
160
track length (cm)



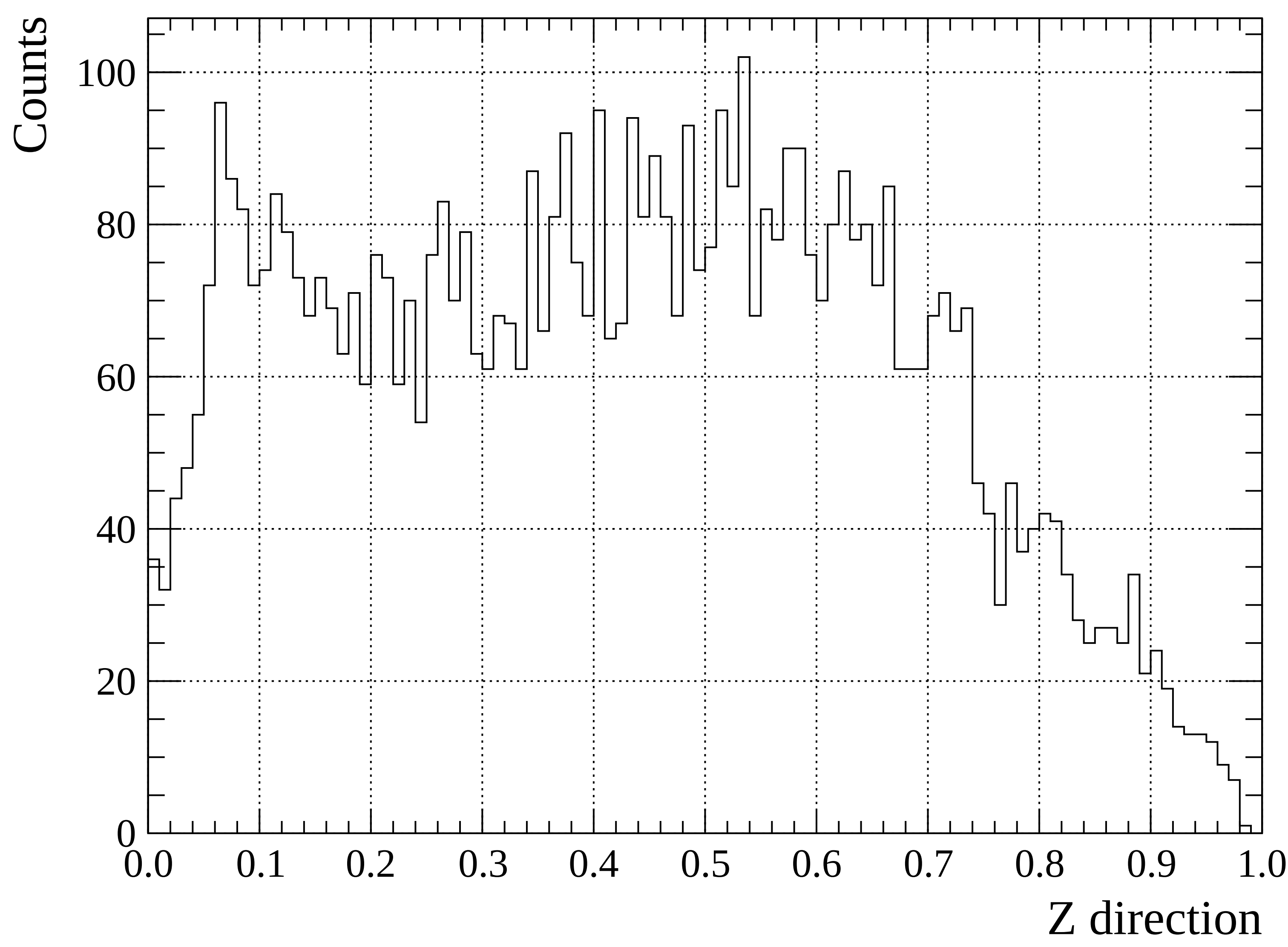
Count
0
10
20
30
40
50
0
5
10
15
20
25
30
35
40
45
50
χ^2

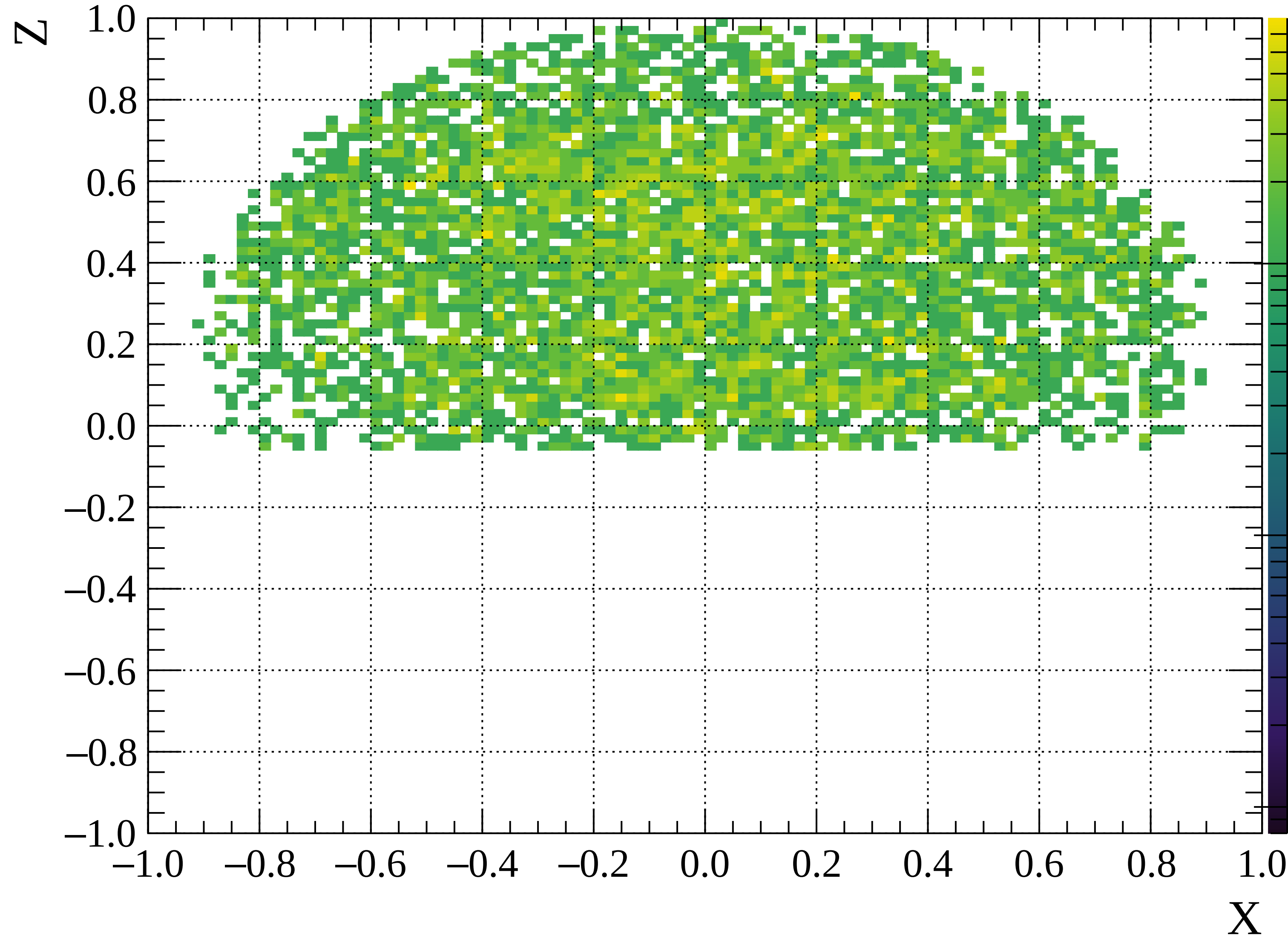


Counts
70
60
50
40
30
20
10
0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
X direction

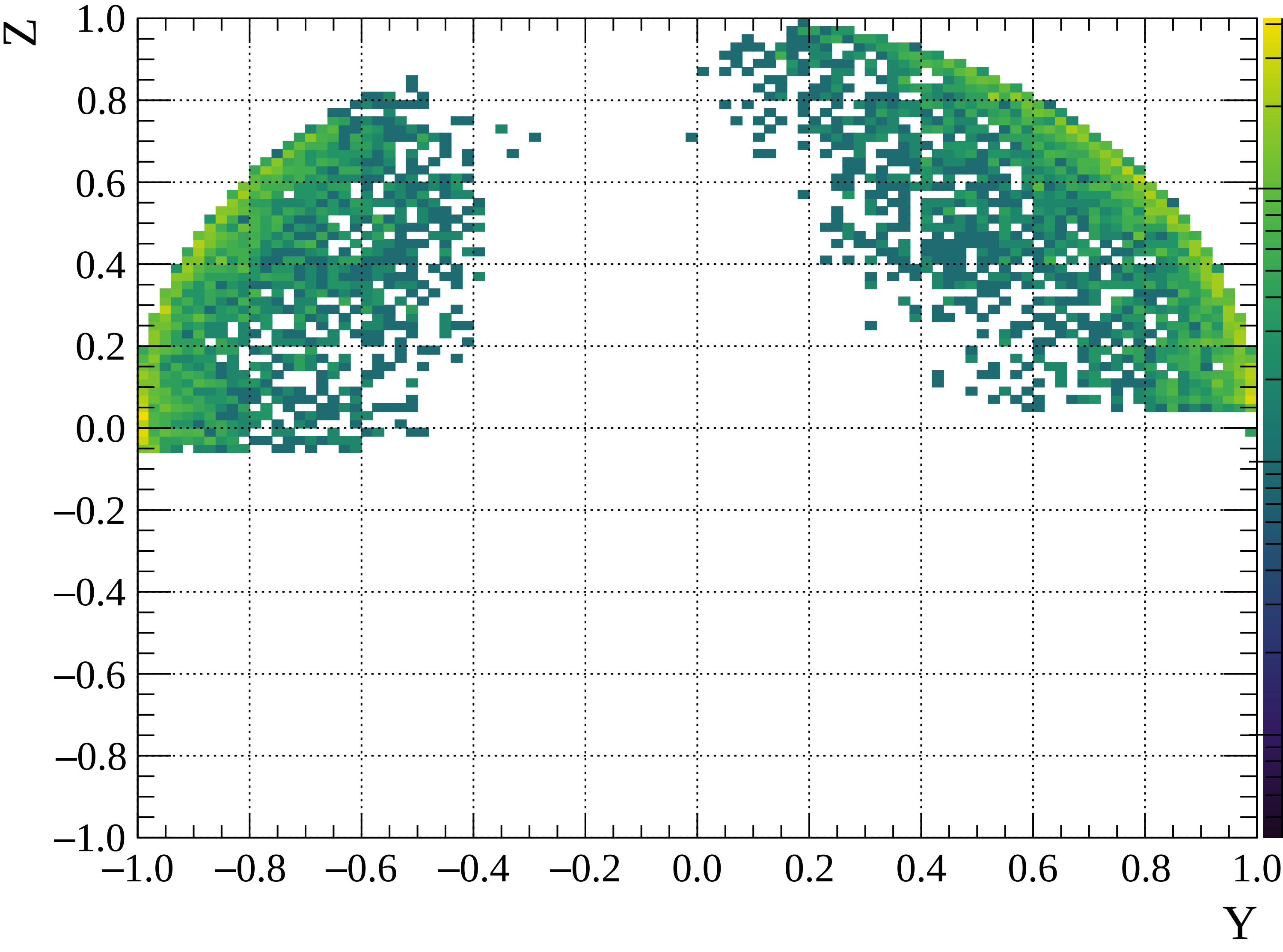


Counts
100
80
60
40
20
0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Y direction

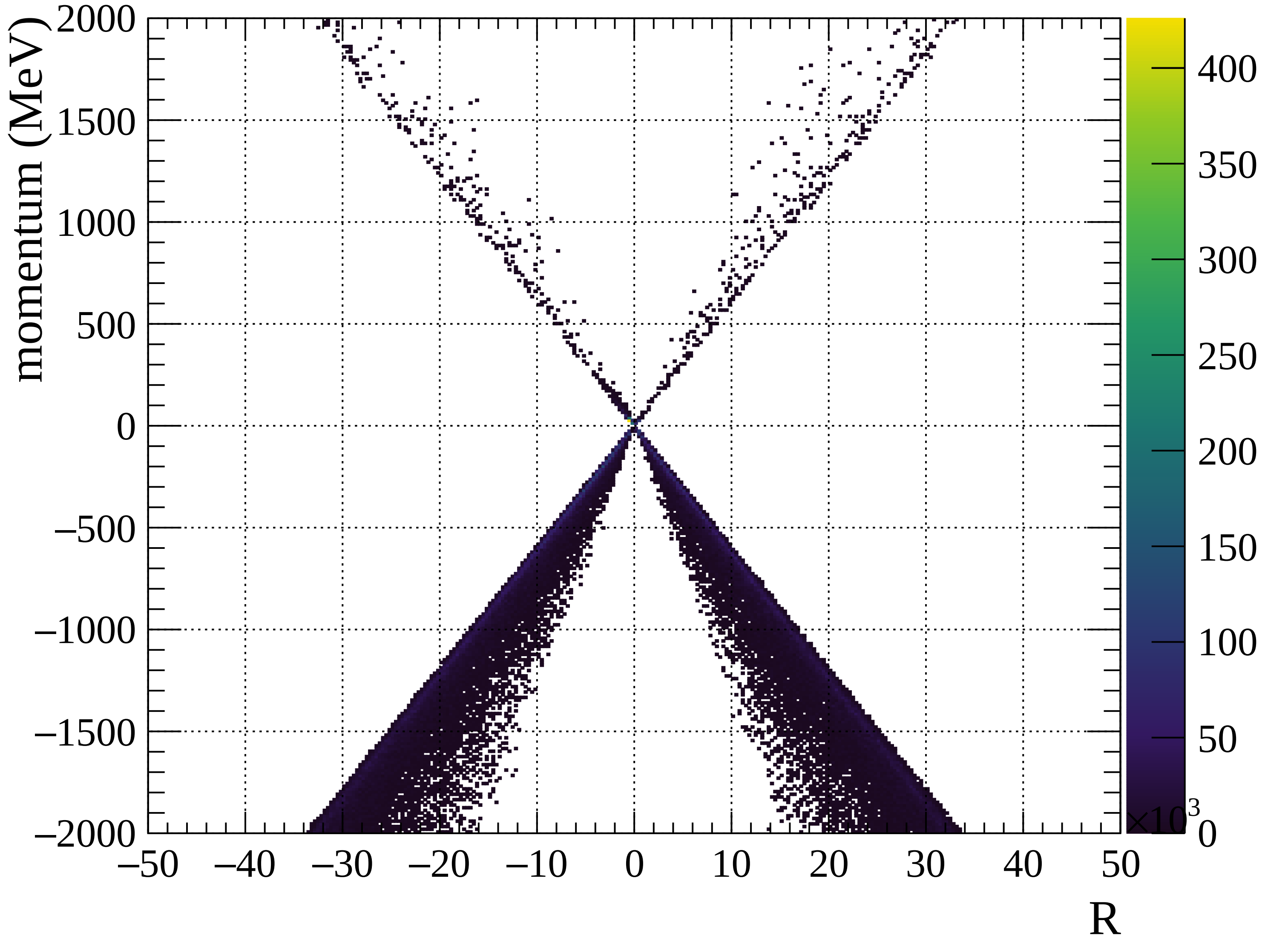

Counts
100
80
60
40
20
0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Z direction



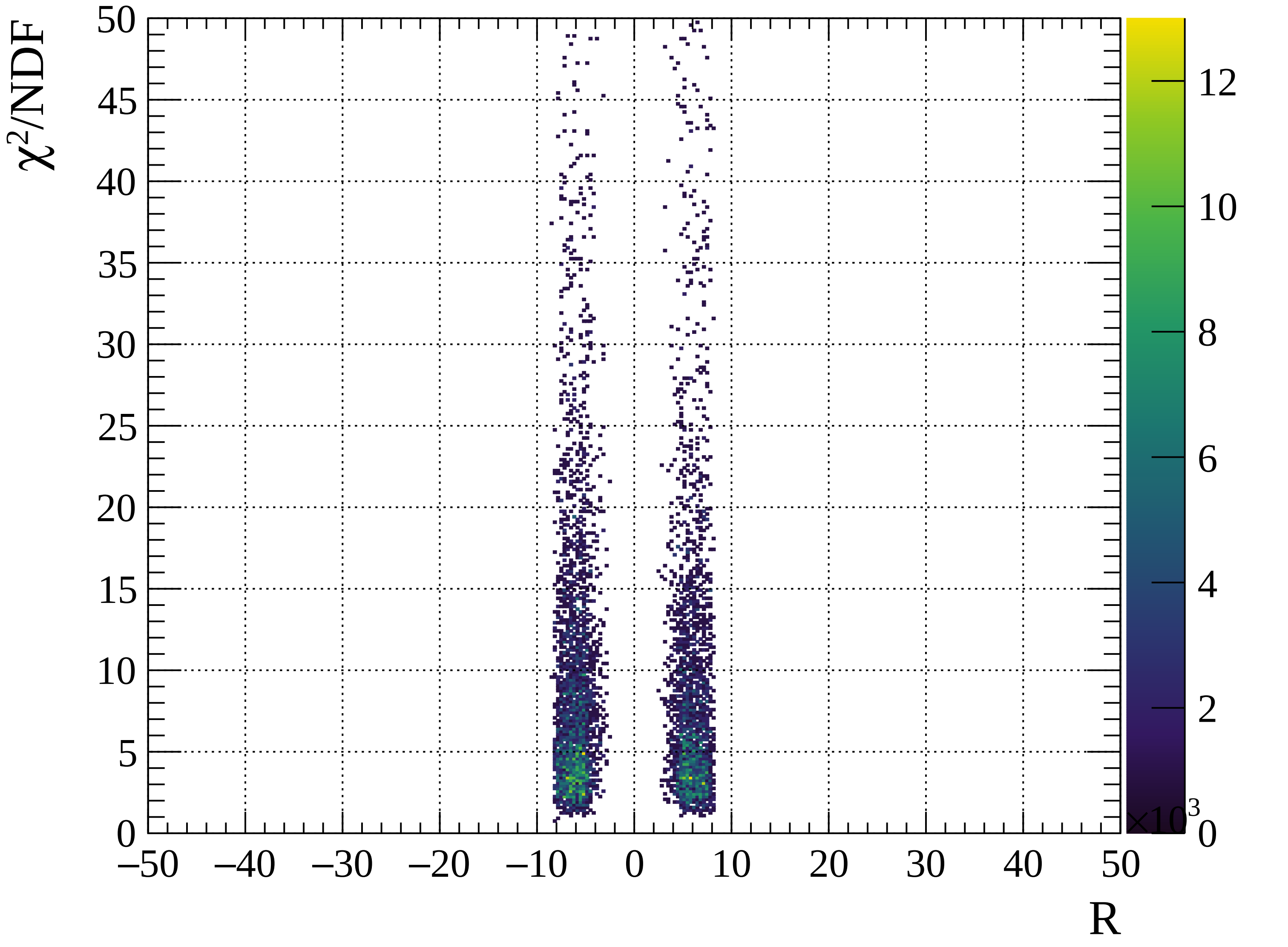
Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
X



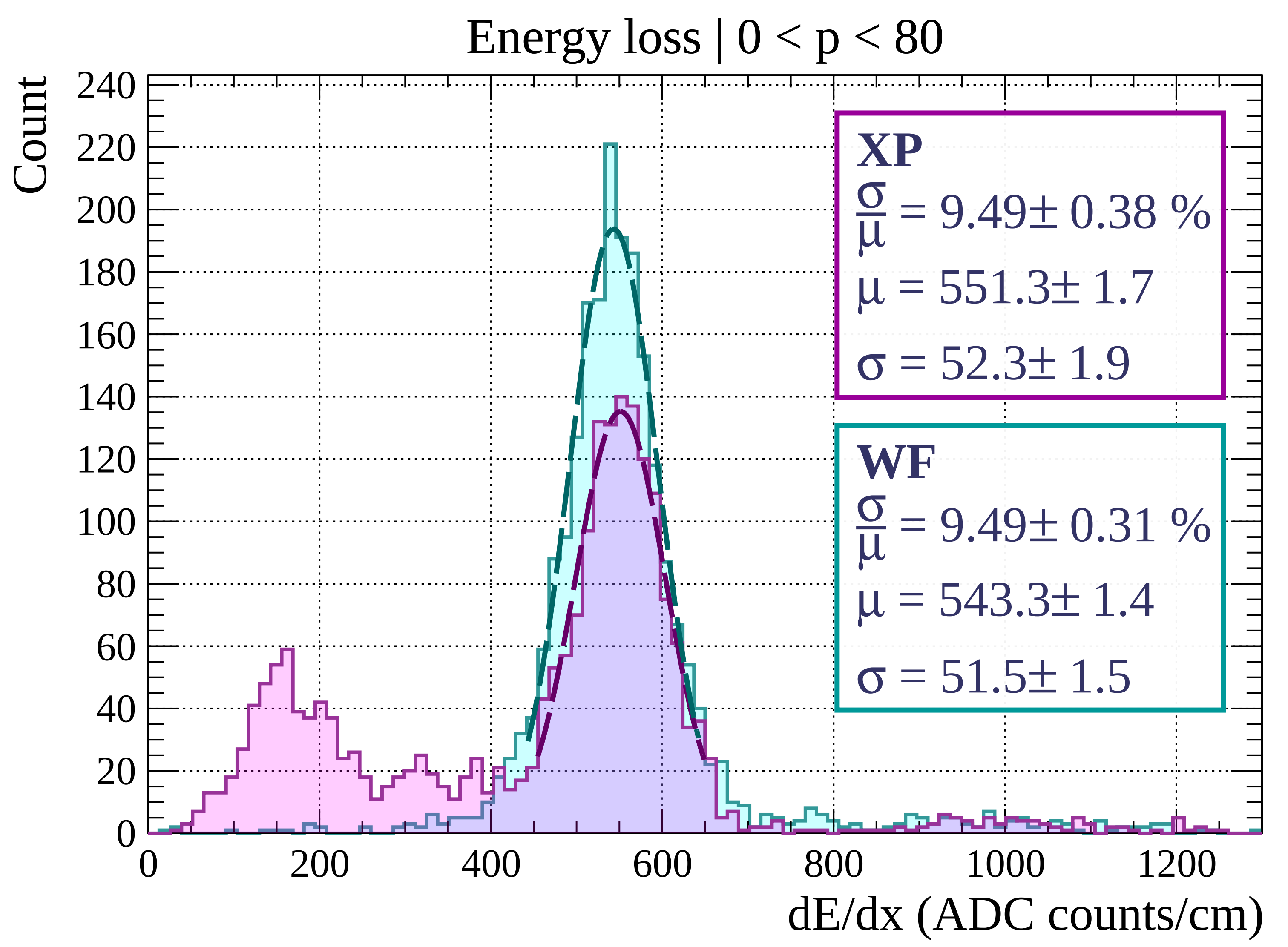
Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
Y



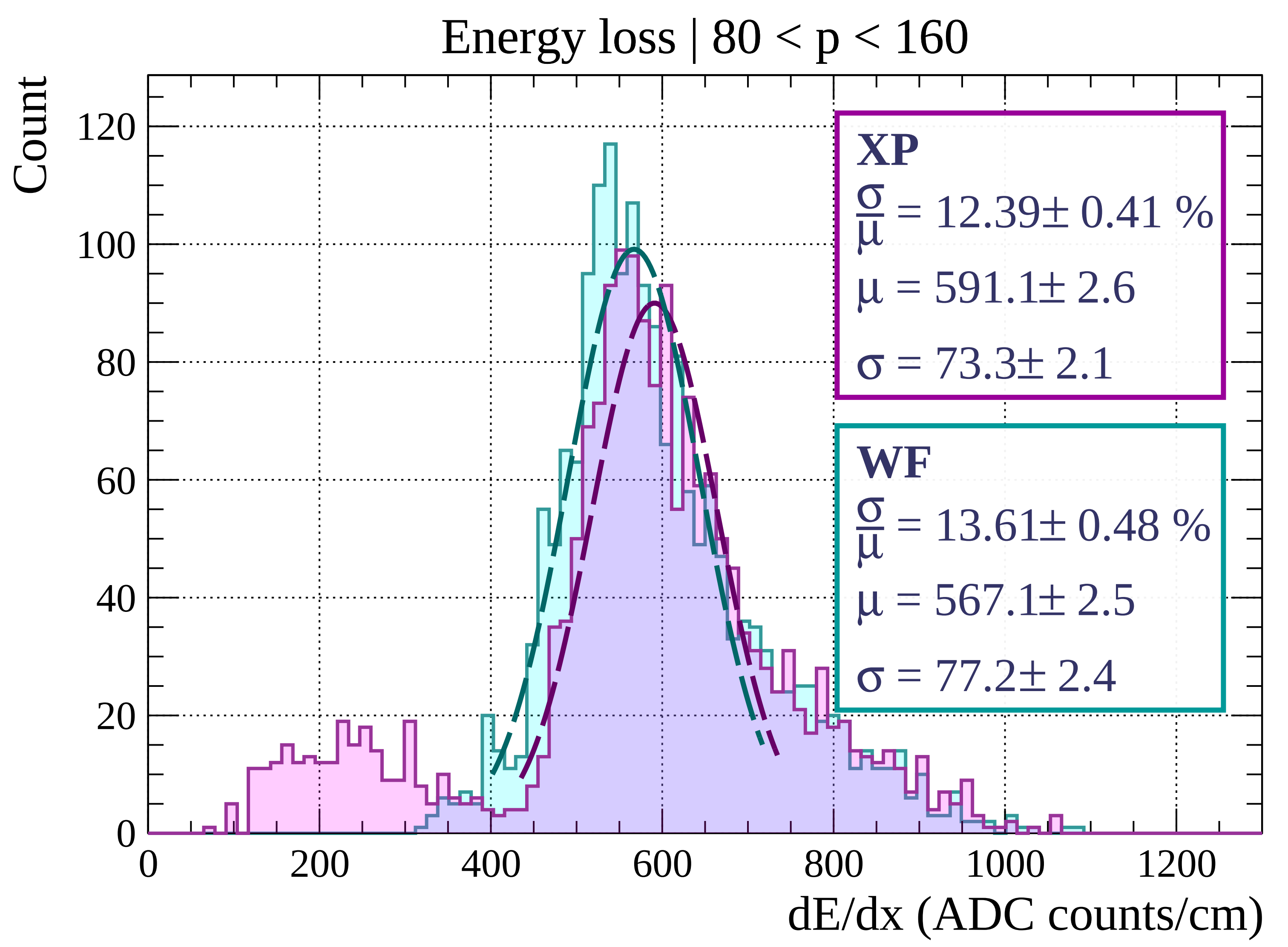
momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−50
−40
−30
−20
−10
0
10
20
30
40
50
R
400
350
300
250
200
150
100
50
0
×10^3

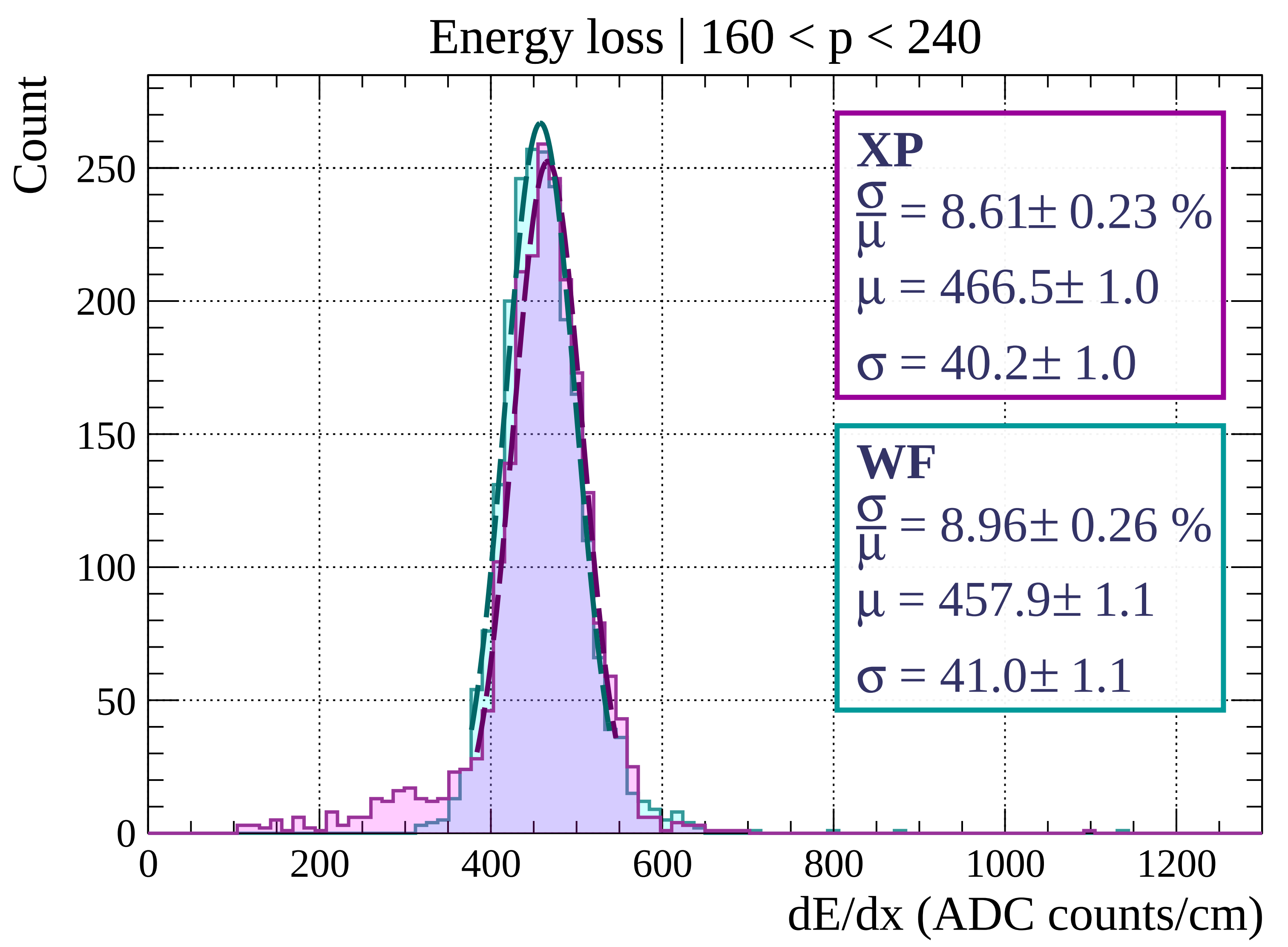


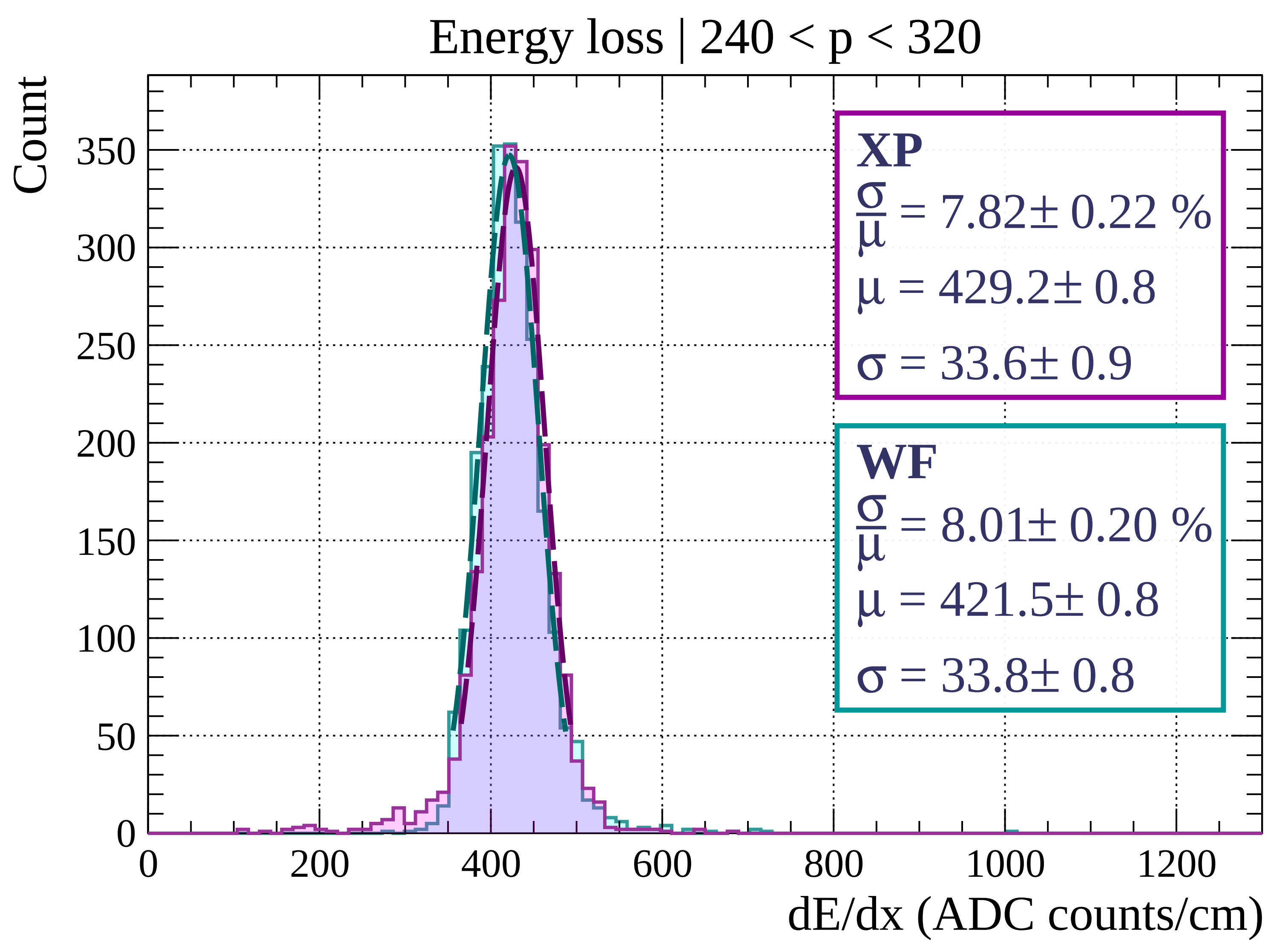
χ^2/NDF
R
$\times 10^3$

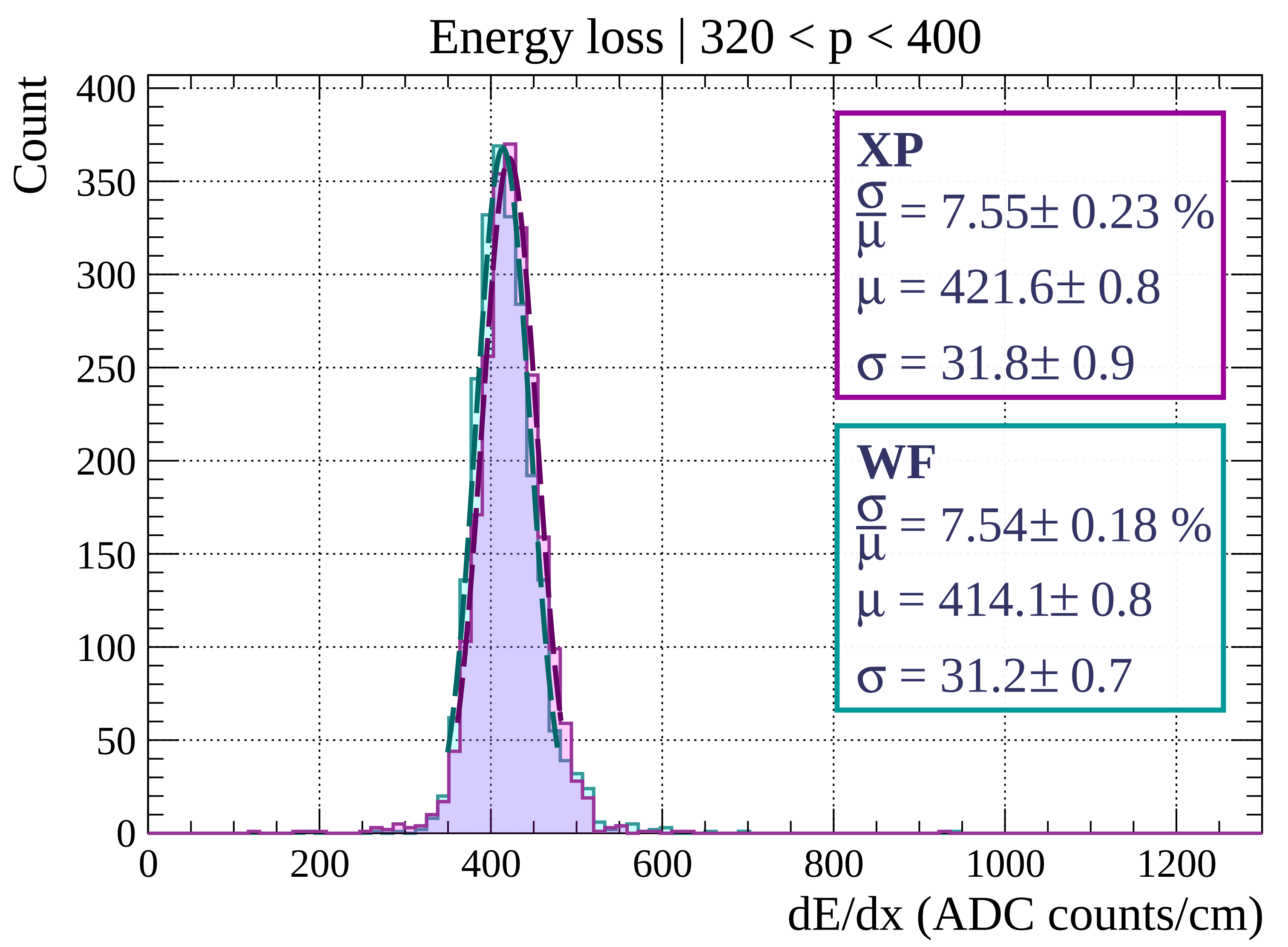


Energy loss | 0 < p < 80
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
220
240
XP
$\frac{\sigma}{\mu}$ = 9.49± 0.38 %
μ = 551.3± 1.7
σ = 52.3± 1.9
WF
$\frac{\sigma}{\mu}$ = 9.49± 0.31 %
μ = 543.3± 1.4
σ = 51.5± 1.5

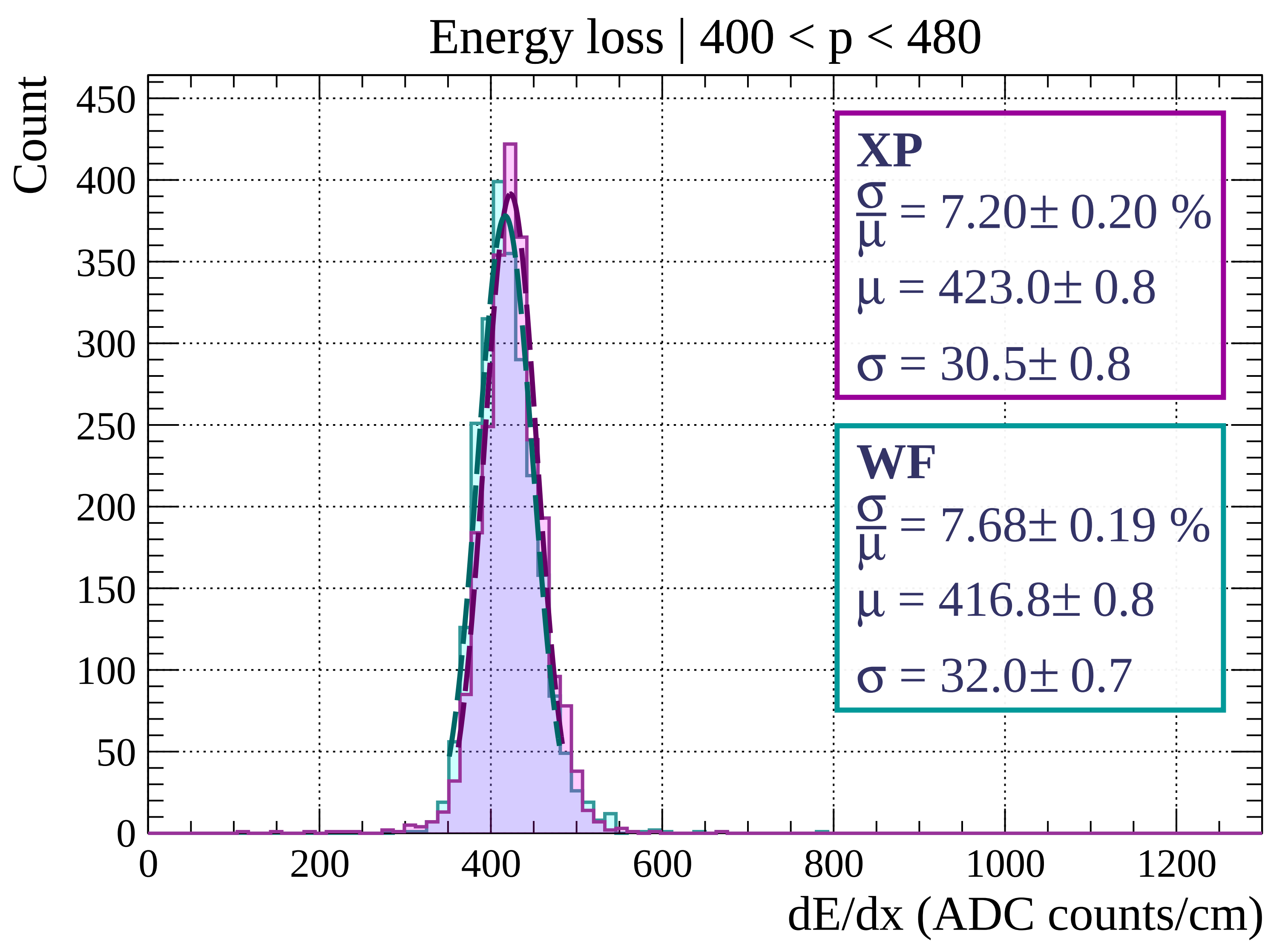

Energy loss | 80 < p < 160
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
120
XP
$\frac{\sigma}{\mu}$ = 12.39± 0.41 %
μ = 591.1± 2.6
σ = 73.3± 2.1
WF
$\frac{\sigma}{\mu}$ = 13.61± 0.48 %
μ = 567.1± 2.5
σ = 77.2± 2.4

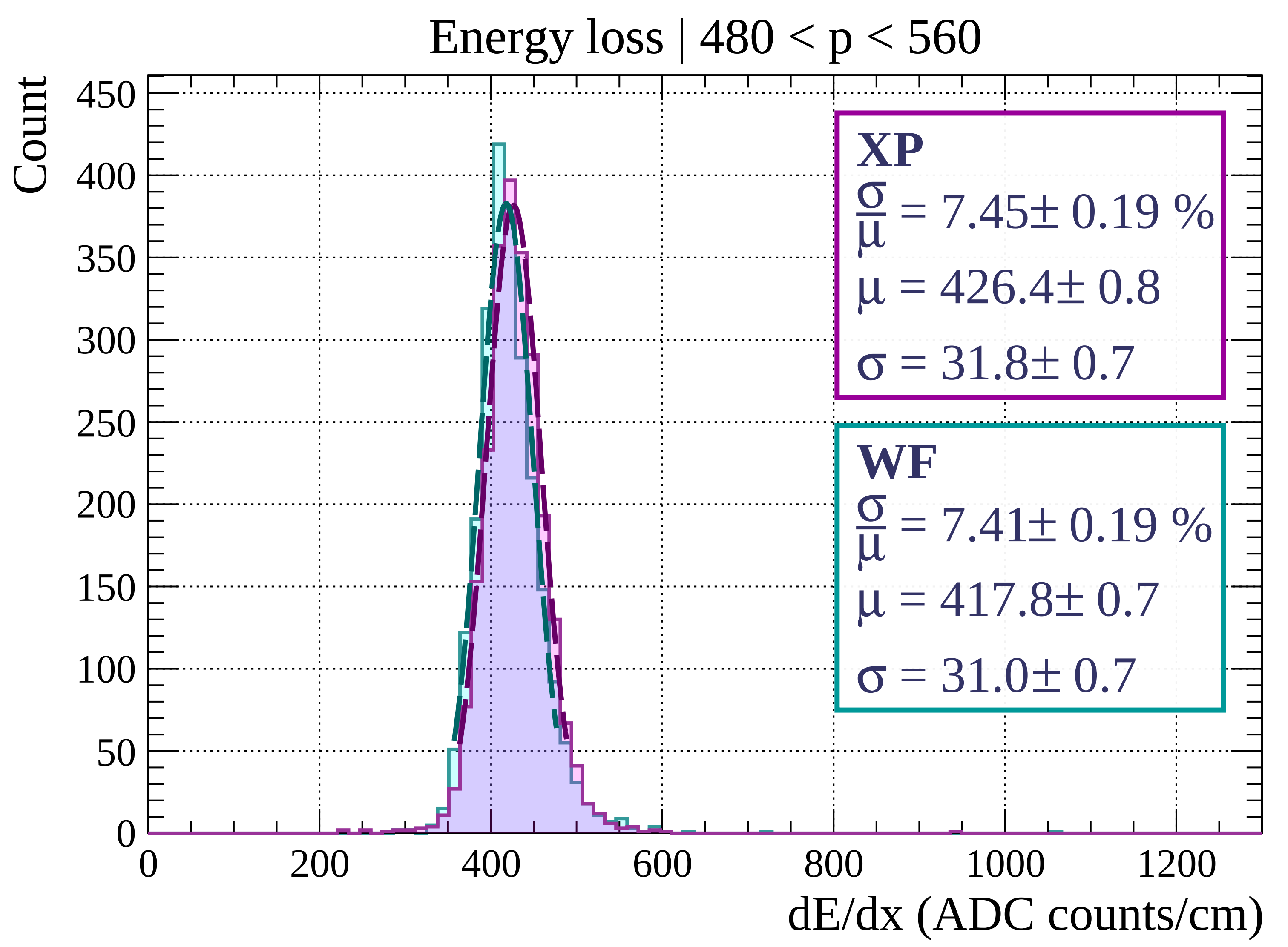

Energy loss | 160 < p < 240
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 8.61± 0.23 %
μ = 466.5± 1.0
σ = 40.2± 1.0
WF
σ/μ = 8.96± 0.26 %
μ = 457.9± 1.1
σ = 41.0± 1.1


Energy loss | 240 < p < 320
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
σ/μ = 7.82 ± 0.22 %
μ = 429.2 ± 0.8
σ = 33.6 ± 0.9
WF
σ/μ = 8.01 ± 0.20 %
μ = 421.5 ± 0.8
σ = 33.8 ± 0.8

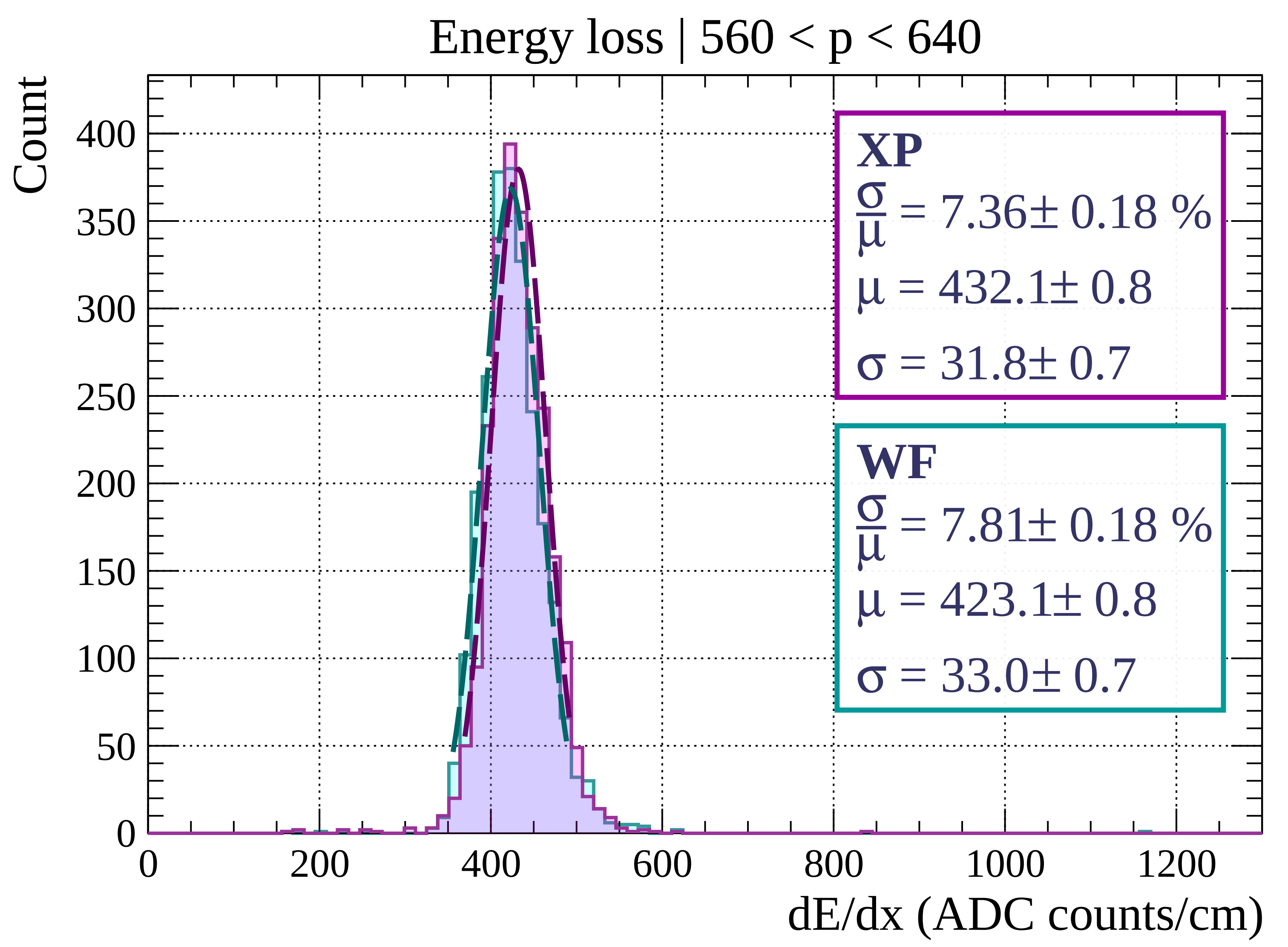


Energy loss | 320 < p < 400
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
σ/μ = 7.55± 0.23 %
μ = 421.6± 0.8
σ = 31.8± 0.9
WF
σ/μ = 7.54± 0.18 %
μ = 414.1± 0.8
σ = 31.2± 0.7

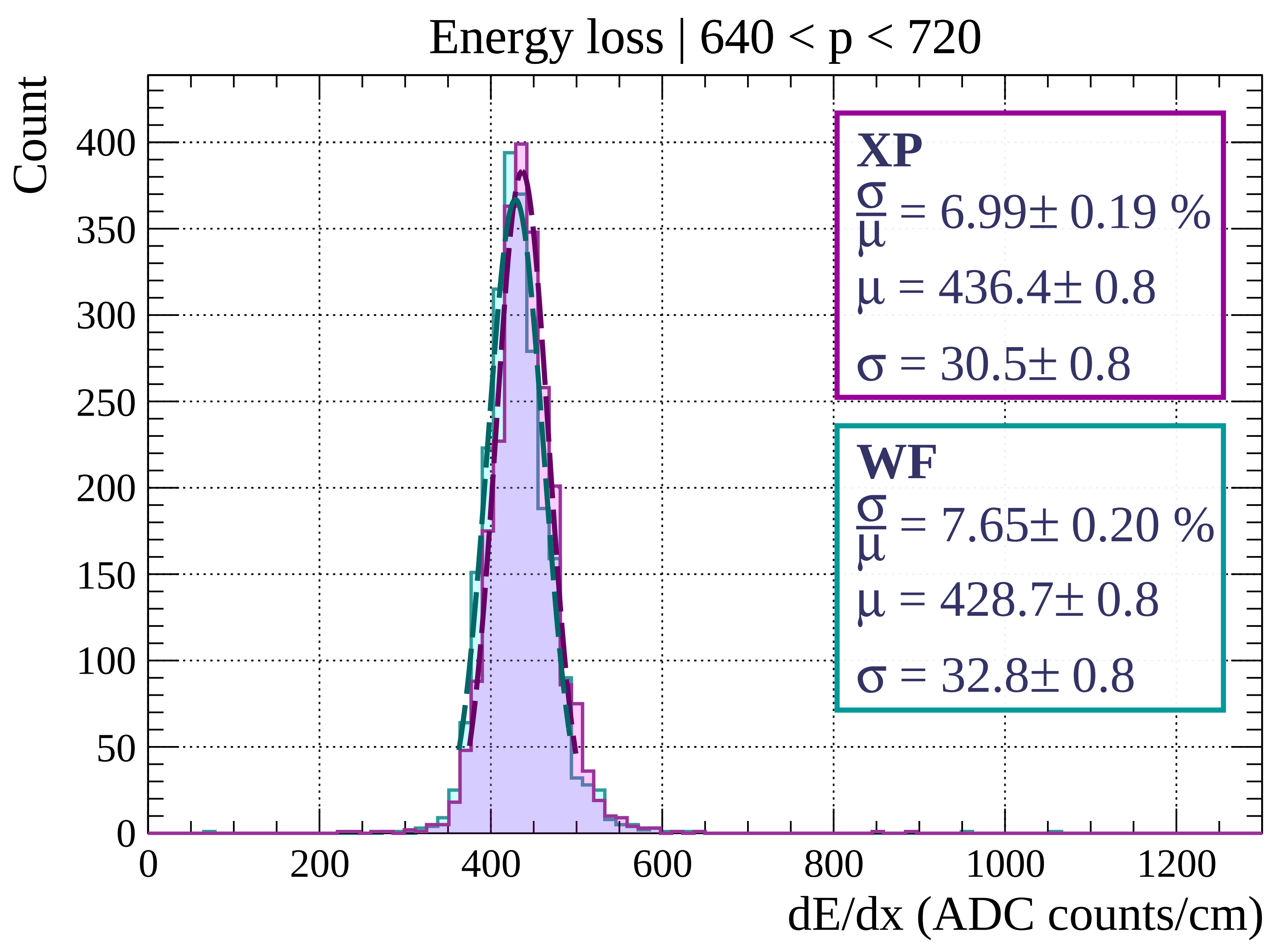

Energy loss | 400 < p < 480
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 7.20± 0.20 %
μ = 423.0± 0.8
σ = 30.5± 0.8
WF
σ/μ = 7.68± 0.19 %
μ = 416.8± 0.8
σ = 32.0± 0.7
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200



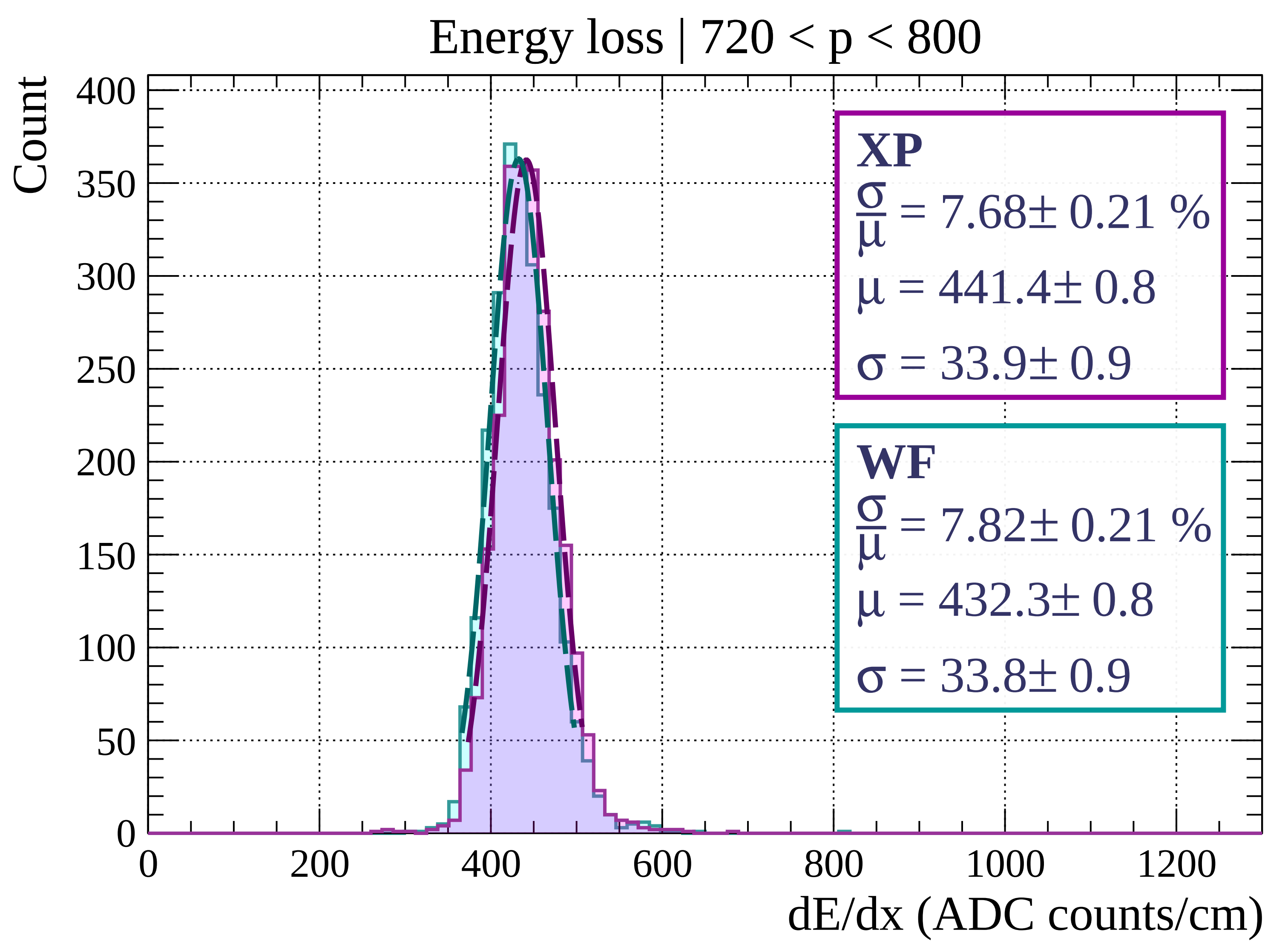
Energy loss | 480 < p < 560
Count
450
400
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 7.45± 0.19 %
μ = 426.4± 0.8
σ = 31.8± 0.7
WF
σ/μ = 7.41± 0.19 %
μ = 417.8± 0.7
σ = 31.0± 0.7



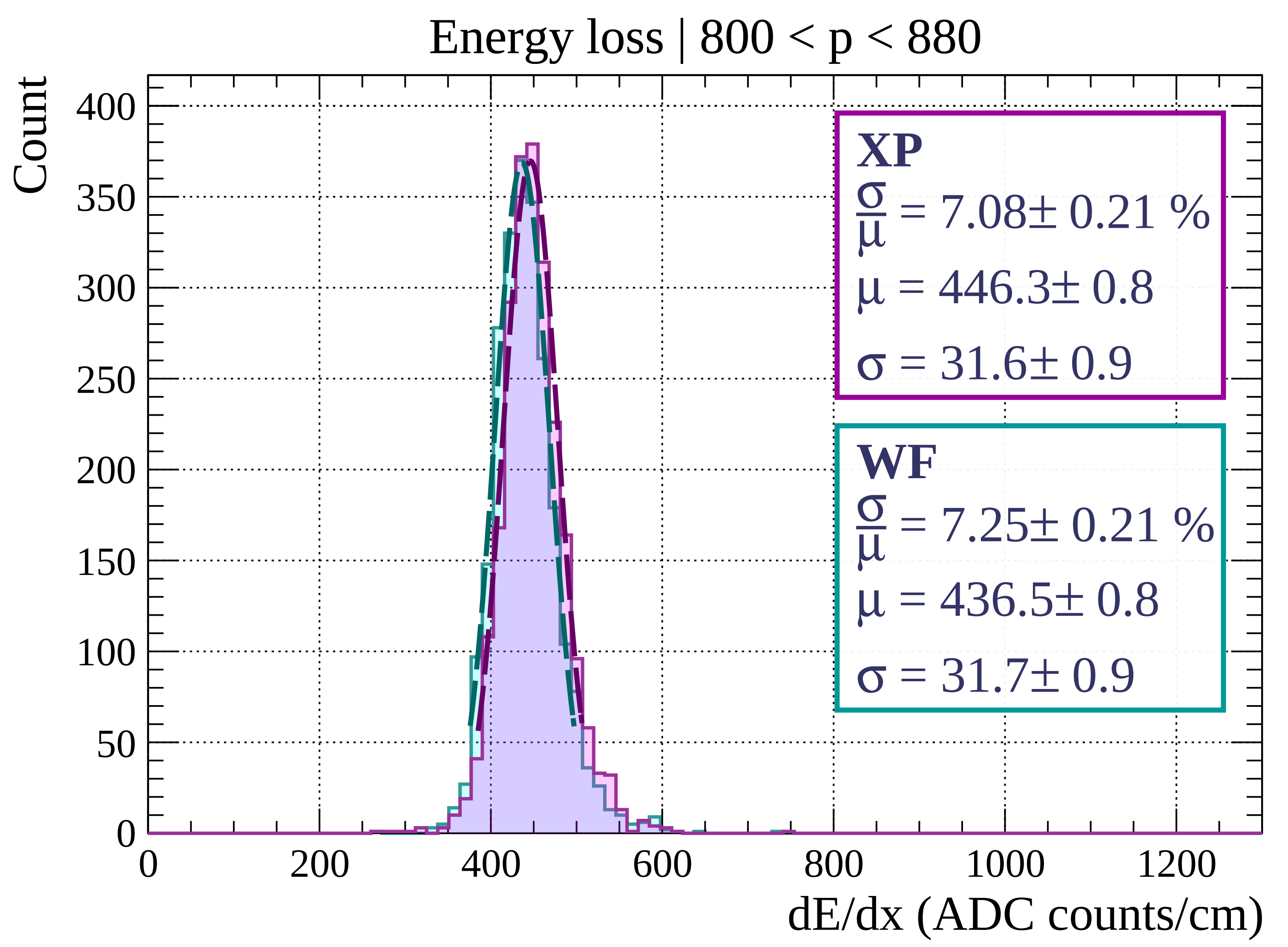
Energy loss | 560 < p < 640
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 7.36± 0.18 %
μ = 432.1± 0.8
σ = 31.8± 0.7
WF
$\frac{\sigma}{\mu}$ = 7.81± 0.18 %
μ = 423.1± 0.8
σ = 33.0± 0.7

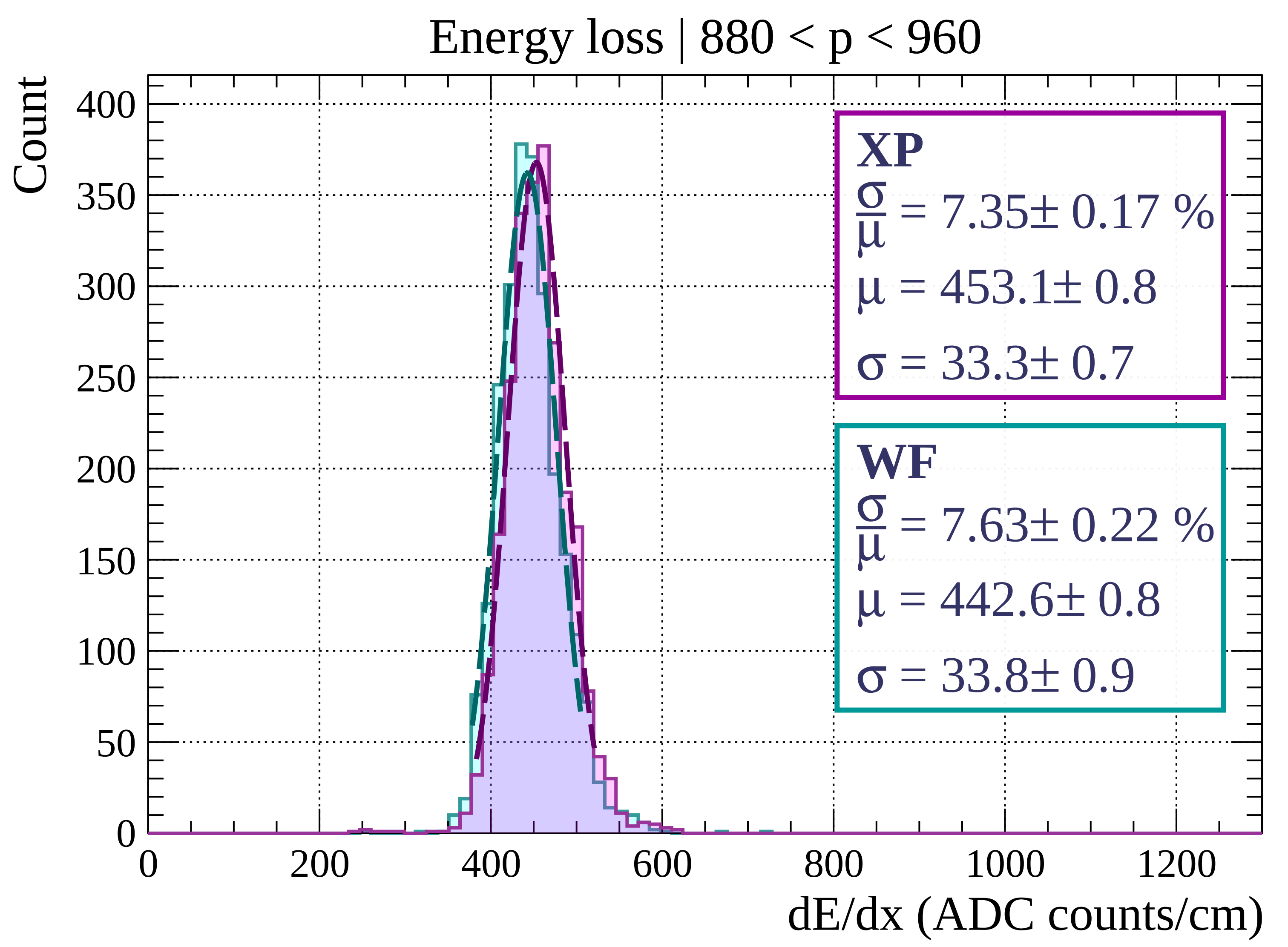


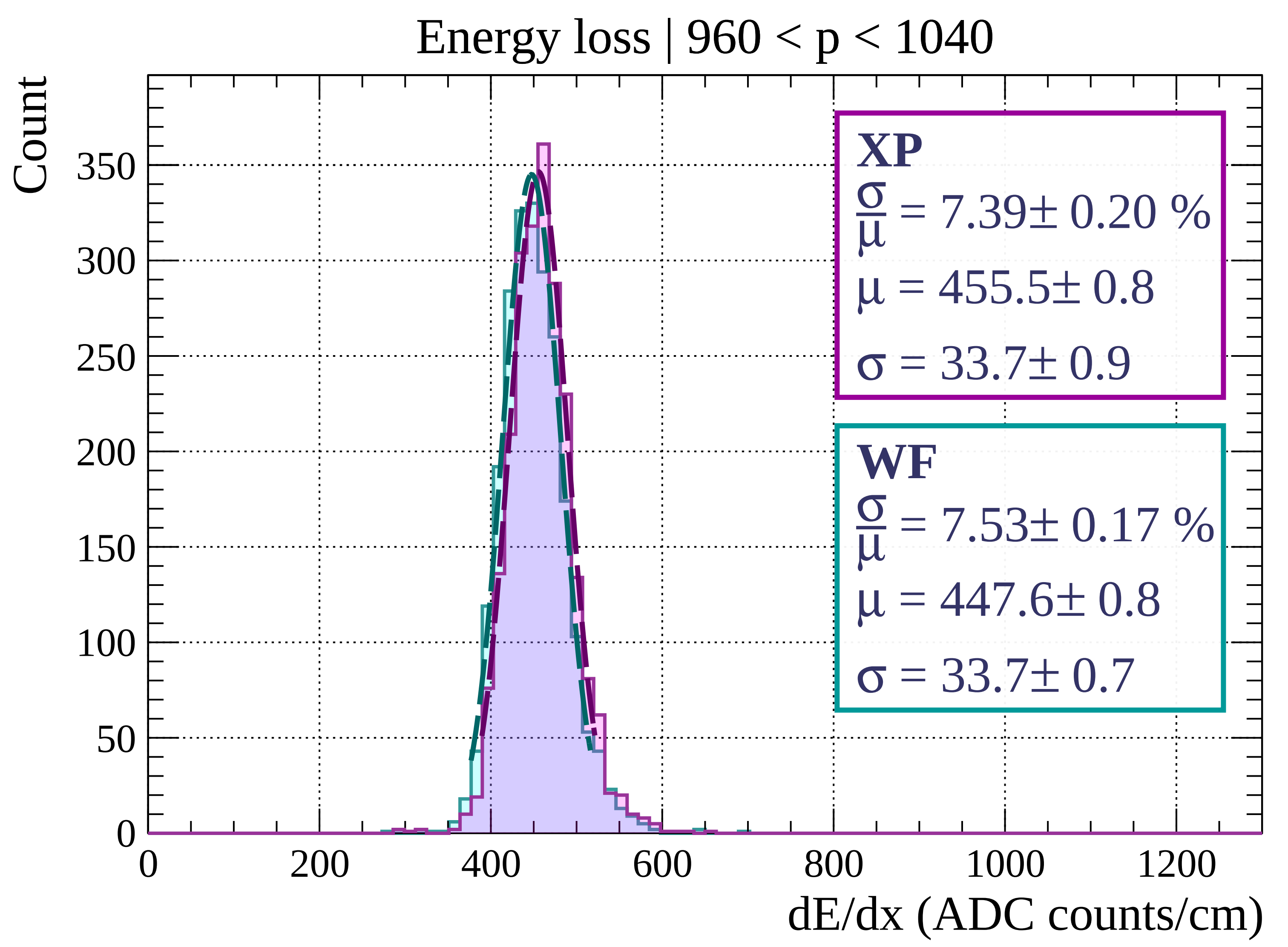
Energy loss | 640 < p < 720
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 6.99± 0.19 %
μ = 436.4± 0.8
σ = 30.5± 0.8
WF
$\frac{\sigma}{\mu}$ = 7.65± 0.20 %
μ = 428.7± 0.8
σ = 32.8± 0.8

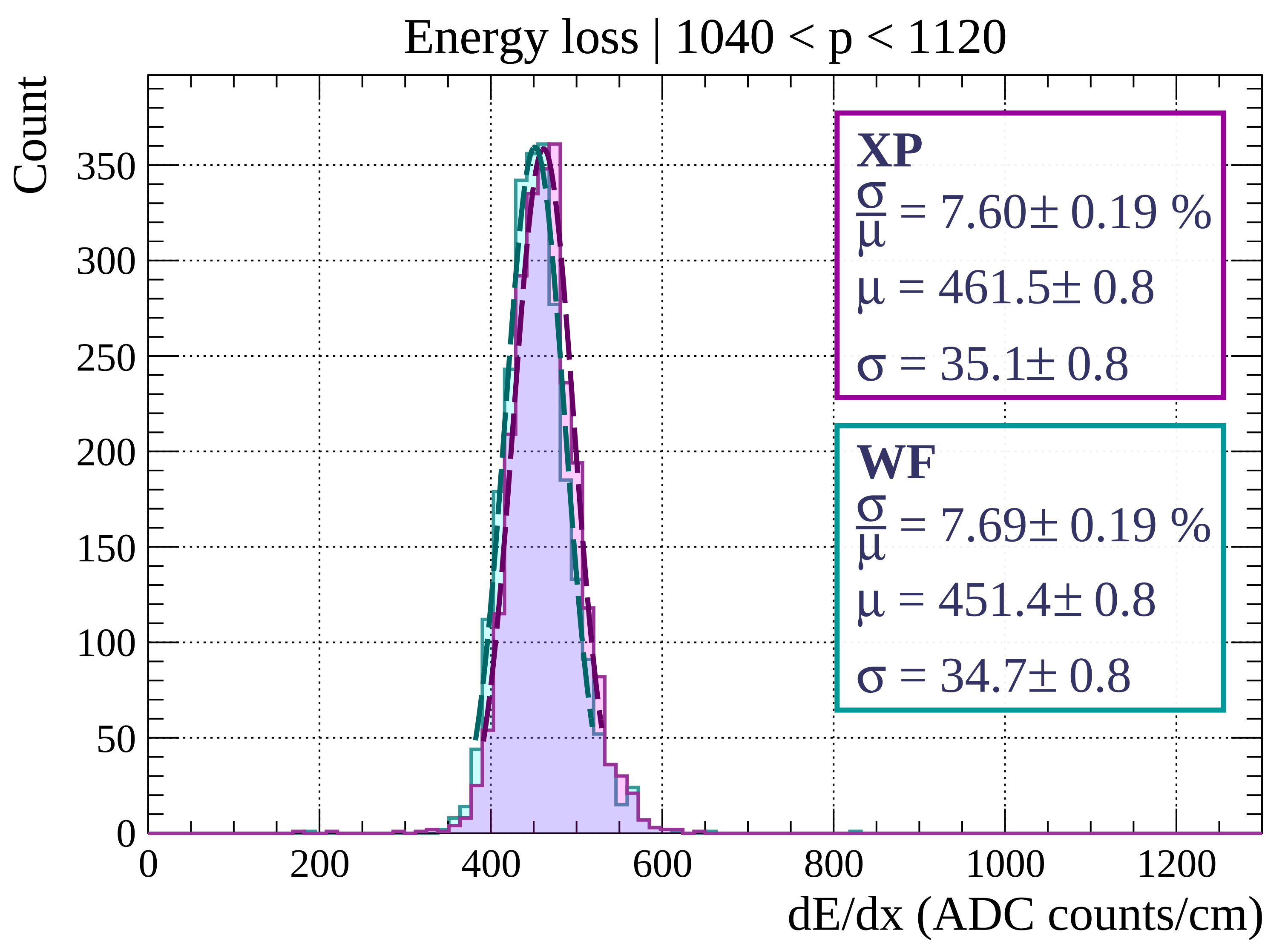


Energy loss | 720 < p < 800
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 7.68± 0.21 %
μ = 441.4± 0.8
σ = 33.9± 0.9
WF
σ/μ = 7.82± 0.21 %
μ = 432.3± 0.8
σ = 33.8± 0.9
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200

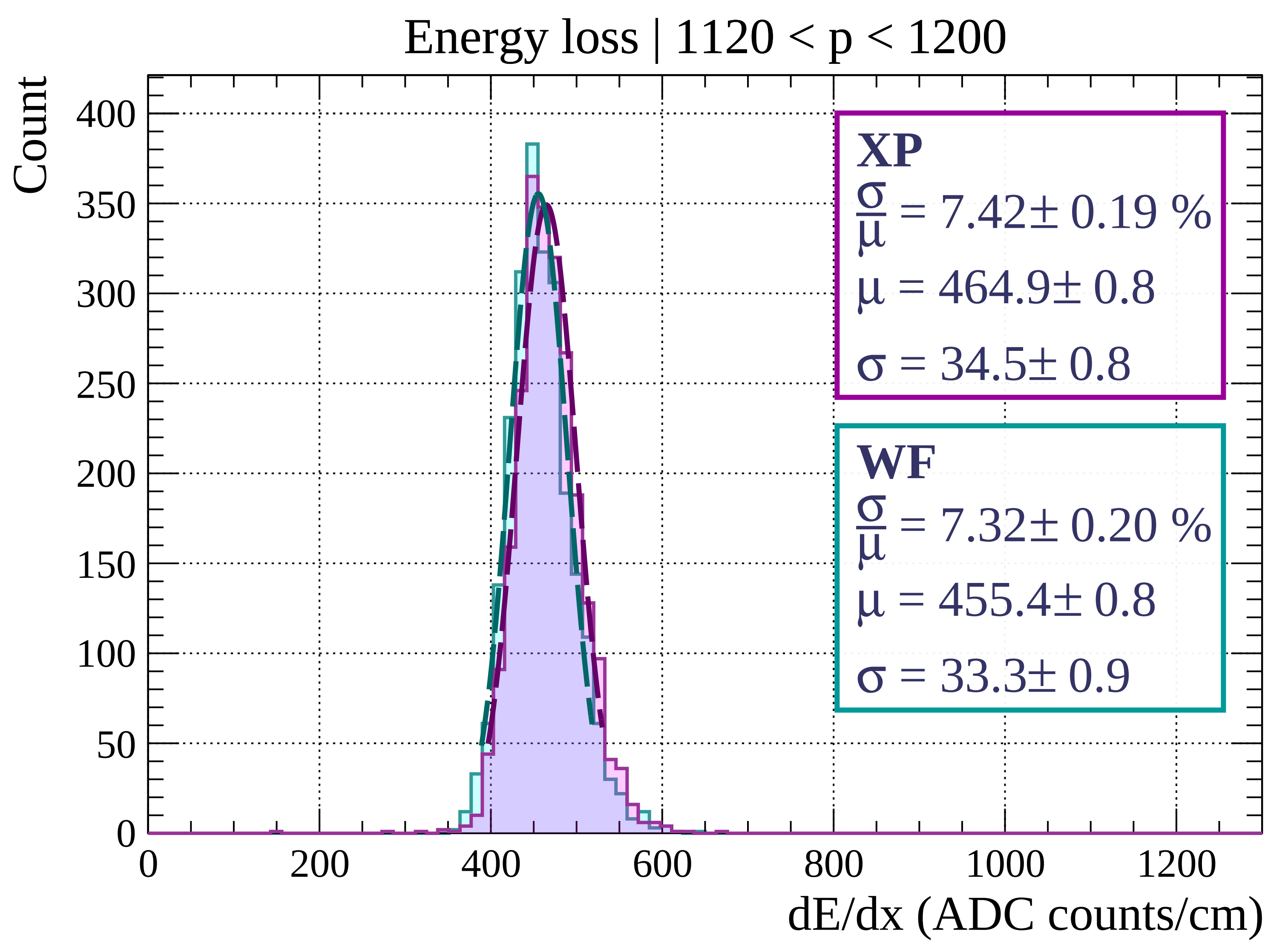

Energy loss | 800 < p < 880
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 7.08± 0.21 %
μ = 446.3± 0.8
σ = 31.6± 0.9
WF
$\frac{\sigma}{\mu}$ = 7.25± 0.21 %
μ = 436.5± 0.8
σ = 31.7± 0.9


Energy loss | 880 < p < 960
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
σ/μ = 7.35± 0.17 %
μ = 453.1± 0.8
σ = 33.3± 0.7
WF
σ/μ = 7.63± 0.22 %
μ = 442.6± 0.8
σ = 33.8± 0.9

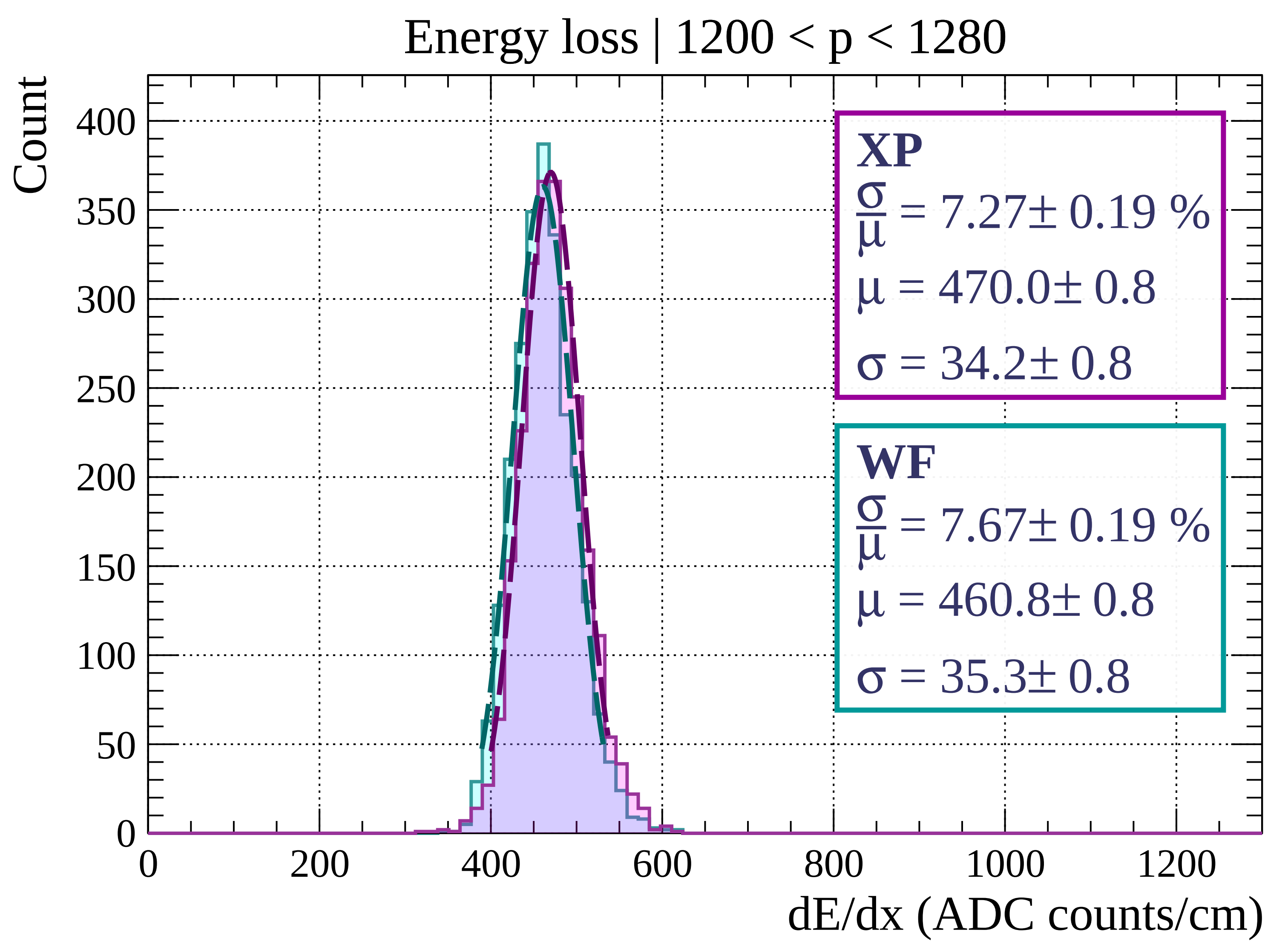

Energy loss | 960 < p < 1040
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 7.39± 0.20 %
μ = 455.5± 0.8
σ = 33.7± 0.9
WF
$\frac{\sigma}{\mu}$ = 7.53± 0.17 %
μ = 447.6± 0.8
σ = 33.7± 0.7

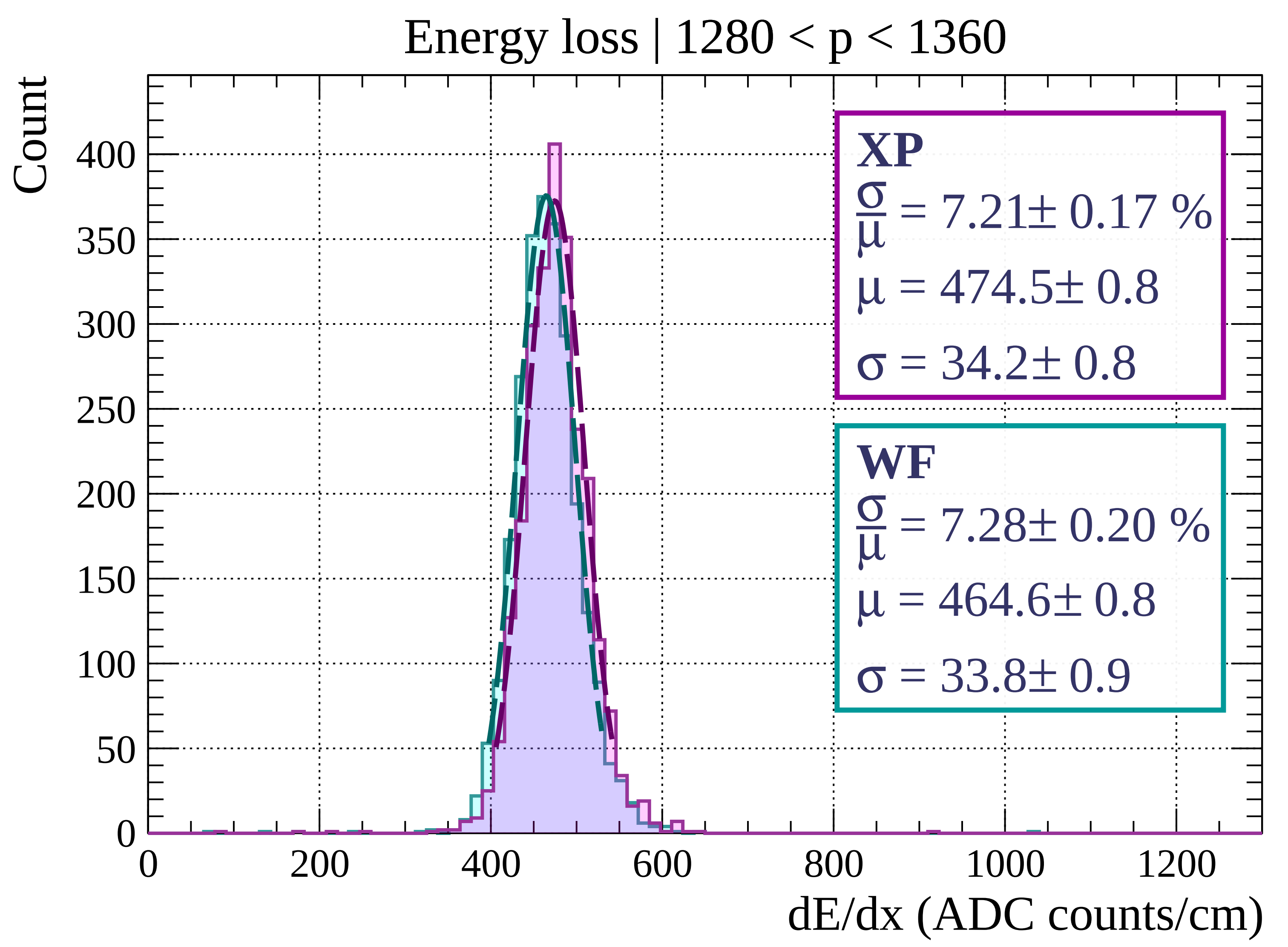


Energy loss | 1040 < p < 1120
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
σ/μ = 7.60± 0.19 %
μ = 461.5± 0.8
σ = 35.1± 0.8
WF
σ/μ = 7.69± 0.19 %
μ = 451.4± 0.8
σ = 34.7± 0.8

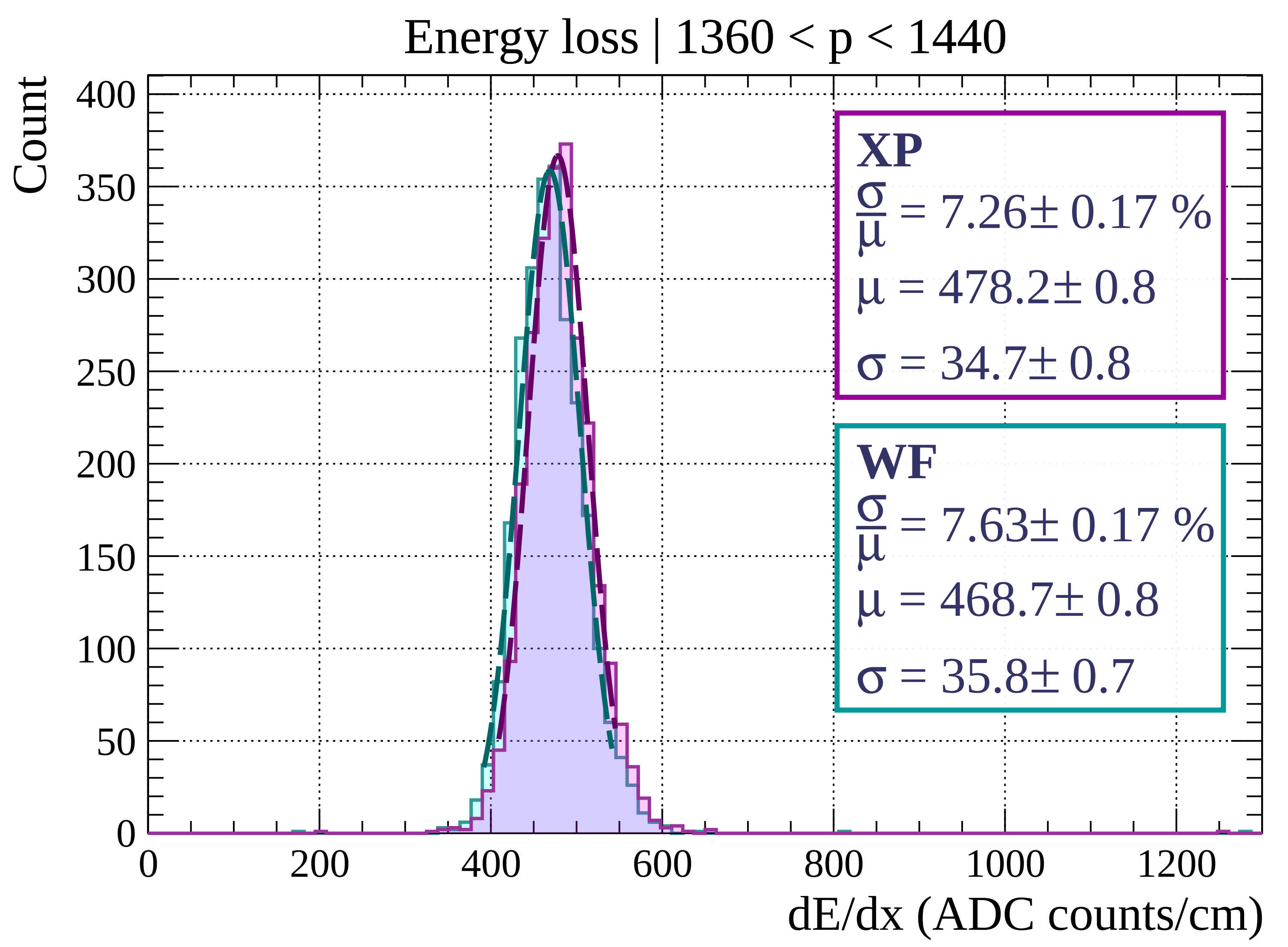


Energy loss | 1120 < p < 1200
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
350
400
XP
$\frac{\sigma}{\mu}$ = 7.42± 0.19 %
μ = 464.9± 0.8
σ = 34.5± 0.8
WF
$\frac{\sigma}{\mu}$ = 7.32± 0.20 %
μ = 455.4± 0.8
σ = 33.3± 0.9

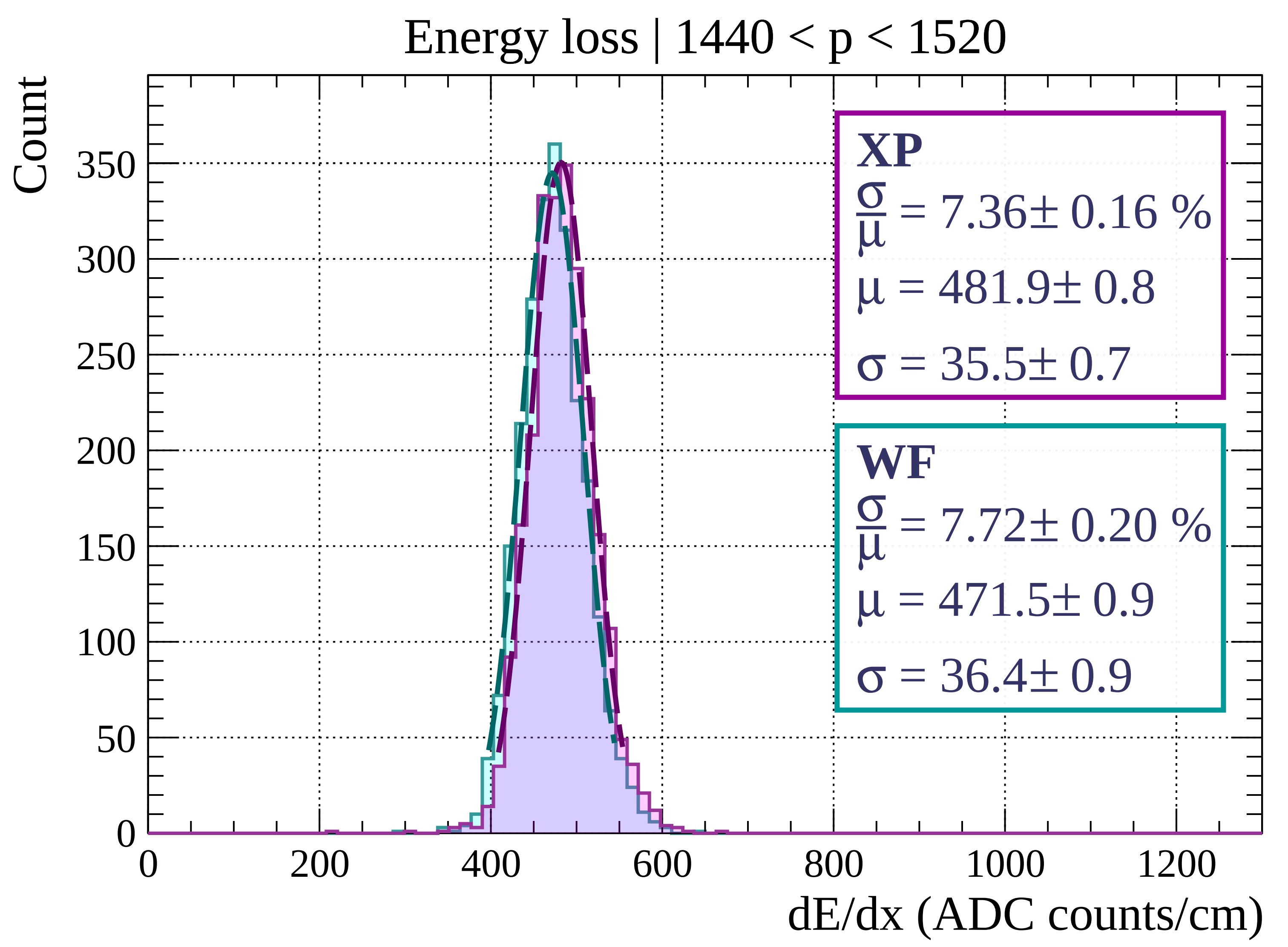

Energy loss | 1200 < p < 1280
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
σ/μ = 7.27± 0.19 %
μ = 470.0± 0.8
σ = 34.2± 0.8
WF
σ/μ = 7.67± 0.19 %
μ = 460.8± 0.8
σ = 35.3± 0.8



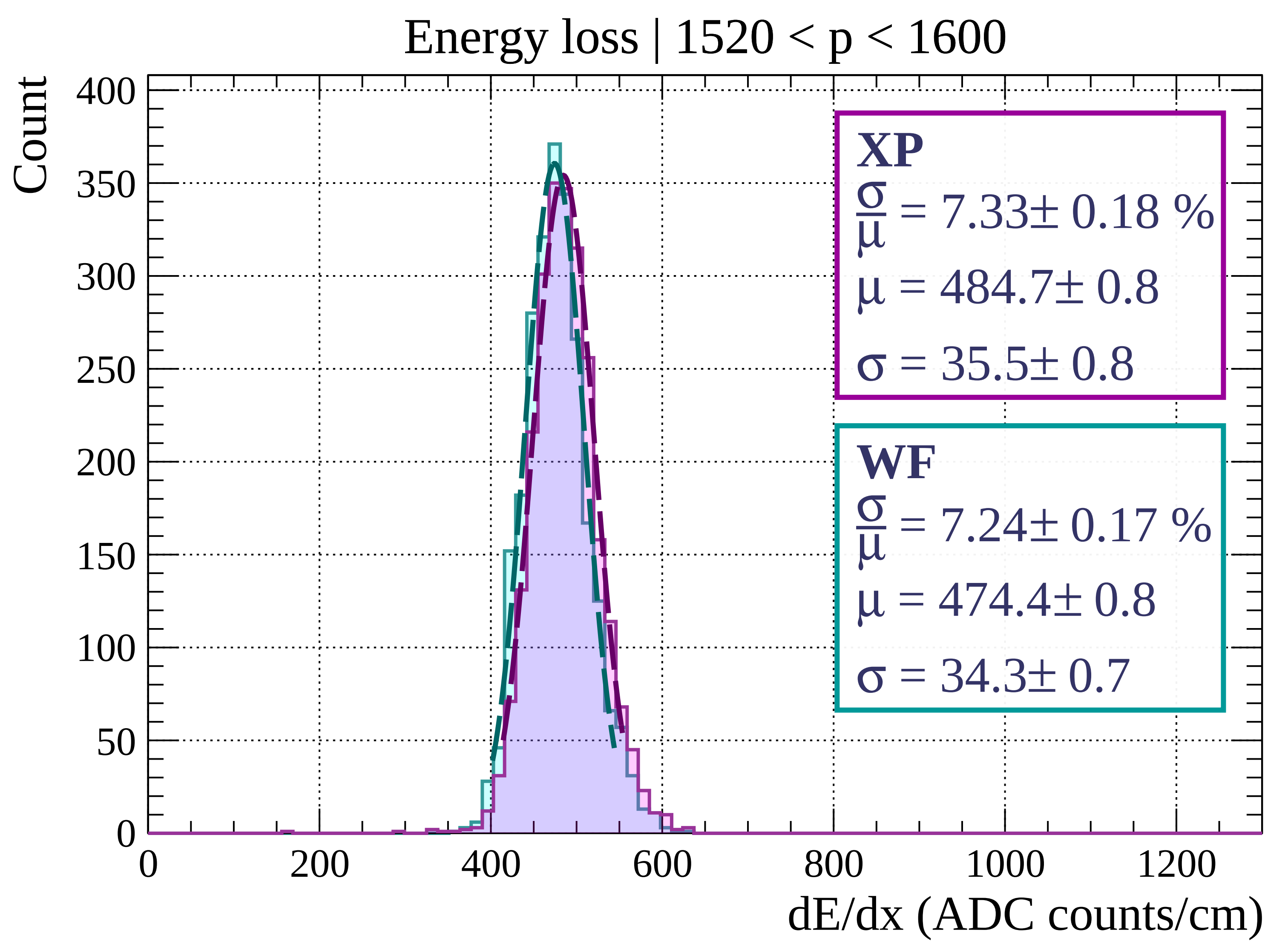
Energy loss | 1280 < p < 1360
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
350
400
XP
$\frac{\sigma}{\mu}$ = 7.21± 0.17 %
μ = 474.5± 0.8
σ = 34.2± 0.8
WF
$\frac{\sigma}{\mu}$ = 7.28± 0.20 %
μ = 464.6± 0.8
σ = 33.8± 0.9



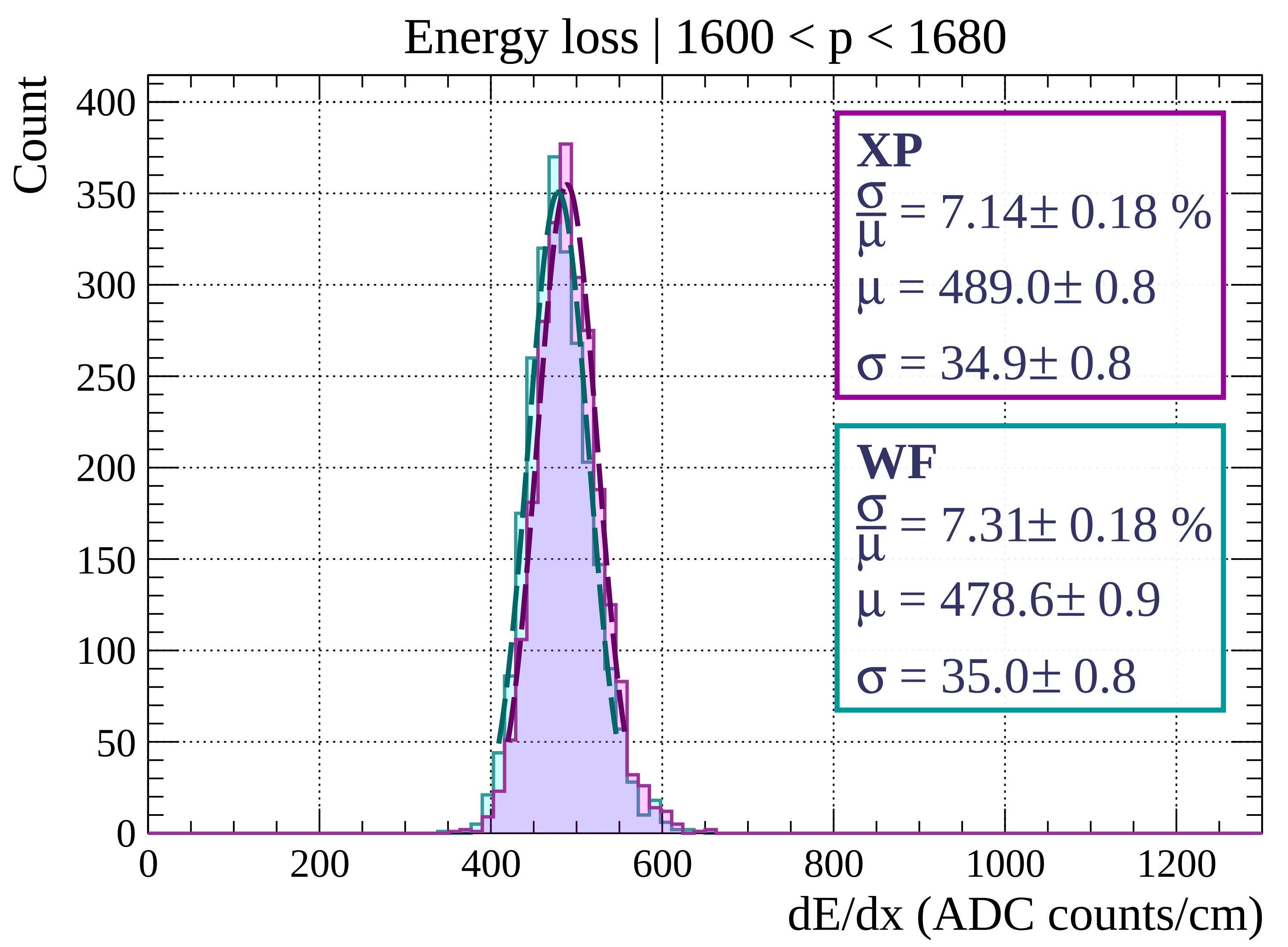
Energy loss | 1360 < p < 1440
Count
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 7.26± 0.17 %
μ = 478.2± 0.8
σ = 34.7± 0.8
WF
σ/μ = 7.63± 0.17 %
μ = 468.7± 0.8
σ = 35.8± 0.7

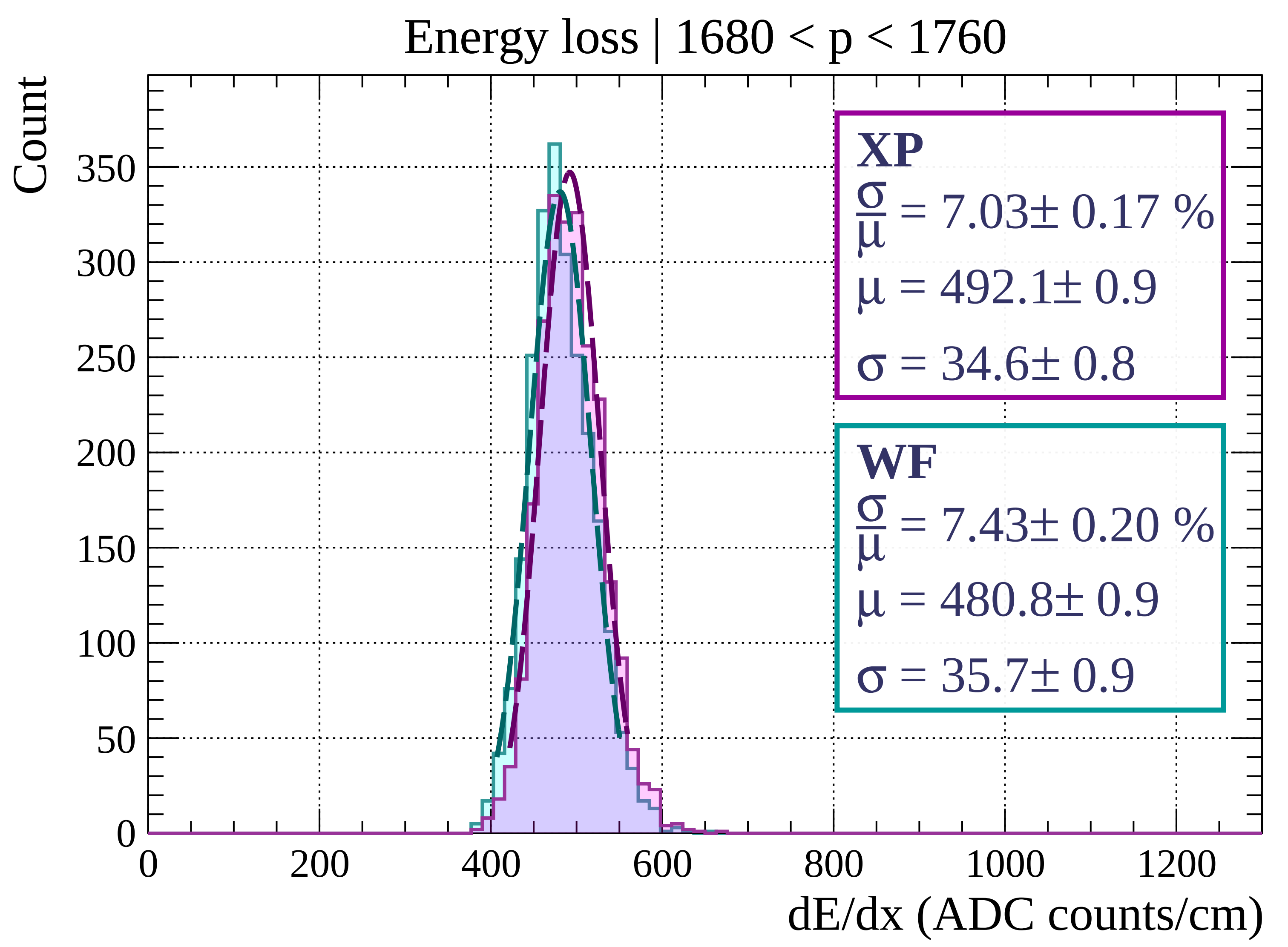


Energy loss | 1440 < p < 1520
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 7.36± 0.16 %
μ = 481.9± 0.8
σ = 35.5± 0.7
WF
$\frac{\sigma}{\mu}$ = 7.72± 0.20 %
μ = 471.5± 0.9
σ = 36.4± 0.9

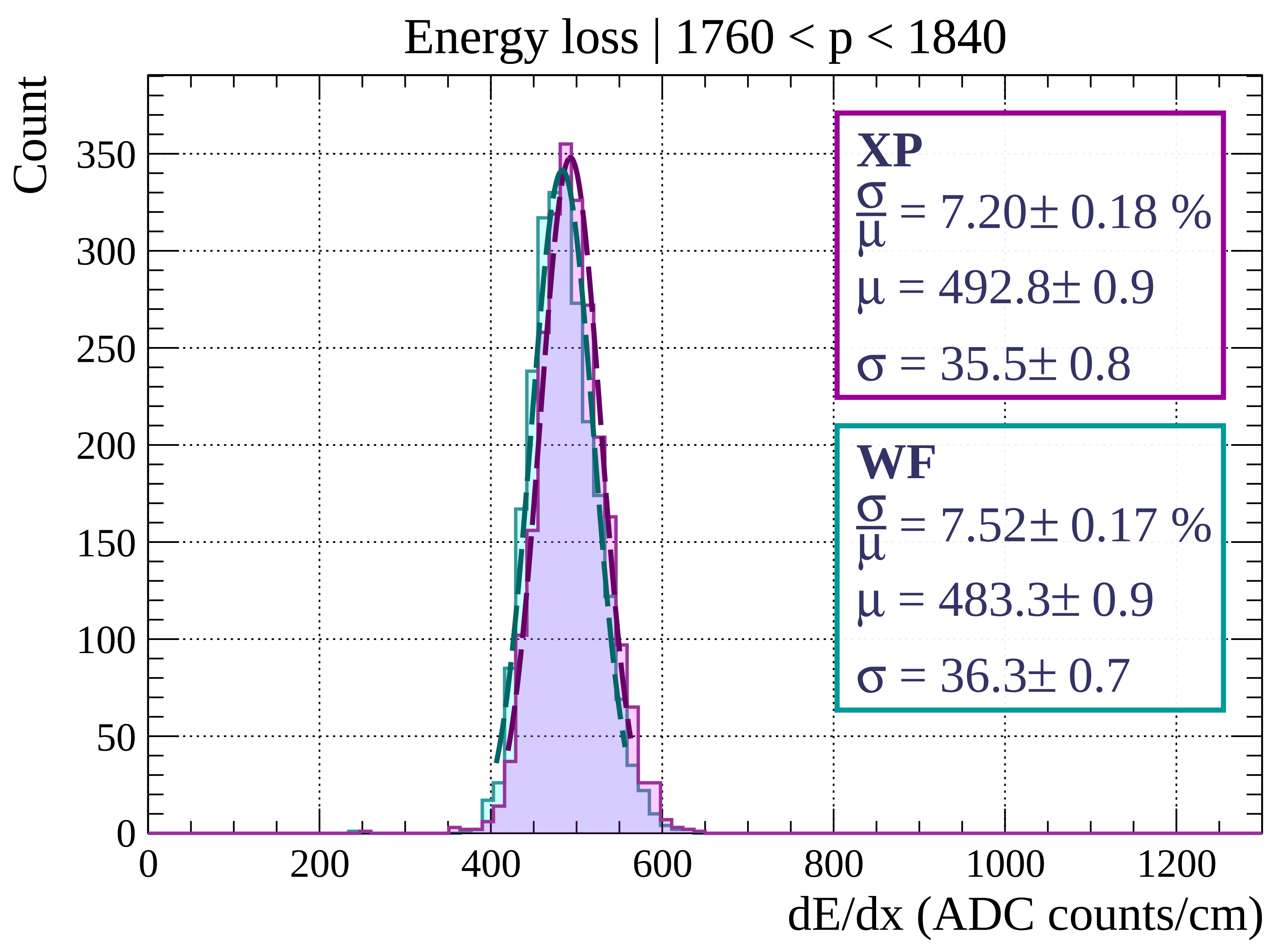


Energy loss | 1520 < p < 1600
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 7.33± 0.18 %
μ = 484.7± 0.8
σ = 35.5± 0.8
WF
σ/μ = 7.24± 0.17 %
μ = 474.4± 0.8
σ = 34.3± 0.7

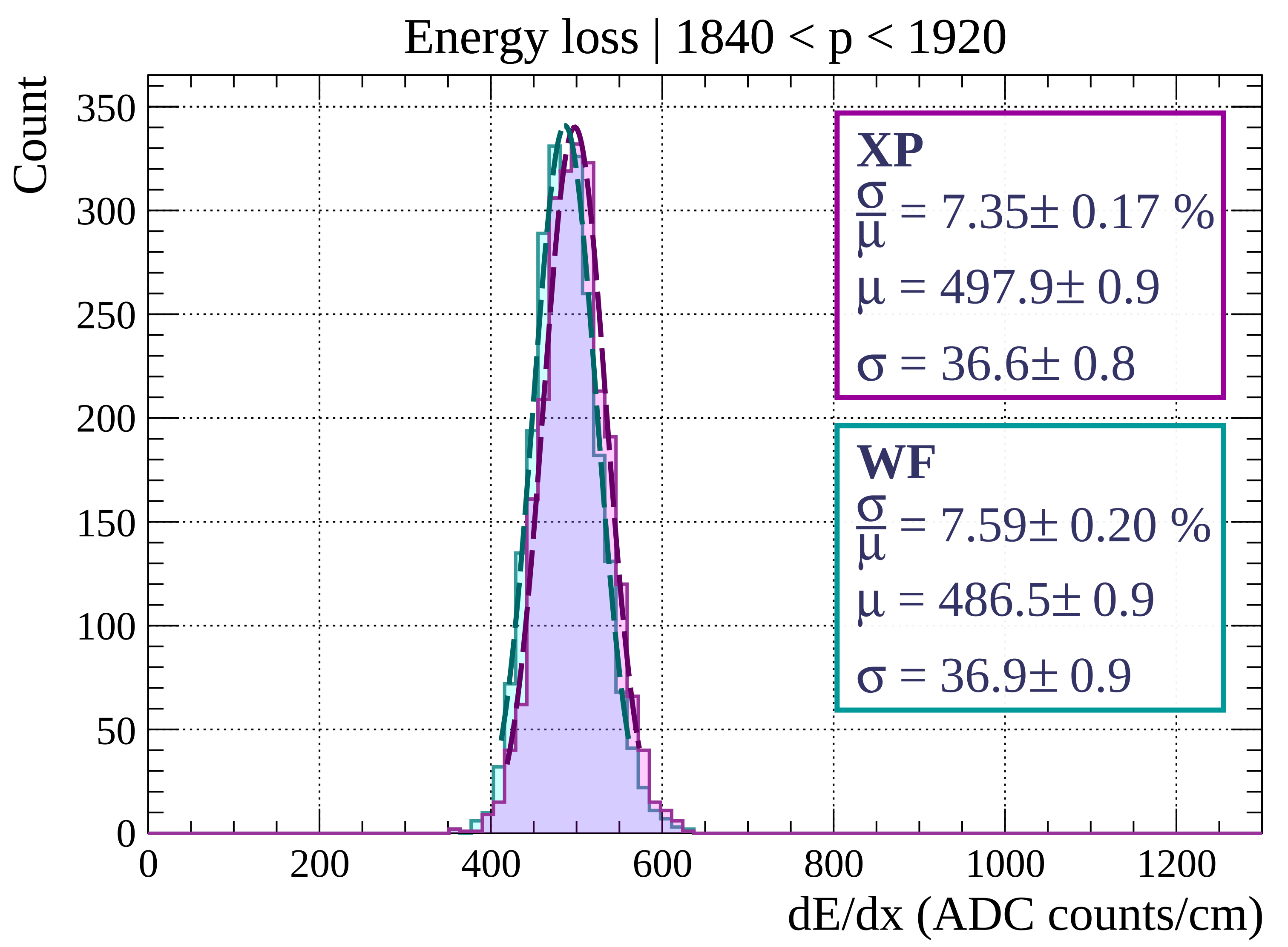

Energy loss | 1600 < p < 1680
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 7.14± 0.18 %
μ = 489.0± 0.8
σ = 34.9± 0.8
WF
$\frac{\sigma}{\mu}$ = 7.31± 0.18 %
μ = 478.6± 0.9
σ = 35.0± 0.8



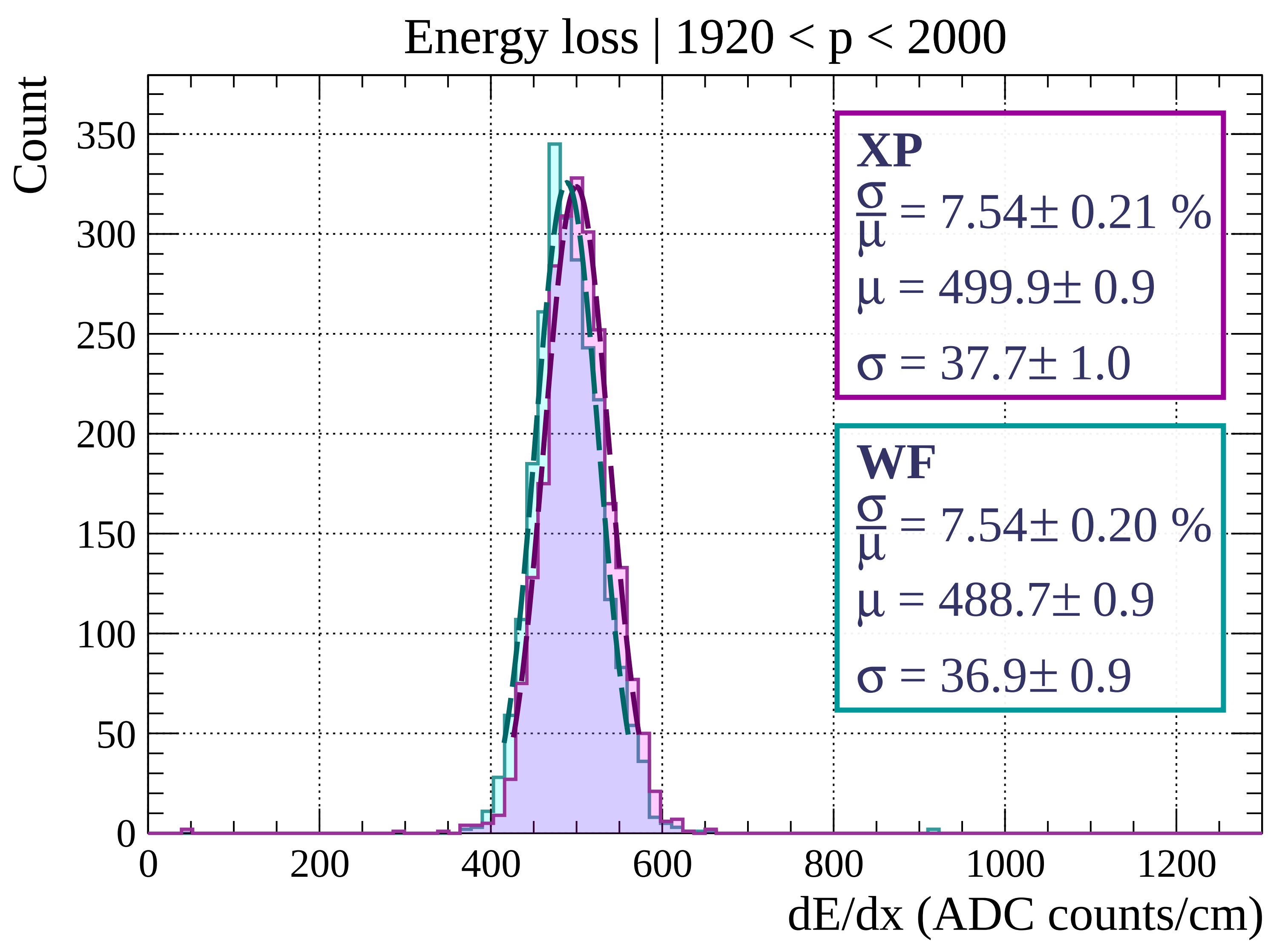
Energy loss | 1680 < p < 1760
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 7.03± 0.17 %
μ = 492.1± 0.9
σ = 34.6± 0.8
WF
$\frac{\sigma}{\mu}$ = 7.43± 0.20 %
μ = 480.8± 0.9
σ = 35.7± 0.9

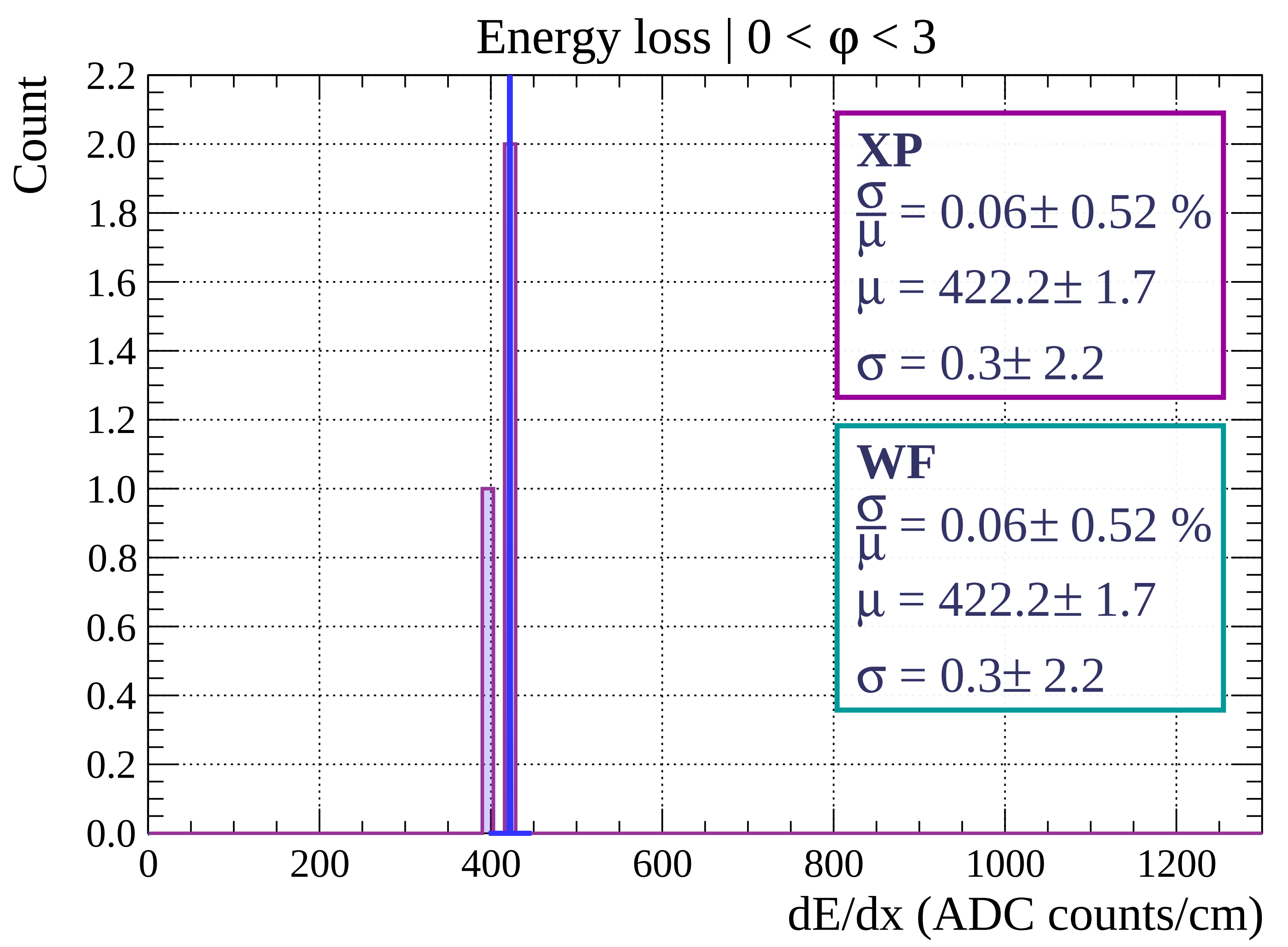


Energy loss | 1760 < p < 1840
Count
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 7.20± 0.18 %
μ = 492.8± 0.9
σ = 35.5± 0.8
WF
$\frac{\sigma}{\mu}$ = 7.52± 0.17 %
μ = 483.3± 0.9
σ = 36.3± 0.7

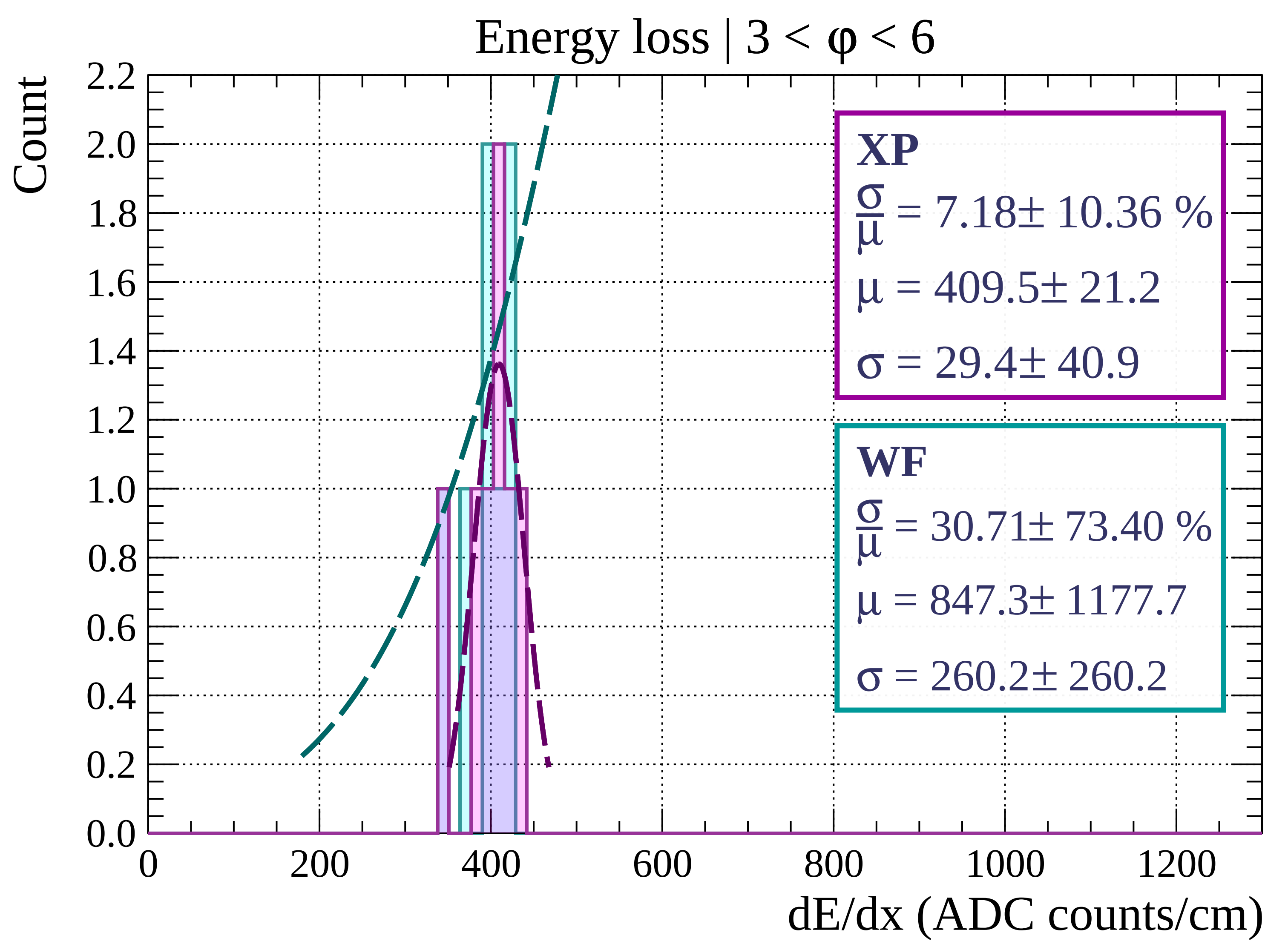


Energy loss | 1840 < p < 1920
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 7.35± 0.17 %
μ = 497.9± 0.9
σ = 36.6± 0.8
WF
$\frac{\sigma}{\mu}$ = 7.59± 0.20 %
μ = 486.5± 0.9
σ = 36.9± 0.9

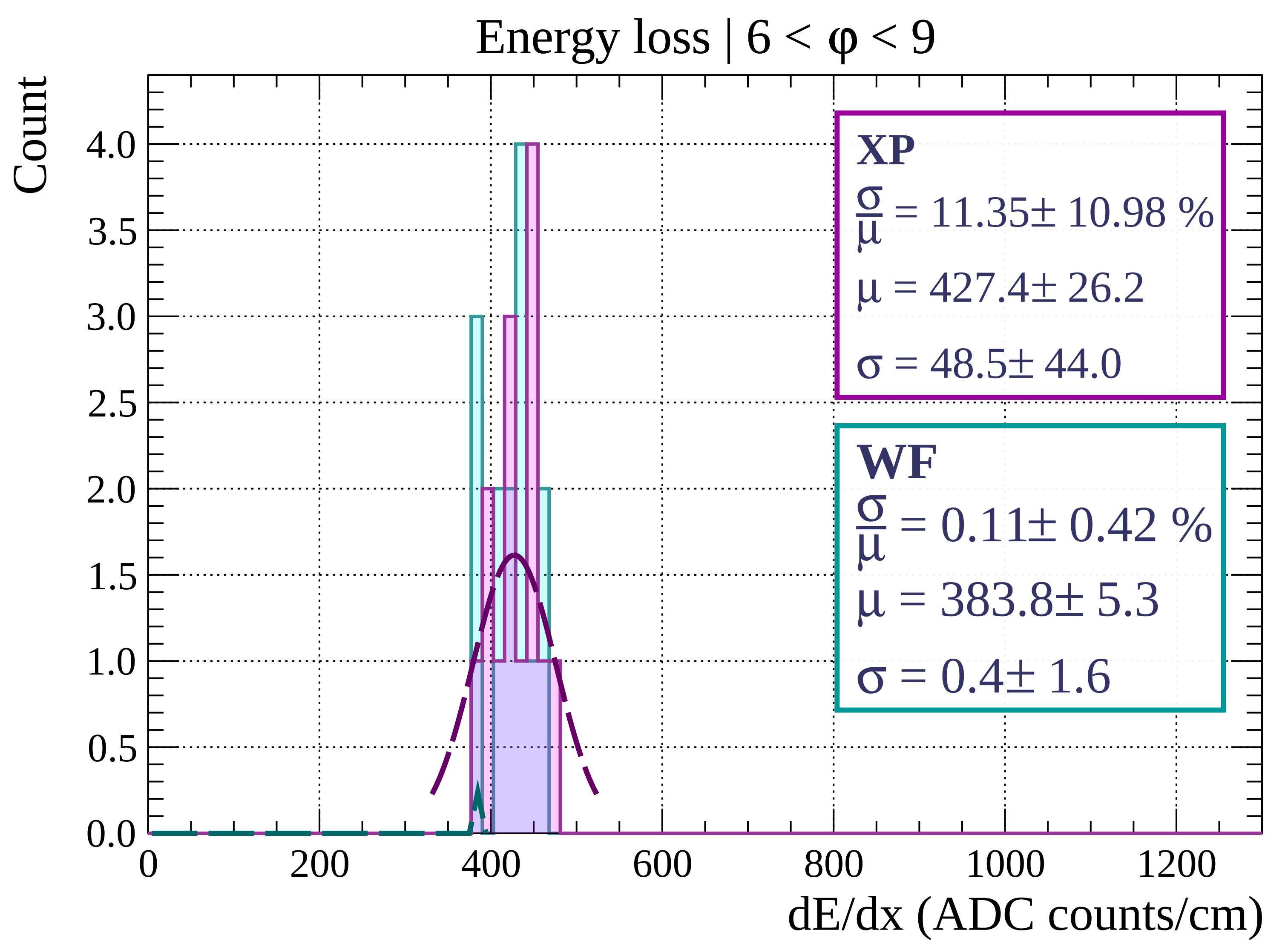

Energy loss | 1920 < p < 2000
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
σ/μ = 7.54± 0.21 %
μ = 499.9± 0.9
σ = 37.7± 1.0
WF
σ/μ = 7.54± 0.20 %
μ = 488.7± 0.9
σ = 36.9± 0.9

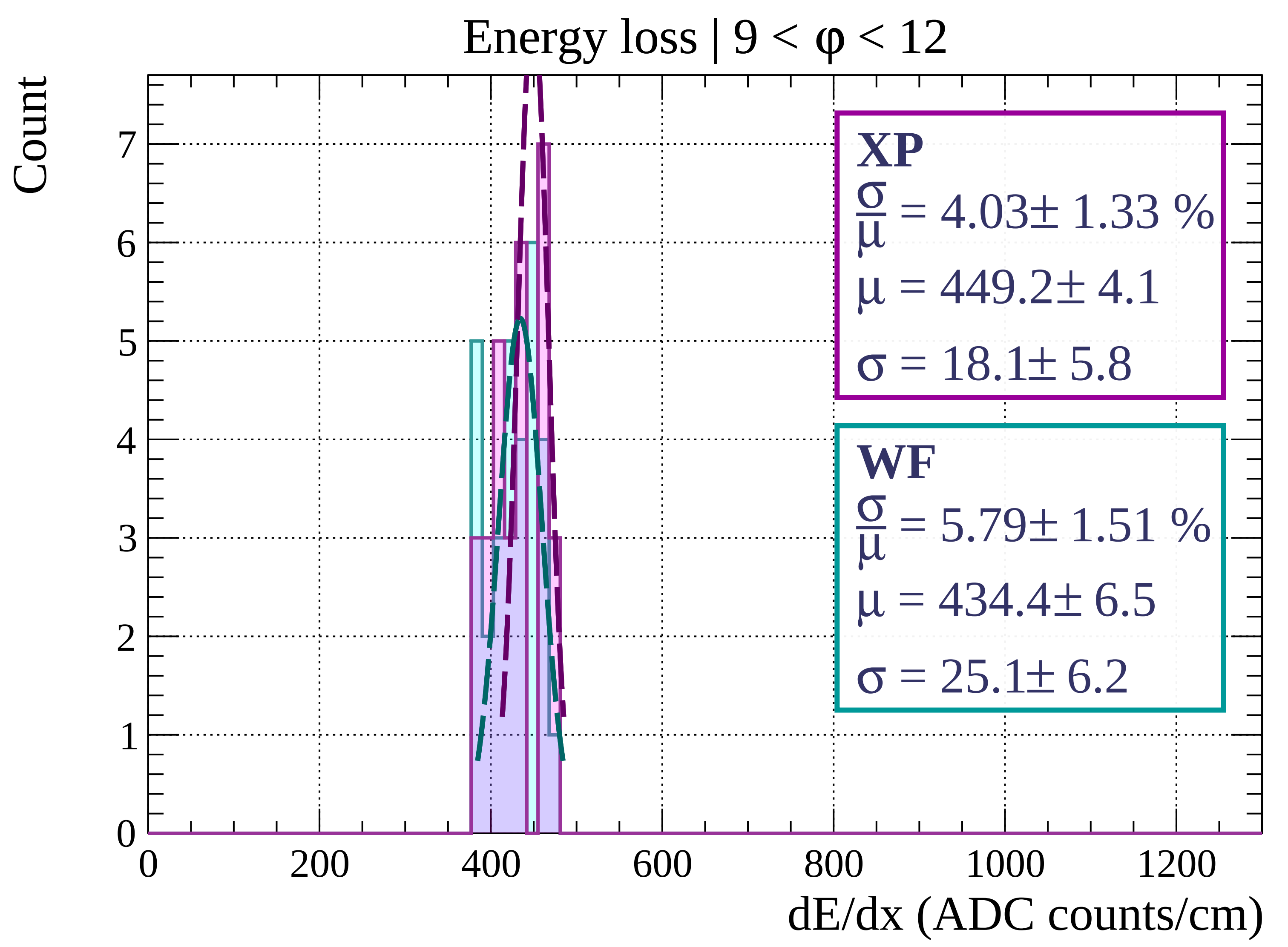


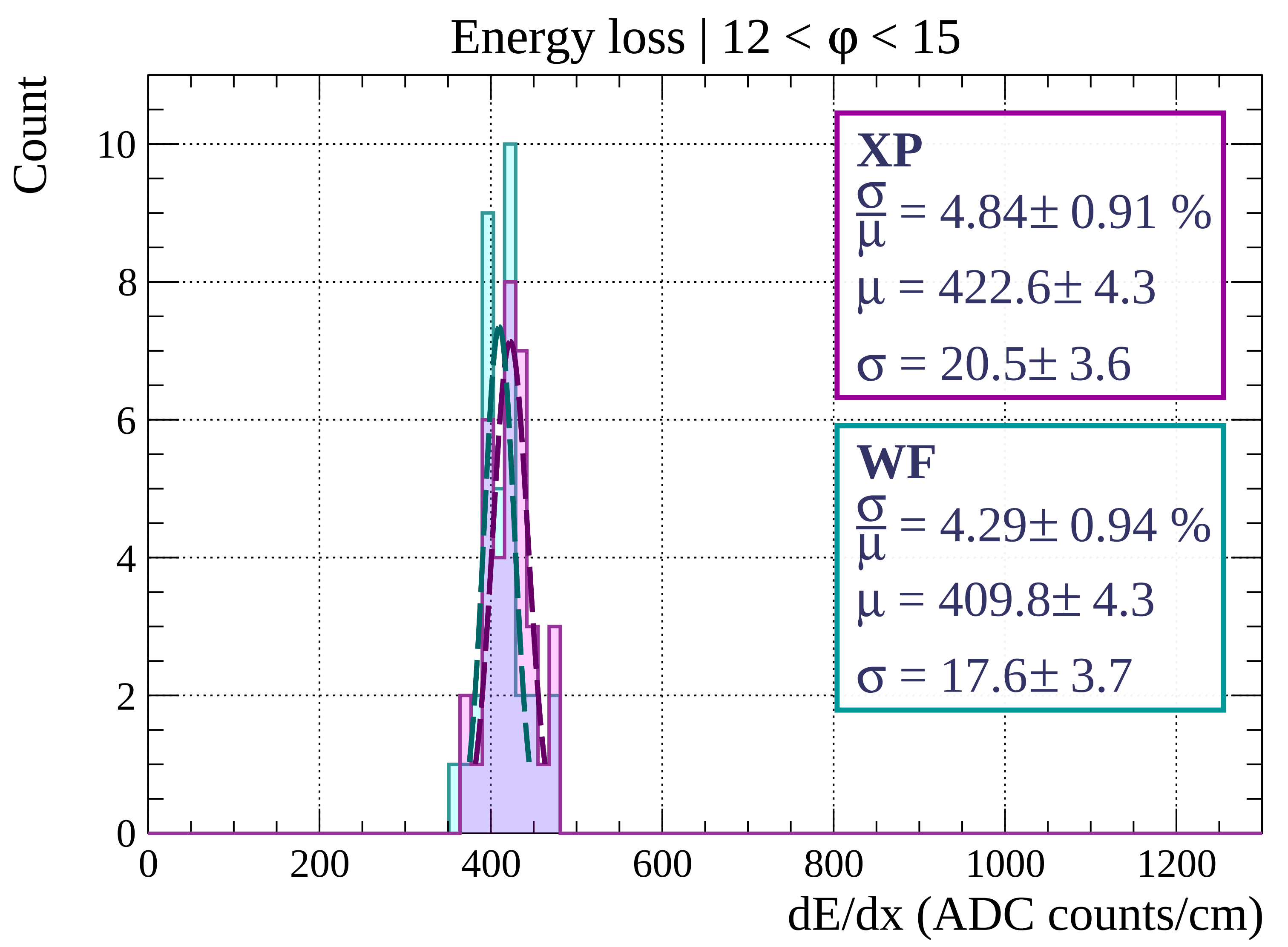
Energy loss | 0 < φ < 3
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
XP
σ/μ = 0.06± 0.52 %
μ = 422.2± 1.7
σ = 0.3± 2.2
WF
σ/μ = 0.06± 0.52 %
μ = 422.2± 1.7
σ = 0.3± 2.2



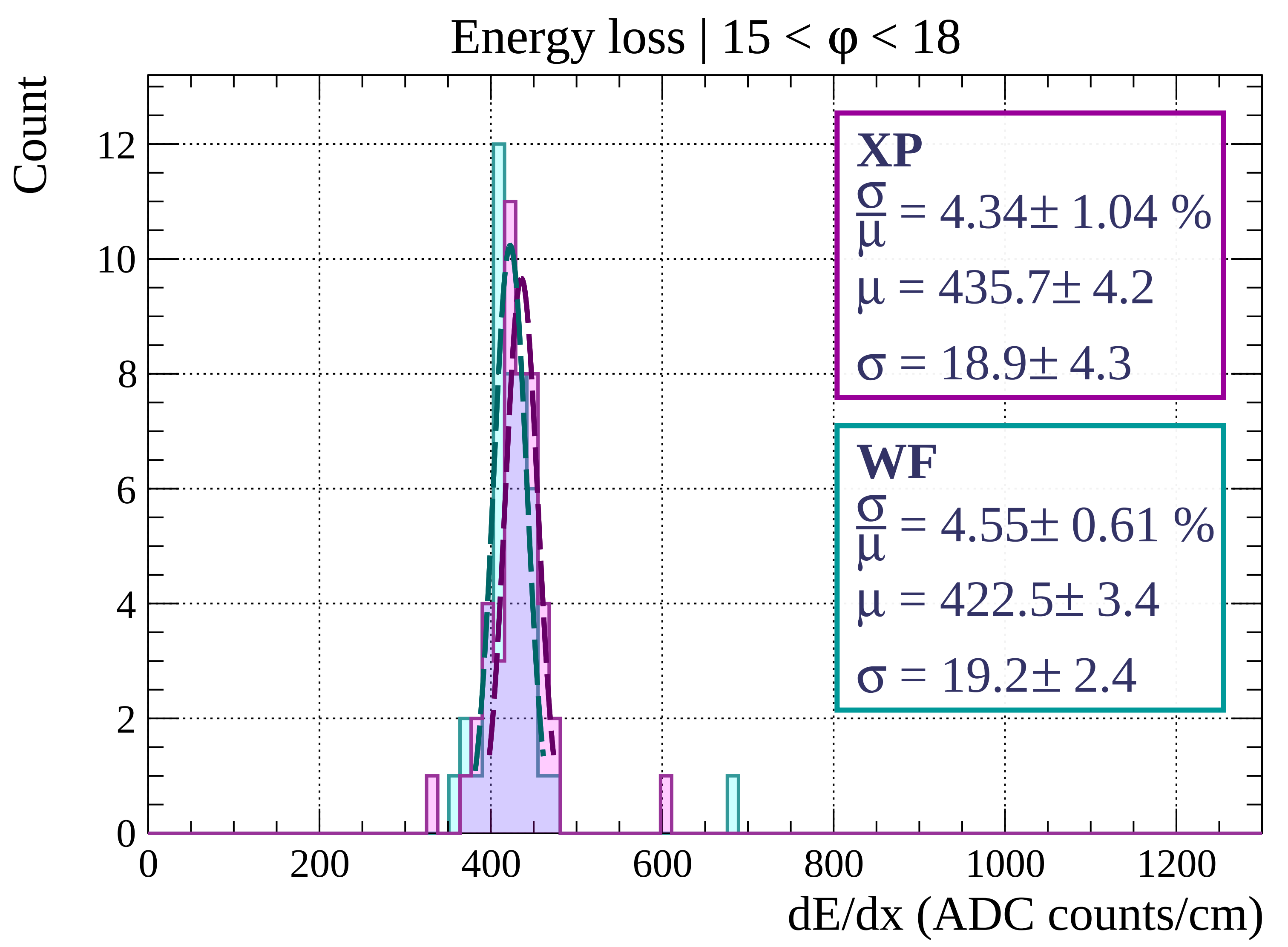
Energy loss | 3 < φ < 6
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
XP
σ/μ = 7.18± 10.36 %
μ = 409.5± 21.2
σ = 29.4± 40.9
WF
σ/μ = 30.71± 73.40 %
μ = 847.3± 1177.7
σ = 260.2± 260.2

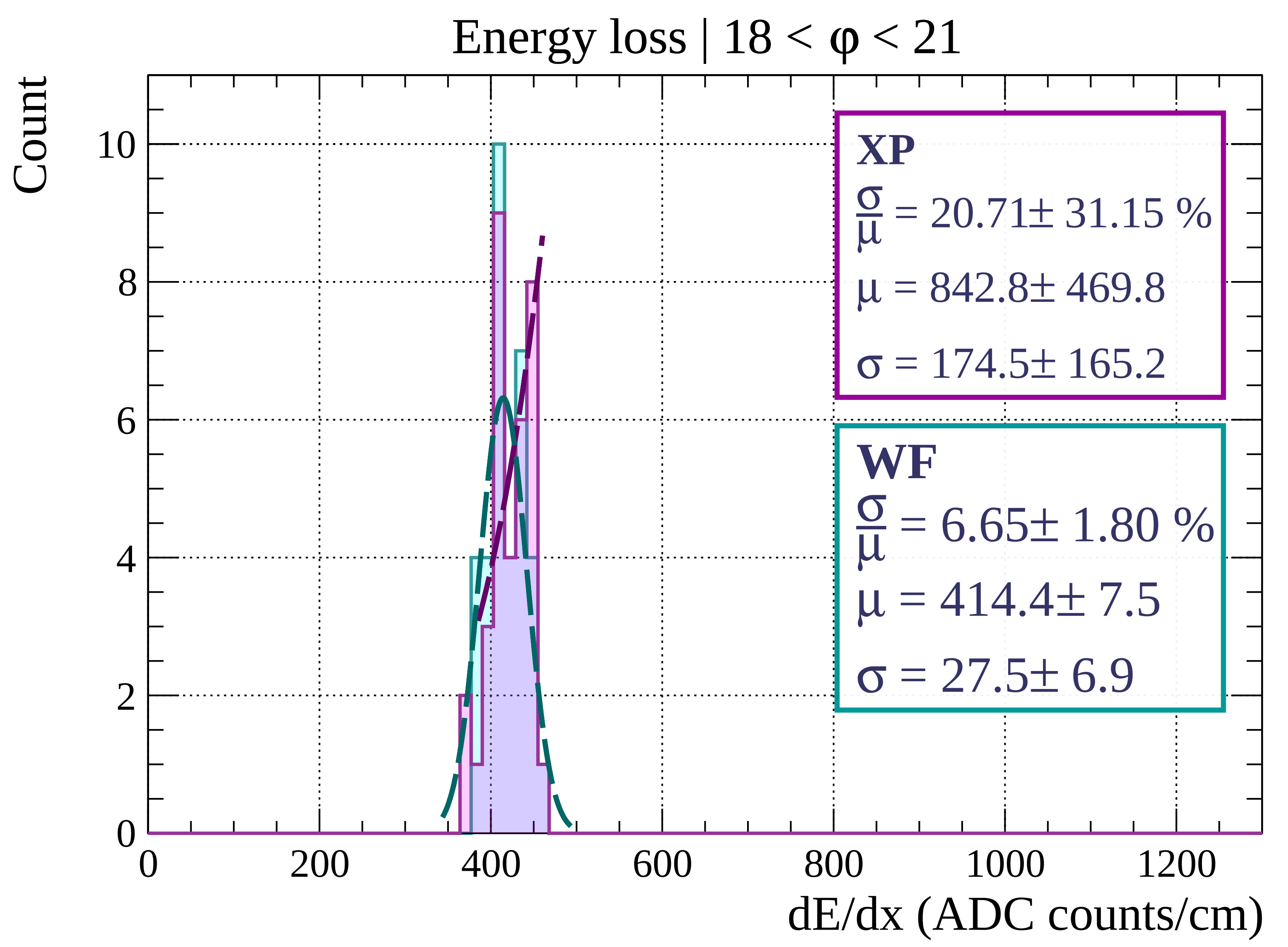

Energy loss | 6 < φ < 9
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 11.35± 10.98 %
μ = 427.4± 26.2
σ = 48.5± 44.0
WF
σ/μ = 0.11± 0.42 %
μ = 383.8± 5.3
σ = 0.4± 1.6


Energy loss | 9 < φ < 12
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
7
XP
σ/μ = 4.03± 1.33 %
μ = 449.2± 4.1
σ = 18.1± 5.8
WF
σ/μ = 5.79± 1.51 %
μ = 434.4± 6.5
σ = 25.1± 6.2

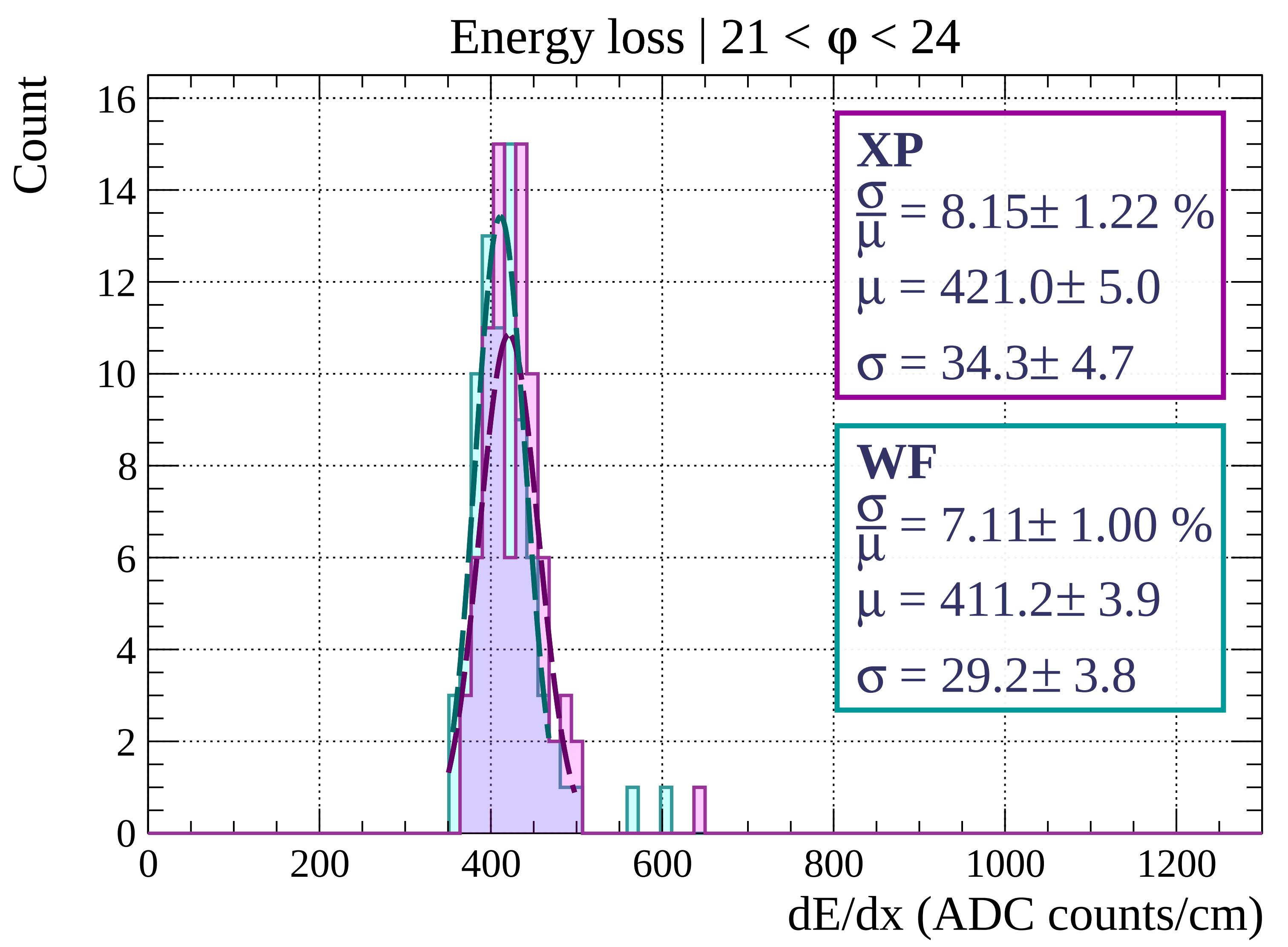


Energy loss | 12 < φ < 15
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
2
4
6
8
10
XP
σ/μ = 4.84± 0.91 %
μ = 422.6± 4.3
σ = 20.5± 3.6
WF
σ/μ = 4.29± 0.94 %
μ = 409.8± 4.3
σ = 17.6± 3.7

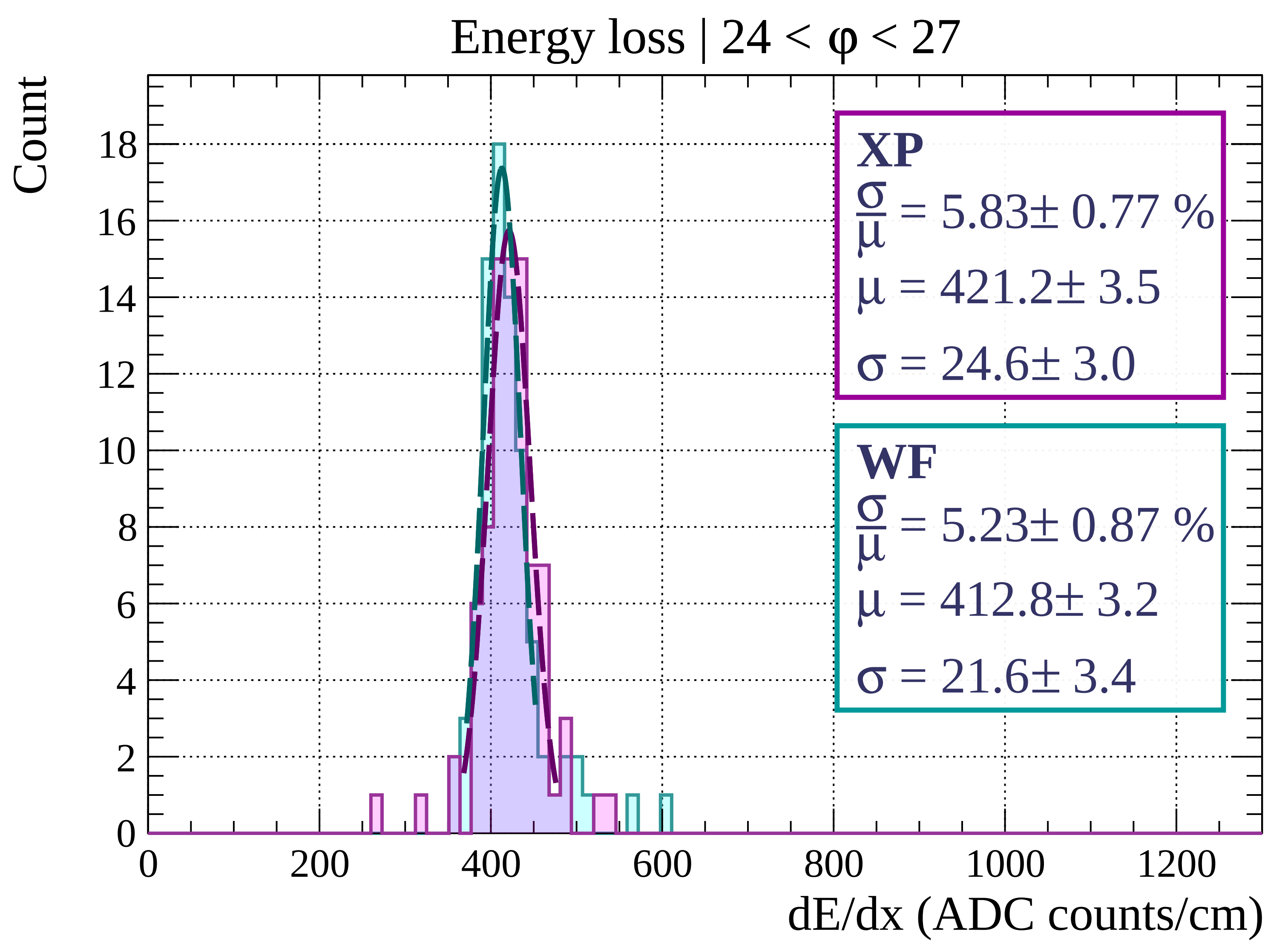

Energy loss | 15 < φ < 18
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
XP
σ/μ = 4.34± 1.04 %
μ = 435.7± 4.2
σ = 18.9± 4.3
WF
σ/μ = 4.55± 0.61 %
μ = 422.5± 3.4
σ = 19.2± 2.4



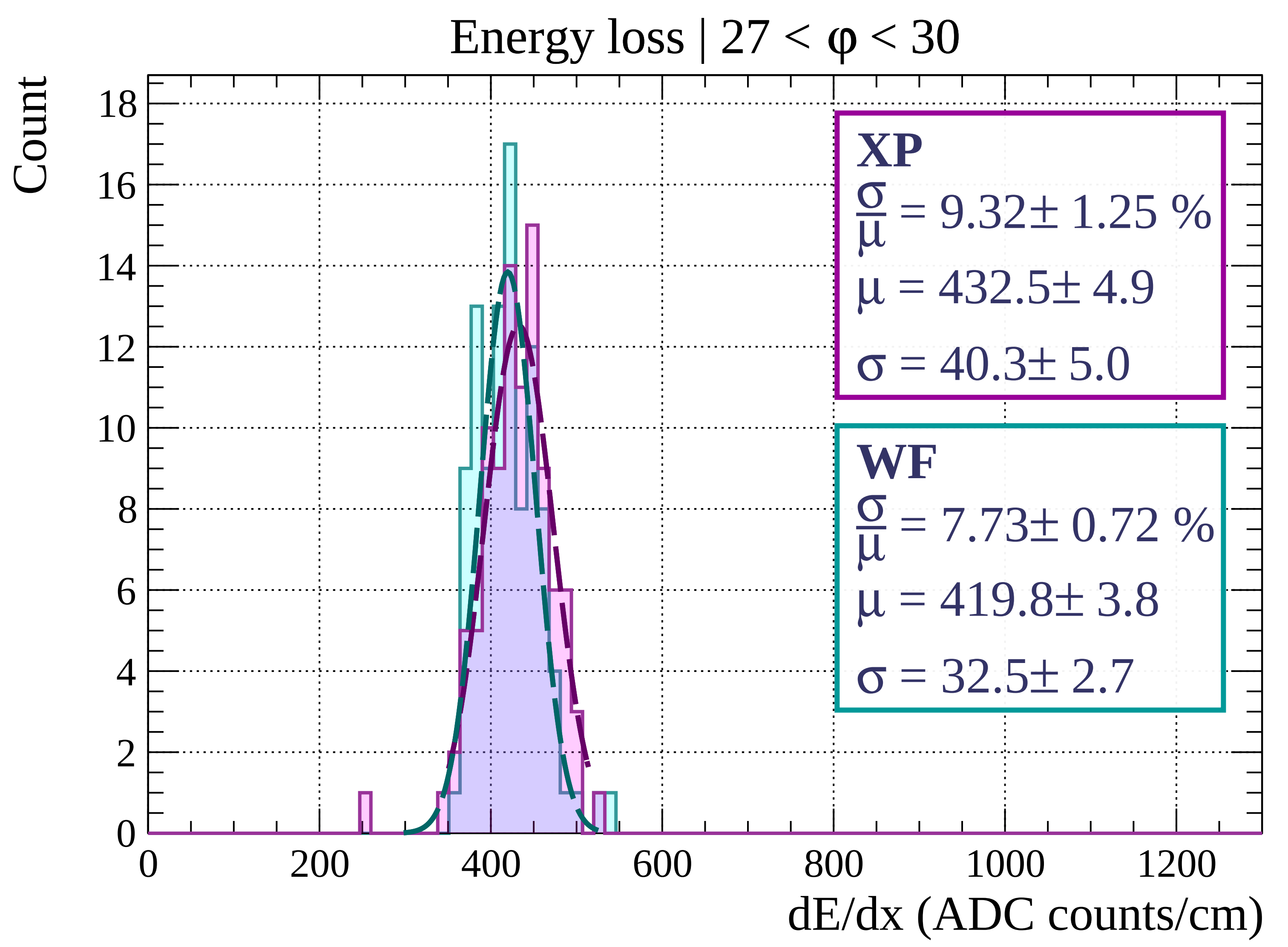
Energy loss | 18 < φ < 21
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 20.71± 31.15 %
μ = 842.8± 469.8
σ = 174.5± 165.2
WF
$\frac{\sigma}{\mu}$ = 6.65± 1.80 %
μ = 414.4± 7.5
σ = 27.5± 6.9



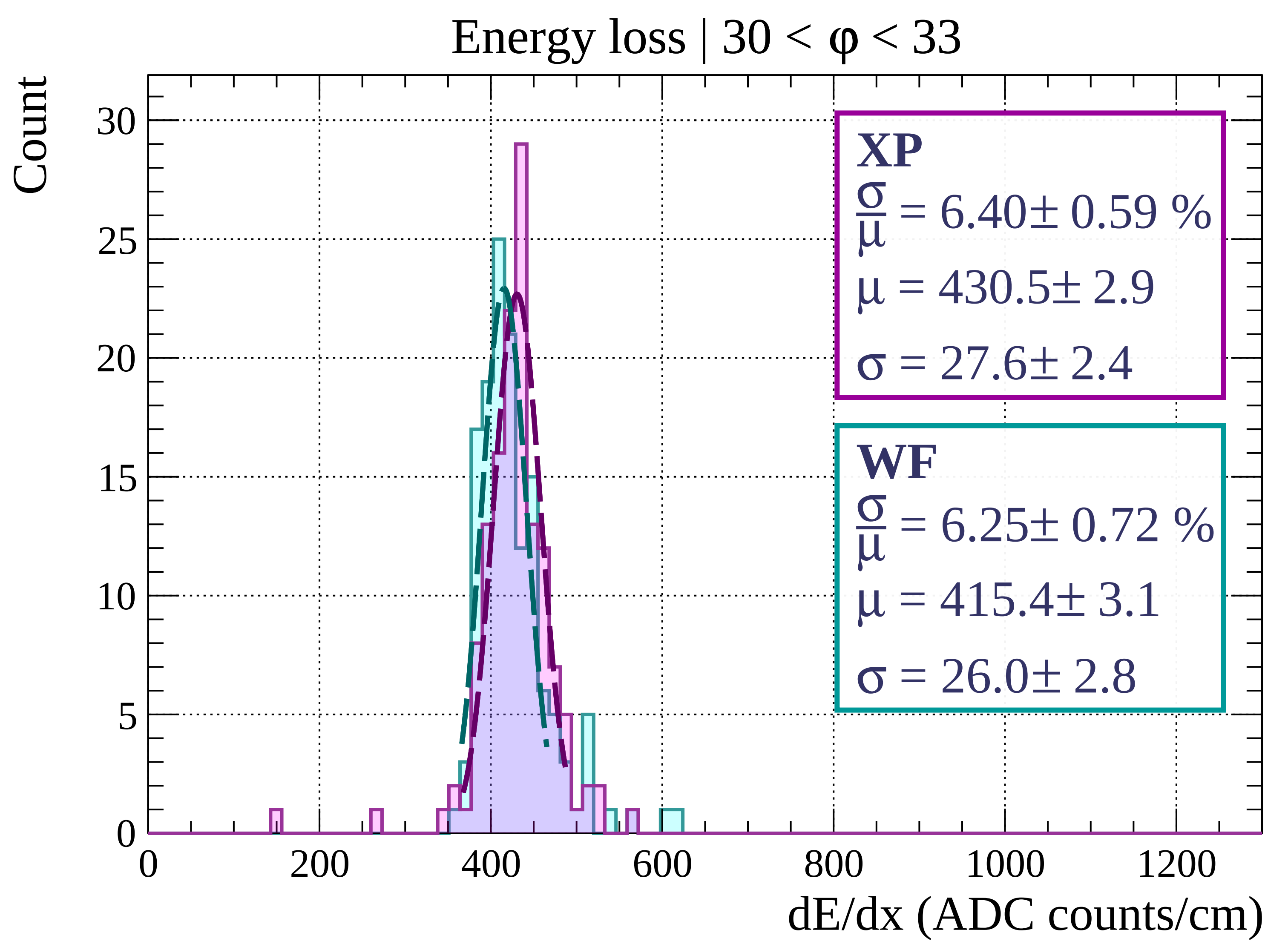
Energy loss | 21 < φ < 24
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
16
0
200
400
600
800
1000
1200
XP
σ/μ = 8.15± 1.22 %
μ = 421.0± 5.0
σ = 34.3± 4.7
WF
σ/μ = 7.11± 1.00 %
μ = 411.2± 3.9
σ = 29.2± 3.8

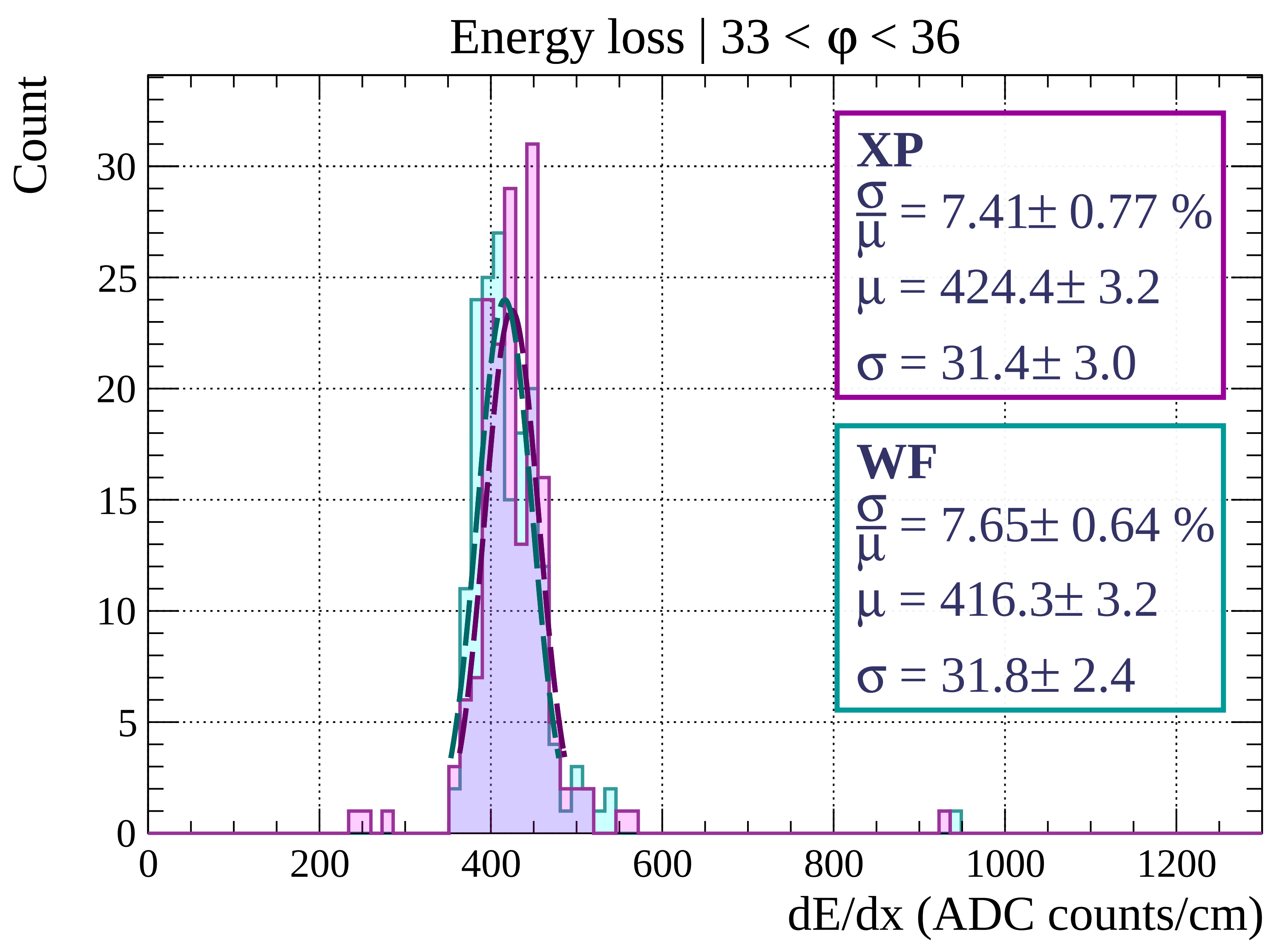


Energy loss | 24 < φ < 27
Count
18
16
14
12
10
8
6
4
2
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 5.83± 0.77 %
μ = 421.2± 3.5
σ = 24.6± 3.0
WF
σ/μ = 5.23± 0.87 %
μ = 412.8± 3.2
σ = 21.6± 3.4

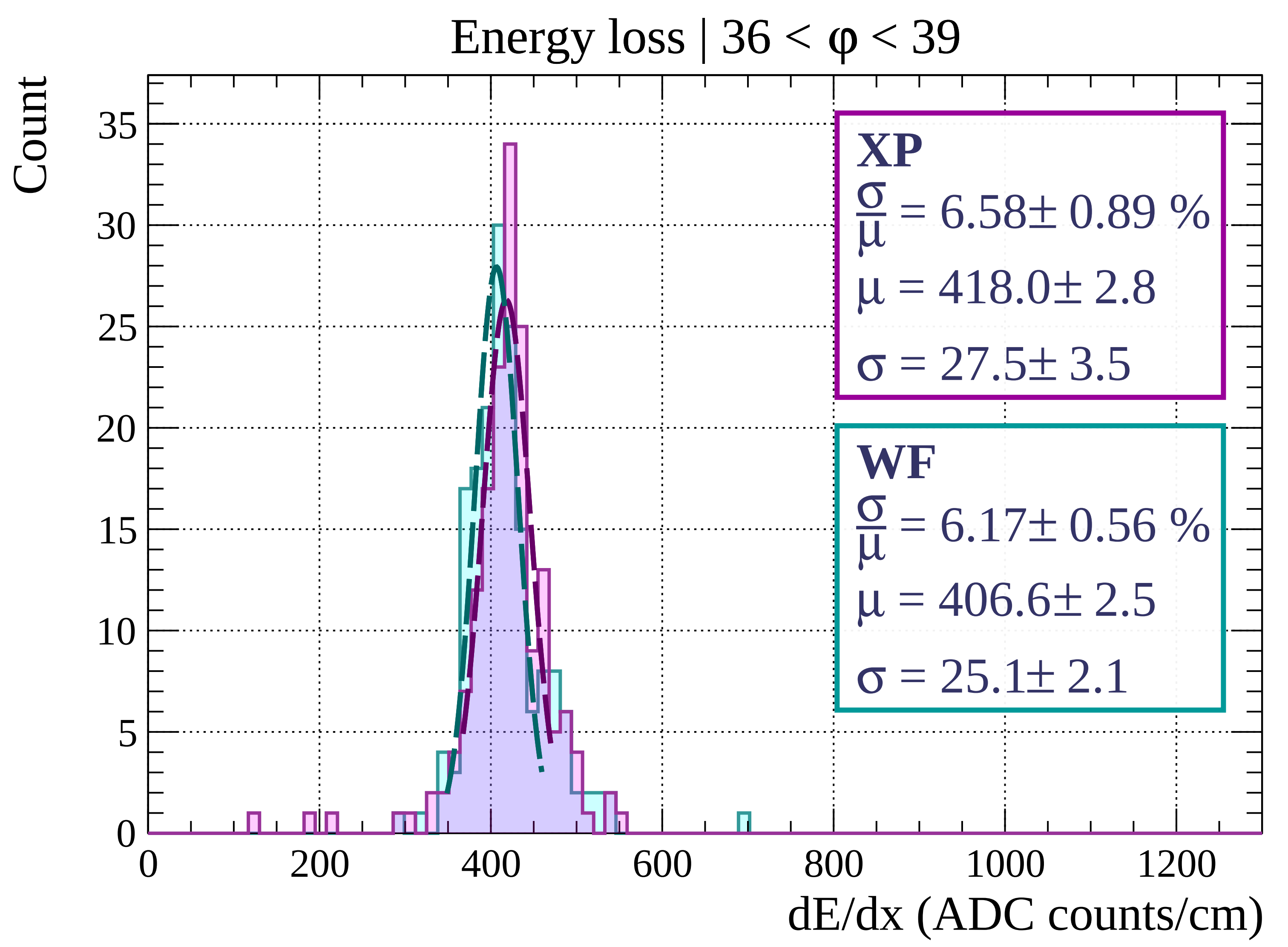


Energy loss | 27 < φ < 30
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 9.32± 1.25 %
μ = 432.5± 4.9
σ = 40.3± 5.0
WF
σ/μ = 7.73± 0.72 %
μ = 419.8± 3.8
σ = 32.5± 2.7

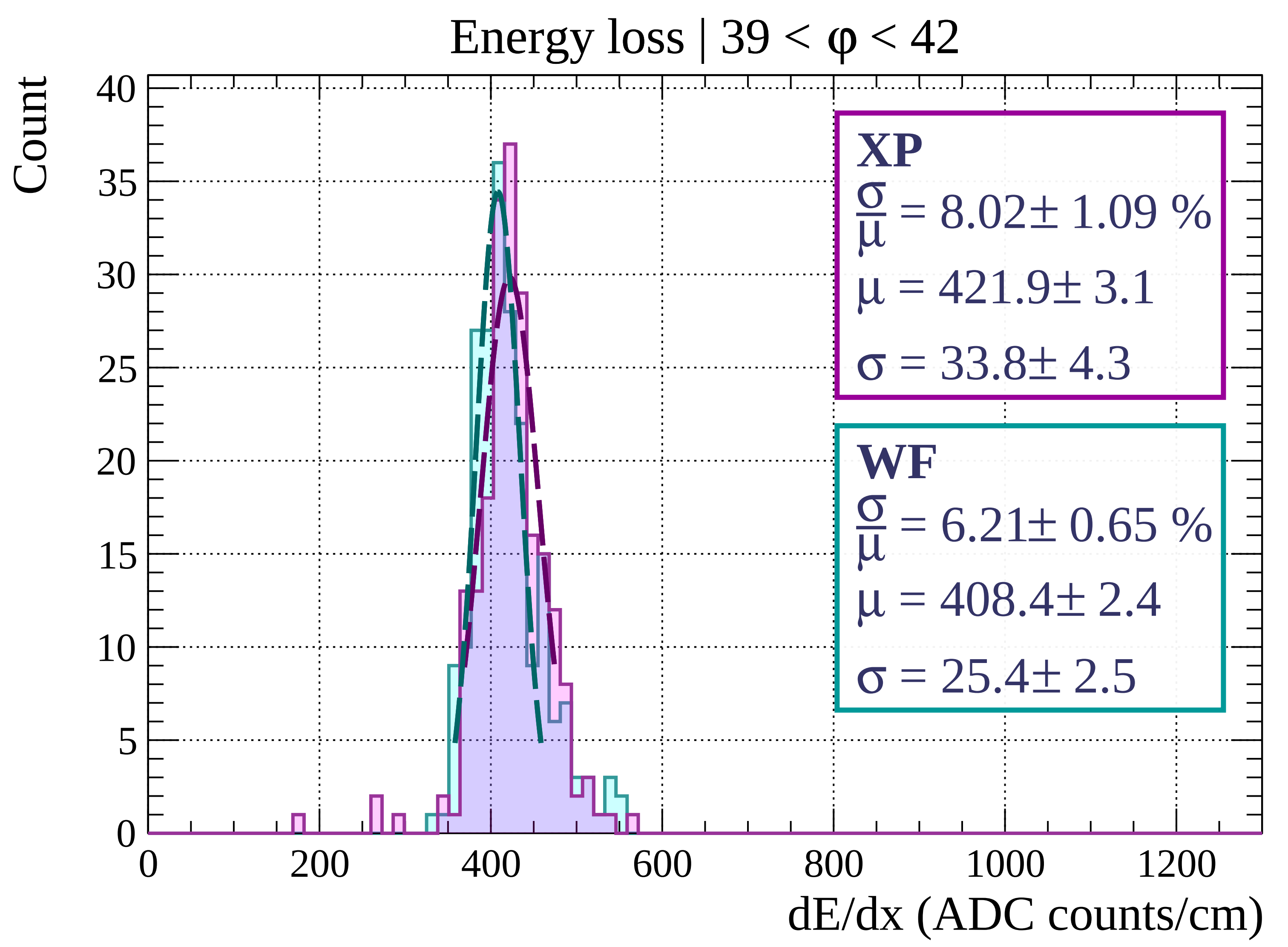

Energy loss | 30 < φ < 33
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
5
10
15
20
25
30
XP
σ/μ = 6.40± 0.59 %
μ = 430.5± 2.9
σ = 27.6± 2.4
WF
σ/μ = 6.25± 0.72 %
μ = 415.4± 3.1
σ = 26.0± 2.8



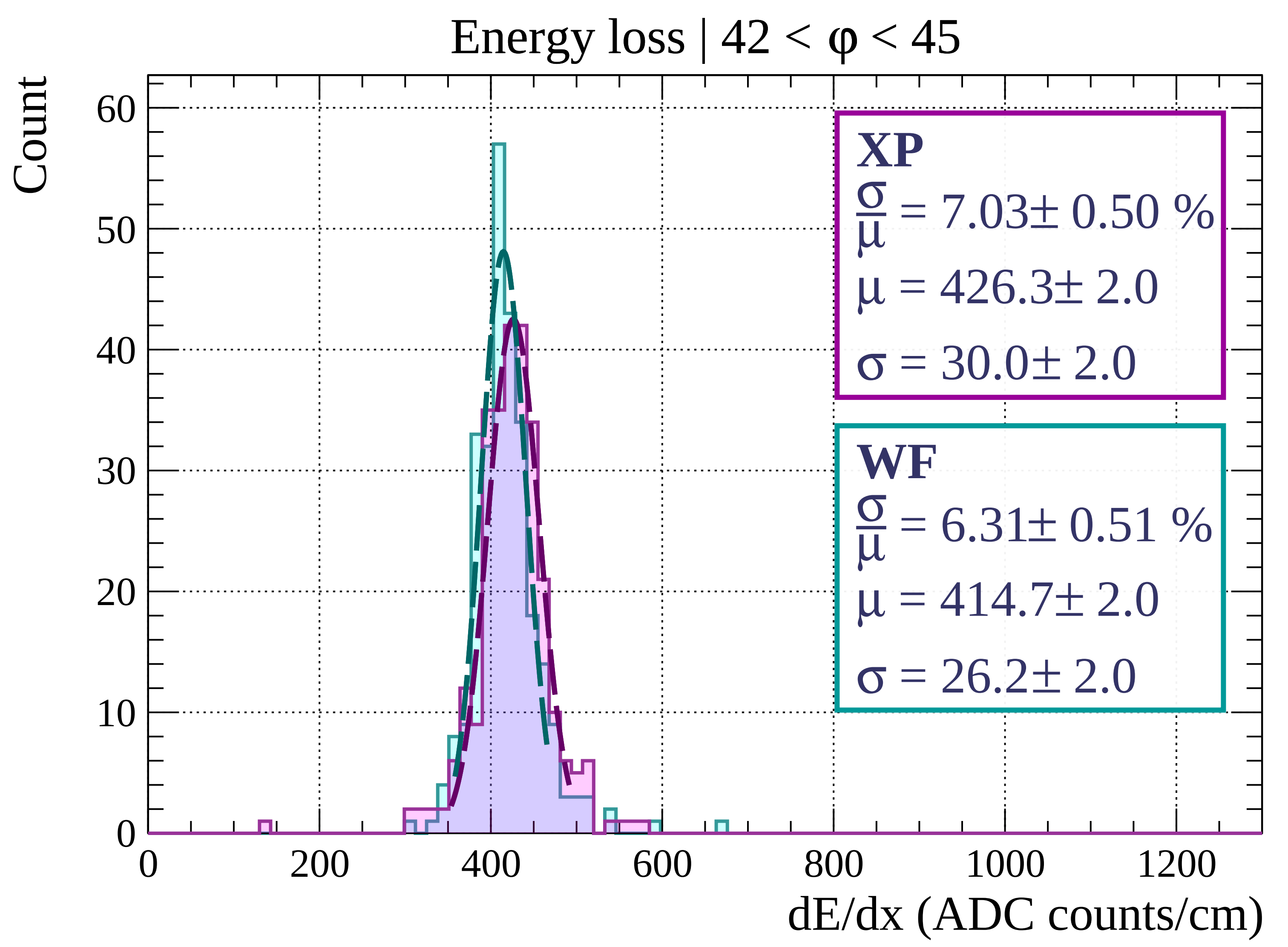
Energy loss | 33 < φ < 36
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
XP
σ/μ = 7.41± 0.77 %
μ = 424.4± 3.2
σ = 31.4± 3.0
WF
σ/μ = 7.65± 0.64 %
μ = 416.3± 3.2
σ = 31.8± 2.4



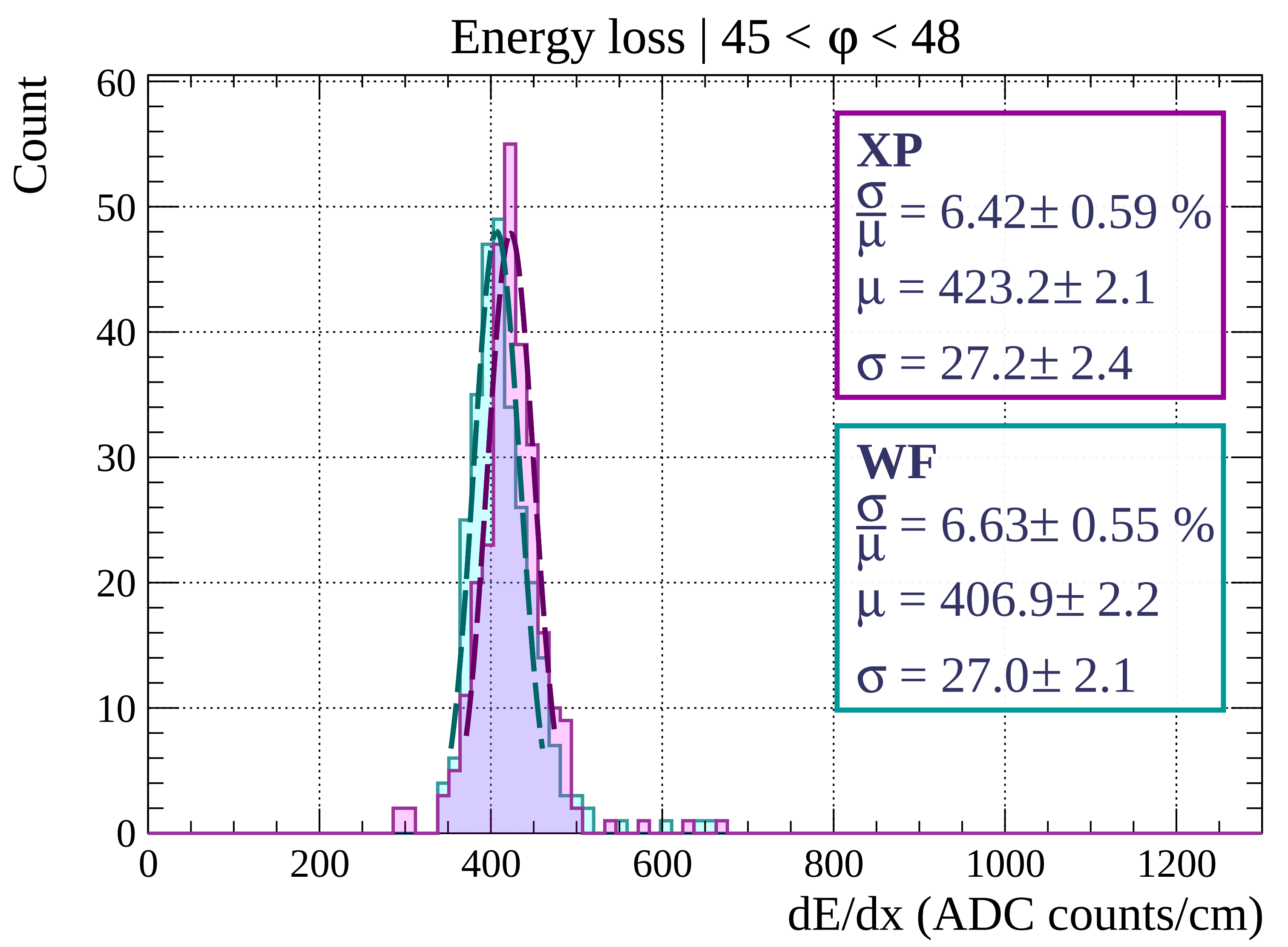
Energy loss | 36 < φ < 39
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 6.58± 0.89 %
μ = 418.0± 2.8
σ = 27.5± 3.5
WF
$\frac{\sigma}{\mu}$ = 6.17± 0.56 %
μ = 406.6± 2.5
σ = 25.1± 2.1



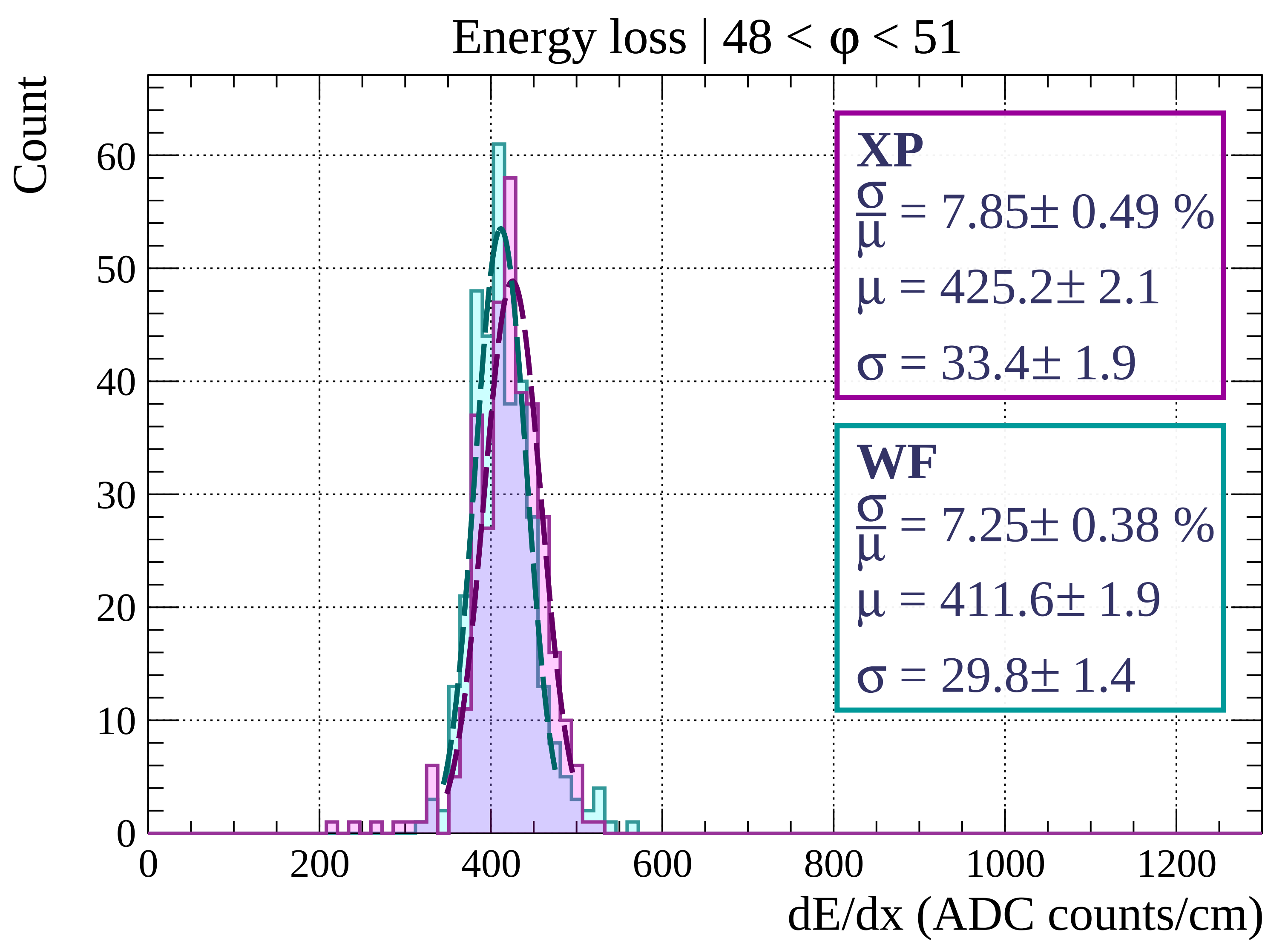
Energy loss | 39 < φ < 42
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.02± 1.09 %
μ = 421.9± 3.1
σ = 33.8± 4.3
WF
$\frac{\sigma}{\mu}$ = 6.21± 0.65 %
μ = 408.4± 2.4
σ = 25.4± 2.5
0
200
400
600
800
1000
1200
5
10
15
20
25
30
35
40



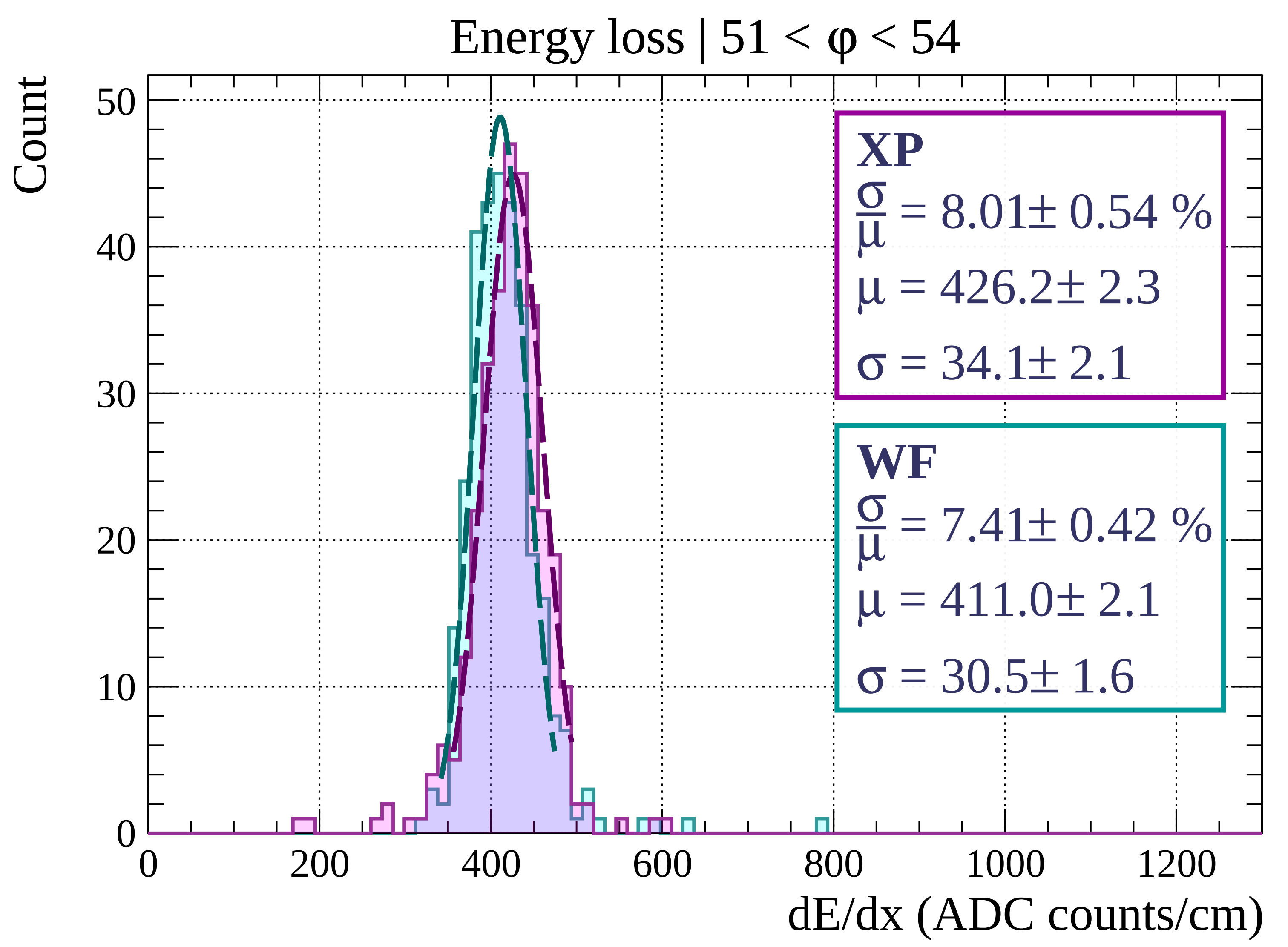
Energy loss | 42 < φ < 45
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 7.03± 0.50 %
μ = 426.3± 2.0
σ = 30.0± 2.0
WF
$\frac{\sigma}{\mu}$ = 6.31± 0.51 %
μ = 414.7± 2.0
σ = 26.2± 2.0



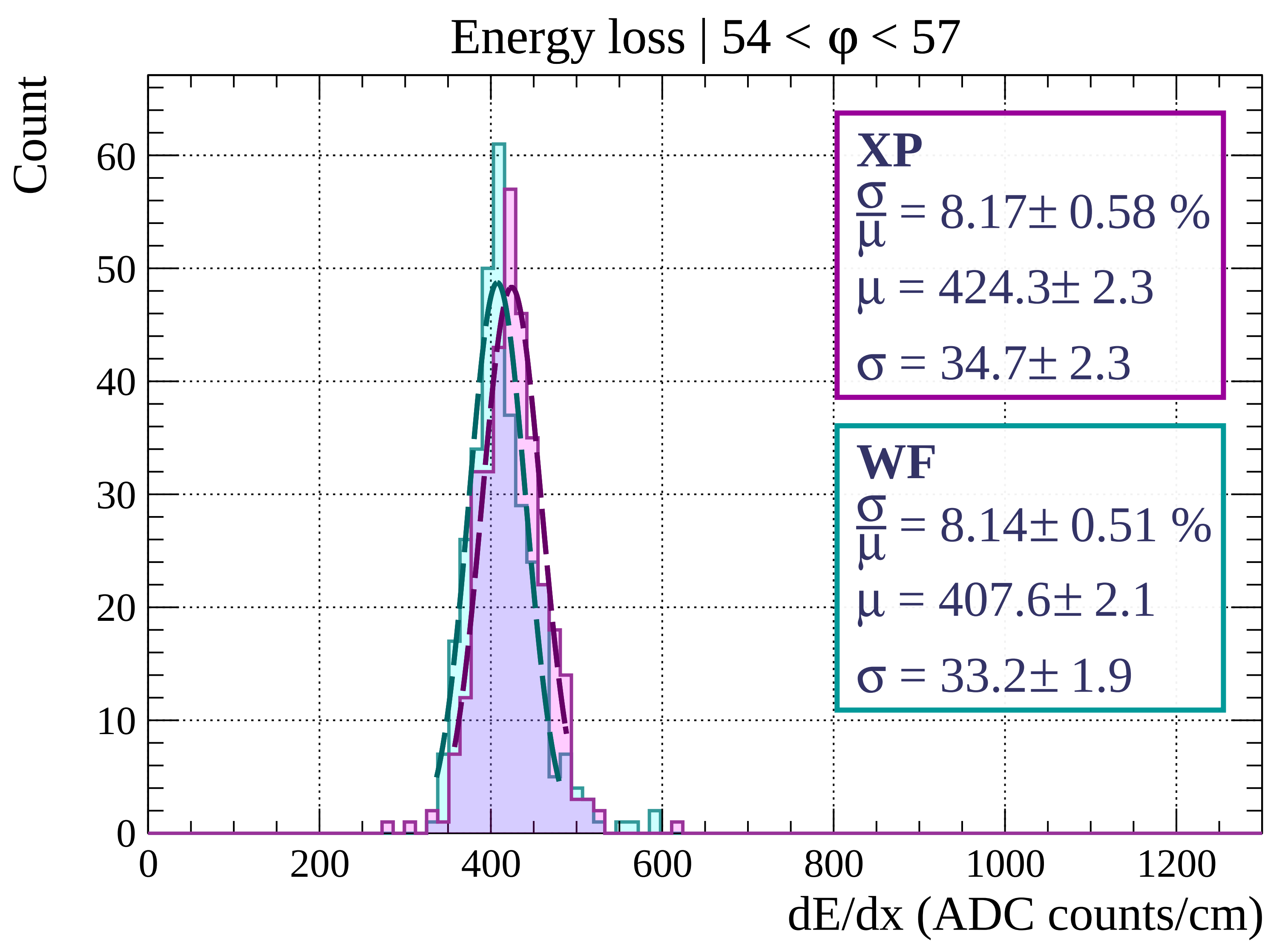
Energy loss | 45 < φ < 48
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP
σ/μ = 6.42± 0.59 %
μ = 423.2± 2.1
σ = 27.2± 2.4
WF
σ/μ = 6.63± 0.55 %
μ = 406.9± 2.2
σ = 27.0± 2.1



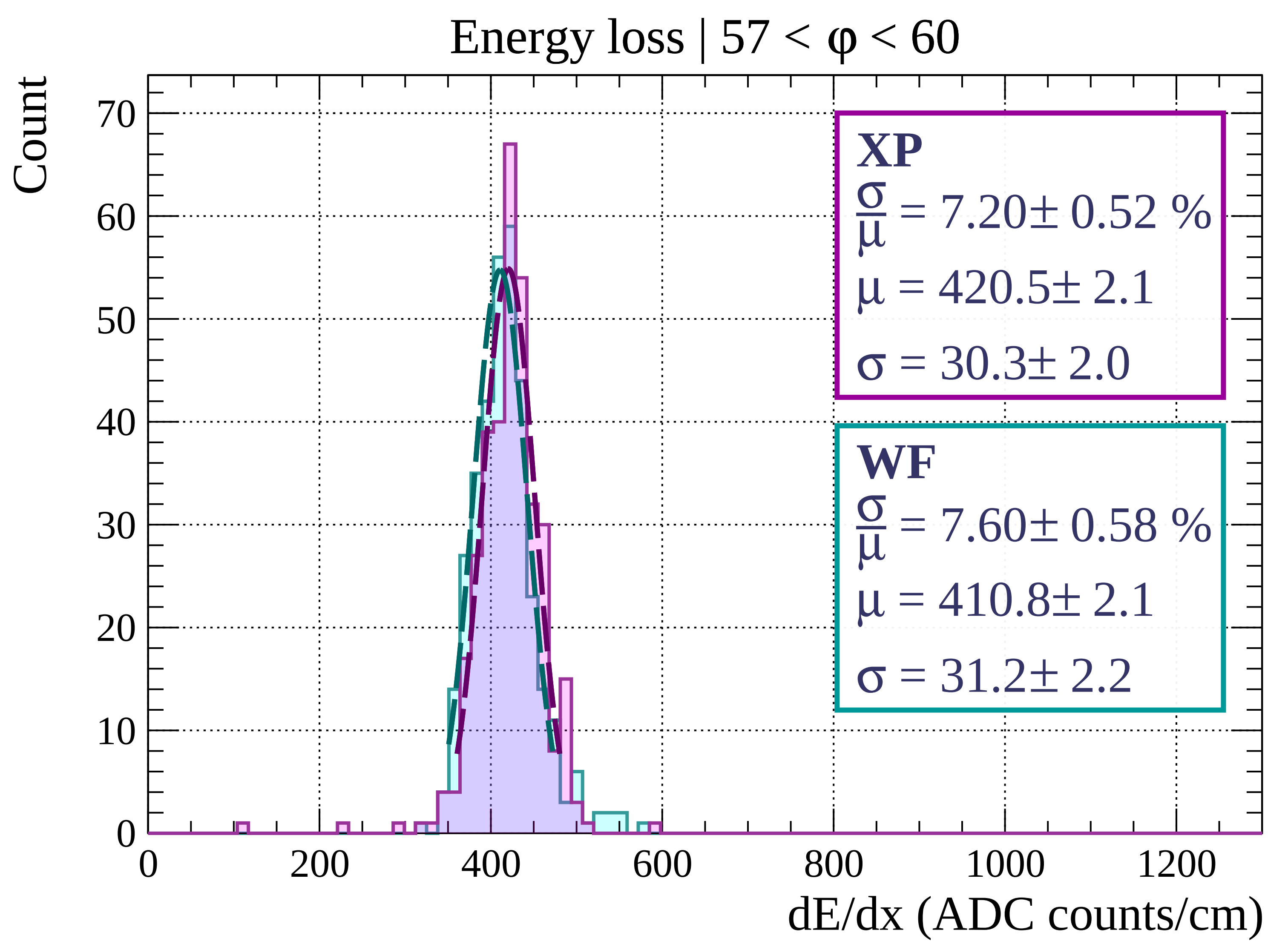
Energy loss | 48 < φ < 51
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
XP
σ/μ = 7.85± 0.49 %
μ = 425.2± 2.1
σ = 33.4± 1.9
WF
σ/μ = 7.25± 0.38 %
μ = 411.6± 1.9
σ = 29.8± 1.4



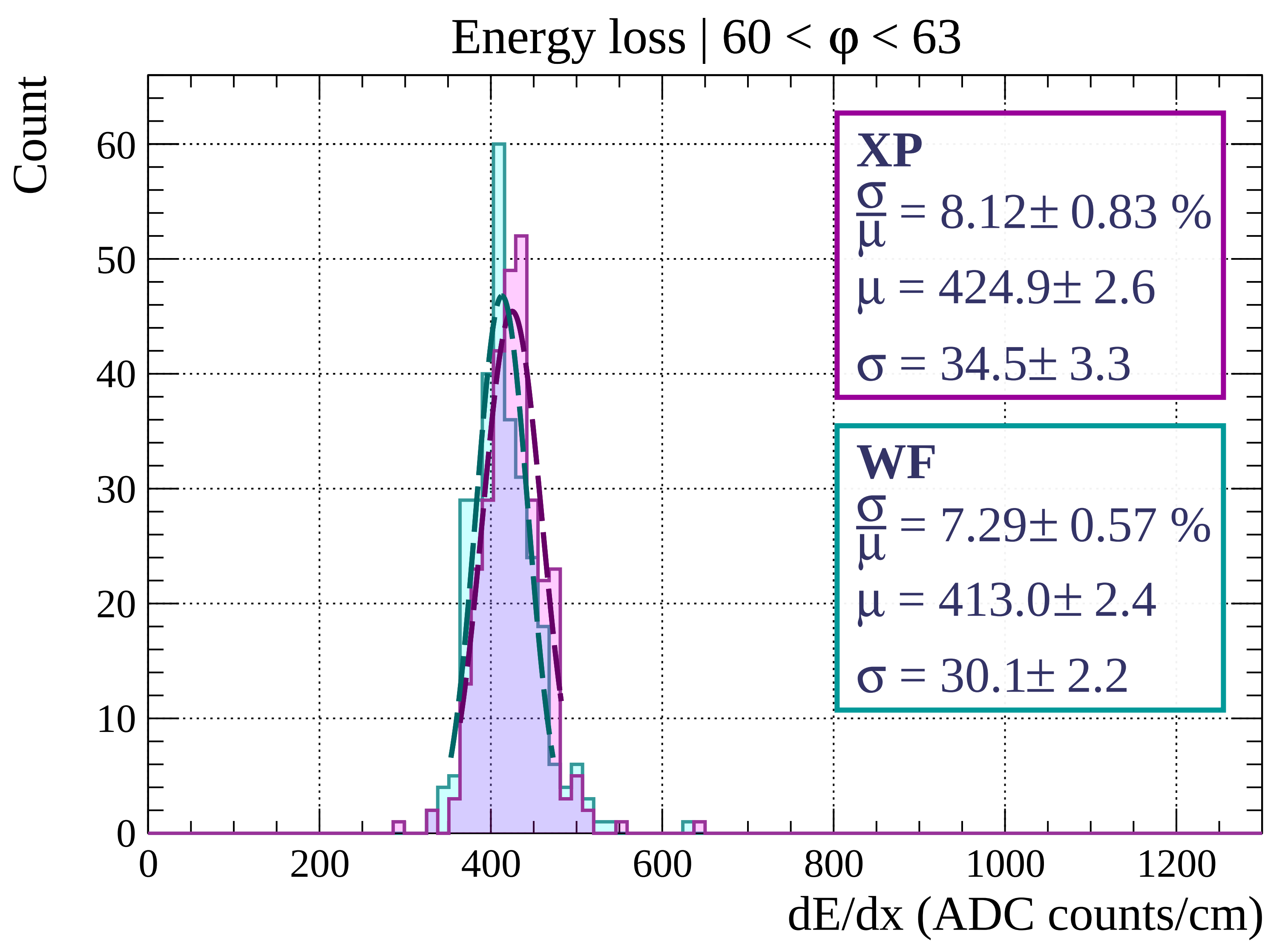
Energy loss | 51 < φ < 54
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.01± 0.54 %
μ = 426.2± 2.3
σ = 34.1± 2.1
WF
$\frac{\sigma}{\mu}$ = 7.41± 0.42 %
μ = 411.0± 2.1
σ = 30.5± 1.6

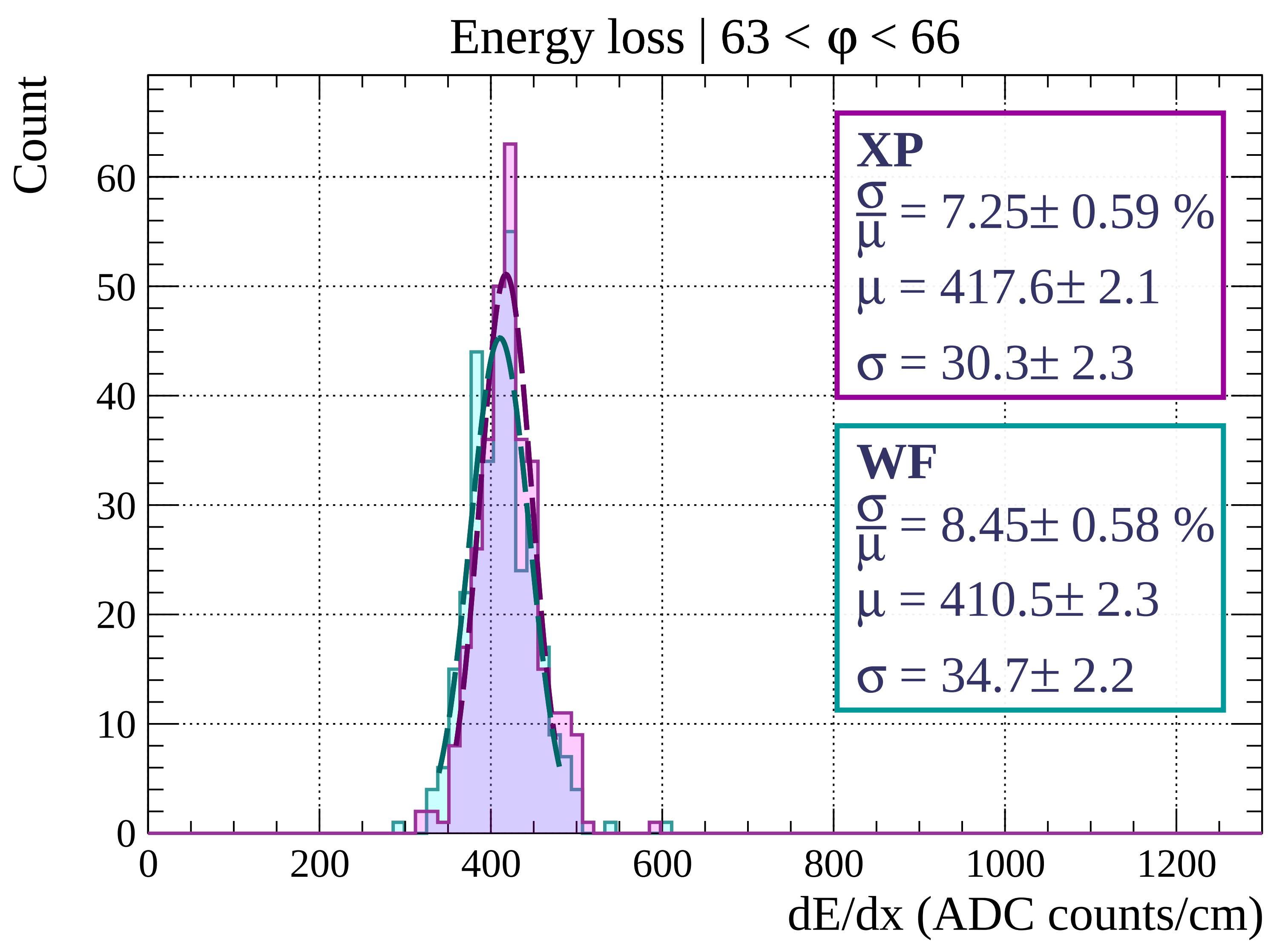


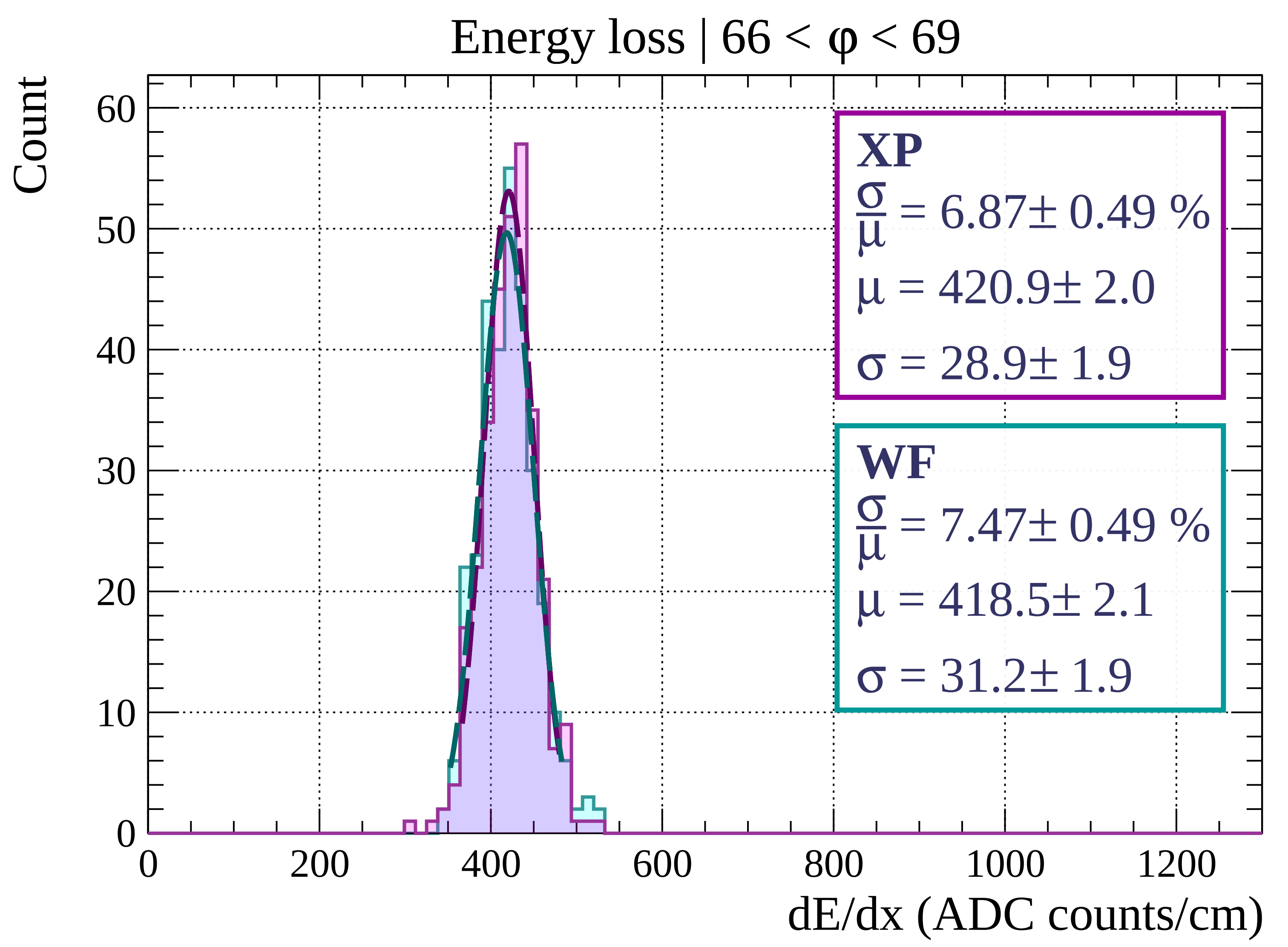
Energy loss | 54 < φ < 57
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP
σ/μ = 8.17± 0.58 %
μ = 424.3± 2.3
σ = 34.7± 2.3
WF
σ/μ = 8.14± 0.51 %
μ = 407.6± 2.1
σ = 33.2± 1.9



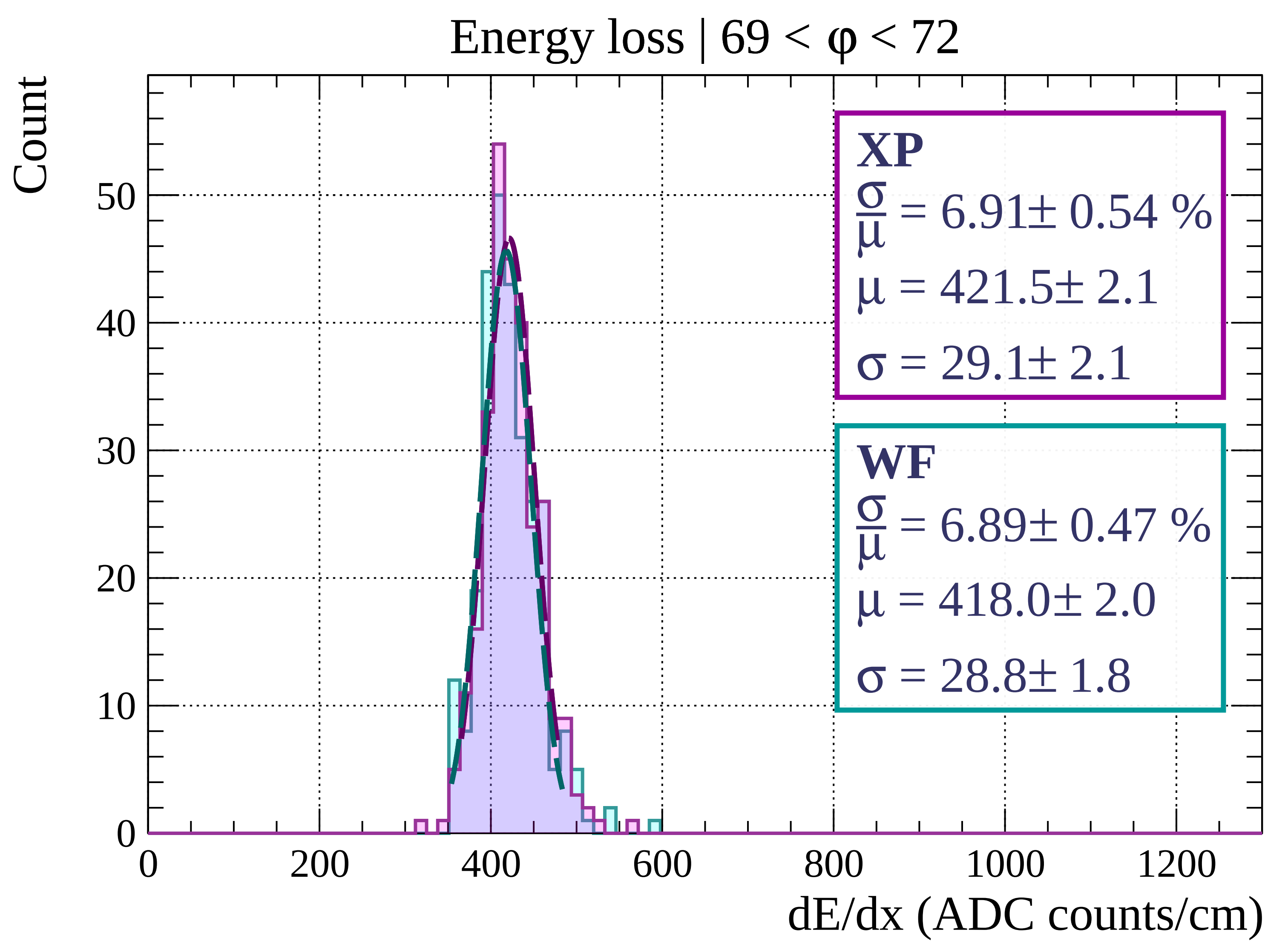
Energy loss | 57 < φ < 60
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
XP
σ/μ = 7.20± 0.52 %
μ = 420.5± 2.1
σ = 30.3± 2.0
WF
σ/μ = 7.60± 0.58 %
μ = 410.8± 2.1
σ = 31.2± 2.2


Energy loss | 60 < φ < 63
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 8.12± 0.83 %
μ = 424.9± 2.6
σ = 34.5± 3.3
WF
σ/μ = 7.29± 0.57 %
μ = 413.0± 2.4
σ = 30.1± 2.2
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200

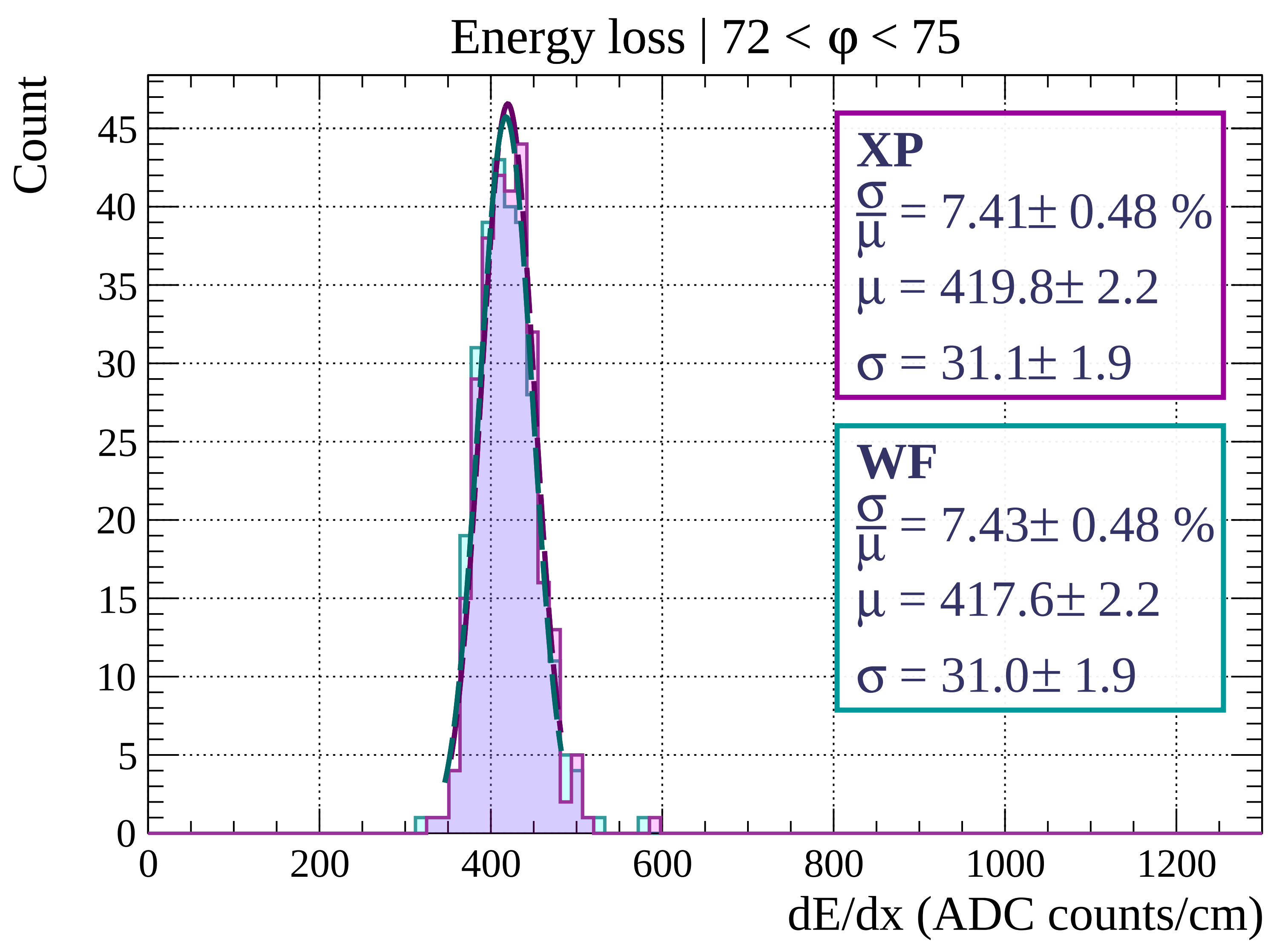

Energy loss | 63 < φ < 66
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
XP
σ/μ = 7.25± 0.59 %
μ = 417.6± 2.1
σ = 30.3± 2.3
WF
σ/μ = 8.45± 0.58 %
μ = 410.5± 2.3
σ = 34.7± 2.2



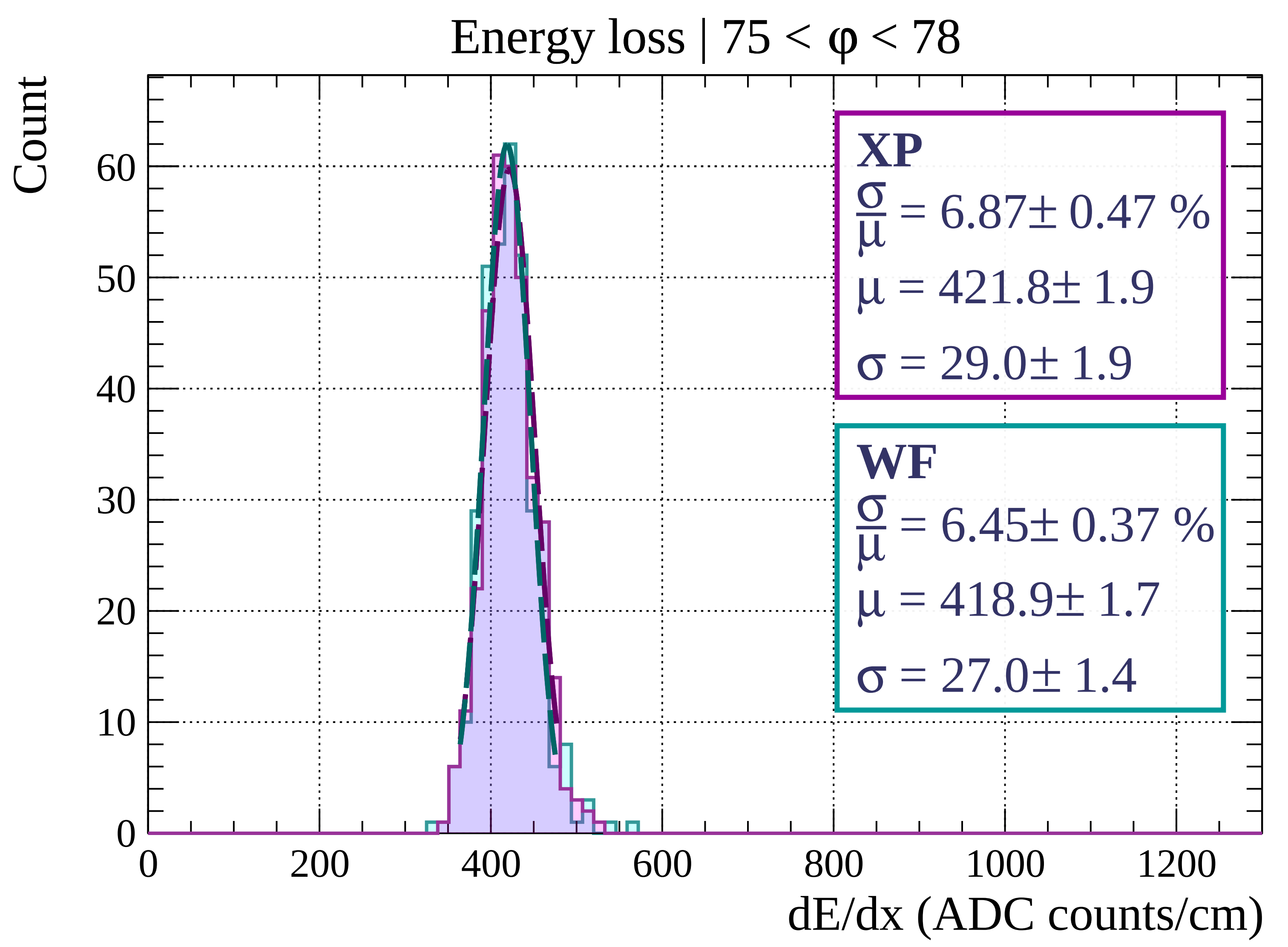
Energy loss | 66 < φ < 69
Count
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 6.87± 0.49 %
μ = 420.9± 2.0
σ = 28.9± 1.9
WF
σ/μ = 7.47± 0.49 %
μ = 418.5± 2.1
σ = 31.2± 1.9

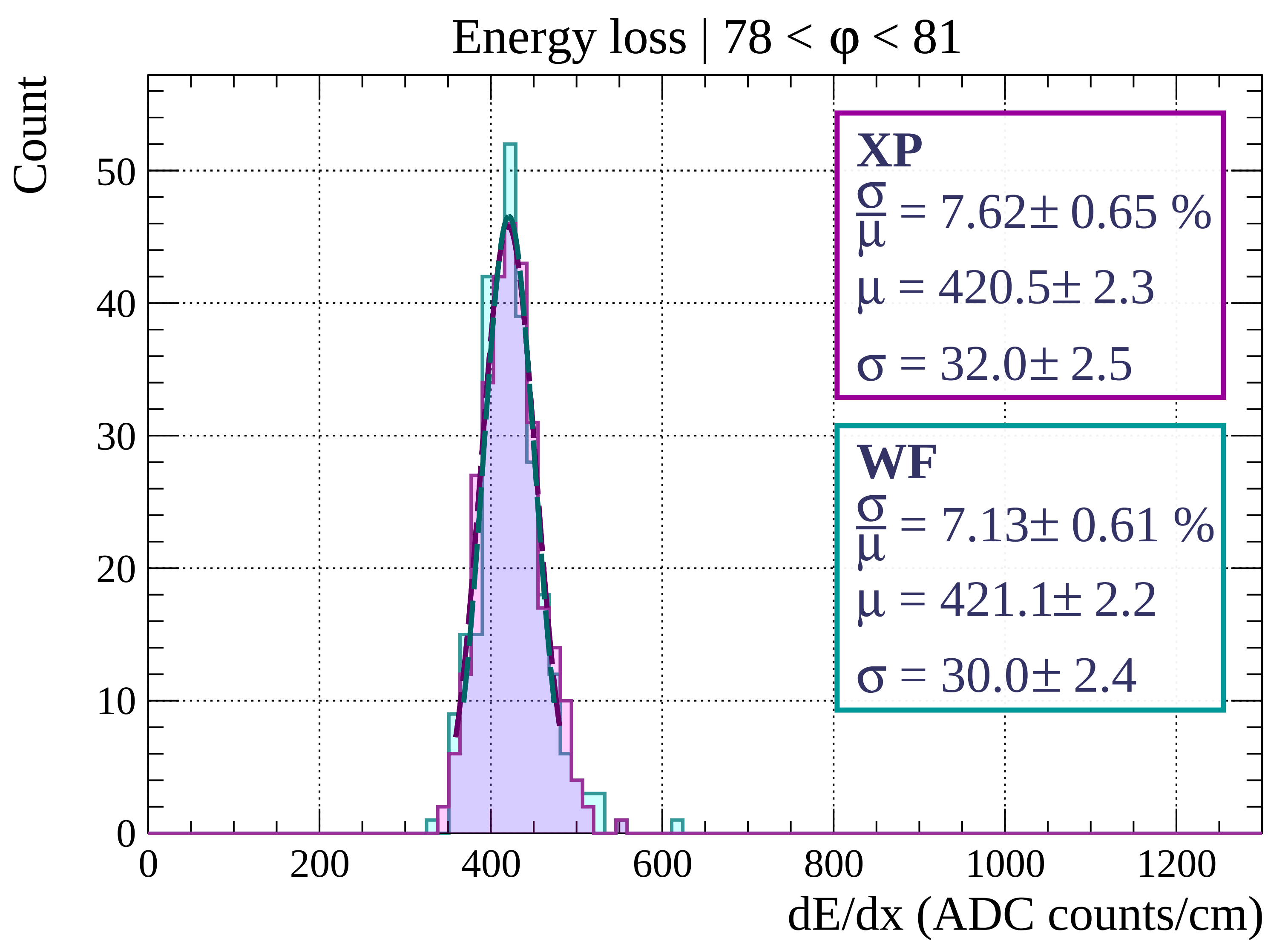


Energy loss | 69 < φ < 72
Count
XP
σ/μ = 6.91± 0.54 %
μ = 421.5± 2.1
σ = 29.1± 2.1
WF
σ/μ = 6.89± 0.47 %
μ = 418.0± 2.0
σ = 28.8± 1.8
0
10
20
30
40
50
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)

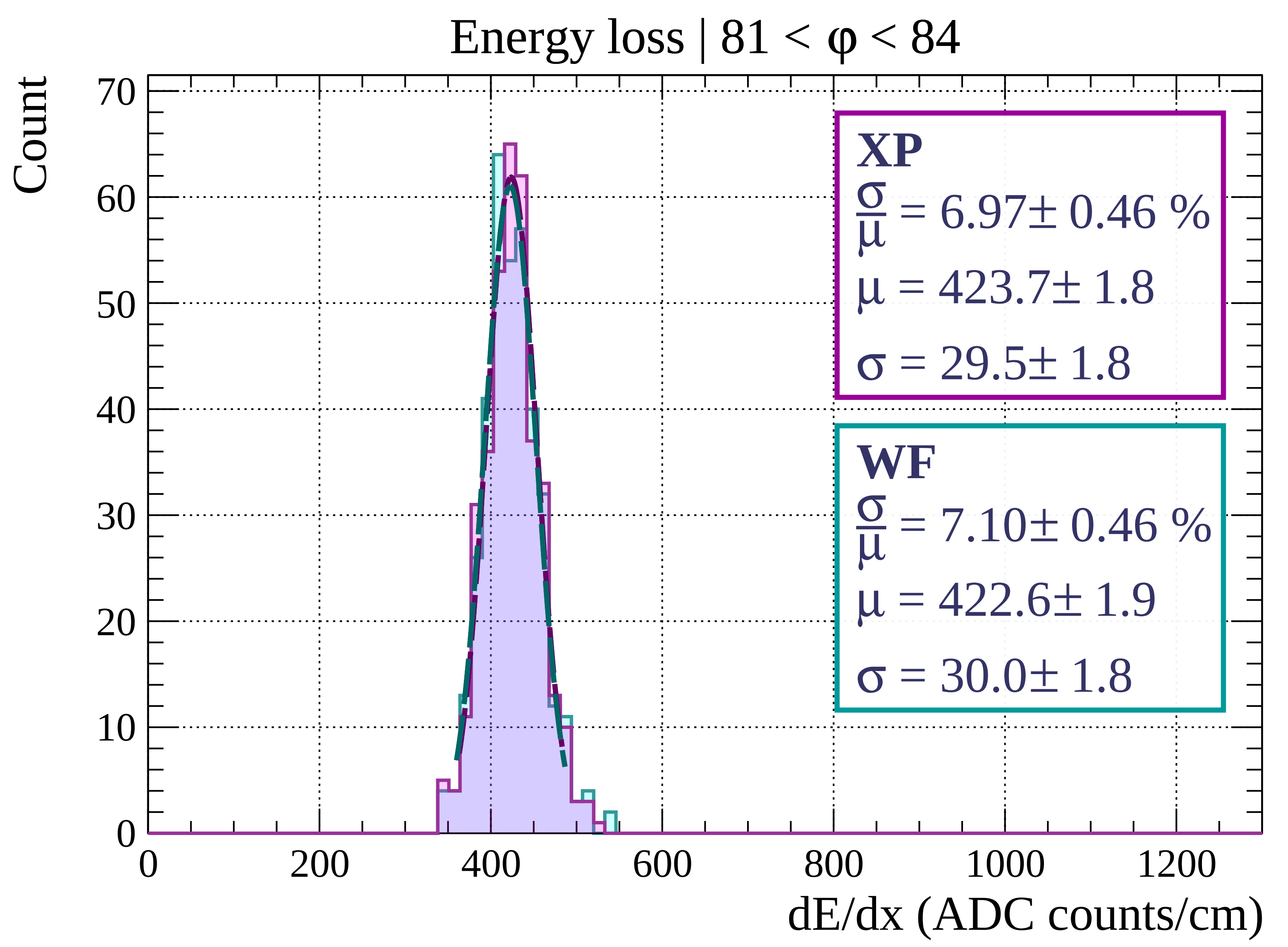


Energy loss | 72 < φ < 75
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
5
10
15
20
25
30
35
40
45
XP
σ/μ = 7.41± 0.48 %
μ = 419.8± 2.2
σ = 31.1± 1.9
WF
σ/μ = 7.43± 0.48 %
μ = 417.6± 2.2
σ = 31.0± 1.9

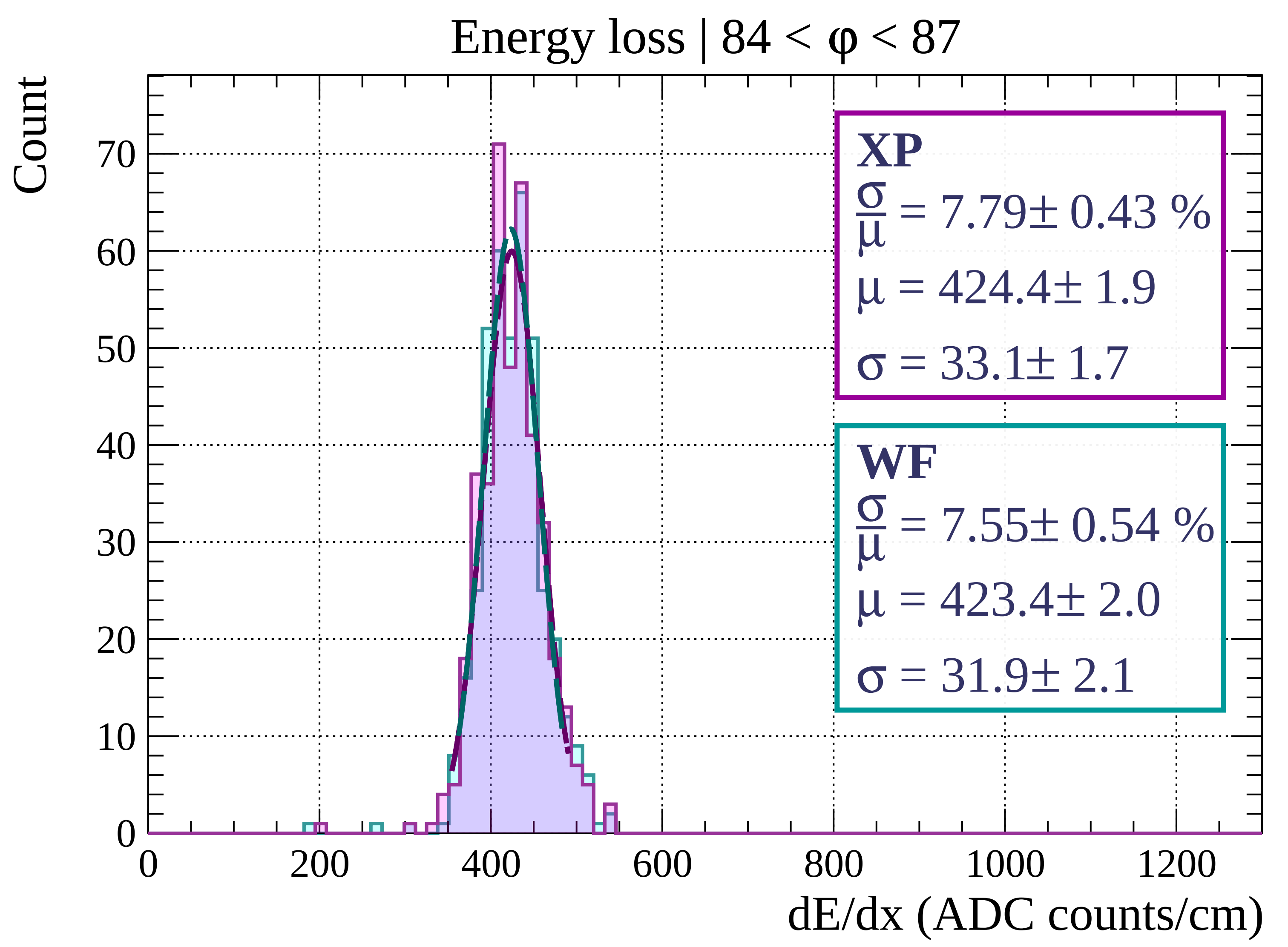

Energy loss | 75 < φ < 78
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
XP
σ/μ = 6.87± 0.47 %
μ = 421.8± 1.9
σ = 29.0± 1.9
WF
σ/μ = 6.45± 0.37 %
μ = 418.9± 1.7
σ = 27.0± 1.4

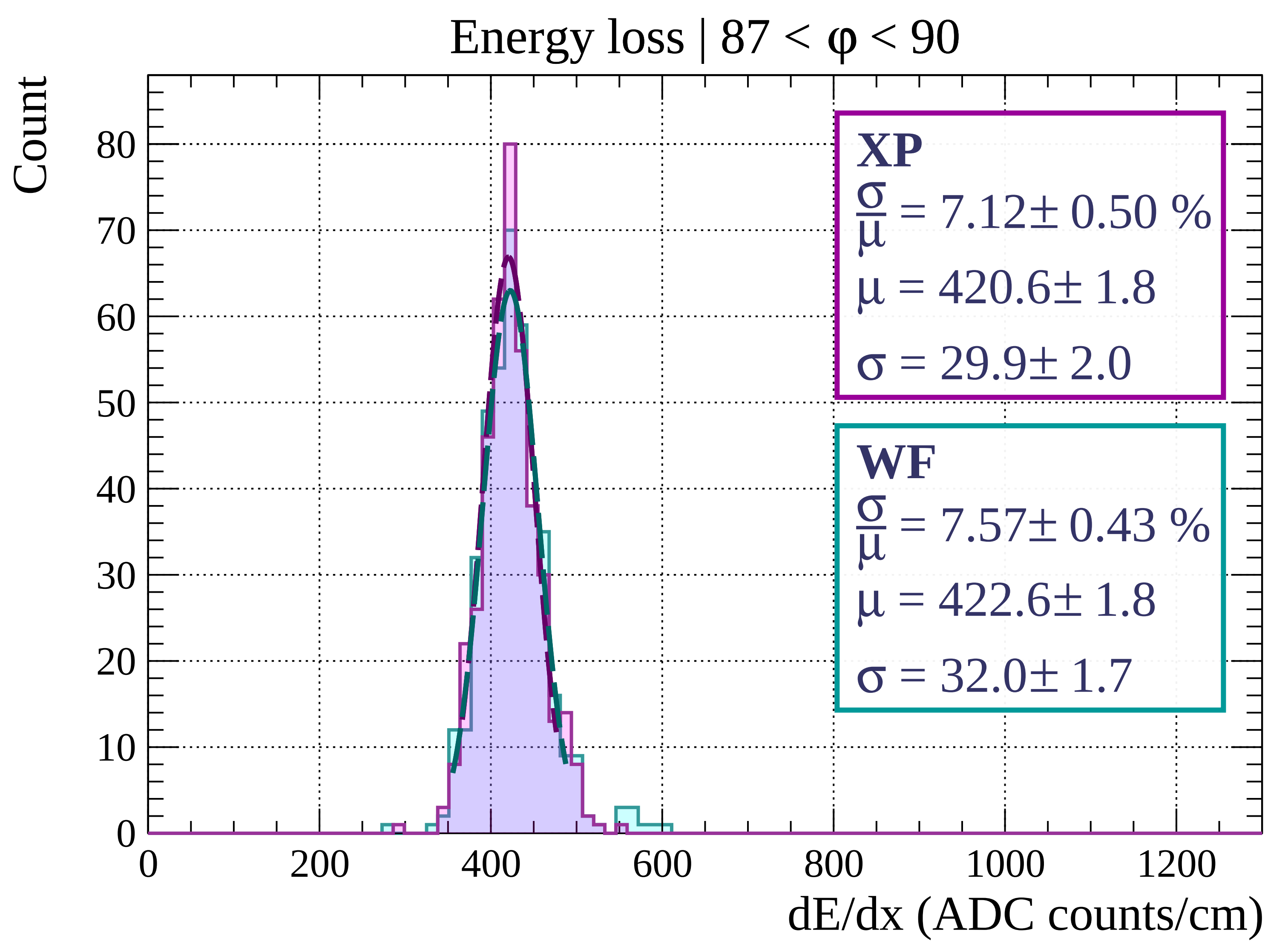


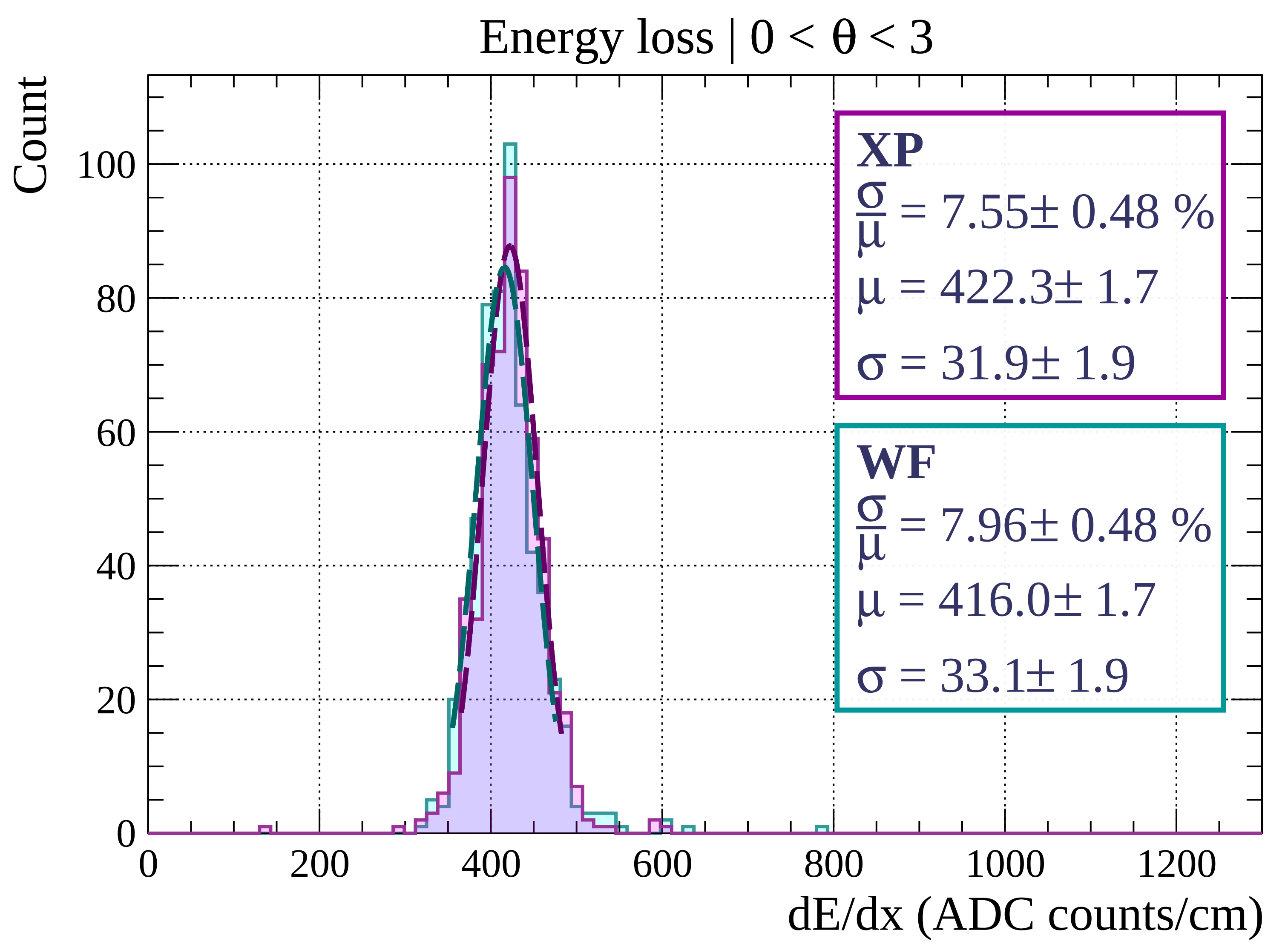
Energy loss | 78 < φ < 81
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
XP
σ/μ = 7.62± 0.65 %
μ = 420.5± 2.3
σ = 32.0± 2.5
WF
σ/μ = 7.13± 0.61 %
μ = 421.1± 2.2
σ = 30.0± 2.4



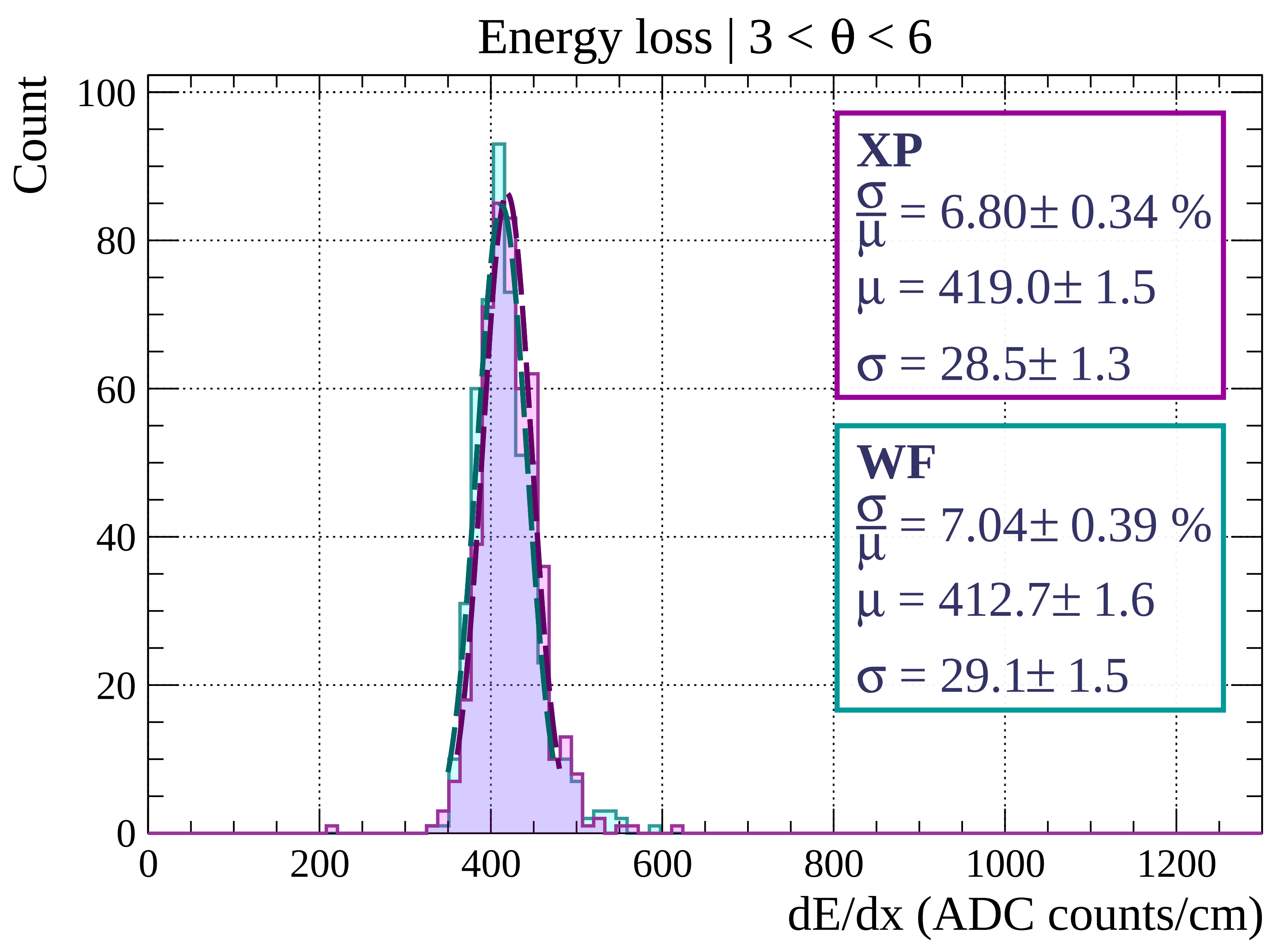
Energy loss | 81 < φ < 84
Count
XP
σ/μ = 6.97± 0.46 %
μ = 423.7± 1.8
σ = 29.5± 1.8
WF
σ/μ = 7.10± 0.46 %
μ = 422.6± 1.9
σ = 30.0± 1.8
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200

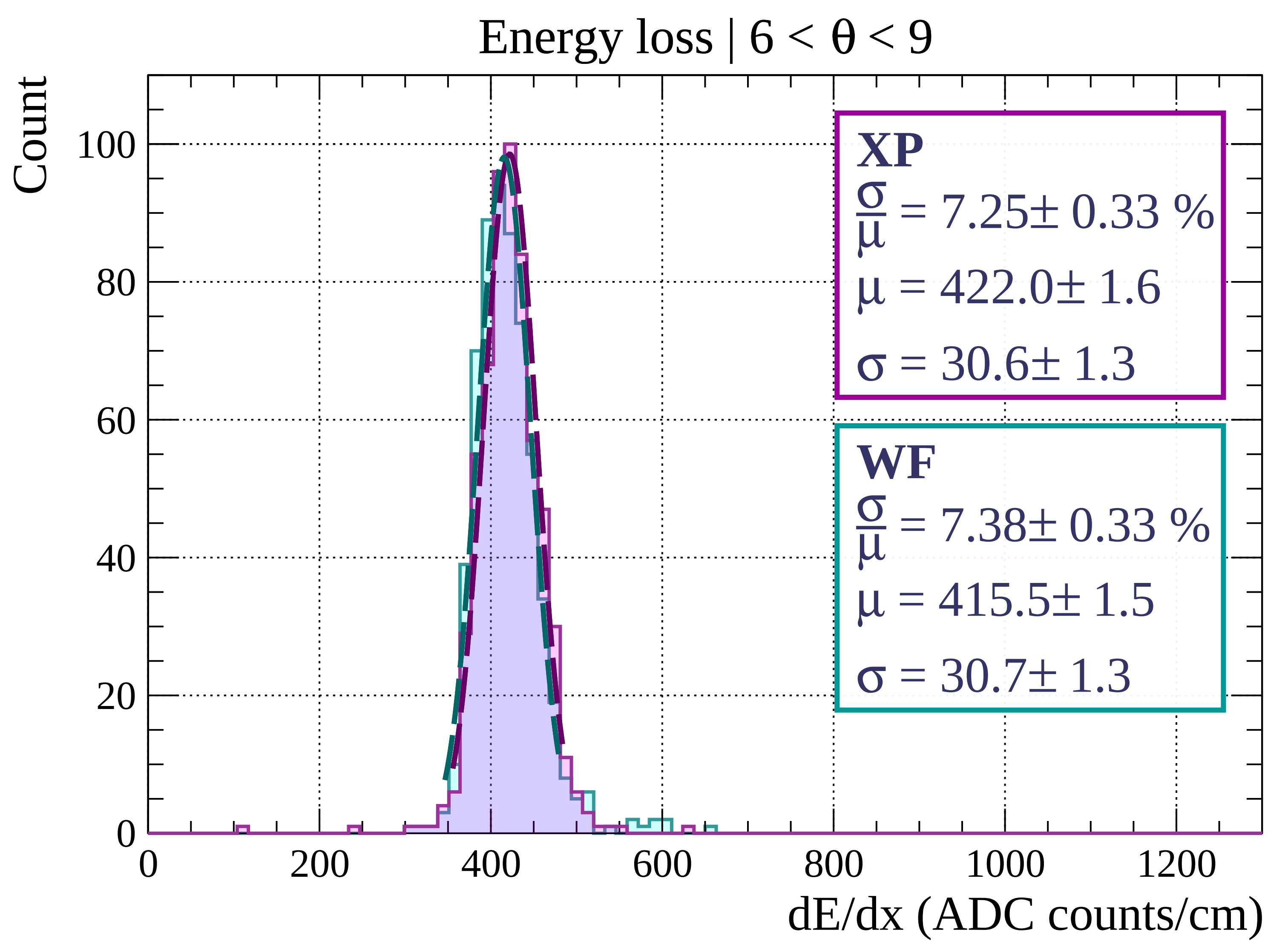

Energy loss | 84 < φ < 87
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
XP
σ/μ = 7.79± 0.43 %
μ = 424.4± 1.9
σ = 33.1± 1.7
WF
σ/μ = 7.55± 0.54 %
μ = 423.4± 2.0
σ = 31.9± 2.1


Energy loss | 87 < φ < 90
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 7.12± 0.50 %
μ = 420.6± 1.8
σ = 29.9± 2.0
WF
σ/μ = 7.57± 0.43 %
μ = 422.6± 1.8
σ = 32.0± 1.7
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200

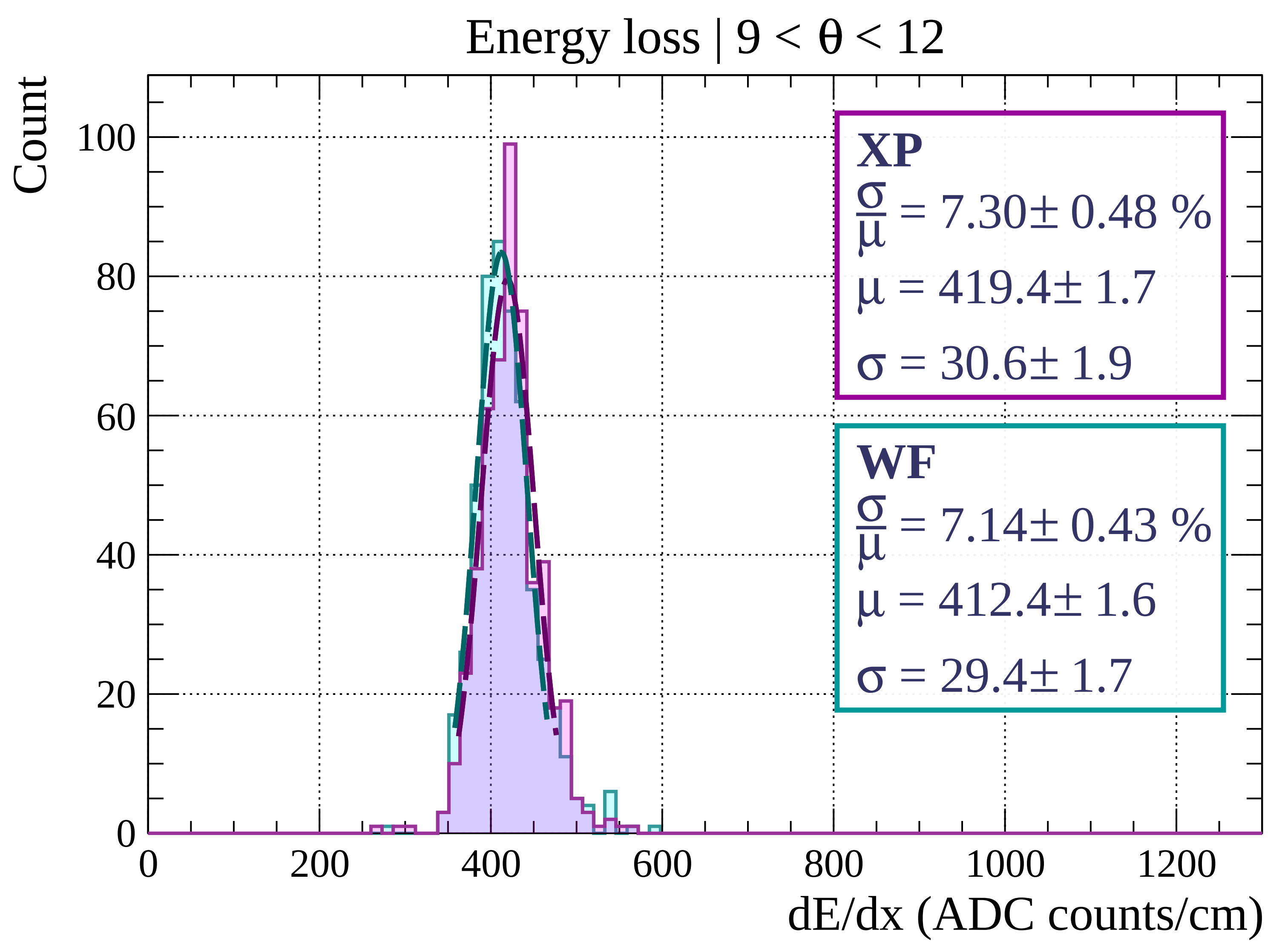


Energy loss | 0 < θ < 3
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
XP
σ/μ = 7.55± 0.48 %
μ = 422.3± 1.7
σ = 31.9± 1.9
WF
σ/μ = 7.96± 0.48 %
μ = 416.0± 1.7
σ = 33.1± 1.9

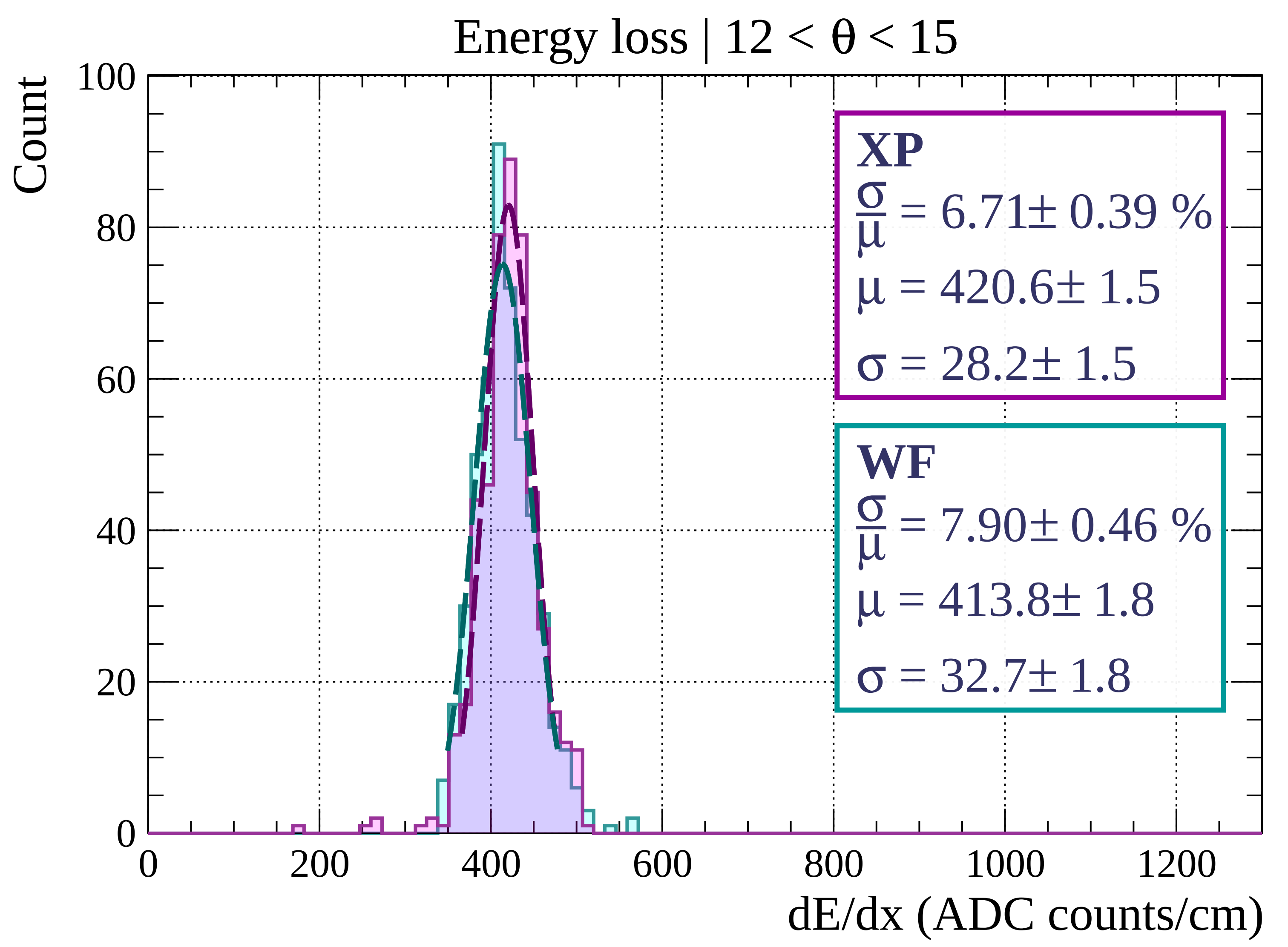

Energy loss | 3 < θ < 6
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
XP
$\frac{\sigma}{\mu}$ = 6.80± 0.34 %
μ = 419.0± 1.5
σ = 28.5± 1.3
WF
$\frac{\sigma}{\mu}$ = 7.04± 0.39 %
μ = 412.7± 1.6
σ = 29.1± 1.5



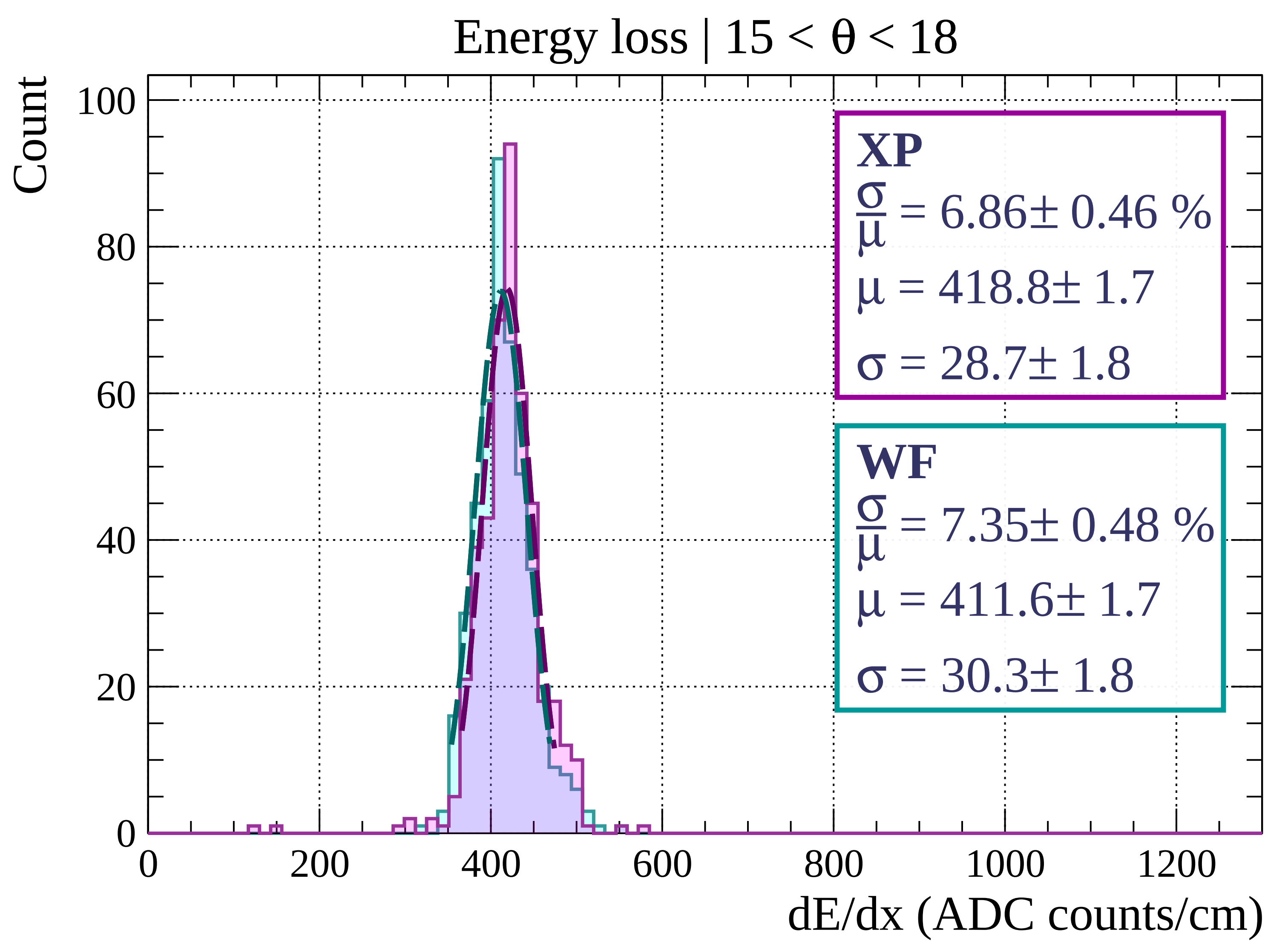
Energy loss | 6 < θ < 9
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
XP
σ/μ = 7.25± 0.33 %
μ = 422.0± 1.6
σ = 30.6± 1.3
WF
σ/μ = 7.38± 0.33 %
μ = 415.5± 1.5
σ = 30.7± 1.3



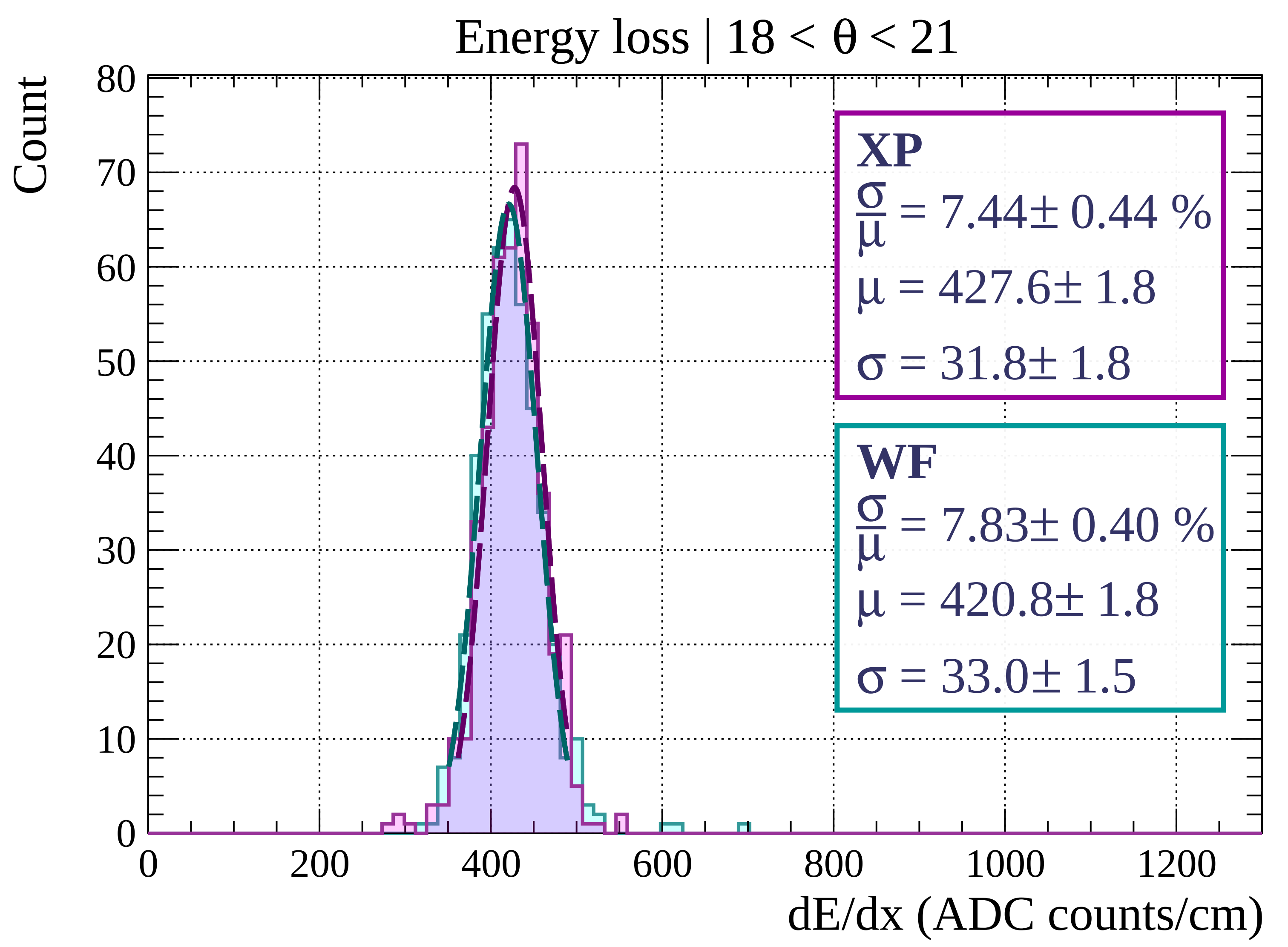
Energy loss | 9 < θ < 12
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP
σ/μ = 7.30± 0.48 %
μ = 419.4± 1.7
σ = 30.6± 1.9
WF
σ/μ = 7.14± 0.43 %
μ = 412.4± 1.6
σ = 29.4± 1.7

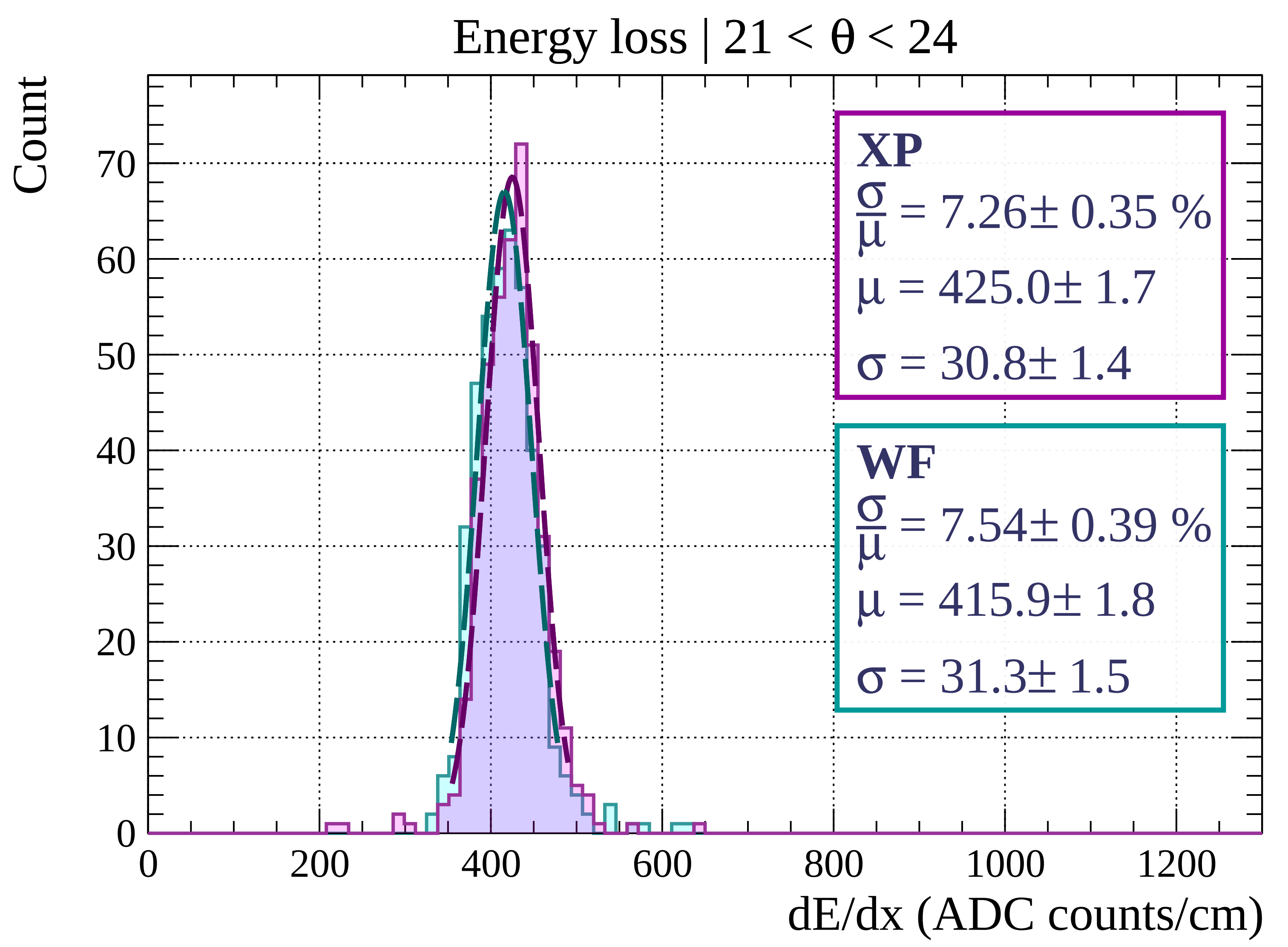


Energy loss | 12 < θ < 15
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 6.71± 0.39 %
μ = 420.6± 1.5
σ = 28.2± 1.5
WF
σ/μ = 7.90± 0.46 %
μ = 413.8± 1.8
σ = 32.7± 1.8

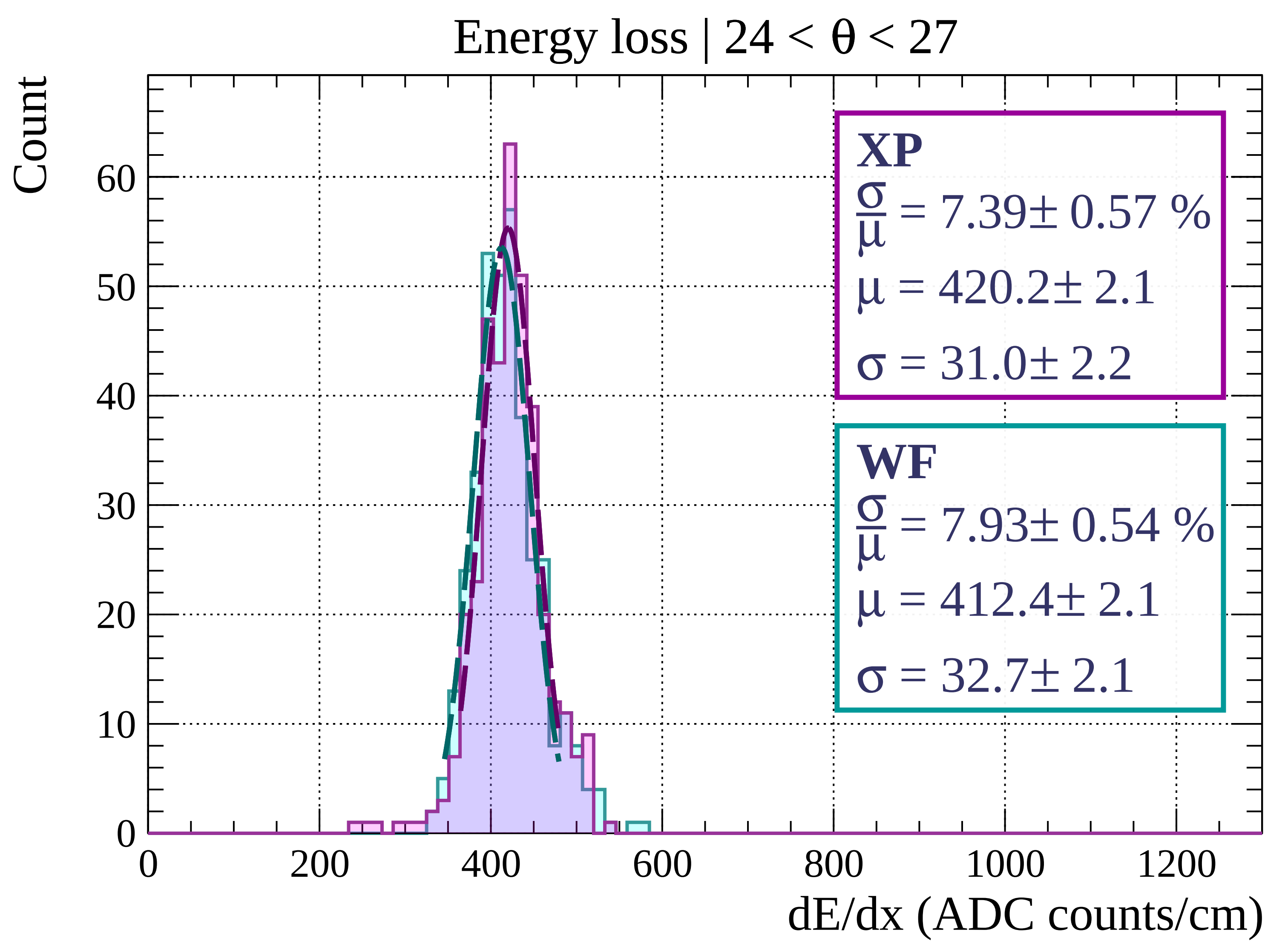


Energy loss | 15 < θ < 18
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
XP
σ/μ = 6.86± 0.46 %
μ = 418.8± 1.7
σ = 28.7± 1.8
WF
σ/μ = 7.35± 0.48 %
μ = 411.6± 1.7
σ = 30.3± 1.8

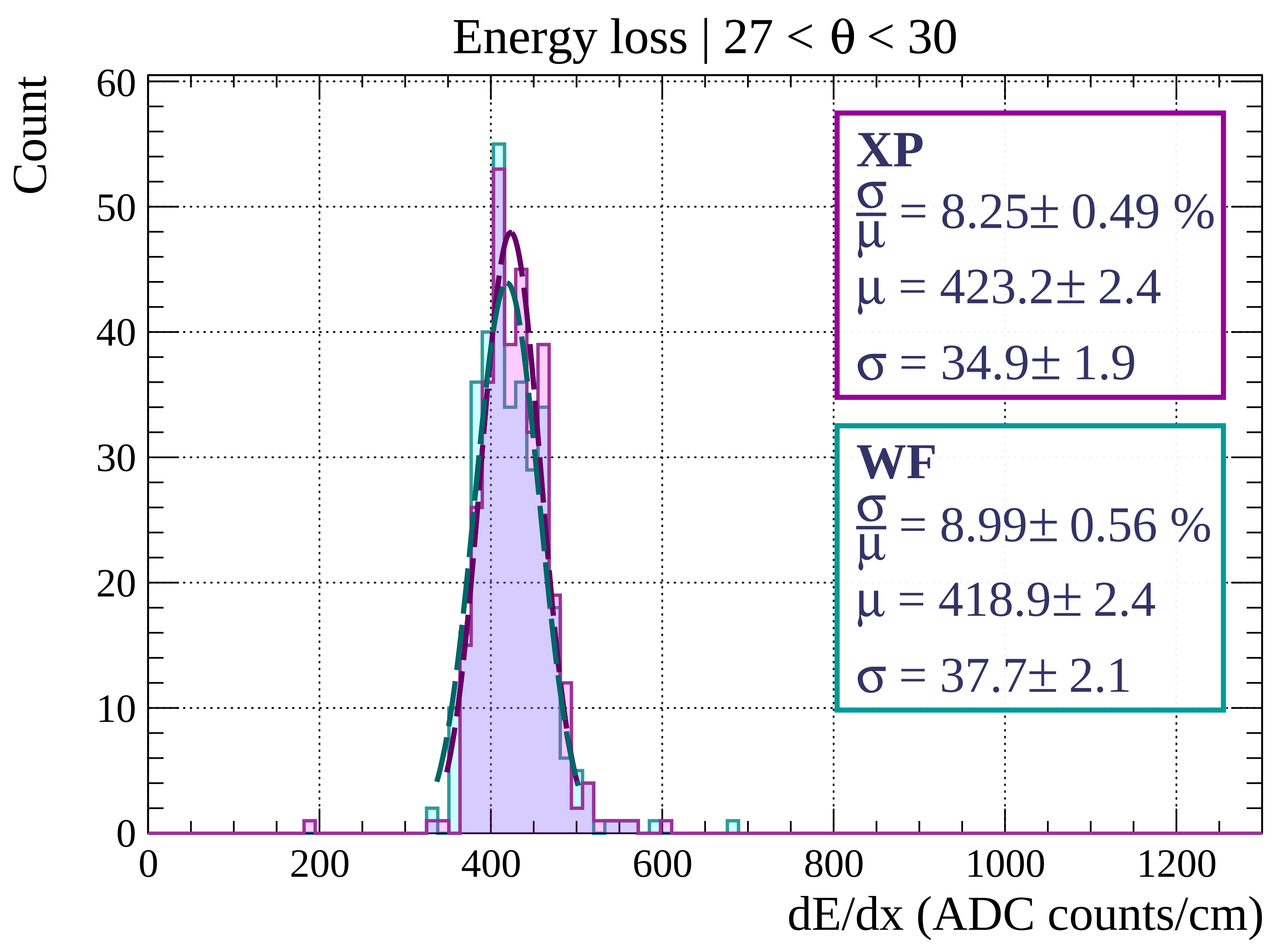

Energy loss | 18 < θ < 21
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 7.44± 0.44 %
μ = 427.6± 1.8
σ = 31.8± 1.8
WF
σ/μ = 7.83± 0.40 %
μ = 420.8± 1.8
σ = 33.0± 1.5
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200



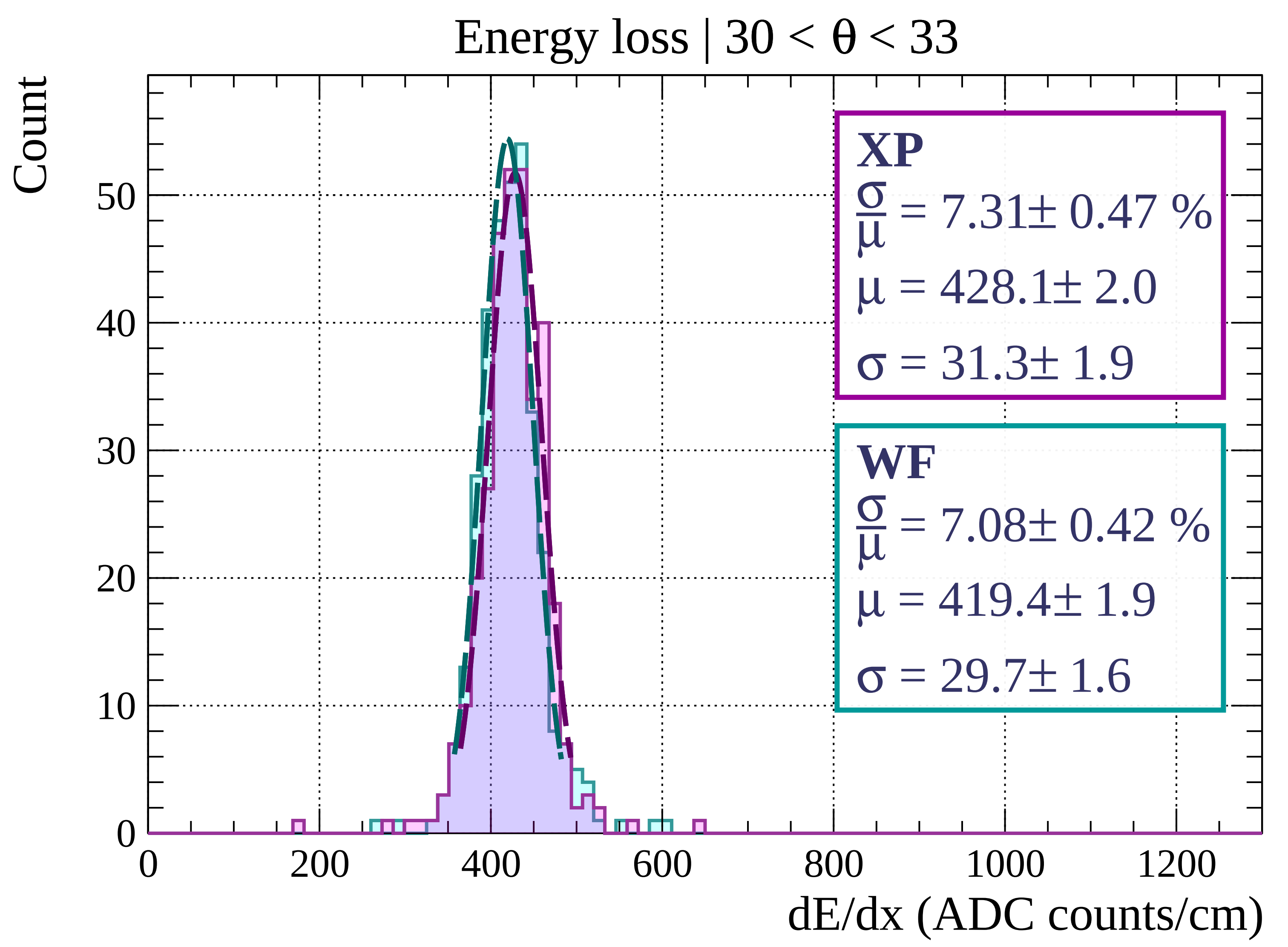
Energy loss | 21 < θ < 24
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
XP
$\frac{\sigma}{\mu}$ = 7.26± 0.35 %
μ = 425.0± 1.7
σ = 30.8± 1.4
WF
$\frac{\sigma}{\mu}$ = 7.54± 0.39 %
μ = 415.9± 1.8
σ = 31.3± 1.5



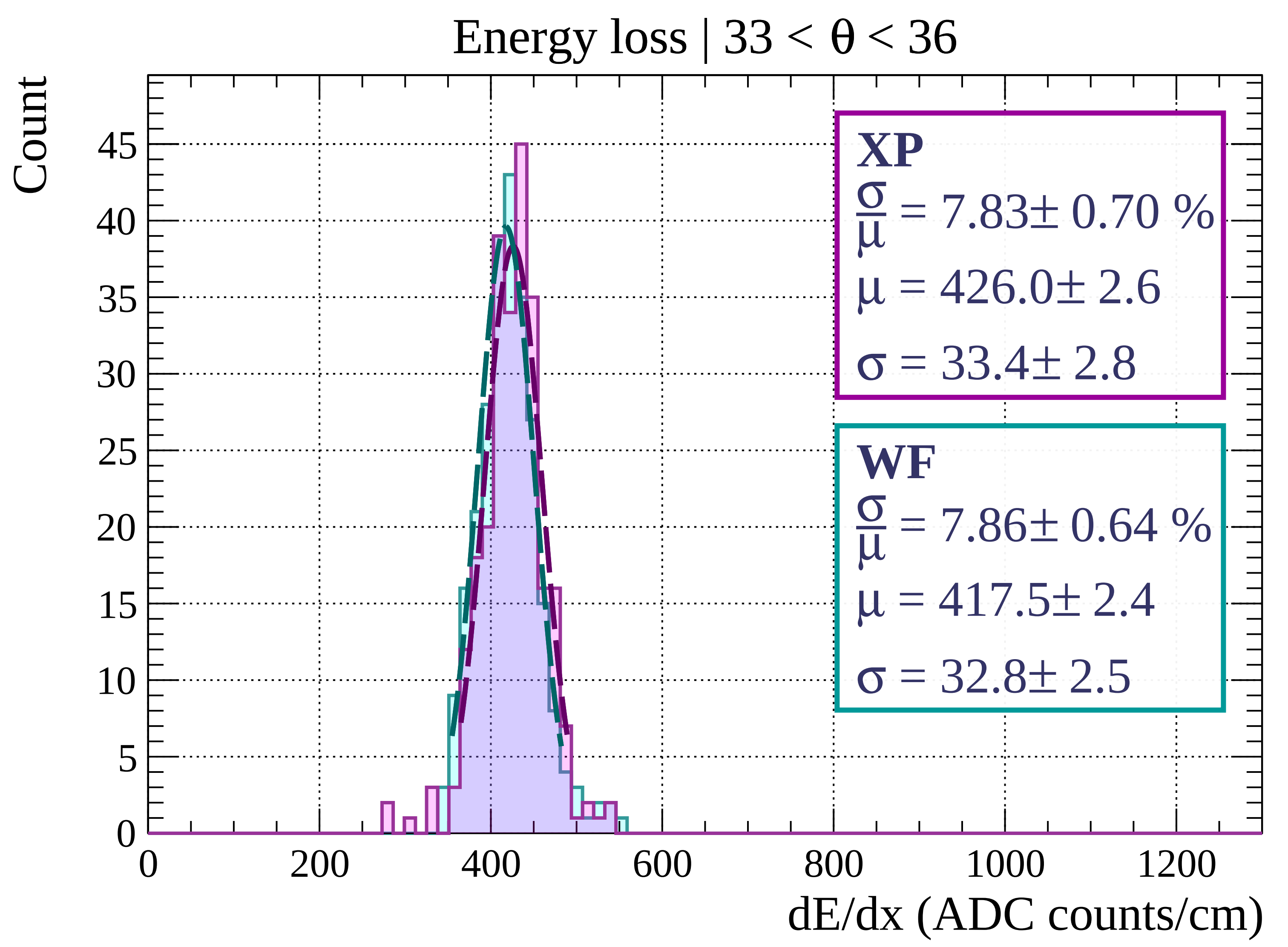
Energy loss | 24 < θ < 27
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
XP
$\frac{\sigma}{\mu}$ = 7.39± 0.57 %
μ = 420.2± 2.1
σ = 31.0± 2.2
WF
$\frac{\sigma}{\mu}$ = 7.93± 0.54 %
μ = 412.4± 2.1
σ = 32.7± 2.1

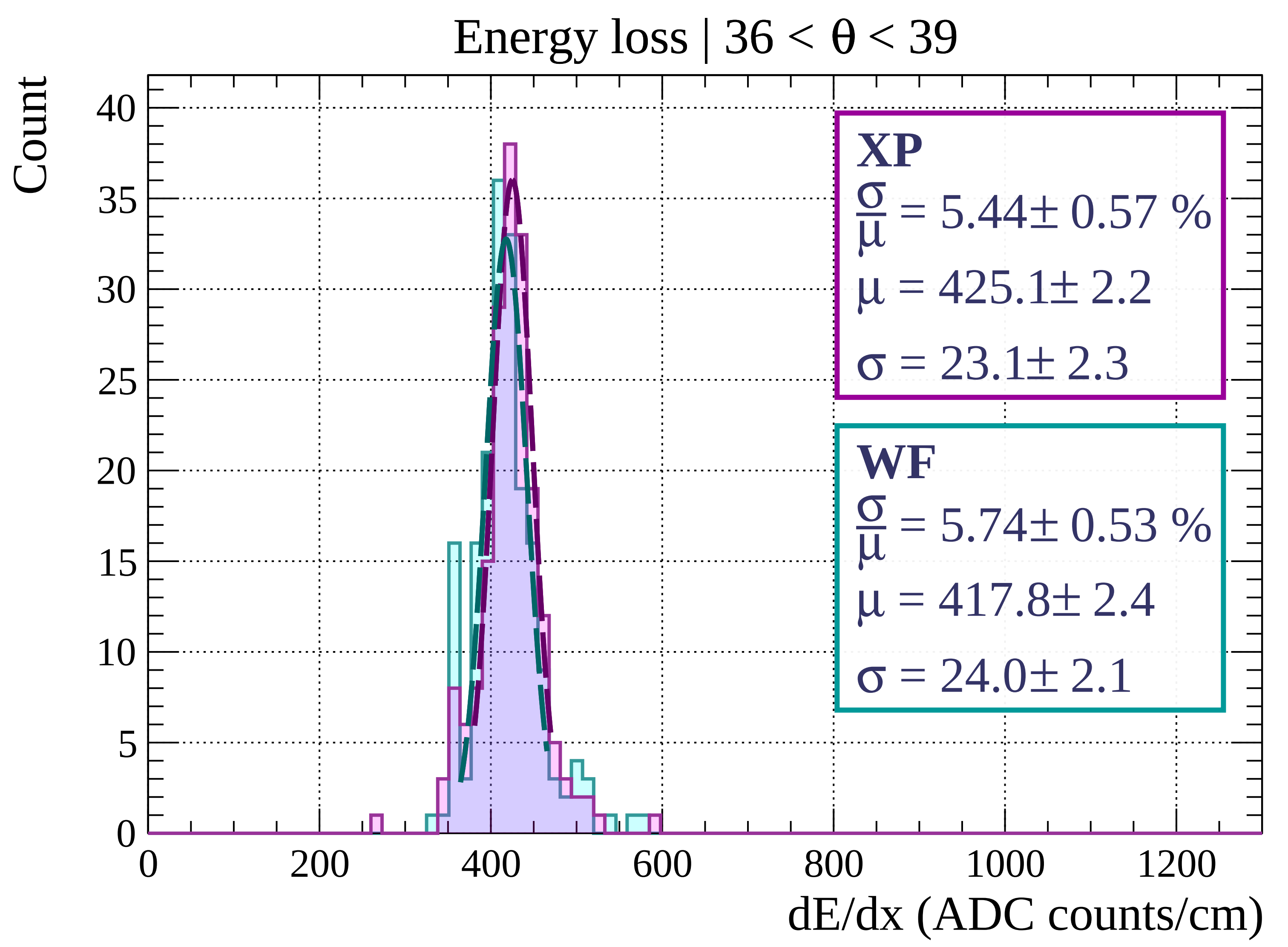


Energy loss | 27 < θ < 30
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
XP
σ/μ = 8.25± 0.49 %
μ = 423.2± 2.4
σ = 34.9± 1.9
WF
σ/μ = 8.99± 0.56 %
μ = 418.9± 2.4
σ = 37.7± 2.1

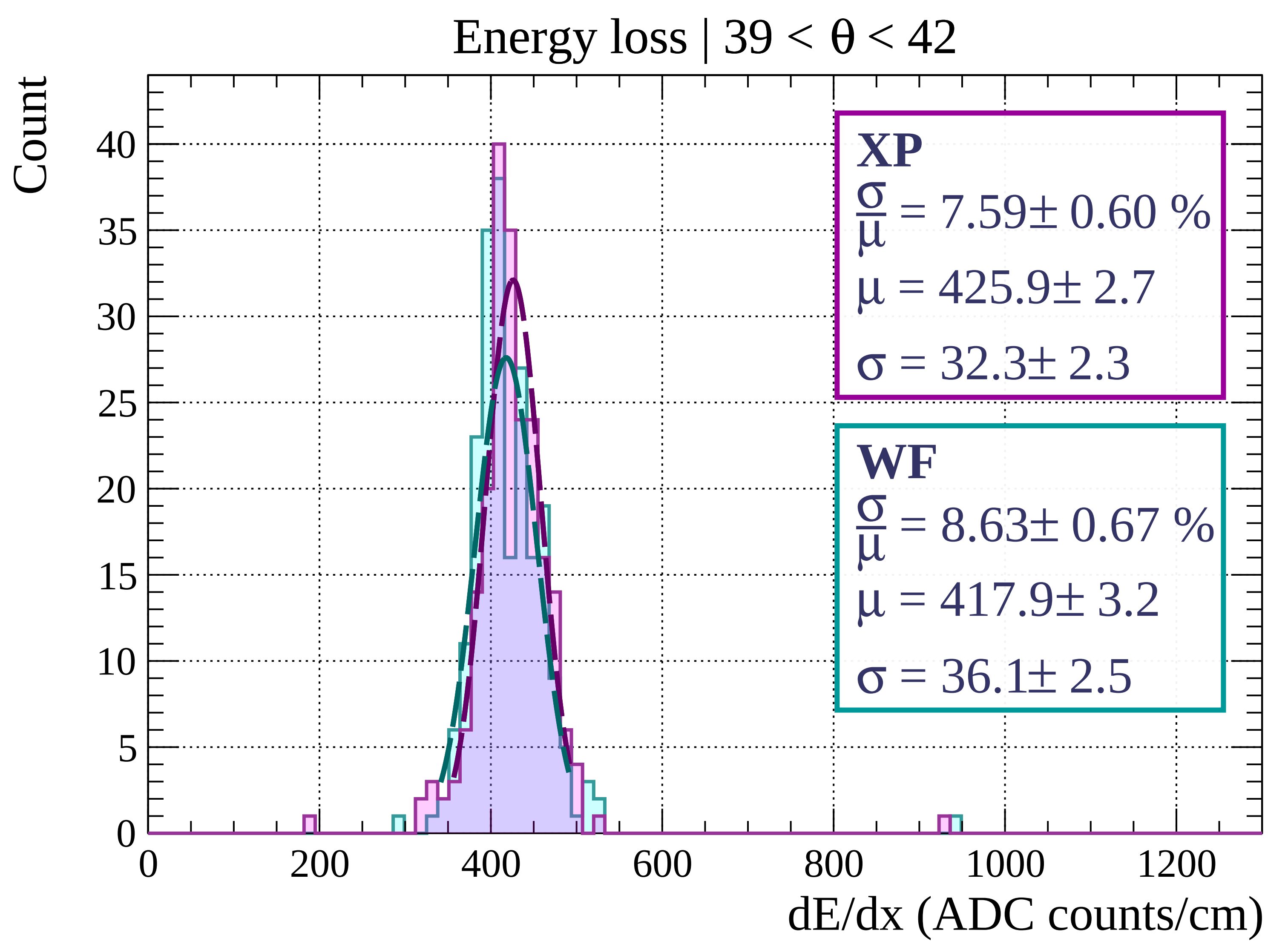


Energy loss | 30 < θ < 33
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
XP
σ/μ = 7.31± 0.47 %
μ = 428.1± 2.0
σ = 31.3± 1.9
WF
σ/μ = 7.08± 0.42 %
μ = 419.4± 1.9
σ = 29.7± 1.6

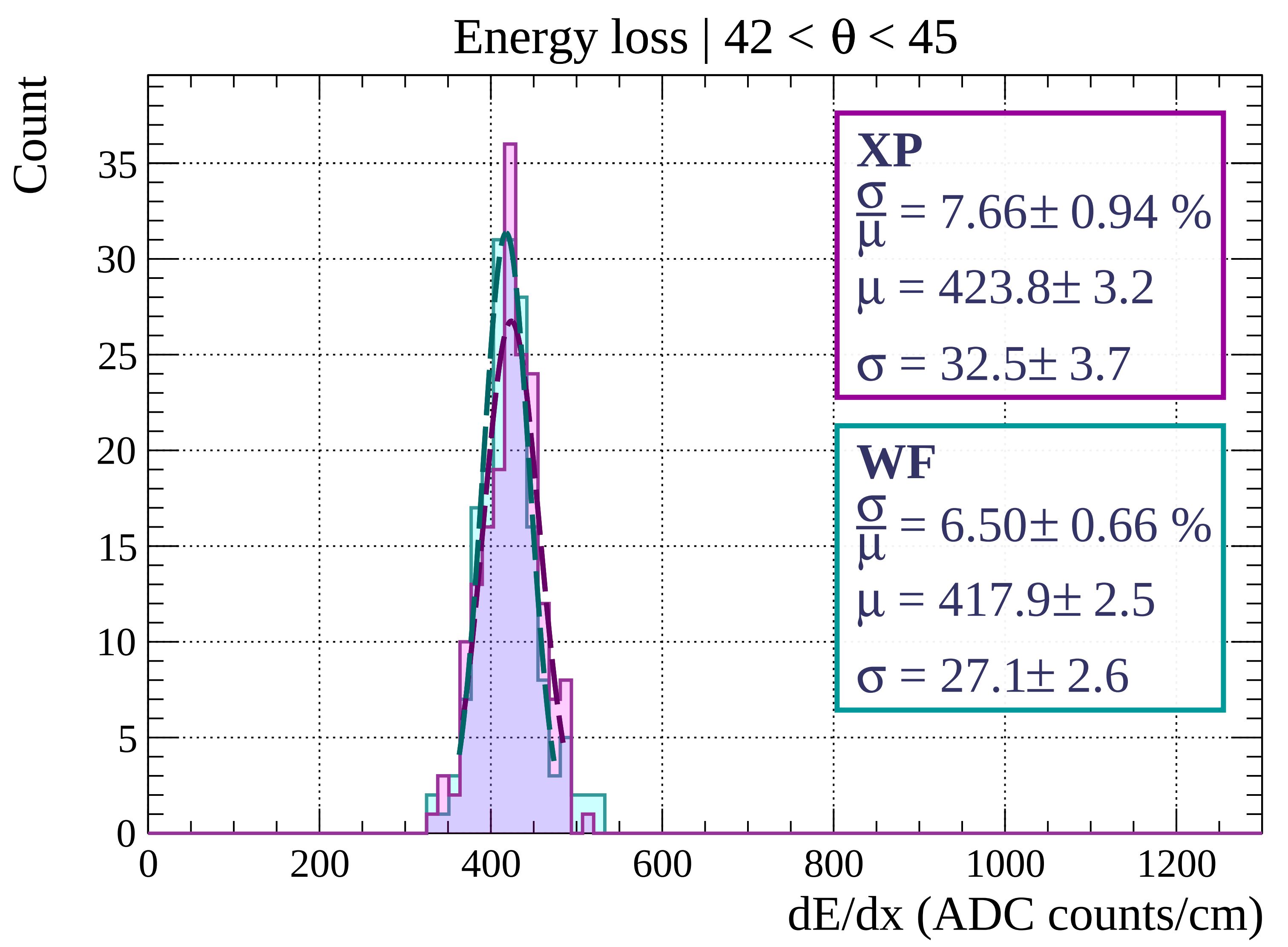

Energy loss | 33 < θ < 36
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
5
10
15
20
25
30
35
40
45
XP
$\frac{\sigma}{\mu}$ = 7.83± 0.70 %
μ = 426.0± 2.6
σ = 33.4± 2.8
WF
$\frac{\sigma}{\mu}$ = 7.86± 0.64 %
μ = 417.5± 2.4
σ = 32.8± 2.5



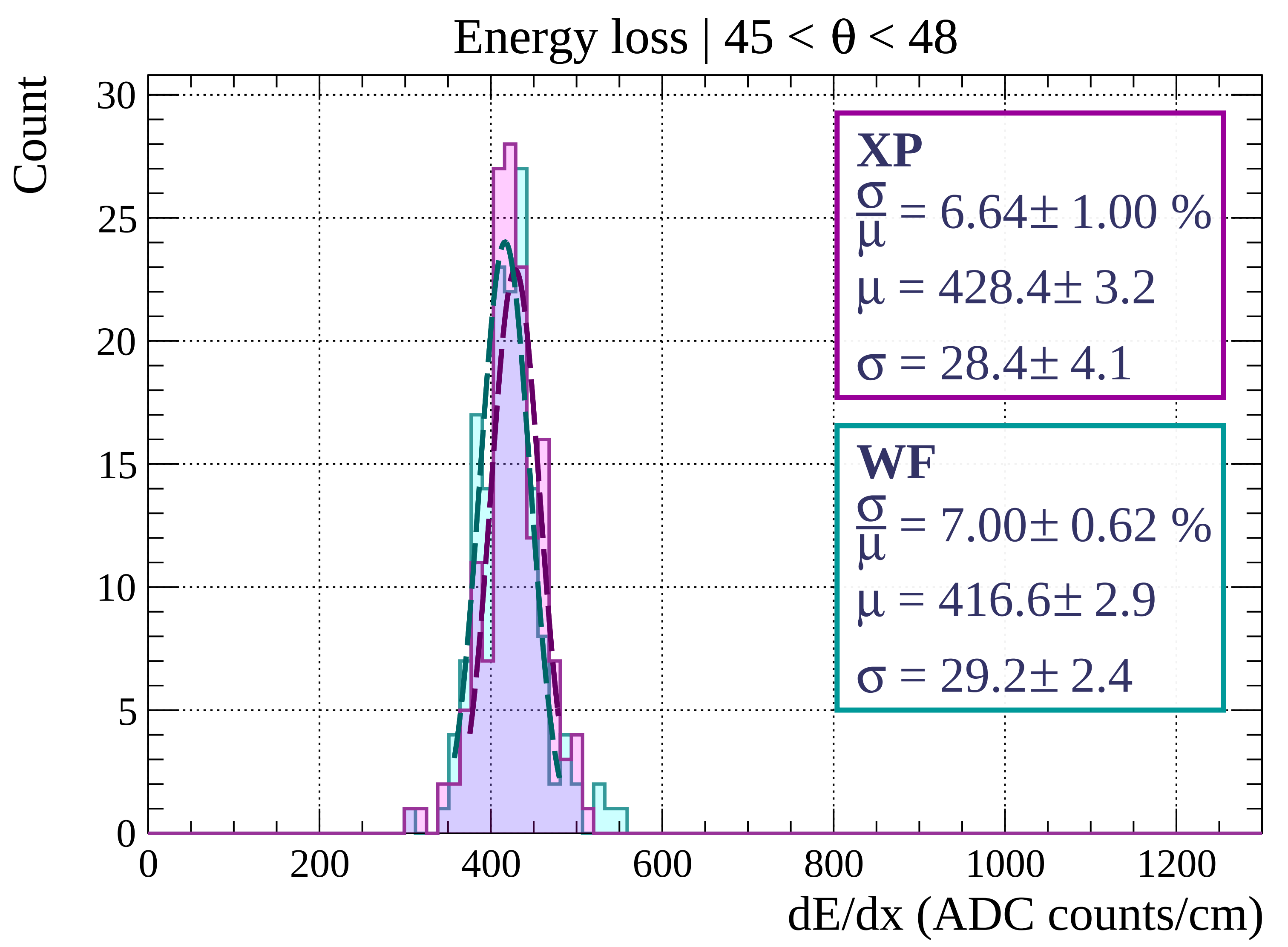
Energy loss | 36 < θ < 39
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
35
40
XP
σ/μ = 5.44± 0.57 %
μ = 425.1± 2.2
σ = 23.1± 2.3
WF
σ/μ = 5.74± 0.53 %
μ = 417.8± 2.4
σ = 24.0± 2.1



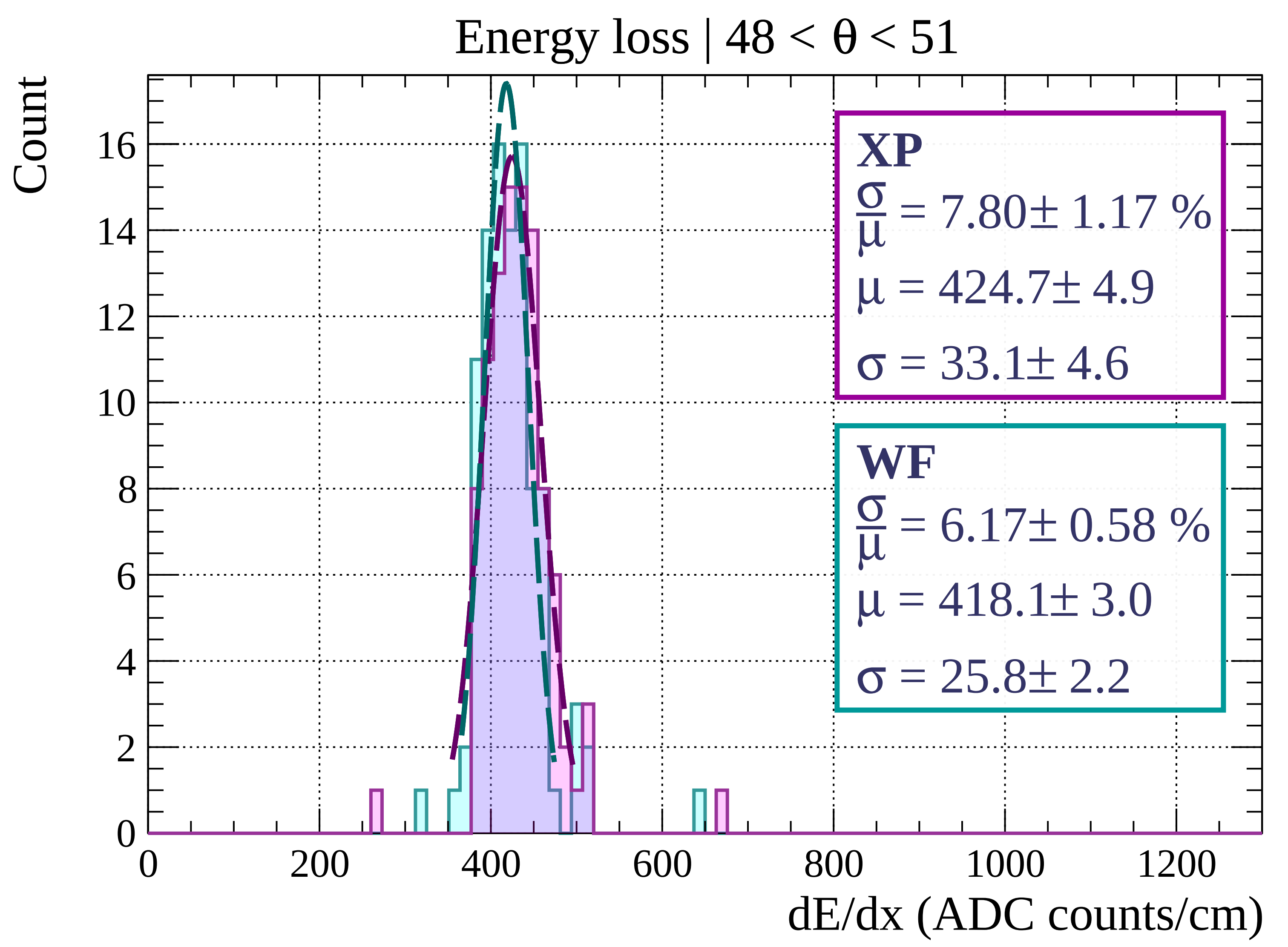
Energy loss | 39 < θ < 42
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
40
0
200
400
600
800
1000
1200
XP
σ/μ = 7.59± 0.60 %
μ = 425.9± 2.7
σ = 32.3± 2.3
WF
σ/μ = 8.63± 0.67 %
μ = 417.9± 3.2
σ = 36.1± 2.5



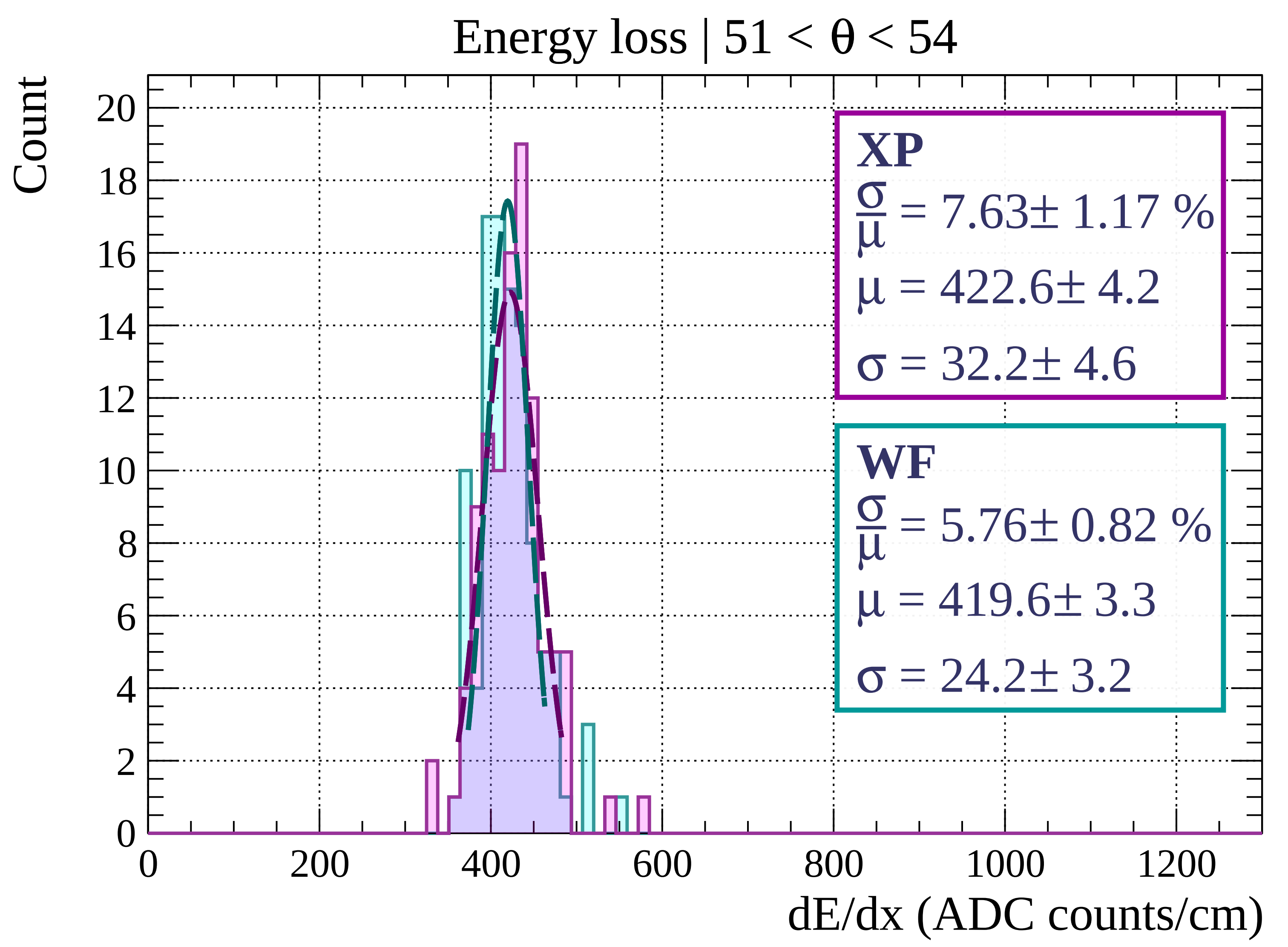
Energy loss | 42 < θ < 45
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200
XP
σ/μ = 7.66± 0.94 %
μ = 423.8± 3.2
σ = 32.5± 3.7
WF
σ/μ = 6.50± 0.66 %
μ = 417.9± 2.5
σ = 27.1± 2.6

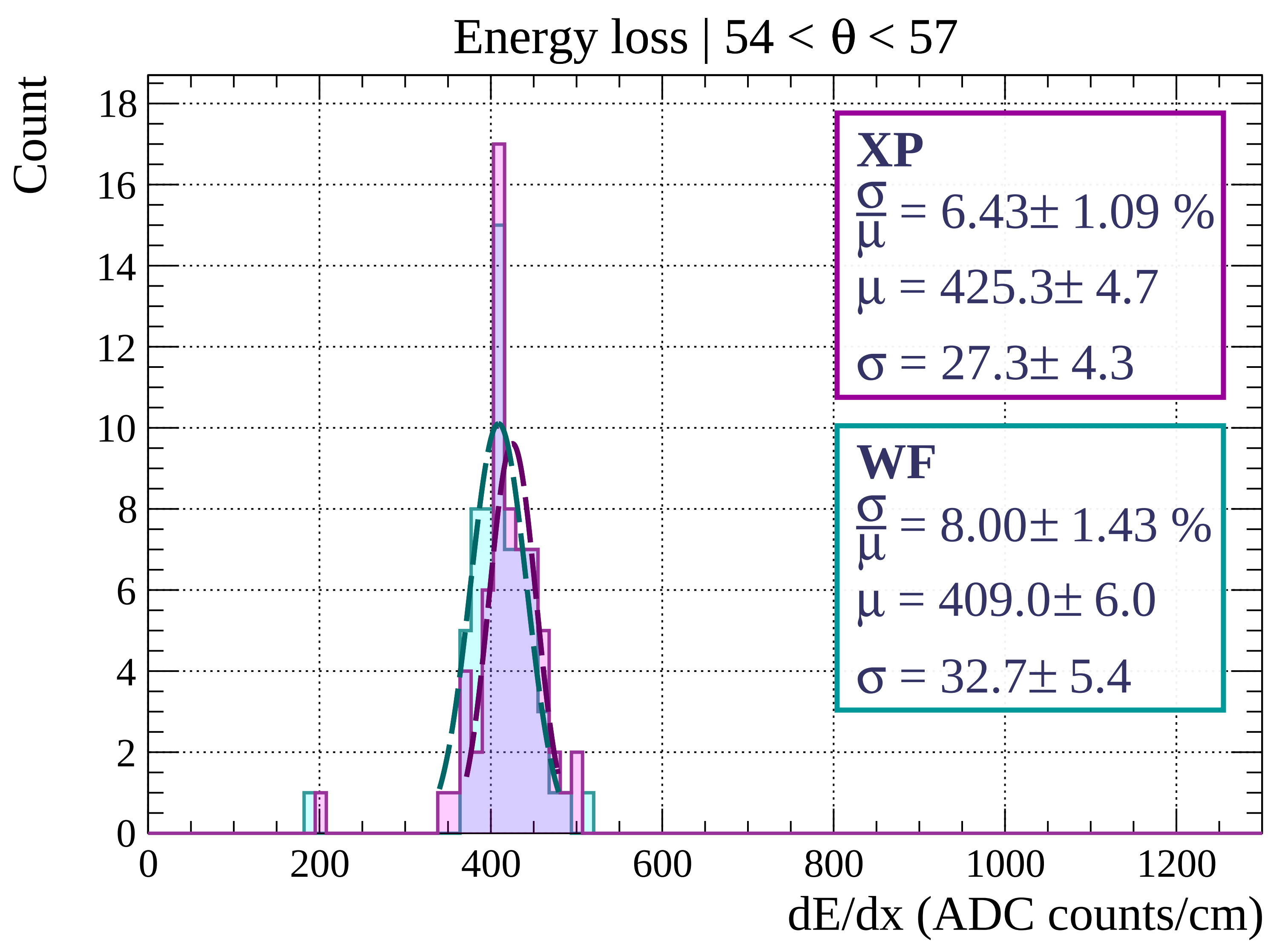


Energy loss | 45 < θ < 48
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200
XP
σ/μ = 6.64± 1.00 %
μ = 428.4± 3.2
σ = 28.4± 4.1
WF
σ/μ = 7.00± 0.62 %
μ = 416.6± 2.9
σ = 29.2± 2.4

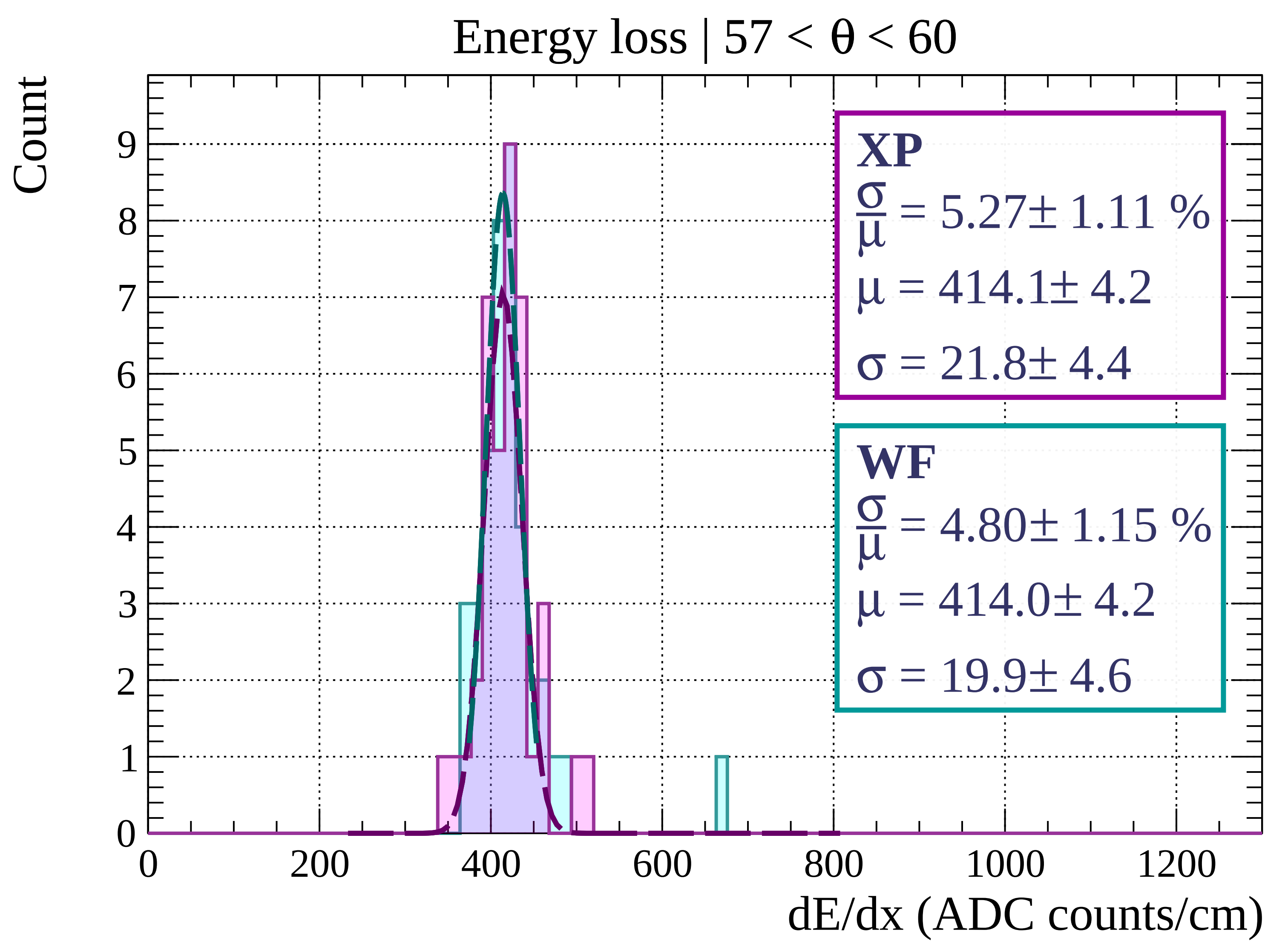


Energy loss | 48 < θ < 51
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
16
0
200
400
600
800
1000
1200
XP
σ/μ = 7.80± 1.17 %
μ = 424.7± 4.9
σ = 33.1± 4.6
WF
σ/μ = 6.17± 0.58 %
μ = 418.1± 3.0
σ = 25.8± 2.2

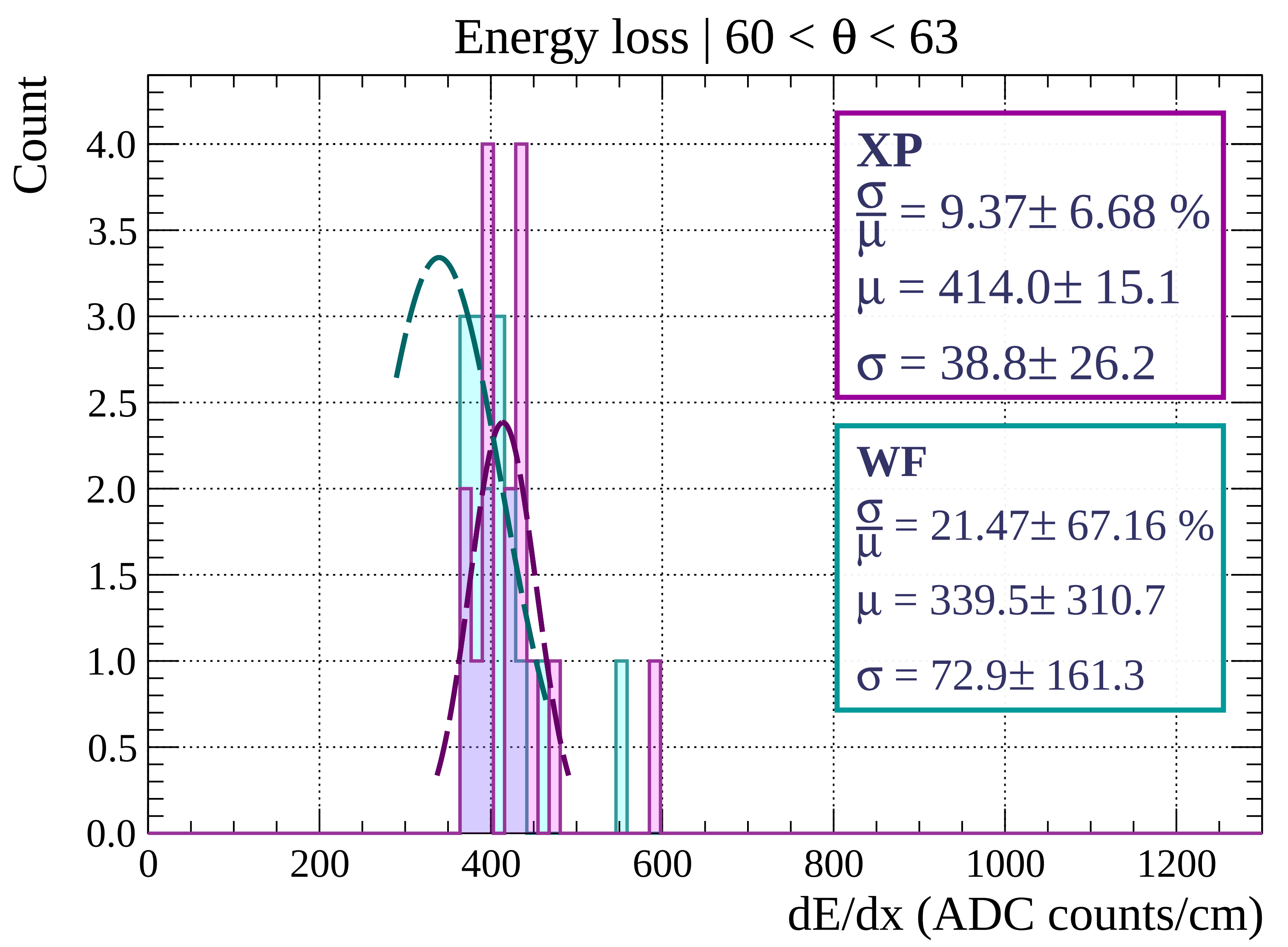

Energy loss | 51 < θ < 54
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
2
4
6
8
10
12
14
16
18
20
XP
σ/μ = 7.63± 1.17 %
μ = 422.6± 4.2
σ = 32.2± 4.6
WF
σ/μ = 5.76± 0.82 %
μ = 419.6± 3.3
σ = 24.2± 3.2



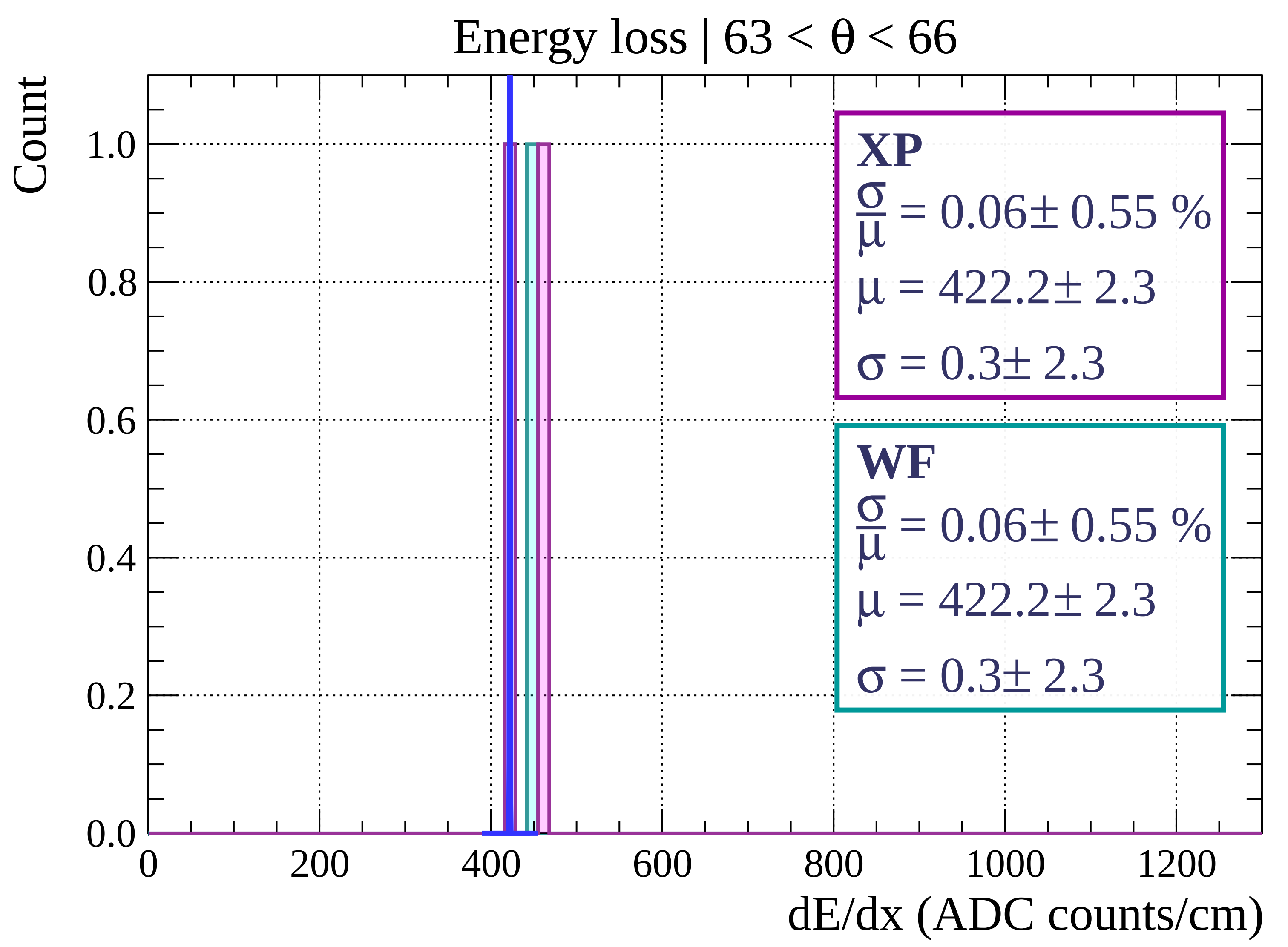
Energy loss | 54 < θ < 57
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
2
4
6
8
10
12
14
16
18
XP
$\frac{\sigma}{\mu}$ = 6.43± 1.09 %
μ = 425.3± 4.7
σ = 27.3± 4.3
WF
$\frac{\sigma}{\mu}$ = 8.00± 1.43 %
μ = 409.0± 6.0
σ = 32.7± 5.4



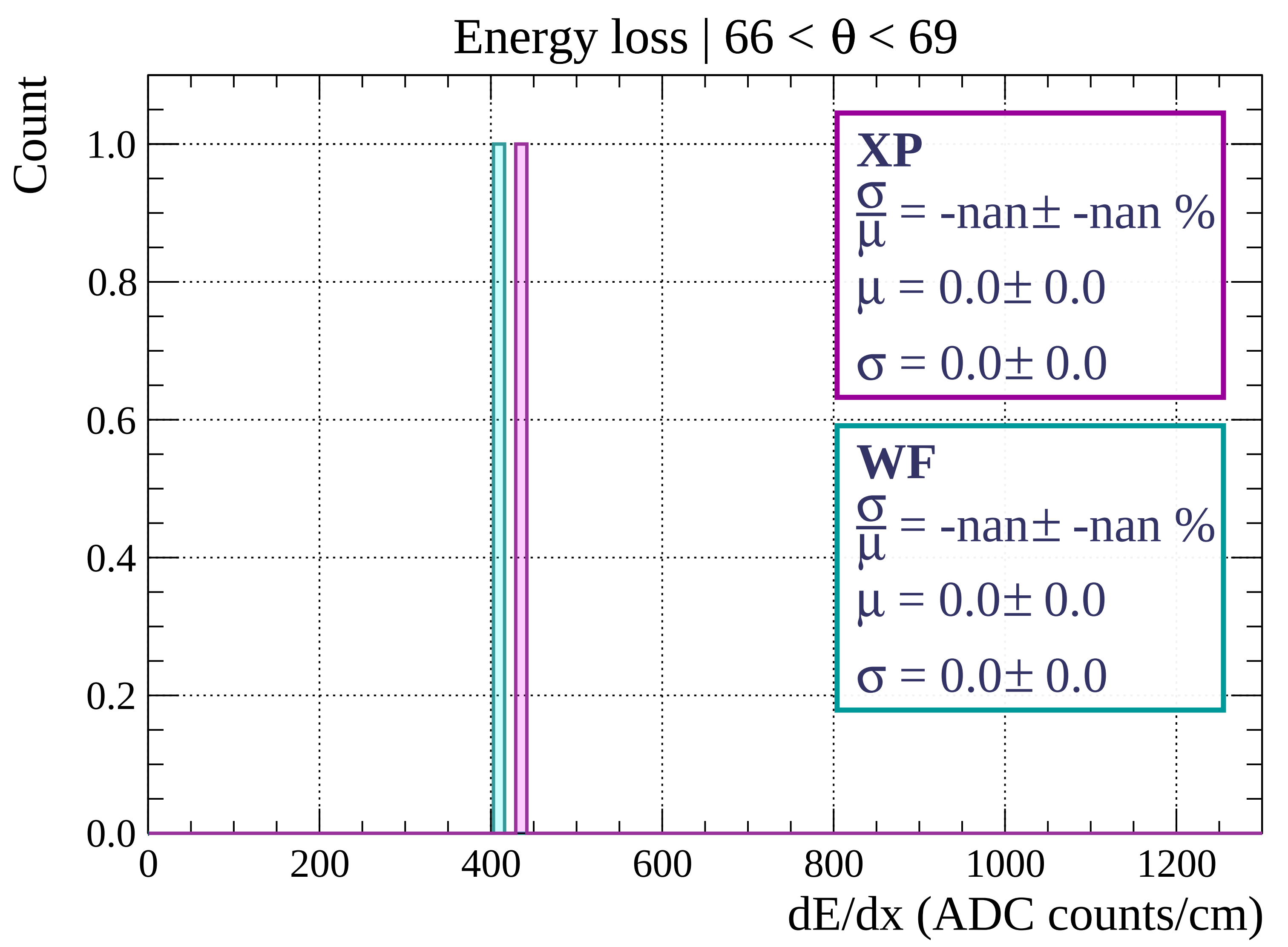
Energy loss | 57 < θ < 60
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 5.27± 1.11 %
μ = 414.1± 4.2
σ = 21.8± 4.4
WF
σ/μ = 4.80± 1.15 %
μ = 414.0± 4.2
σ = 19.9± 4.6

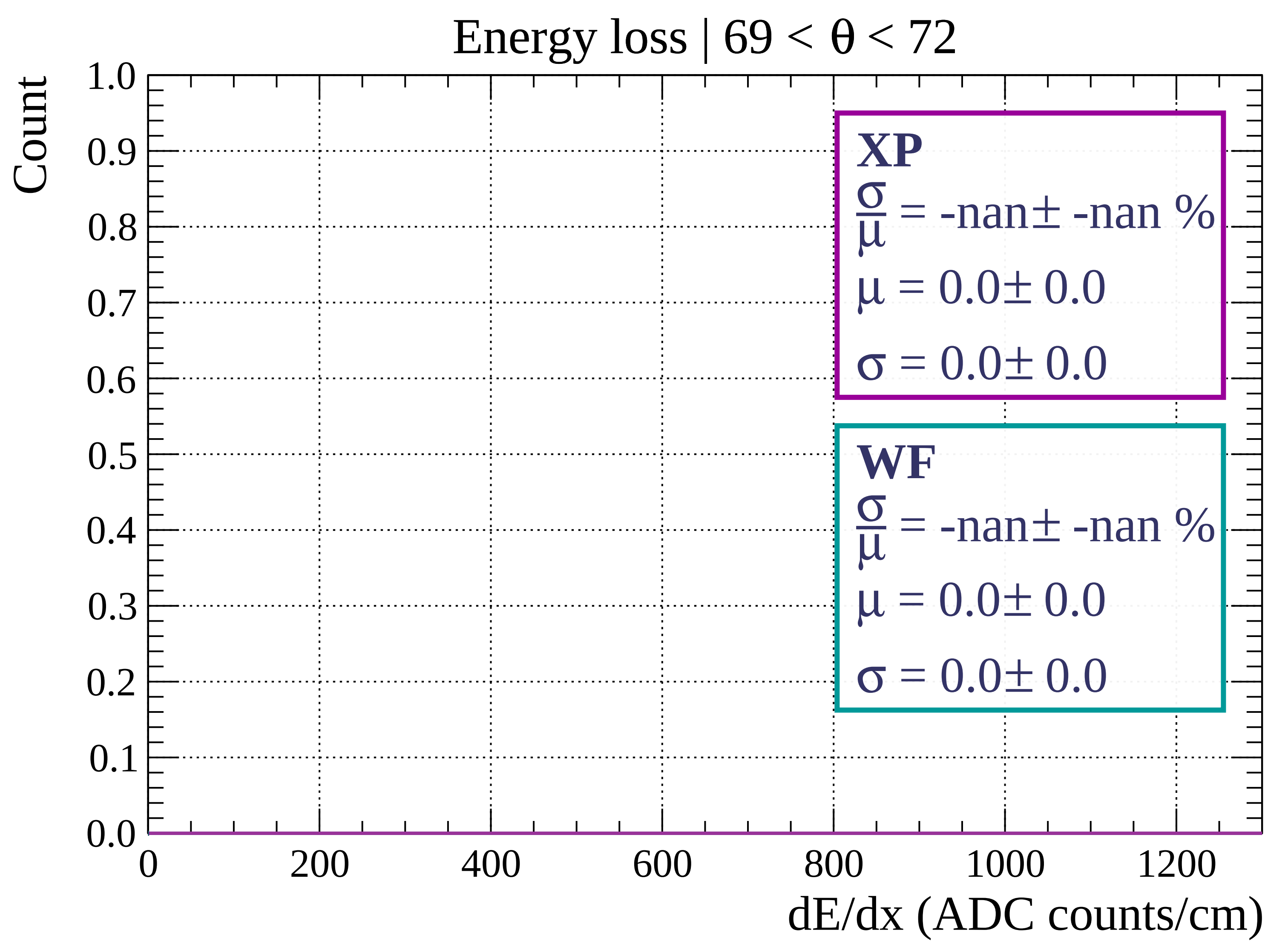


Energy loss | 60 < θ < 63
Count
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.37± 6.68 %
μ = 414.0± 15.1
σ = 38.8± 26.2
WF
σ/μ = 21.47± 67.16 %
μ = 339.5± 310.7
σ = 72.9± 161.3

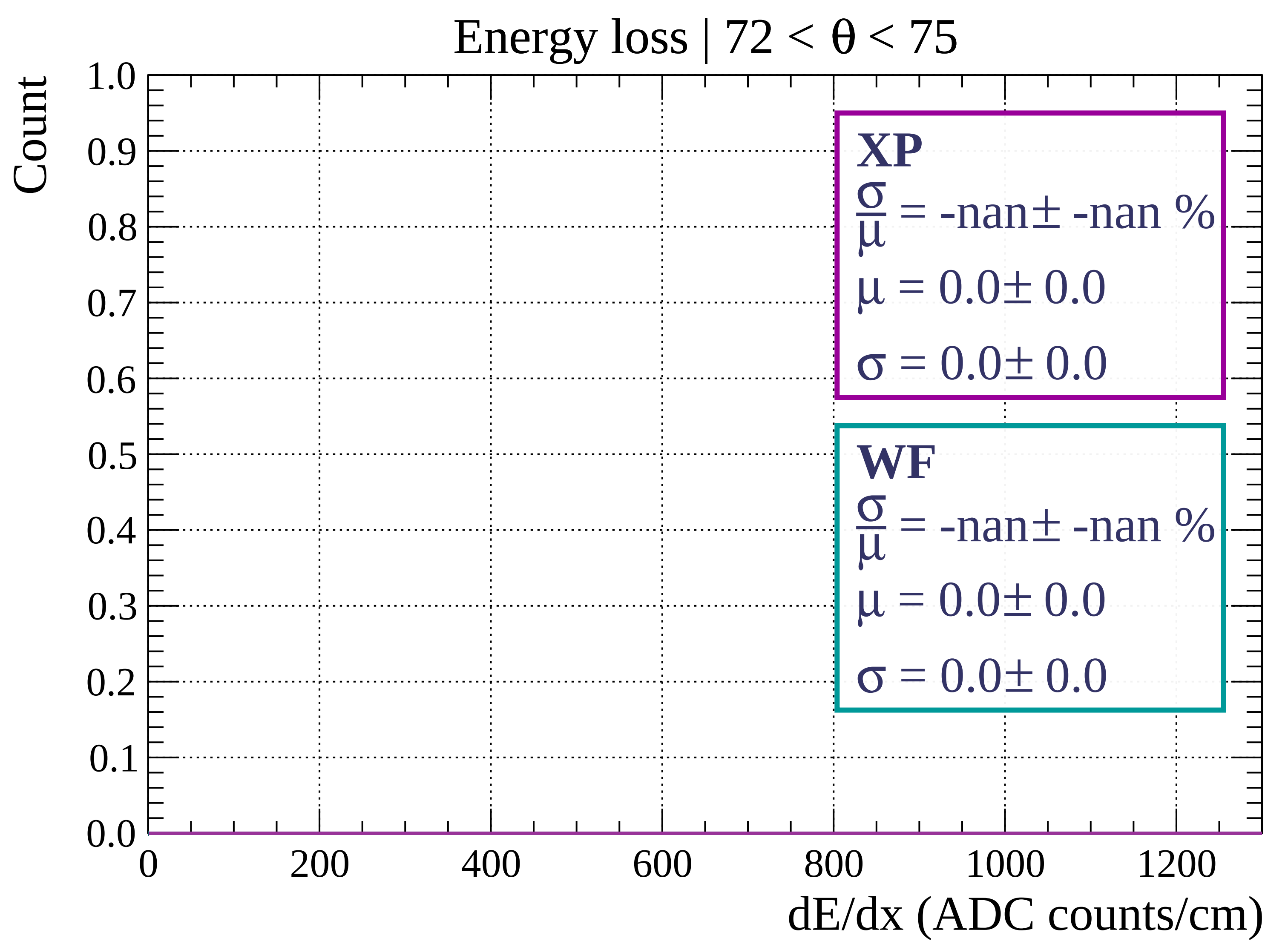


Energy loss | 63 < θ < 66
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 0.06± 0.55 %
μ = 422.2± 2.3
σ = 0.3± 2.3
WF
σ/μ = 0.06± 0.55 %
μ = 422.2± 2.3
σ = 0.3± 2.3

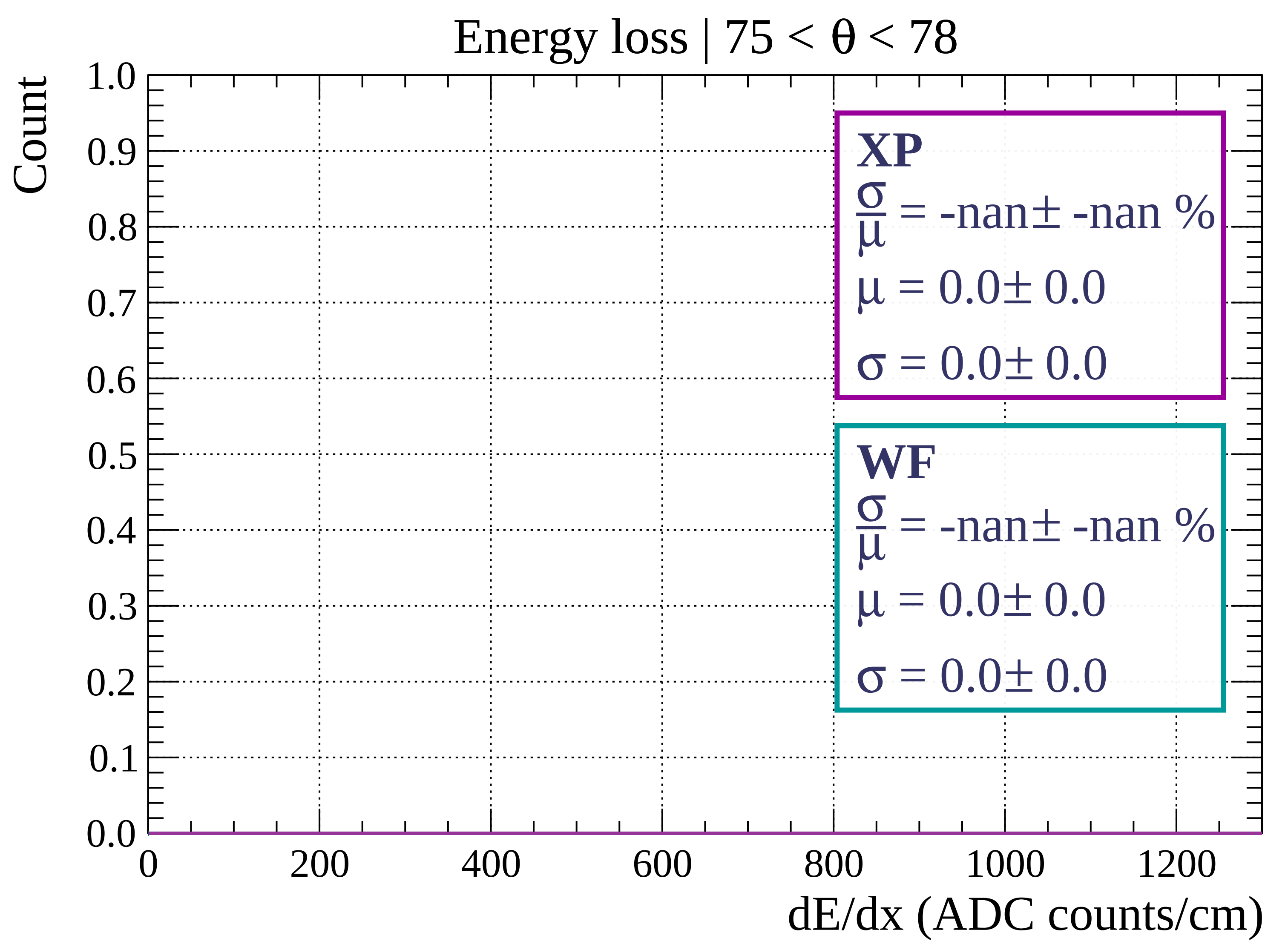

Energy loss | 66 < θ < 69
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

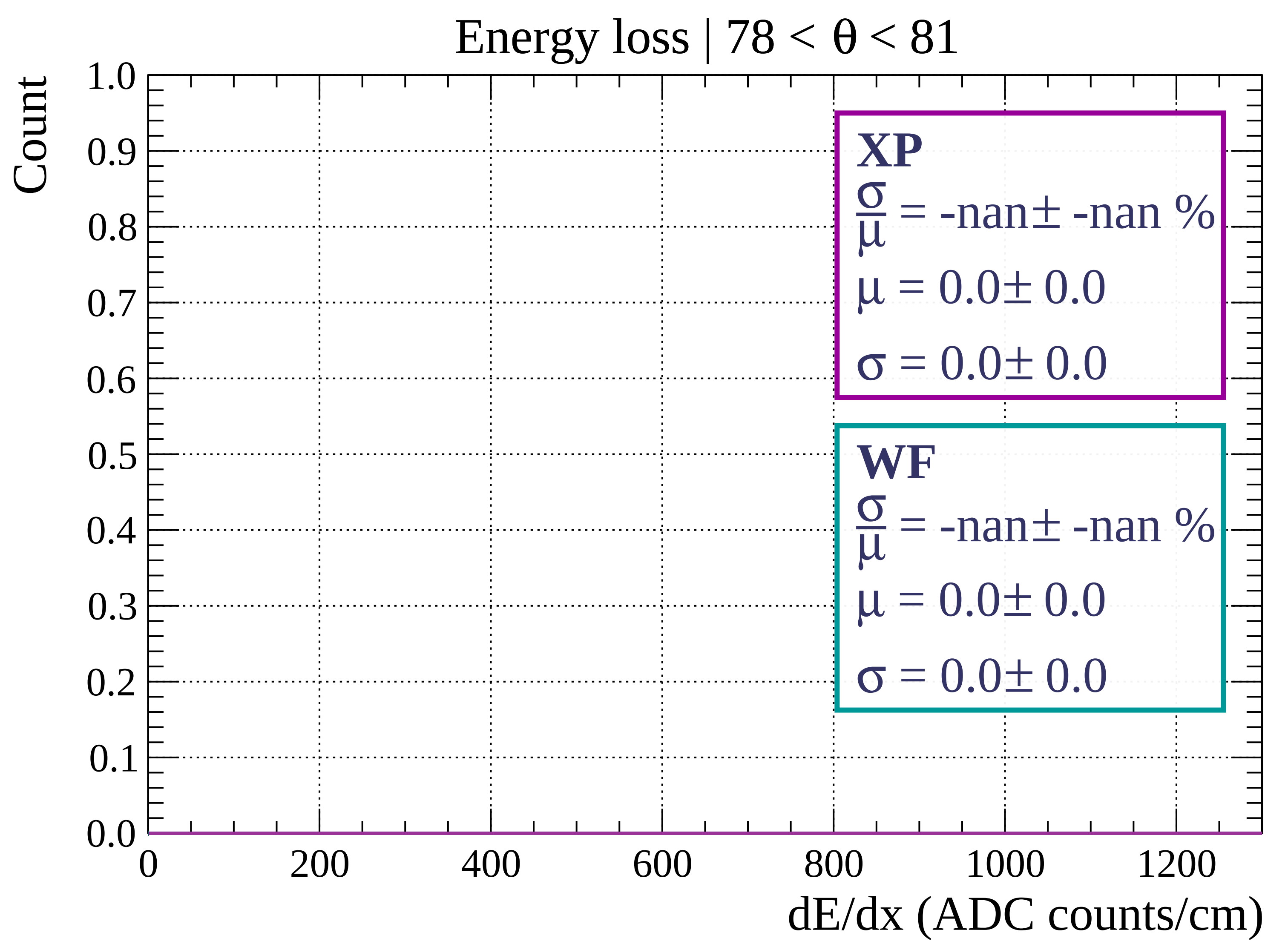


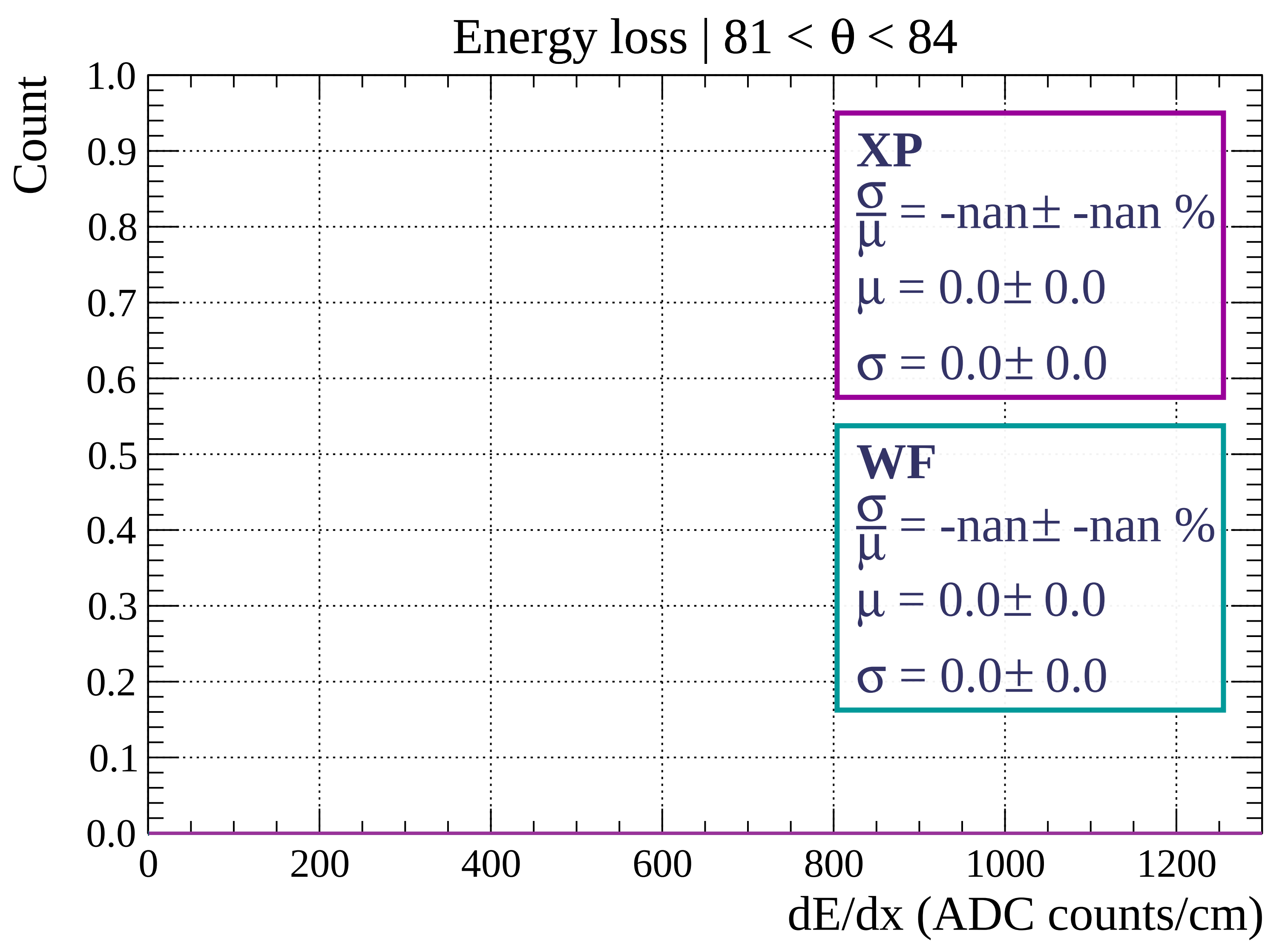
Energy loss | 69 < θ < 72
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

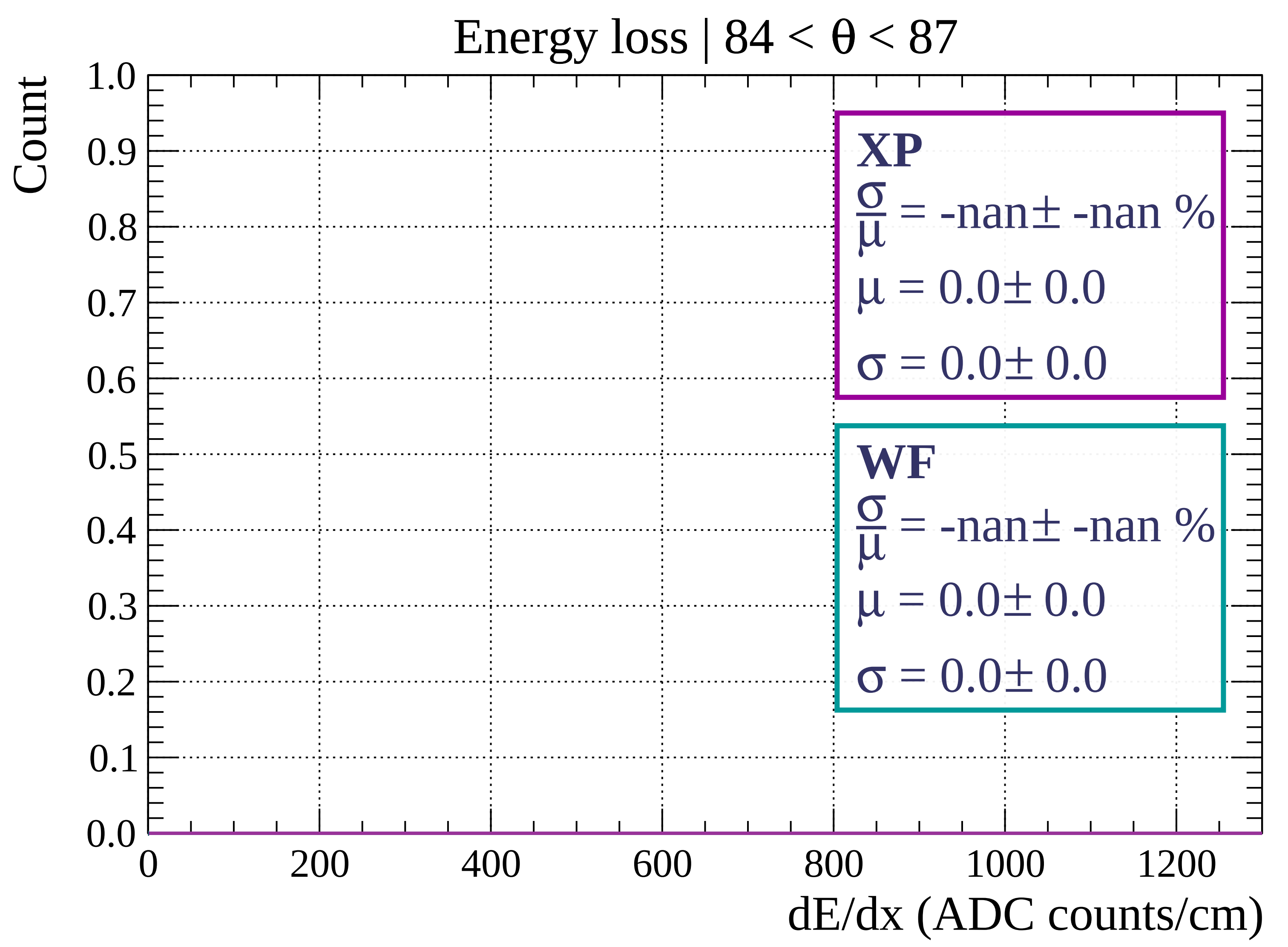


Energy loss | 72 < θ < 75
Count
dE/dx (ADC counts/cm)
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

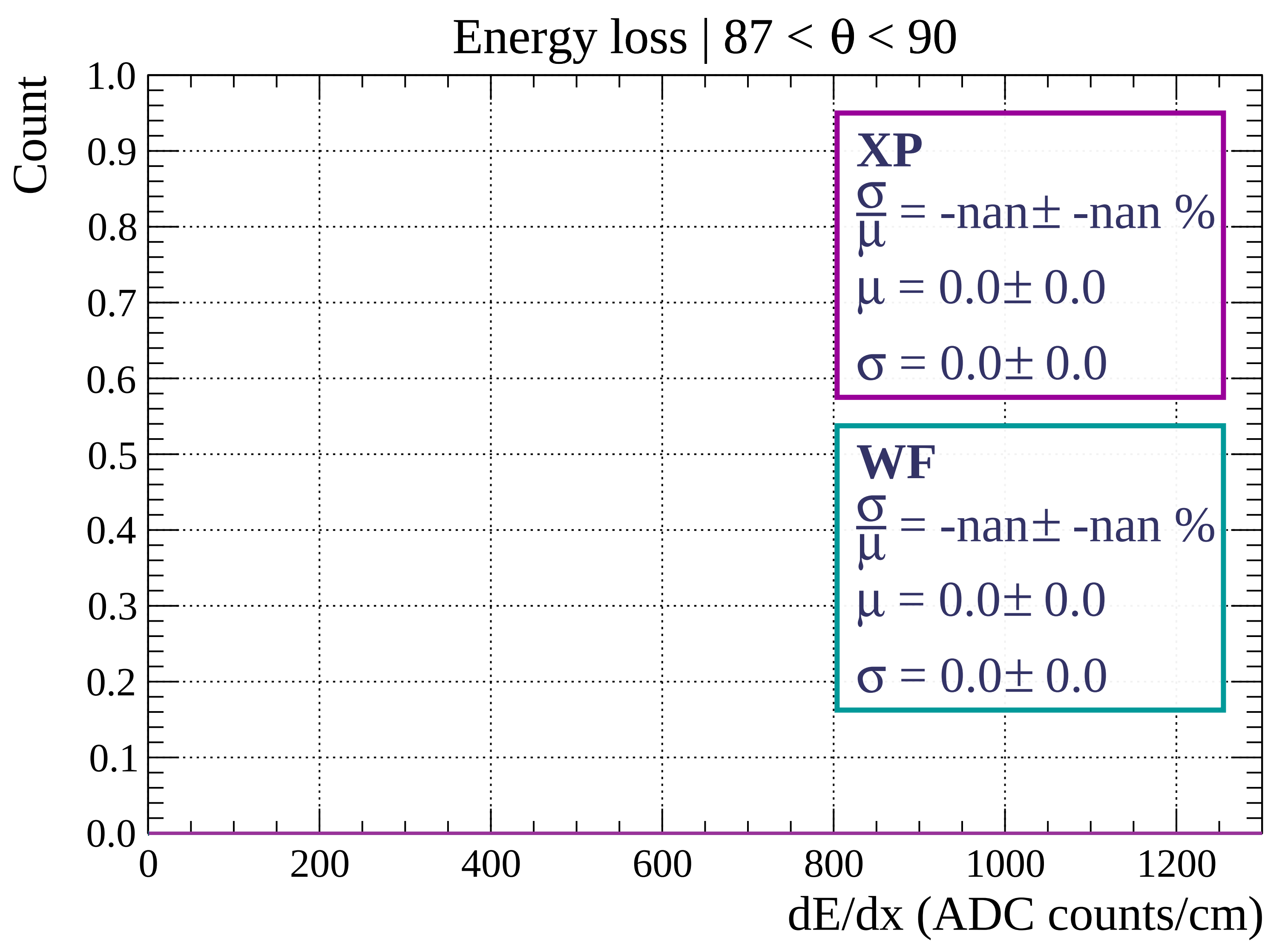

Energy loss | 75 < θ < 78
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

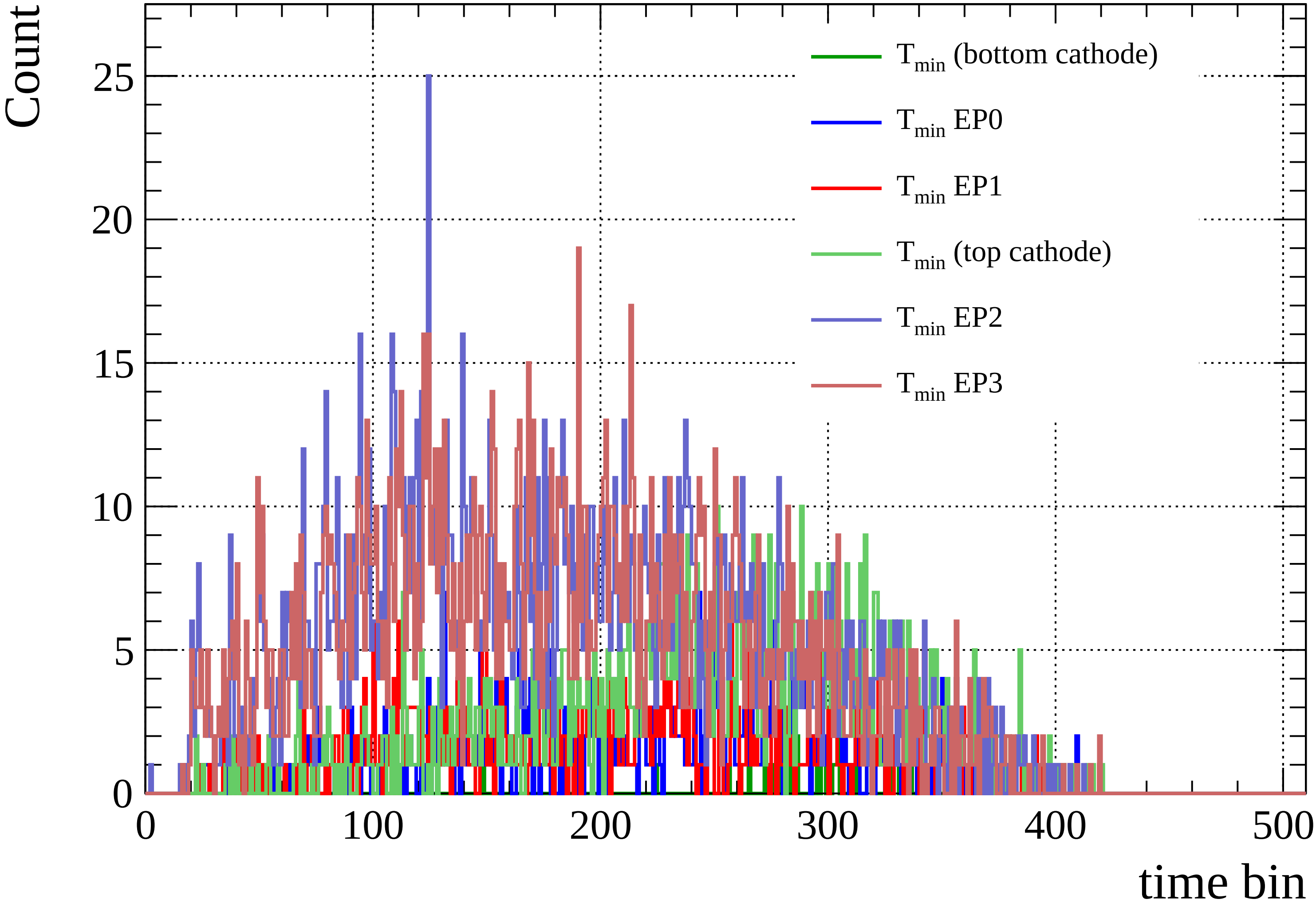

Energy loss | 78 < θ < 81
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 81 < θ < 84
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

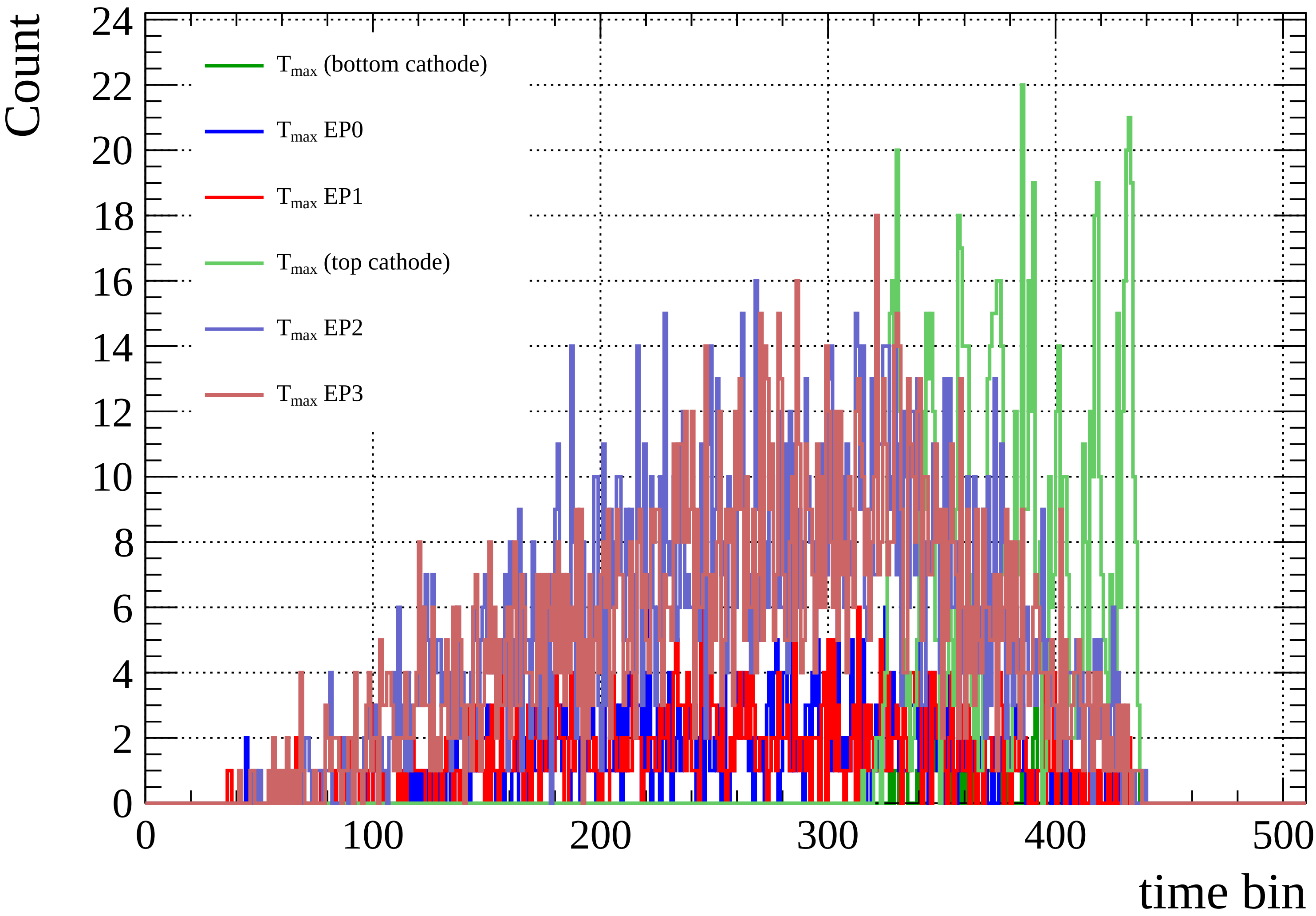


Energy loss | 84 < θ < 87
Count
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 87 < θ < 90
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



Count
25
20
15
10
5
0
0
100
200
300
400
500
time bin
T_{min} (bottom cathode)
T_{min} EP0
T_{min} EP1
T_{min} (top cathode)
T_{min} EP2
T_{min} EP3


Count
24
22
20
18
16
14
12
10
8
6
4
2
0
T_{max} (bottom cathode)
T_{max} EP0
T_{max} EP1
T_{max} (top cathode)
T_{max} EP2
T_{max} EP3
0
100
200
300
400
500
time bin